



TECHNICAL REPORT ON THE MOUNT MILLIGAN MINE NORTH-CENTRAL BRITISH COLUMBIA

CENTERRA GOLD INC.

Technical Report pursuant to NI 43-101

Qualified Persons:

Bruno Borntraeger, B.A.Sc. Geological Engineering, P. Eng.

Gordon Zurowski, P. Eng. PEO

Cheyenne Sica, P.Geo., EGBC

Tengfei Yue, P.Eng.

Curtis Clarke, MMSAQP Mining Engineer

David Luzi, Ph.D., P. Geo.

Brian Thomas, P.Geo.

Jennifer Simper, P.Geo.

Date: November 7, 2022

Effective Date: December 31, 2021

CERTIFICATE OF QUALIFIED PERSON - BRUNO BORNTRAEGER

I, Bruno Borntraeger, state that:

I am a Specialist Geotechnical Engineer | Associate at:
Knight Piesold Ltd.
Suite 1400 750 West Pender St.
Vancouver, BC., Canada, V6C 2T8

This certificate applies to the technical report titled Technical Report on the Mount Milligan Mine with an effective date of: December 31, 2021.

I am a “qualified person” for the purposes of National Instrument 43-101 (the “Instrument”). My qualifications as a qualified person are as follows. I am a graduate from the University of British Columbia with a Bachelor of Applied Science in Geological Engineering, 1990. I am a member in good standing of the Engineers and Geoscientists of British Columbia (Licence #20926). My relevant experience after graduation for the purpose of the Technical Report includes 30 years geotechnical engineering, mine waste and water management, environmental compliance, mine development with practical experience in feasibility studies, detailed engineering, permitting, construction, operations, and closure.

My most recent personal inspection of each property described in the Technical Report occurred in September 2022 and was for a duration of three days.

I am responsible for Items 1.14 and 18.2 of the Technical Report.

I am independent of the issuer as described in Section 1.5 of the Instrument.

My prior involvement with the property that is the subject of the Technical Report is as follows. Tailings Dam Engineer of Record (2010-present).

I have read National Instrument 43-101. The part of the Technical Report for which I am responsible has been prepared in compliance with this Instrument; and

At the effective date of the Technical Report, to the best of my knowledge, information, and belief, the part of the Technical Report for which I am responsible, contains all scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Dated at Vancouver, BC, this 7th of November 2022.

(original signed and sealed)

Bruno Borntraeger

Bruno Borntraeger, P. Eng. (BC, 20926)

CERTIFICATE OF QUALIFIED PERSON - GORDON ZUROWSKI

To accompany the technical report titled "Technical Report on the Mount Milligan Mine, North-Central British Columbia" (the "Technical Report") for Centerra Gold Inc. dated 7 November 2022, with an effective date of 31 December 2021.

I, Gordon Zurowski, P.Eng., do hereby certify that:

I am a Principal Mining Engineer with AGP Mining Consultants Inc., with a business address at #246-132K Commerce Park Drive, Barrie ON, L4N 0Z7 Canada.

I am a graduate of the University of Saskatchewan with a B.Sc. Geological Engineering degree from 1989.

I am a member in good standing of the Professional Engineers Ontario (PEO) in Canada, membership #100077750.

I have practiced my profession in the mining industry continuously since graduation. My relevant experience includes over 30 years in Canada, the United States of America, Central and South America, Europe, Asia, Africa, and Australia.

I have read the definition of "qualified person" set out in National Instrument 43-101 Standards of Disclosure for Mineral Projects ("NI 43-101") and certify that by virtue of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I fulfill the requirements to be a "qualified person" for the purposes of NI 43-101.

I am independent of the issuer, Centerra Gold Inc., as defined in Section 1.5 of NI 43-101.

I am responsible for Sections 1.11, 1.12, 1.20, 1.21, 15, 16, 25.2, and 26.4 of the Technical Report and accept professional responsibility for those sections of the Technical Report.

I have had no previous involvement with the Mount Milligan Project.

My most recent site visit to the Mount Milligan Mine Project was on 20 April 2022 for one day.

As of the effective date of the Technical Report, to the best of my knowledge, information and belief, the portions of the Technical Report for which I am responsible contain all scientific and technical information that is required to be disclosed to make the portions of the Technical Report for which I am responsible not misleading.

I have read NI 43-101 and Form 43-101F1, and the Technical Report has been prepared in accordance with NI 43-101 and Form 43-101F1.

Dated this 7th day of November 2022, in Stouffville, Ontario, Canada.

"signed electronically"

Gordon Zurowski, P.Eng.

EGBC Permit to Practice: 1003765

CERTIFICATE OF QUALIFIED PERSON - CHEYENNE SICA

I, Cheyenne Sica, P. Geo., as an author of this report entitled “Technical Report on the Mount Milligan Mine” prepared for Centerra Gold Inc. with an effective date of December 31, 2021, do hereby certify that:

I am currently employed as Exploration Manager by Centerra Gold Services Inc., 299 Victoria Street, Suite 200, Prince George, BC, V2L 5B8.

I am a Registered Professional Geologist with Engineers & Geoscientists British Columbia. Registration # 230355.

I am a Registered Professional Geologist with the Association of Professional Geoscientists of Ontario. Registration # 3005.

I have worked as an Exploration Geologist in industry for 12 years since graduation from university, and as an Exploration Manager for over a year.

I graduated from the University of Toronto with a M.Sc. of Earth Sciences in 2016; and graduated from Queen’s University with a Bachelor of Science Honours, Geological Sciences in 2010.

I have read the definition of “qualified person” as set out in the National Instrument 43-101 (the “Instrument”) and certify that because of my education, association with a professional association (as defined in the Instrument) and past relevant work experience, I fulfill the requirements to be a “qualified person” for the purposes of the Instrument.

I most recently visited the Mount Milligan mine from September 7-13th, 2022. I have visited the property on a regular basis during exploration programs since my employment with Centerra Gold Inc. starting in 2018.

I am responsible for Items 1.3-1.7, 1.20, 4.1, 6-11 and 27.1.

I am not independent of the issuer as described in Section 1.5 of the Instrument.

I have read National Instrument 43-101 guidelines and the parts of the Technical Report for which I am responsible have been prepared in compliance with the Instrument;

At the effective date of the Technical Report, to the best of my knowledge, information and belief, the parts of the Technical Report for which I am responsible contains all scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Dated at Victoria, B.C., this 7th day of November 2022

(original signed and sealed) Cheyenne Sica

Cheyenne Sica, P. Geo.

CERTIFICATE OF QUALIFIED PERSON - TENGFEI YUE

I, Tengfei Yue, state that:

I am the Director, Metallurgy at:
Centerra Gold Inc.
1 University Avenue
Toronto, ON, M5J 2P1

This certificate applies to the technical report titled "Technical Report on the Mount Milligan Mine" with an effective date of: December 31, 2021.

I am a "qualified person" for the purposes of National Instrument 43-101 (the "Instrument"). My qualifications as a qualified person are as follows.

I am a graduate of the University of British Columbia with a Bachelor of Applied Science in Mining Engineering (2012 graduation) and an active, practicing Professional Engineer registered, and in good standing, under Engineers and Geoscientists of British Columbia (EGBC). My relevant experience includes on site process optimization via mathematical modelling, bench scale and pilot scale testing. My last 10 years of experience also includes working in the process plant as a metallurgist, progressively with increasing responsibilities.

I most recently visited Mt Milligan Mine from Sep 28th to Oct 4th of 2022.

I am responsible for Items 1.9, 1.13, 1.20, 1.21, 13, 17, 25.3, and 26.5 of the Technical Report.

I am not independent of the issuer as described in Section 1.5 of the Instrument.

My prior involvement with the property that is the subject of the Technical Report is as follows. I was the Superintendent, Metallurgy at Mount Milligan from May 2020 to end of 2021.

I have read National Instrument 43-101. The parts of the Technical Report for which I am responsible have been prepared in compliance with this Instrument; and

At the effective date of the Technical Report, to the best of my knowledge, information, and belief, the parts of Technical Report for which I am responsible, contains all scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Dated at Burnaby, British Columbia, this 7th of November 2022.

(original signed and sealed) Tengfei Yue

Tengfei Yue, P.Eng. (48733)

CERTIFICATE OF QUALIFIED PERSON – DAVID LUZI

I, David Luzi, state that:

I am a Senior Principal at:
Stantec Consulting Ltd.
500-4730 Kingsway,
Burnaby, BC, V5H 0C6

This certificate applies to the technical report titled Technical Report on the Mount Milligan Mine, North-Central British Columbia, Centerra Gold Inc., Technical Report pursuant to NI 43-101 (the "Technical Report"), November 7, 2022.

I am a "qualified person" for the purposes of National Instrument 43-101 (the "Instrument"). My qualifications as a qualified person are as follows. I am a graduate of University of British Columbia with a M.Sc (2000) and a Ph.D. (2014) in Vancouver, British Columbia, Canada. I am a Professional Geoscientist with Engineers and Geoscientists of British Columbia, license 45648. My academic and professional experience includes over 20 years of hydrology and water management experience that is relevant for the purpose of the Technical Report.

I have frequently visited the Mt. Milligan Mine since 2017 and have been to site at least once a year since 2017. My most recent site visit occurred in August 2022.

I am responsible for Item 18.3 of the Technical Report.

I am independent of the issuer as described in section 1.5 of the Instrument.

I have read National Instrument 43-101. The part of the Technical Report for which I am responsible has been prepared in compliance with this Instrument; and

At the effective date of the Technical Report, to the best of my knowledge, information, and belief, the part of Technical Report for which I am responsible, contains all scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Dated at Vancouver, British Columbia this 25th day of October 2022.

(Original signed and sealed) David Luzi

David Luzi EGBC License Number 45648

CERTIFICATE OF QUALIFIED PERSON - BRIAN THOMAS

I, Brian Thomas, state that:

I am a Principal Resource Geologist at:
WSP Golder
33 McKenzie Street
Sudbury, ON

This certificate applies to the technical report titled be "Technical Report on the Mount Milligan Mine" with an effective date of December 31, 2021.

I am a "qualified person" for the purposes of National Instrument 43-101 (the "Instrument"). My qualifications as a qualified person are as follows:

I am a graduate of Laurentian University, Sudbury, ON, with a B.Sc. in Geology from 1994. I am a member in good standing of the Professional Geoscientists of Ontario (#1366). My relevant experience after graduation includes over twenty-seven years of experience in geological consulting, mine geology and mineral resource evaluation of mineral projects nationally and internationally in a variety of commodities including 8 years of direct gold operational experience with Placer Dome Ltd.

I did not conduct a personal investigation of the Mount Milligan site due to COVID-19 related travel restrictions in place at the time.

I am responsible for Item (s) 1.10, 1.20, 1.21, 14, 25.1, 26.2 of the Technical Report.

I am independent of the issuer as described in Section 1.5 of the Instrument.

My prior involvement with the property that is the subject of the Technical Report is as follows. I have previously conducted a third-party review of the 2019 mineral resource estimate and assisted with the compilation of the 2019 NI 43-101 Technical Report.

I have read National Instrument 43-101. The parts of the Technical Report on the Mount Milligan Mine for which I am responsible have been prepared in compliance with this Instrument; and

At the effective date of the Technical Report, to the best of my knowledge, information, and belief, the Technical Report on the Mount Milligan Mine and/or the Items 1.10, 14, 1.20, 25.1, 26.2 of the Technical Report for which I am responsible, contains all scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Dated at Sudbury, ON, Canada, this 7th of November 2022.

(original signed and sealed) Brian Thomas

Brian Thomas; P.Geo.

Professional Geoscientists of Ontario, PGO, (#1366)



CERTIFICATE OF QUALIFIED PERSON - JENNIFER SIMPER

I, Jennifer Simper, state that:

I am a Senior Resource Geologist at:
WSP Golder
3300, 237 – 4th Avenue SW,
Calgary, Alberta, Canada
T2P 4K3

This certificate applies to the technical report titled “Technical Report on the Mount Milligan Mine” with an effective date of: December 31, 2021.

I am a “qualified person” for the purposes of National Instrument 43-101 (the “Instrument”). My qualifications as a qualified person are as follows. I am a graduate of the University of Calgary with a BSc in Geology from 2006. I am a member in good standing with the Association of Professional Engineers and Geoscientists of Alberta (#79249). My relevant experience after graduation and over the past sixteen years for the purpose of the Technical Report includes five years of exploration experience, including three years of gold exploration, and ten years as a consulting geologist reviewing exploration data and preparing resource estimates nationally and internationally in a variety of commodities.

My most recent personal inspection of the Mount Milligan Property occurred from July 28 - 29, 2021.

I am responsible for Items 1.8, 12, 25.1 and 26.1 of the Technical Report.

I am independent of the issuer as described in Section 1.5 of the Instrument.

I have had no prior involvement with the property that is the subject of the Technical Report.

I have read National Instrument 43-101 guidelines. The parts of the Technical Report for which I am responsible have been prepared in compliance with this Instrument; and

At the effective date of the Technical Report, to the best of my knowledge, information, and belief, the parts of Technical Report] for which I am responsible, contain all scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Dated at Calgary, Alberta, this 7th day of November 2022.

(original signed and sealed) Jennifer Simper

Jennifer Simper, P.Geol.

APEGA, P.Geol., (# 79249)

CERTIFICATE OF QUALIFIED PERSON – CURTIS CLARKE

I, Curtis Clarke, state that:

I am a Senior Mining Consultant at:
Golder Associates Ltd.
6925 Century Blvd.
Mississauga, ON
L5N 7K2

This certificate applies to the technical report titled “Technical Report on the Mount Milligan Mine” with an effective date of December 31, 2021 (the “Technical Report”).

I am a “qualified person” for the purposes of National Instrument 43-101 (“NI 43-101”). My qualifications as a qualified person are as follows: I am a graduate of Michigan Technological University with a Bachelor of Science in Mining Engineering degree granted in 1986. I am QP Member in good standing with the Mining & Metallurgical Society of America, #1354QP. My relevant experience after graduation for the purpose of the Technical Report includes:

- Eighteen years’ of underground mining experience in copper/zinc, lead/silver, gold, and nickel mines as a miner, mining engineer and technical representative.
- Eighteen years’ experience as a consulting mining engineer conducting feasibility studies, operational efficiency studies, mine evaluations, project implementations and project management for mining projects and operations.

I have not visited the mine described in the Technical Report.

I am responsible for Items 1.1, 2, 3, 4, 5, 6, 23, 24, and parts of Items 19, 25 and 26 of the Technical Report.

I am independent of the issuer as described in Section 1.5 of NI 43-101.

I have not had any prior involvement with the property that is the subject of the Technical Report.

I have read NI 43-101 and the part of the Technical Report for which I am responsible has been prepared in compliance with NI 43-101; and

At the effective date of the Technical Report, to the best of my knowledge, information, and belief, the parts of the Technical Report for which I am responsible, contain all scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Dated at Toronto, ON, this 7th day of November, 2022.

(original signed and sealed) Curtis Clarke

Curtis Clarke, QP,
MMSA Registration Number 1354QP

Cautionary Note Regarding Forward-Looking Information

Information contained in this Technical Report and the documents referred to herein which are not statements of historical facts, may be “forward-looking information” for the purposes of Canadian securities laws. Such forward looking information involves risks, uncertainties and other factors that could cause actual results, performance, prospects, and opportunities to differ materially from those expressed or implied by such forward looking information. The words “expect”, “target”, “estimate”, “may”, “will”, and similar expressions identify forward-looking information. These forward-looking statements relate to, among other things, mineral reserve and mineral resource estimates; grades and recoveries; development plans; mining methods and metrics including strip ratio; recovery process; production expectations including expected cash flows, capital cost estimates and expected life of mine operating costs; and expected outcomes of continuous improvement projects and opportunities.

Forward-looking information is necessarily based upon a number of estimates and assumptions that, while considered reasonable by Centerra Gold Inc. (“Centerra”) are inherently subject to significant political, business, economic and competitive uncertainties and contingencies. There may be factors that cause results, assumptions, performance, achievements, prospects or opportunities in future periods not to be as anticipated, estimated or intended. These factors include the following risks relating to the Mount Milligan project, Centerra and/or TCM: (A) strategic, legal, political and regulatory risks, including delays or refusals to grant required permits and licences; the status of relationships with local communities; Indigenous claims and consultative issues, increases in contributory demands; management of external stakeholder expectations; litigation; potential defects of title not known as of the date hereof; the impact of changes in, or to the more aggressive enforcement of laws, regulations and government practices; the inability of Centerra or TCM to enforce its respective legal rights in certain circumstances; risks related to anti-corruption legislation; and potential risks related to kidnapping or acts of terrorism; (B) financial risks, including sensitivity of the business to the volatility of metal prices; the imprecision of mineral reserves and mineral resources estimates, and the assumptions they rely on; the accuracy of the production and cost estimates; reliance on a few key customers for the gold-copper concentrate at Mount Milligan; the impact of currency fluctuations; continued compliance with financial covenants in Centerra’s credit agreement that is secured by certain assets used at the Mount Milligan Mine, Centerra’s access to cash flow from its subsidiaries; and changes to taxation laws; risks arising from the large streaming arrangement with RGLD GOLD AG and Royal Gold Inc.; and (C) operational and geotechnical risks, including the adequacy of insurance to mitigate operational risks; unanticipated ground and water conditions; shortages of water for processing activities; mechanical breakdowns; the occurrence of

any labour unrest or disturbance; the ability to accurately predict decommissioning and reclamation costs; the ability to attract and retain qualified personnel; geological problems, including earthquakes and other natural disasters; metallurgical and other processing problems; unusual or unexpected mineralogy or rock formations; tailings design or operational issues, including dam breaches or failures; delays in the Company's supply chain and transportation routes, including cessation or disruption in rail and shipping networks, whether caused by decisions of third-party providers, or force majeure events (including, but not limited to, flooding, wildfires, COVID-19, or other global events such as wars).

There can be no assurances that forward-looking information and statements will prove to be accurate, as many factors and future events, both known and unknown could cause actual results, performance, or achievements to vary or differ materially from the results, performance or achievements that are or may be expressed or implied by such forward-looking statements contained herein or incorporated by reference. Accordingly, all such factors should be considered carefully when making decisions with respect to Centerra and prospective investors should not place undue reliance on forward-looking information. Forward-looking information in this technical report is as of the issue date, November 7, 2022. Centerra and the Qualified Persons who authored this Technical Report assume no obligation to update or revise forward-looking information to reflect changes in assumptions, changes in circumstances or any other events affecting such forward-looking information, except as required by applicable law.

Non-GAAP Measures

This document contains the following non-GAAP financial measures: all-in sustaining costs per ounce sold on a by-product basis, sustaining capital expenditures, non-sustaining capital expenditures and free cash flow (deficit). These financial measures do not have any standardized meaning prescribed by GAAP and are therefore unlikely to be comparable to similar measures presented by other issuers, even as compared to other issuers who may be applying the World Gold Council ("WGC") guidelines, which can be found at <http://www.gold.org>.

All-in sustaining costs on a by-product basis per ounce sold include adjusted operating costs, the cash component of capitalized stripping costs, corporate general and administrative expenses, accretion expenses, and sustaining capital, net of copper and silver credits. The measure incorporates costs related to sustaining production. Copper and silver credits represent the expected revenue from the sale of these metals. Free cash flow is calculated as cash provided by operations less additions to property, plant, and equipment.

Management believes that the use of these non-GAAP measures will assist analysts, investors, and other stakeholders of the Company in understanding the costs associated with producing gold, understanding the economics of gold mining, assessing our operating performance, our ability to generate free cash flow from current operations and to generate free cash flow on an overall Company basis, and for planning and forecasting of future periods. However, the measures do have limitations as analytical tools as they may be influenced by the point in the life cycle of a specific mine and the level of additional exploration or expenditures a company has to make to fully develop its properties. Accordingly, these non-GAAP measures should not be considered in isolation, or as a substitute for, analysis of our results as reported under GAAP. See “Non-GAAP and Other Financial Measures” on pages 38 to 39 in Centerra’s Management’s Discussion & Analysis for the three and nine months ended September 30, 2022 for a discussion of non-GAAP measures used in this document.

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1. SUMMARY

This Technical Report summarizes the current and planned operations, Mineral Resources and Mineral Reserves for the operating Mount Milligan copper-gold mine (the Mount Milligan Mine, the Project, or the Property) located between Fort St. James and Mackenzie, British Columbia, Canada. The Technical Report was prepared by and for Centerra Gold Inc. (Centerra) by qualified persons (QPs), as listed in Item 2. This Technical Report conforms to National Instrument 43-101 *Standards of Disclosure for Mineral Projects* (NI 43-101) and follows the format set out in Form 43-101F1 for Technical Reports.

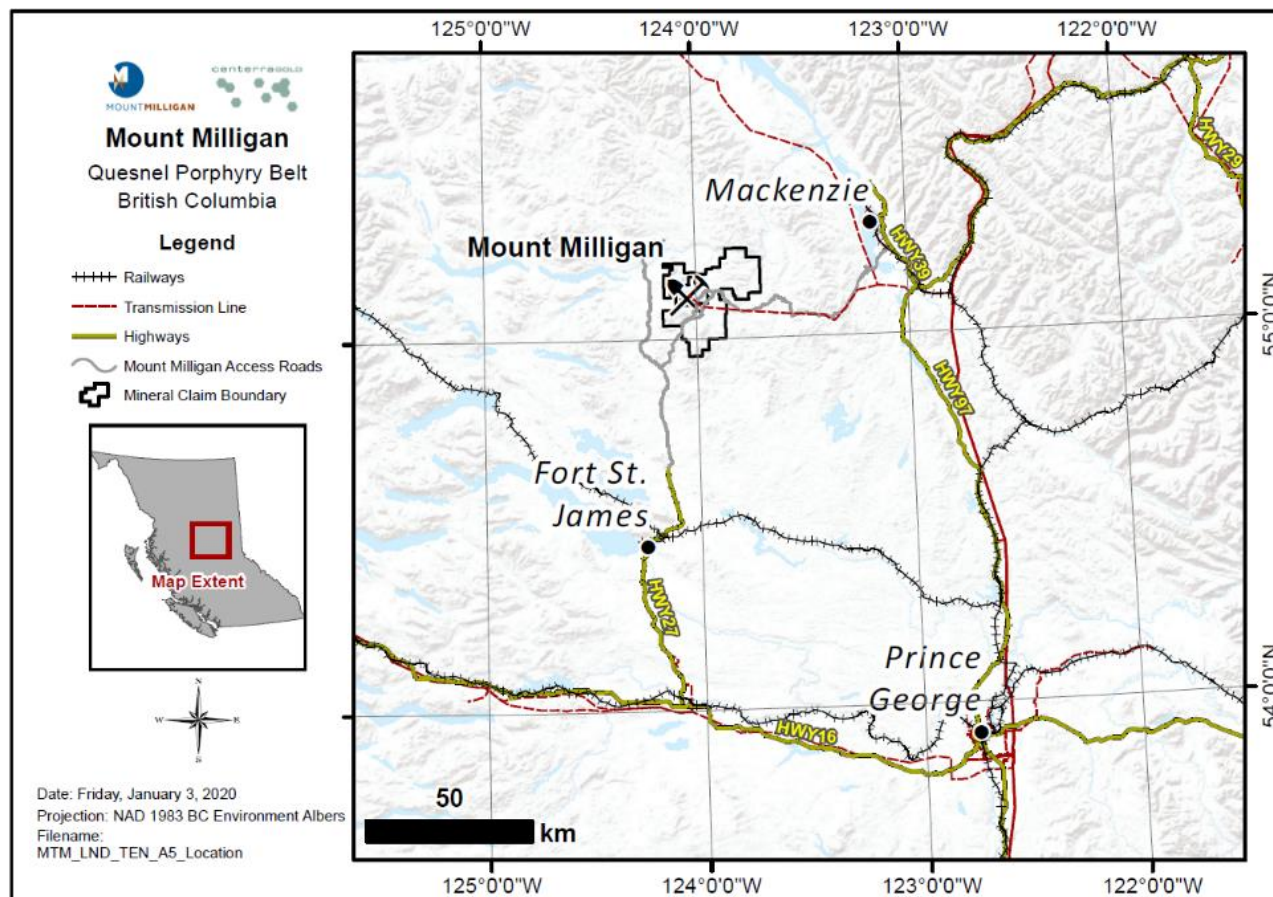
The previous Technical Report for the Mount Milligan Mine, “Technical Report on the Mount Milligan Mine”, published March 26, 2020, with an effective date December 31, 2019, is referred to in this Technical Report as the 2019 Technical Report.

All currencies in this Technical Report are presented in US dollars, unless otherwise noted.

1.1. PROJECT DESCRIPTION AND LOCATION

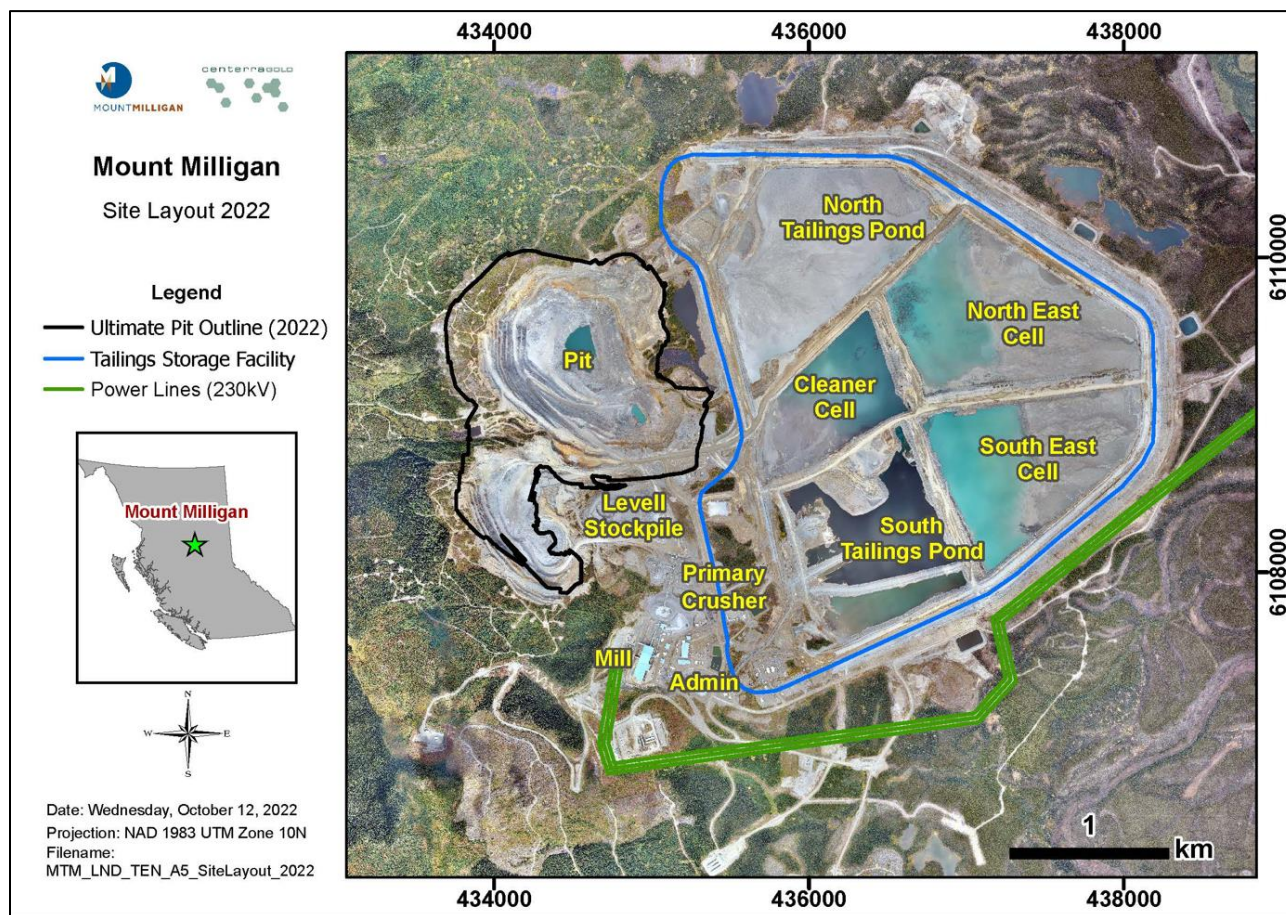
The Mine is located 155 kilometres (km) northwest of Prince George (population approximately 79,000) in north-central British Columbia. Forestry-based communities Mackenzie (population approximately 3,200) and Fort St. James (population approximately 1,600) are within daily commuting distance of the Project site. Both communities are serviced by rail. Figure 1-1 shows the location of the mine and local communities.

Figure 1-1: Mount Milligan Mine Location



The mine site comprises an open pit mine, tailings storage facility (TSF), mineralized stockpiles, a processing plant, workshop, warehouse, administration buildings, and camp. Figure 1-2 provides a plan view of the Project.

Figure 1-2: Plan View of Mount Milligan Mine Site



1.2. PROJECT HISTORY AND OWNERSHIP

Limited exploration activity was first recorded in 1937. In 1984, prospector Richard Haslinger (Haslinger) and BP Resources Canada Limited (BP Resources) located claims on the current site. In 1986, Lincoln Resources Inc. (Lincoln) optioned the claims and in 1987 completed a diamond drilling program that led to the discovery of significant copper-gold mineralization. In the late 1980s, Lincoln amalgamated with Continental Gold Corp. (Continental Gold) and continued ongoing drilling in a joint venture with BP Resources.

In 1991, Placer Dome Inc. (Placer Dome) acquired the Project from the joint-venture partners, resumed exploration drilling and completed a pre-feasibility study (PFS) for the development of a 60,000 tonnes per day (tpd) open pit mine and flotation process plant.

Barrick Gold Corporation (Barrick) purchased Placer Dome in 2006 and sold its Canadian assets to Goldcorp Inc. (Goldcorp), who then in turn sold the Project to Atlas Cromwell Ltd. (Atlas Cromwell).

Atlas Cromwell changed its name to Terrane Metals Corp. (Terrane) and initiated a comprehensive work program.

In October 2010, Thompson Creek Metals Company Inc. (TCM) acquired the Mount Milligan development project through its acquisition of Terrane, entered a streaming agreement with Royal Gold and subsequently constructed the Mount Milligan Mine, which commenced commercial production in February 2014.

In October 2016, TCM was acquired by a subsidiary of Centerra, in connection with that acquisition, Terrane and certain other subsidiary entities of TCM were amalgamated into TCM. The Mount Milligan Mine is now fully owned by TCM, an indirect subsidiary of Centerra. Table 1-1 depicts the historical production to December 31, 2021.

Table 1-1: Historical Production to December 31, 2021

Years	Milled Ore Tonnage ('000 t)	Head Grade		Metal Recovery		Concentrate Production			Waste Tonnage ('000 t)
		Cu (%)	Au (g/t)	Cu Rec (%)	Au Rec (%)	Concentrate ('000 dmt)	Cu (M lb)	Au ('000 oz)	
2013	2,055	0.29	0.56	79.20%	54.30%	18.7	10.4	20.1	24,753
2014	14,290	0.27	0.63	80.40%	63.10%	125.4	68	184	11,224
2015	16,138	0.26	0.64	80.20%	68.60%	140.7	75.2	226	14,413
2016	19,277	0.19	0.58	74.70%	58.90%	125.6	61.6	212	20,363
2017	17,743	0.18	0.64	78.90%	62.40%	121.5	56.4	228.1	20,557
2018	13,556	0.2	0.71	81.40%	64.50%	106	49.6	199.5	19,764
2019	16,350	0.26	0.53	81.30%	67.40%	159.5	75	187.8	23,730
2020	20,067	0.26	0.41	77.40%	62.90%	185.3	87.3	166	22,043
2021	20,900	0.21	0.46	78.30%	65.80%	162.3	77	201.5	25,265
Total	140,376	0.23	0.56	78.98%	63.94%	1145	560.5	1625	182,112

Figures are shown on a 100% produced basis.

Royalties and metals streams associated with the Project are discussed in Item 4.

1.3. GEOLOGY

The Mount Milligan deposit is located within the Quesnel terrane, a composite belt of volcanic arc and oceanic terranes that accreted to the North American Cordillera during the Middle Jurassic period. The Mount Milligan property is mostly underlain by Upper Triassic volcanic rocks of the Witch Lake succession. The Witch Lake succession is composed of an augite-phyric volcanoclastic unit and lesser coherent basaltic andesite to andesite units with subordinate epiclastic beds. In the

northwestern portion of the property the volcanic rocks are intruded by Early Jurassic to Cretaceous monzonitic rocks.

Porphyry style Au-Cu mineralization at the Mount Milligan deposit comprises two styles, early-stage porphyry gold-copper (Au-Cu), and late-stage high-gold low-copper (HGLC). The early-stage porphyry Au-Cu mineralization comprises mainly chalcopyrite and pyrite, occurs with potassic alteration and early-stage vein types, and is spatially associated with composite monzonite porphyry stocks (especially at their hanging-wall and footwall margins), hydrothermal breccia, and narrow dyke and breccia complexes. Late-stage, structurally controlled pyritic HGLC style mineralization is associated with carbonate-phyllitic alteration and intermediate- to late-stage vein types, and is spatially associated with faults, fault breccias, and faulted lithological contacts (i.e., faulted monzonite porphyry dyke margins). It crosscuts and overprints the earlier stage porphyry Au-Cu mineralization.

Chalcopyrite, the main copper-bearing mineral, is associated with potassic alteration at the contact margin between volcanic and intrusive rocks. It occurs most commonly as fine-grained disseminations and fracture fillings, and less commonly as veinlets and in veinlet selvages. Gold occurs as grains from 1 to 100 µm in size, as observed in process samples. Grains occur as microfracture fillings and are attached to pyrite, chalcopyrite, or bornite (Ditson, 1997). Gold also forms inclusions within pyrite, chalcopyrite, and magnetite grains. Scanning Electron Microscope (SEM) work indicates electrum throughout the deposit with varying gold to silver ratios.

1.4. DEPOSITS

The Mount Milligan deposit is categorized as a silica-saturated alkalic Cu-Au porphyry deposit associated with alkaline monzodioritic-to-syenitic igneous rocks and recognized in only a few mineral provinces worldwide (Deyell and Tosdal, 2004). Porphyry copper ± gold deposits commonly consist of vein stockworks, vein sets, veinlets, and disseminations of pyrite and chalcopyrite ± bornite that occur in large zones of economic bulk-mineable mineralization within porphyritic igneous intrusions, their contact margins, and adjoining host rocks. The mineralization is spatially, temporally, and genetically associated with hydrothermal alteration of the intrusive bodies and host rocks.

Similar alkalic Cu-Au porphyry deposits in British Columbia include Galore Creek, Mount Polley, Copper Mountain, New Afton, and Lorraine. These deposits occur in both the Quesnel and Stikine island arc terranes and range in age from Late Triassic to Early Jurassic. Global examples include Ok Tedi in Papua New Guinea as well as Northparkes and Cadia in Australia.

1.5. EXPLORATION

Since the 2016 acquisition, Centerra has focussed on compiling all historical geologic and exploration data (approximately 227,000 files and 404 GB of data), building an Exploration department and domaining exploration programs into near-field, brownfield, and greenfield. Exploration since 2016 has included ground and airborne geophysical surveys, soil sampling, trenching, and core drilling.

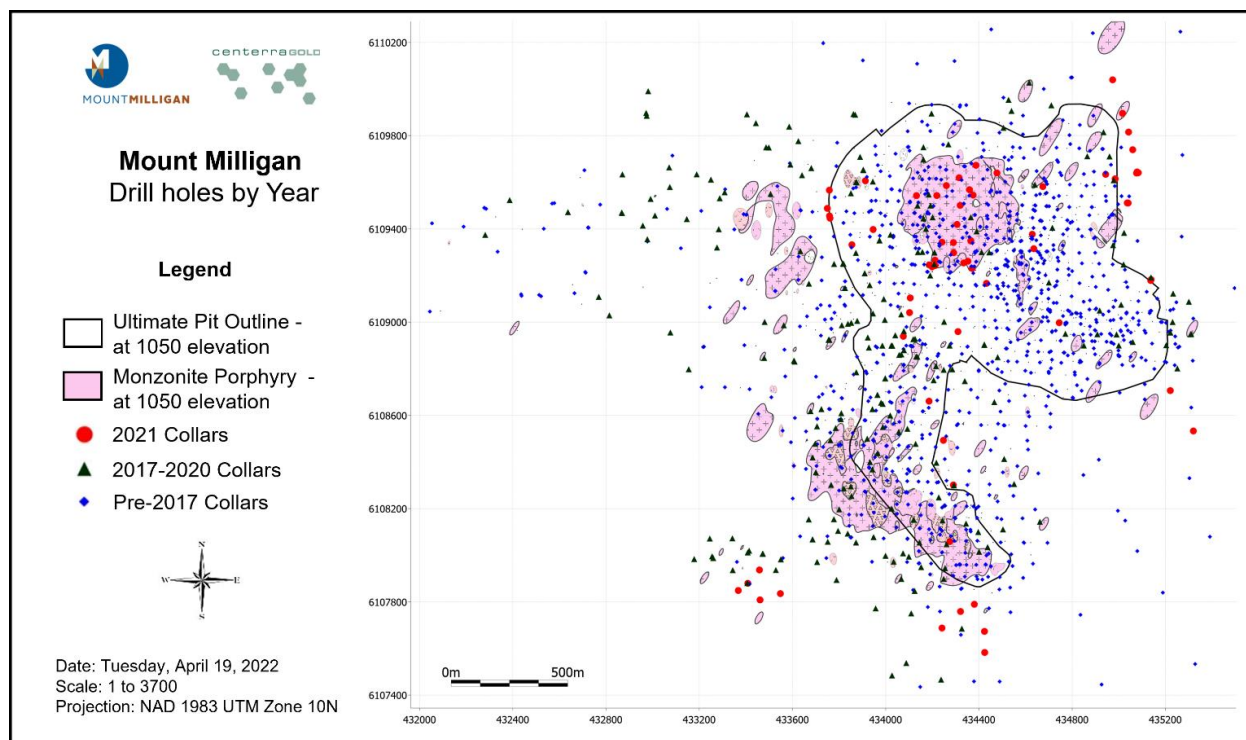
Historically, five exploration target zones were identified in the near field resource area (DWBX, WBX, MBX, 66 and Southern Star); three in the brownfield area within the mine lease (North Slope, Goldmark and South Boundary); and three in the greenfield area within the claims area (Heidi, Mitzi, and Snell). Exploration since 2017 has continued to test most of these zones and refine understanding of their geological relationships and mineral potential. In addition, new target zones have been developed and continue to be tested.

Exploration efforts have identified two distinct mineralization styles; early-stage porphyry Au-Cu mineralization and late-stage, structurally controlled high-gold low-copper (HGLC) mineralization. Current exploration targets are focussed on using these mineralization styles to expand the current resource.

1.6. DRILLING

Numerous drilling programs have been conducted since the deposit was first drilled in 1987. Except for early programs, the majority of core drilled has been predominately NQ and to a lesser extent HQ size. In total, there have been 1,351 holes drilled at Mount Milligan, recovering over 388 km of core. Figure 1-3 is a map showing collar locations of drill holes of the various campaigns.

Figure 1-3: Drill Hole Locations at Mount Milligan, 1987 - 2021



Detailed geological (lithology, alteration, and mineralization) and geotechnical (core recovery, rock quality designation [RQD], hardness or compressive strength [CS], degree of breakage, degree of weathering or oxidation, fracture and joint frequency, and specific gravity [SG] information) has been routinely recorded for all diamond drilling programs. Core recovery routinely exceeds 90% and averages 96%.

1.7. SAMPLES

A formal analytical Quality Assurance and Quality Control (QA/QC) program was introduced after 1992. Prior to that date, external check assays were commissioned from independent laboratories.

Centerra follows industry standard sampling procedures for NQ and HQ diamond drill core. Core is photographed then sawn in half; one half is placed in a sample bag with a uniquely referenced sample number and securely transported to Bureau Veritas (ISO 9001 Certified Laboratory) for appropriate analysis, while the remaining half is returned to the core box and securely stored for future sample reference. Blanks, duplicates, and certified reference material are inserted at regular intervals into the sample stream by Centerra staff. Sample subsets, totalling approximately 5% of the sample volume, are sent quarterly or annually to third party certified Laboratories as a method of QA.

Although production blast holes are routinely sampled and analysed by an onsite analytical laboratory, the results are not used to support the Resource, but instead are used for production grade control purposes.

1.8. DATA VERIFICATION

Drilling, Sampling, and Analysis

Ms. Jennifer Simper (P.Geol.), is an employee of Golder, and is an independent Qualified Person, as defined by National Instrument (NI) 43-101, conducted a personal inspection and database verification at Mount Milligan on July 28-29, 2021. Confirmation of core logging, independent assay and specific gravity verification, inspection of drill hole collars, and data collection procedures and processes were all reviewed. Previous audits (2019) have detailed efforts undertaken by Centerra to verify, validate, and correct the historical database. The 2021 site audit found no material issues with the underlying database and Centerra procedures were found to be consistent with Canadian Institute of Mining, Metallurgy and Petroleum (CIM) Mineral Exploration Best Practice Guidelines, 2014. The Qualified Person endorses the use of the database to support the 2021 Resource Estimate.

1.9. MINERAL PROCESSING AND METALLURGICAL TESTWORK

Metallurgical investigations conducted by various research laboratories prior to commencement of operations conclusively showed that froth flotation is the optimum process for the recovery of copper and gold; with this processing approach being adopted. These investigations were the basis of the performance models used in previous resource modelling. The 2019 Technical Report addressed previous assumptions in the copper and gold recovery models together with identified issues in the plant to produce new performance equations.

Since disclosure of the 2019 Technical Report, further investigations and projects have been undertaken to improve the recovery process and update the accuracy of the copper and gold recovery models. Using these new performance models, the LOM average recoveries are estimated at 80.2% for copper and 68.0% for gold, targeting a concentrate grade with a LOM average of 21.5% copper. Test results indicated that impurity element contents in the concentrate were below the penalty levels normally imposed by most smelters; therefore, no significant penalties are expected.

Further improvements to metallurgical recovery are being assessed, including the recent installation of Staged Flotation Reactors (SFRs). An initial assessment for the Mount Milligan flowsheet and ore

has shown increases to both gold and copper recoveries by 3.9% and 2.7%, respectively, using this flotation equipment at the time of writing this Technical Report.

1.10. MINERAL RESOURCES

Caution to readers: *In this Item, all estimates and descriptions related to mineral resource estimates are forward-looking information. There are many material factors that could cause actual results to differ materially from the conclusions, forecasts or projections set out in this item. Some of the material factors include differences from the assumptions regarding the following: estimates of Cut-off Grade (COG) and geological continuity at the selected cut-off, metallurgical recovery, commodity prices or product value, mining and processing methods, and General and Administration (G&A) costs. The material factors or assumptions that were applied in drawing the conclusions, forecasts and projections set forth in this Item are summarized in other Items of this Technical Report.*

Centerra completed an update of the Mineral Resource estimate in the fourth quarter of 2021 for the Project. Centerra validated and verified the database, and updated the geological interpretations based on available data.

The effective date of the Mineral Resource estimate is December 31, 2021. The Mount Milligan Au-Cu porphyry deposit is estimated to contain a combined Measured and Indicated Mineral Resource of 189 million tonnes (Mt) at 0.18% Cu and 0.30 g/t Au containing 742 million pounds (lbs) of copper and 1.8 million ounces (oz) of gold and an Inferred Mineral Resource of 4.6 Mt at 0.07% Cu and 0.47 g/t Au.

The Mineral Resource estimate reported within the 2021 (\$1,550/oz gold and \$3.50/lb copper) constraining resource pit shell was based on a cut-off grade of 0.2% copper-equivalent (CuEq). The Mineral Resource estimate was defined using the December 31, 2021, mining surface and is reported exclusive of the Mineral Reserve in Table 1-2.

Table 1-2: Mineral Resource Statement, Effective Date December 31, 2021 (exclusive of Mineral Reserves)

Category	Cut-off CuEq (%)	Tonnes (kt)	Copper Grade (%)	Gold Grade (g/t)	Contained Copper (Mlb)	Contained Gold (koz)
Measured (M)	0.2	36,529	0.21	0.26	169	305
Indicated (I)	0.2	152,796	0.17	0.31	573	1,523
Total M+I	0.2	189,325	0.18	0.30	742	1,828
Inferred	0.2	4,638	0.07	0.47	7	70

Notes:

- (1) CIM definitions were followed for the classification of Mineral Resources.
- (2) Mineral Resources are reported within a conceptual pit shell at a 0.2% CuEq cut-off value using assumed metal prices of \$3.50 per pound copper and \$1,550 per ounce gold, and a US\$/CA\$ exchange rate of \$1.00/CA\$1.30. Refer to Item 14.8 for CuEq Formula and Item 14.10 for conceptual pit shell parameters.
- (3) All figures have been rounded to reflect the relative accuracy of the estimates.
- (4) Mineral Resources that are not Mineral Reserves do not have a demonstrated economic viability.
- (5) Mineral Resources are reported exclusive of Mineral Reserves within a conceptual Mineral Resource Pit and considering mining progress as of December 31, 2021.
- (6) The QP for the Mineral Resource estimate is Mr. Brian Thomas, P.Geo., an independent QP as defined under NI43-101 and an employee of Golder Associates Ltd.

The resource modelling work was completed by Centerra. The Qualified Person for the Mineral Resource estimate is Mr. Brian Thomas P.Geo. (PGO #1366). Mr. Thomas is an employee of Golder, and is an independent Qualified Person, as defined by NI 43-101.

The Mineral Resource is based upon a geological block model that incorporated 179,983 individual assays from 397,191 m of core drilling in 1,511 drill holes. Assay data density having a drill hole spacing of 25 m to 50 m was determined to be suitable for this type of deposit to be classified within the Measured and Indicated category and within each of the interpreted mineralized zones. The drill hole database is supported by a QA/QC program.

Based on the verification of the drill hole database, and review of the geological interpretations, Mineral Resource block model, and 2021 reconciliation results, it is the QP's opinion that the supporting data and geological interpretations are representative of the Mt. Milligan deposit at the time of reporting, and that the 2021 Mineral Resource estimate has been estimated in conformity with CIM Estimation of Mineral Resource and Mineral Reserves Best Practices Guidelines (Nov 2019), and has been reported in accordance with the Canadian Securities Administrators' National Instrument 43-101.

1.11. MINERAL RESERVE ESTIMATE

The Mineral Reserve estimate was derived through the development of an ultimate open pit design using the Mineral Resource model and pit optimization parameters as a basis for mine design.

The open pit was optimized using long-term metal price estimates of \$3.25/lb copper and \$1,350/oz gold, an exchange rate of US\$1.00/CA\$1.30, and costs related to mining, processing, and G&A (including site services) and sustaining capital costs. Other factors considered include metallurgical recoveries, concentrate grades, transportation costs, smelter treatment charges, the RGLD Gold AG and Royal Gold Inc. (collectively, "Royal Gold") royalty and the Royal Gold stream in determining economic viability. The Mineral Reserve estimate reported in Table 1-3 has been classified as approximately 30% Proven and 70% Probable on a tonnage basis.

A Net Smelter return (NSR) cut-off comprising the costs for processing and G&A operating costs (opex) and sustaining capital unit costs (capex) was calculated to be \$7.40/t or CA\$9.62/t. Mining opex is excluded from this calculation as the definition of ore (and waste) is made at the pit rim with mining opex considered in the definition of the optimized pit shell. One-time processing, or G&A sustaining capex items, were also excluded from the NSR cut-off.

Note that the 2019 Technical Report for the Mount Milligan Mine calculated an NSR cut-off of \$7.64/t CA\$9.55/t.

The Proven and Probable Mineral Reserve totals 246.2 Mt at 0.18% copper and 0.37g/t gold containing one billion pounds of copper and 2.9 million ounces of gold (Table 1-3).

Table 1-3: Mineral Reserve Statement, Effective Date December 31, 2021

Mineral Reserve Category	Tonnes (kt)	Gold (g/t)	Copper(%)	Contained Gold (koz)	Contained Copper (Mlbs)
Proven	76,477	0.37	0.20	914	337
Probable	169,681	0.37	0.18	2,011	659
Total	246,158	0.37	0.18	2,925	996

Notes:

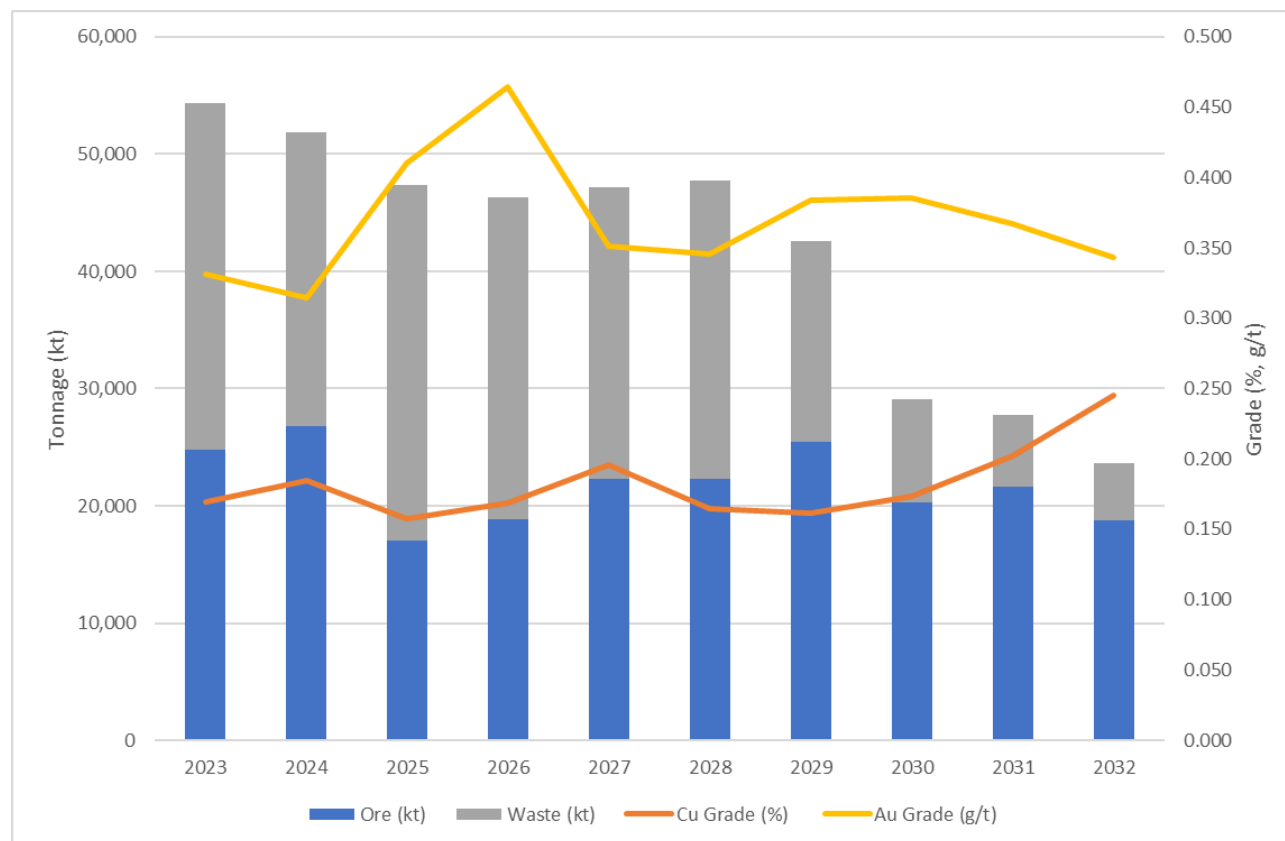
1. CIM definitions were followed for Mineral Reserves estimation.
2. Mineral Reserves are estimated at \$7.40/t (CA\$9.62/t) NSR cut-off value using metal prices of \$3.25 per pound copper and \$1,350 per ounce gold, and a \$/CA\$ exchange rate of \$1.00/CA\$1.30.
3. Figures may not total exactly due to rounding.
4. As of June 30, 2022, Au Contained and Cu Contained have been depleted from the above figures by approximately 82 koz. and 38 Mlbs from the above December 31, 2021, Mineral Reserve balances, respectively.
5. Production at Mount Milligan is subject to a streaming agreement with RGLD Gold AG and Royal Gold, Inc. (collectively, "Royal Gold") which entitles Royal Gold to 35% of gold sales from the Mount Milligan Mine. Under the stream arrangement, Royal Gold will pay \$435 per ounce (/oz) of gold delivered. Mineral Reserves and Resources for the Mount Milligan property are presented on a 100% basis.

The Mineral Reserve estimate was prepared under the supervision of Gordon Zurowski, P.Eng., of AGP Mining Consultants Inc. (AGP) who is a Qualified Person as defined under NI 43-101. The Mineral Reserve estimate is effective December 31, 2021.

1.12. MINING

The mining operation is a conventional shovel and truck open pit mine feeding a 60,000 tpd (permitted throughput on an annualised basis) processing plant. The planned mine life is just over 11 years (2022 – 2033) with a Proven and Probable Reserve of 246.2 Mt, grading 0.37 g/t gold and 0.23% copper. The pit has been planned as a series of seven discrete pushbacks and scheduled to maximize the production of ore. Total ore and waste will be mined at a rate of 54.3 Mt/a in 2023, 51.9 Mt/a in 2024 and decreasing to 19.6 Mt/a by 2032 with the last reclaim of 1.9 Mt from the stockpile in 2033. This yields an overall LOM waste:ore ratio of 0.88:1.0. The mining sequence has been developed to allow for provision of suitable waste material for annual TSF construction requirements. Figure 1-4 provides the LOM mining schedule for ore and waste rock with the average estimated copper and gold grades on an annual basis.

Figure 1-4: Annual Material Mined



The mine fleet comprises four blast hole drills, two rope shovels, one hydraulic excavator, two rubber-tired front-end loaders, fifteen haul trucks and various other dozers, loaders, graders, and excavators. A fleet of articulated trucks are used in dam construction and project activities.

1.13. RECOVERY METHODS

The average process plant feed grade of 0.23% Cu is delivered throughout the LOM period at an average daily rate of 60,000 tonnes to yield a marketable 21.5% Cu concentrate. Process plant ore feed quality is maintained to honour metallurgical constraints, such as ORE/HGLC ratio, Py:Cpy ratio and mercury (Hg) content. Average recovery to concentrate projected to be achieved during the LOM period is 80.2% for copper and 68.0% for gold.

The Mount Milligan process plant was originally designed to process ore at a nominal rate of 60,000 tpd, producing a marketable concentrate containing copper, gold, and silver. A secondary crushing circuit, installed in 2016, together with process plant optimization projects, increased the potential throughput to a nominal rate of 62,500 tpd. Key process equipment includes:

- Primary crushing plant with a 1.525-m x 2.794-m gyratory crusher
- Secondary crushing plant with two cone crushers prior to the grinding circuit, each powered by one 1,000-kW motor
- SAG/ball mill/pebble crusher grinding circuit, comprising one SAG mill, two ball mills, and two cone crushers
- A flotation circuit comprises a total of 4 rougher, 6 scavenger, and 17 cleaner cells that include 3 SFRs
- Regrinding and gravity concentration circuits comprising one tower mill, two IsaMills™, and one centrifugal gold concentrator

1.14. MINE INFRASTRUCTURE

The infrastructure at Mount Milligan Mine includes a concentrator, a TSF and reclaim water ponds, as described elsewhere in this Technical Report, an administrative building and change house, a workshop/warehouse, a permanent operations residence, a first aid station, an emergency vehicle storage, a laboratory, and sewage and water treatment facilities. The power supply is provided by B.C. Hydro via a 91-km hydroelectric power line. Concentrate is transported by truck from the Project site to Mackenzie, transferred onto railcars of the Canadian National Railway, railed to existing port storage facilities of Vancouver Wharves in North Vancouver, and loaded as lots into bulk ore carriers. Concentrate is then shipped to customers via ocean transport.

Tailings Storage Facility

The TSF at the Mount Milligan Mine is designed to store tailings solids and potentially acid generating (PAG) and oxide/weathered waste rock materials in designated areas. The TSF embankment is constructed as a centreline dam using open pit overburden and non-acid generating (NAG) waste rock materials. Construction of each of the embankment stages is scheduled to correspond with material availability from the Open Pit and the projected rate of rise. There will be sufficient volume of waste material produced over the LOM to raise the tailings dam to the required final elevation of 1,112m.

From the process plant, two tailing streams — the rougher/scavenger tailings and the first cleaner/scavenger tailings — are deposited and stored in separate tailing storage areas within the TSF. The rougher-scavenger tailings contain mostly non-sulphide gangue minerals, while the cleaner scavenger tailings contain most of the sulphide gangue minerals. The latter is kept in a lined pond and underwater to prevent acid generation from the oxidation of the sulphide minerals.

The TSF at the Mount Milligan Mine is designed to store tailings solids and PAG and oxide/weathered waste rock materials in designated areas. The TSF embankment is constructed as a centreline dam using open pit overburden and NAG waste rock materials. The TSF comprises two dams to include the Main Embankment and the West Separator Berm (WSB). The dams will eventually join as additional raises are completed. The highest portion of the TSF embankment is in the King Richard Creek valley and is to be approximately 62 metres in height, as measured from crest to downstream toe after Stage 8 construction is complete in 2022.

The Main embankment is subdivided into segments designated: South, Southeast, Northeast, and North Dam. The South Dam is situated across the King Richard Creek valley; the Southeast/Northeast Dams are along the eastern plateau towards the Esker Lakes; the North Dam is constructed through the esker deposit. The WSB is constructed along the western edge of the impoundment providing containment between the TSF and the Open Pit. The WSB has been extended towards the north and south and will continue to be extended until it connects into the Main embankment creating a continuous ring impoundment.

The maximum embankment height will eventually be 85 metres at the South Dam across King Richard Creek valley. The current constructed height at the South Dam is approximately 62 metres. Construction of each of the embankment stages is scheduled to correspond with material availability from the Open Pit and the projected rate of rise. Stage 8 of the TSF is planned for 2022 with Stages 9-19 planned as annual raises based on the current mine schedule and planning. General

arrangements illustrating the staged development together with a typical embankment cross section are shown in Item 18.

1.15. WATER MANAGEMENT

To ensure adequate water supply for the life-of-mine (LOM), the mine is actively investigating additional long-term water sources. Ongoing investigation activities include hydrological and hydrogeological desk-top analysis and field work, environmental impact assessment, community consultation and engineering evaluation. On January 6, 2022, TCM received the approval of an amendment to EAC #M09-01 to utilize LOM surface water withdrawals external to the TSF during the open water season (April 1 to November 30) from either the Nation River as a single surface water source or Rainbow Creek and Philip Lake 1 as a combined surface water source. Following extensive analysis, TCM has submitted a *Water Sustainability Act* licence application for the Rainbow Creek and Philip Creek surface water option. This option, in addition to the currently permitted groundwater withdrawals, will be sufficient to maintain operations at current production targets for the approved LOM.

The economic analysis includes capex and opex, totalling \$17 million (M) related to external water supply costs.

1.16. ENVIRONMENT AND PERMITTING

The environmental assessment and permitting framework for mining in Canada, and British Columbia in particular, is well established, providing a comprehensive mechanism for reviewing major projects to assess potential impacts. The government authorizations held by Mount Milligan Mine including the environmental assessment certificate, the *Mines Act* permit, *Environmental Management Act* permits, and ancillary licences followed rigorous and robust regulatory processes. The mine has been operating under these approvals since commencement with amendments pursued and received as required. Mount Milligan Mine is or will be in the process of applying for additional amendments required for the revised LOM plan.

1.17. COMMUNITY SUSTAINABILITY

In 2006, Terrane initiated a consultation program with local communities and Indigenous groups. In May 2008, Terrane convened a Community Sustainability Committee (CSC) whose membership comprises local Indigenous groups and regional stakeholders from the communities of Fort St. James, Vanderhoof, Mackenzie and Prince George. The Committee acts as TCM's primary

mechanism for community engagement concerning the mine's activities and investments into the region. TCM meets with the CSC on a biannual basis to discuss issues and disburse money from the Community Project Fund to applications meeting criteria

TCM is also party to a Socio-Economic Agreement with the McLeod Lake Indian Band and an Impact Benefit Agreement (IBA) with Nak'azdli Whut'en. Both agreements commit the Company to the provision of financial payments and these amounts have been incorporated into the economic analysis in this Technical Report.

1.18. CAPITAL AND OPERATING COSTS

Total operating and capital costs over Mount Milligan's 11-year LOM are estimated at \$3,502 million, including \$990 million for mining, \$1,212 million for processing, \$492 million for administration (G&A), \$168 million for transportation costs, selling and marketing costs of \$94 million, treatment and refining charges of \$180 million and capital expenditures of \$366 million, as shown in Table 1-4.

The LOM capital expenditures required to exploit the Mineral Reserves in the LOM plan is estimated at \$366 million, which includes the replacement of nine and the addition of two 227t haul trucks and some auxiliary equipment. Major component rebuilds of the mobile fleet has been estimated based on expected operating hours per component.

Also included in the sustaining capital estimate is an external water pumping system and tailings line replacement, but excludes \$127 million TSF construction costs (included in mine opex). Waste mined at Mount Milligan is used for routine TSF raises, the cost of which is capitalized to the TSF rather than as capitalized stripping. The current mine plan does not contemplate any growth capital.

Table 1-4: Operating and Capital Cost Summary

Costs Summary (Total LOM)	Total \$M
Mining	990
Processing (including water supply)	1,212
Admin	492
Transportation	168
Selling and Marketing	94
Treatment and Refining	180
Capital	366
Total	3,502

The all-in sustaining cost per ounce sold, on a by-product basis, which includes sustaining capital and accounts for copper revenue as a credit, averages \$756/oz of gold for the period from 2022 to

the end of the LOM. All-in sustaining cost per ounce sold, on a by-product basis, is a non-GAAP financial performance measure. For further information please see the Non-GAAP Measures portion of this document.

1.19. FINANCIAL EVALUATION

Using a copper price of \$3.25/lb, gold price of \$1,500/oz and CA\$:US\$ exchange rate of 1.30:1.00, as assumed for the Mineral Reserve estimation process, over its 11-year LOM Mount Milligan is expected to generate \$4,150 million of revenue, net of streaming obligations and smelter fees. Net free cash flow on an undiscounted basis for the Mount Milligan Mine from 2022 to the end of its mine life in 2033, but accounting for the cost of closure, is estimated to be \$640 M; this is a non-GAAP financial performance measure. The net present value (NPV) is \$486 M after tax at a 5% discount rate. Table 1-5 provides a summary of economic analysis results.

Table 1-5: Economic Analysis Summary

Item	Total (\$000)
Net Gold Revenue	2,177,197
Net Copper Revenue	1,893,117
Net Silver Revenue	81,187
Total Revenue	4,151,502
Cash Flow from Operating Activities	1,137,812
Capital	-493,737
Free Cash Flow (before financing)	644,075

1.20. INTERPRETATION AND CONCLUSIONS

Based on the information contained herein, the Qualified Persons, as authors of this Technical Report, offer the following interpretations and conclusions.

Geology and Mineral Resources

Mount Milligan is a roughly tabular, near-surface, alkalic copper-gold porphyry deposit currently being mined.

The procedures for drilling, sampling, sample preparation and analyses were determined to be appropriate for the type of mineralization and the estimation of Mineral Resources.

Combined Measured and Indicated Mineral Resources, exclusive of Mineral Reserves, total of 189 Mt at 0.18% Cu and 0.30g/t Au containing 742 million pounds (lbs) of copper and 1.8 million ounces (oz) of gold and an Inferred Mineral Resource of 4.6 Mt at 0.07% Cu and 0.47g/t Au.

The effective date of the Mineral Resource Statement is December 31, 2021. Based on the verification of the drill hole database, and review of the geological interpretations, mineral resource block model, and 2021 reconciliation results, it is the QP's opinion that the supporting data and geological interpretations are representative of the Mt. Milligan deposit at the time of reporting, and that the 2021 Mineral Resource estimate has been estimated in conformity with CIM Estimation of Mineral Resource and Mineral Reserves Best Practices Guidelines (Nov 2019), and has been reported in accordance with the Canadian Securities Administrators' National Instrument 43-101.

Mining and Mineral Reserves

Proven and Probable Mineral Reserves total 246.2 Mt at 0.18% Cu and 0.37g/t Au containing 996 Mlb copper and 2.9 Moz gold, effective December 31, 2021. This estimate was derived through the design of an ultimate pit, conforming to practical mining requirements, based on an optimized pit shell using metal prices of \$3.25/lb Cu and US\$1,350/oz Au and an exchange rate of CA\$:US\$ of 1.30:1.00.

The Mineral Reserve estimate has been prepared using industry standard best practise methodologies with the classification of Proven and Probable Mineral Reserves, conforming to CIM definitions and NI 43-101 guidelines.

Mining is carried out using a conventional drill-blast, load, and haul approach. All waste rock is either used for TSF dam construction, or stored within the TSF, to comply with environmental approvals requirements.

The Mineral Reserve estimate is higher than that reported at end-2019 which was 191.0Mt at 0.23% Cu and 0.39g/t Au containing 959 Mlb copper and 2.4 Moz Au.

Mineral Processing

Recovery models were updated to reflect recent plant performance. Continuous improvement initiatives resulted in higher throughput achieved compared to historic results. Despite not achieving the targeted 21.9 Mt annual throughput levels before, additional upgrades, including the 25%

increase in pebble crushing capacity are expected to reduce current circuit bottlenecks and increase annual throughput. The addition of the Staged Flotation Reactor (SFR) circuit is helping to maintain metal recovery rates at higher throughput levels and will be further optimized. Preliminary results indicate increases of 3.9% and 2.7% for gold and copper recovery, respectively.

Economic Analysis

Capital costs have been estimated for an eleven-year mine life and operating costs have been estimated for an average 60,000-tpd process plant feed rate. Operating cost estimates were developed from first principles using site historic costs up to approximately Q1/2022 as a basis for calibrating the estimates

The base case economic assumptions are metal prices of \$1,500/oz Au and \$3.25/lb Cu, and a CA\$:US\$ exchange rate of 1.30.

The LOM all-in sustaining cost per ounce sold, on a by-product basis, before taxes is estimated at \$756/oz gold sold. This is a non-GAAP financial performance measure.

The NPV for the base case model is \$486M after tax at a 5% discount rate.

Risks and Mitigation

Following a comprehensive risk assessment, six (6) key risks have been identified that could have significant impact on the operation. These include:

- Delay or non-compliance with permits
- Ineffective tailings dam management
- Relationships with local communities including indigenous groups
- Challenges in attracting and retaining employees
- Capital and operating cost management
- Inability to transport concentrate

Section 25.8 provides an overview of these six key risks and their mitigation plans and actions..

1.21. RECOMMENDATIONS

The following recommendations are provided by the Qualified Persons.

Mineral Exploration

It is recommended that mineral exploration continue targeting both mineralization styles that have been identified with the Mount Milligan deposit, early-stage gold and copper porphyry and late stage structurally controlled high-gold-low-copper. Ongoing exploration programs should continue identifying, delineating, and testing target zones within the three strategic mineral exploration domains: at the deposit scale (Near-pit/infill), at the porphyry cluster scale (Brownfield outside the ultimate pit boundary but inside the mine lease), and for standalone deposits (Greenfield outside the mine lease).

In 2022, planned drilling at the Mount Milligan Mine focuses on zones where positive drilling results were returned in previous drilling campaigns from 2017-2021. Drilling programs continue to target gold and copper porphyry mineralization below and adjacent to the current ultimate open-pit boundary, as well as continue to test targets with potential for shallower mineralization peripheral to the current resource and reserve pits. Resource expansion and infill drilling is recommended on the western margin of the open-pit, i.e., in both the shallow Goldmark zone and the underlying DWBX zone, and on the eastern margin of the open-pit, i.e., in both the shallow Great Eastern Fault (GEF) zone and underlying porphyry target associated with the recently discovered Great Eastern stock.

Additional greenfield exploration work is recommended to advance near surface porphyry targets in the Mount Milligan district and adjacent properties. Additional geophysics surveys recommended include airborne mobile Magneto-Telluride (MT) surveys and detailed ground IP surveys, regional geochemistry surveys including stream sediment sampling as well as soil and till sampling grids, regional data compilation, and ongoing helicopter supported diamond drilling programs.

Budgets for programs over the next three years should be on par with Centerra programs in recent years (2018-2021). Annually, these budgets have been in the range of \$5.5-8.5 M comprising diamond drilling (30,000 to 43,000 m), geophysics programs (15 to 60 line-km IP surveys and other surveys) and development of a comprehensive 3D exploration model and drilling database that compiles lithology, alteration, mineralization, structural, geochemical, and geophysical information. Ongoing advancement of the 3D exploration models for both the Mount Milligan deposit and the Mount Milligan Greenfield district are recommended to improve understanding of the geometry of the deposit from fault block to fault block, to find extensions of known mineralization that could potentially be added to resources and reserves and to guide future discoveries.

Mineral Resources

The QP recommends that Centerra evaluate the following potential changes to the block modelling methodologies:

- Evaluate the spatial continuity of Cu and Au grades independently for each Domain.
- Incorporate the 100 (shell) Domain into existing or reinterpreted Domains.
- Review the resource categorization methodology in conjunction with search estimate parameters.
- Apply more focus on structural trends in Domain wireframe reinterpretation.
- Analyze the distribution of potentially deleterious minerals for potential negative effects.

Water Management

Continue to identify water sources for LOM operations and progress related approvals in consultation with Indigenous partners.

Continue to monitor water supply, storage and use very closely to minimize potential process plant downtime due to lack of water.

Mine Planning and Operations

Evaluate using an optimized pit with a lower revenue factor as the basis for designing the ultimate pit, which is expected to reduce the strip ratio and possibly provide a more manageable balance between ore and waste tonnes. However, this is also likely to decrease the mined ore tonnes and may reduce the mine life.

Re-work the mining schedule, including considerations for additional waste rock stockpiles, to optimize scheduling of adequate TSF dam building material as it is required for each construction season. The schedule in the report provides adequate construction materials and accounts for stockpiling. The material balance is acceptable, but there may be optimization opportunity in the dyke design and construction methodology used.

Evaluate opportunities for in-pit waste and tailings storage.

Mineral Processing and Metallurgy

Evaluate enclosure of the secondary crusher circuit screen structure. Historically, cold winter weather prevented continued operation of the secondary crushing circuit, which negatively impacted the plant production rate significantly. The enclosure will allow the surrounding areas of the screen to be heated and hence improve screen availability and operability.

Evaluate additional rougher flotation and regrind capacity in the separation circuit. As throughput rates increased over the last few years and is expected to continue increasing, additional rougher flotation capacity, coupled with regrind, will alleviate decrease in flotation retention time and increase recovery.

2. INTRODUCTION

This Technical Report, which was prepared by and for Centerra, summarizes the current and planned operations and the Mineral Reserves and Mineral Resources for the Mount Milligan Mine, located between Fort St James and Mackenzie, British Columbia, Canada. This Technical Report serves as an update to the Mineral Reserves.

Centerra acquired its indirect 100% interest in the Mount Milligan Mine on October 20, 2016, when it acquired (through a subsidiary) TCM (the Acquisition). Prior to the Acquisition, TCM held the Property through its wholly owned subsidiary Terrane, which it acquired in 2010. As part of the Acquisition, TCM and Terrane amalgamated. The Project was constructed by TCM prior to the Acquisition and commenced commercial production in February 2014.

When used in this Technical Report, “TCM” means the company existing prior to the Acquisition, unless otherwise noted.

Centerra, a global mining company organized under the laws of Canada, is engaged in the acquisition, exploration, development, and operation of mineral properties. Centerra’s shares are listed on the Toronto Stock Exchange under the trading symbol “CG” and the New York Stock Exchange under the trading symbol “CGAU”.

Items of significant change from the 2020 Technical Report include copper and gold price assumptions, metallurgical recovery estimates, capital and operating costs, increased NSR cut-off value and geologic model updates, which have resulted in an updated Mineral Resource and Mineral Reserve estimate and a new ultimate pit design and mining-processing schedule. As per the 2020 Technical Report, the stream arrangement with Royal Gold and H.R.S. Resources royalty are incorporated into this Technical Report, which impacts on the Project’s expected cash flows over the LOM.

All dollar figures in this Technical Report refer to US Dollars (US\$), unless otherwise noted.

2.1. SOURCES OF INFORMATION

This Technical Report is based on published material and data, professional opinions, and unpublished materials available to Centerra or prepared by its employees. In addition, certain information used to support this Technical Report was derived from previous technical reports on the Project and from reports and documents listed in the References Item. Other sources of data include

geologic and block model reports, drill hole assay data, the block model, mine plans, cost estimates, and economic models that were prepared by employees of Centerra.

This Technical Report has been prepared in compliance with NI 43-101 and follows the format set out in Form 43-101F1 for Technical Reports.

2.2. CONTRIBUTING PERSONS AND SITE INSPECTIONS

This Technical Report has been prepared by the persons listed in Table 2-1, each of whom is a Qualified Person (QP), as defined by NI 43-101 and has provided a QP certificate.

Other Centerra employees compiled certain Items of this Technical Report under the supervision of those identified in Table 2-1. These Centerra employees are experienced technical and accounting/finance professionals in their respective areas of expertise.

Standard professional procedures have been followed in preparing the contents of this Technical Report. Data used in this Technical Report have been verified, where possible, and all data is considered to have been collected in a professional manner.

Site inspections by QPs are detailed in the QP certificates earlier in this Technical Report.

Table 2-1: Authors, Qualified Persons, and Responsibilities

Qualified Person/Author	Title	Primary Area(s) of Responsibility
Cheyenne Sica (P.Geo), Centerra Gold Services Inc.	Exploration Manager	Geology, Exploration, Drilling, and Sample Preparation
Tengfei, Yue (P.Eng.), Centerra Gold	Director, Metallurgy	Mineral Processing and Metallurgy
Brian Thomas (P.Geo), WSP-Golder	Principle Resource Geologist	Mineral Resource Estimate
Jennifer Simper (P.Geo.), WSP-Golder	Lead Geologist	Data Verification
Curtis Clarke (MMSA QP) WSP-Golder	Senior Mining Consultant	Lead Author
Catherine Taylor (P.Eng. ARM), Centerra	Vice President, Risk & Insurance	Risk and Mitigation
Gordon Zurowski (P.Eng.) AGP Mining Consultants Inc.	Mine Engineer	Mineral Reserves, Mining Methods
David Luzi, (P.Geo.) Stantec	Hydrologist	Water Management
Angela Waterman	Regional Director ESG	Environmental
Bruno Borntraeger (P. Eng.), Knight Piésold Consulting	Specialist Geotechnical Engineer	Tailings Storage Facility dam

2.3. UNITS

This Technical Report utilizes metric units throughout as set forth in the Glossary included in Item 27.1. Grades are in percent of copper metal by weight and grams per tonne (g/t) for gold. Tonnages are metric tonnes of 2,204.6 pounds. Gold sales in units of Troy ounces with a conversion of 31.1 grams per Troy ounce.

3. RELIANCE ON OTHER EXPERTS

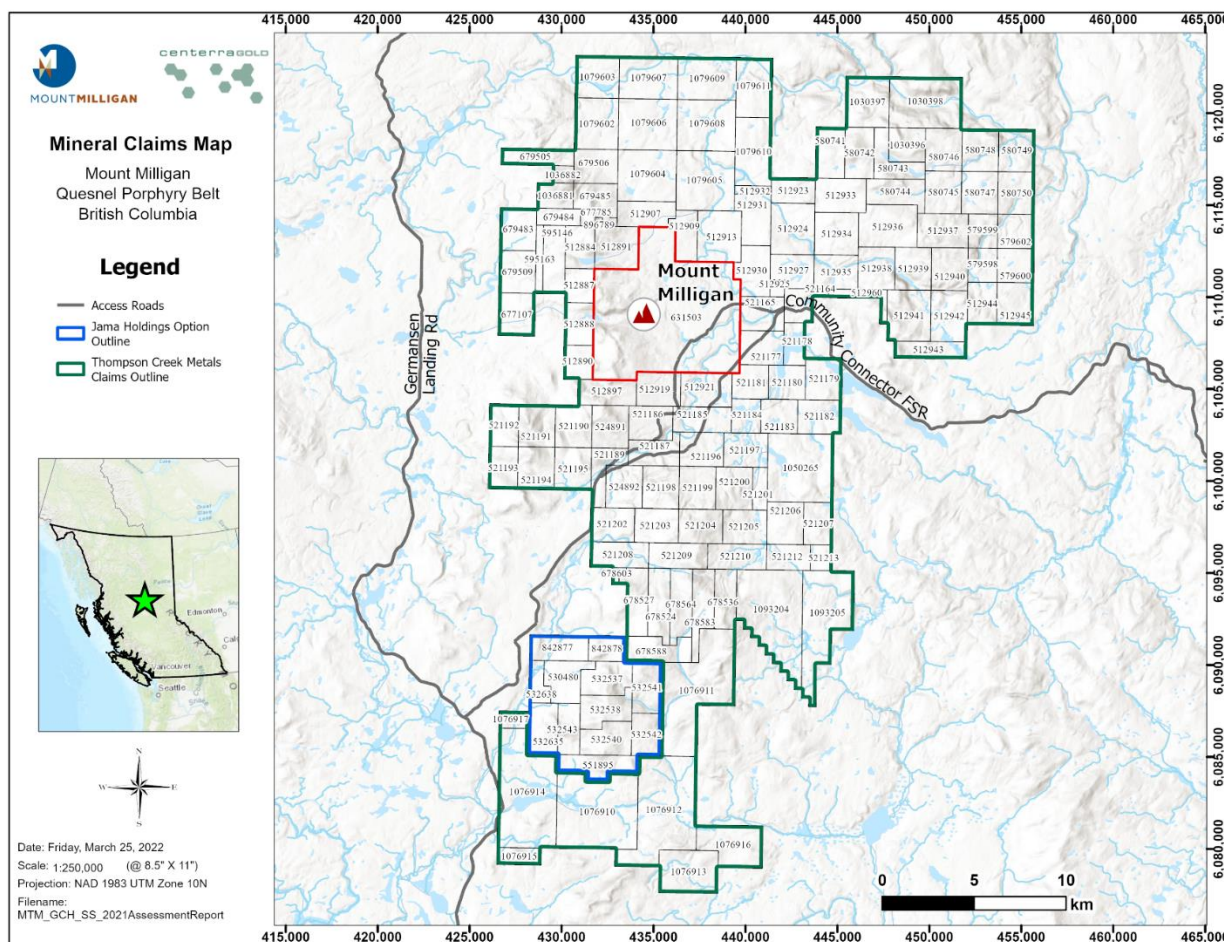
No other external consultants, aside from named QPs in Item 2.2, have provided content for this Technical Report.

4. PROPERTY DESCRIPTION AND LOCATION

4.1. PROPERTY DESCRIPTION

The Mount Milligan mine is situated in Mining Lease 631503 surrounded by many mineral claims which form the Mount Milligan Property. Figure 4-1 is a plan view of Mineral Tenure Map.

Figure 4-1: Mount Milligan Mineral Tenure Map



The Property comprises 121 mineral claims, covering a total area of 61,891.9 ha, the details of which are depicted on Table 4-1.

Table 4-1: List of Mineral Tenures as of March 2022

Project	Title Number	Claim Name	Owner	Title Sub Type	Issue Date	Good Date To	Status	Area (ha)
Mount Milligan	631503		TCM (100%)	Lease	2009-09-09	2022-09-09	GOOD	5138
Mount Milligan	512884		TCM (100%)	Claim	2005-05-18	2028-01-13	GOOD	369.632
Mount Milligan	512887		TCM (100%)	Claim	2005-05-18	2028-01-13	GOOD	295.844
Mount Milligan	512888		TCM (100%)	Claim	2005-05-18	2028-01-13	GOOD	369.979
Mount Milligan	512890		TCM (100%)	Claim	2005-05-18	2028-01-13	GOOD	296.121
Mount Milligan	512891		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	554.449
Mount Milligan	512897		TCM (100%)	Claim	2005-05-18	2028-01-13	GOOD	444.34
Mount Milligan	512907		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	424.903
Mount Milligan	512909		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	351.094
Mount Milligan	512913		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	665.236
Mount Milligan	512919		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	444.319
Mount Milligan	512921		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	518.369
Mount Milligan	512923		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	332.428
Mount Milligan	512924		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	665.165
Mount Milligan	512925		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	73.961
Mount Milligan	512927		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	406.695
Mount Milligan	512930		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	480.648
Mount Milligan	512931		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	480.341
Mount Milligan	512932		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	92.341
Mount Milligan	512933		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	517.134
Mount Milligan	512934		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	554.332
Mount Milligan	512935		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	443.673
Mount Milligan	512936		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	720.559
Mount Milligan	512937		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	517.346
Mount Milligan	512938		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	462.136
Mount Milligan	512939		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	462.135
Mount Milligan	512940		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	462.134
Mount Milligan	512941		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	665.851
Mount Milligan	512942		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	554.875
Mount Milligan	512943		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	370.069
Mount Milligan	512944		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	369.861
Mount Milligan	512945		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	462.324
Mount Milligan	512960		TCM (100%)	Claim	2005-05-18	2026-03-18	GOOD	203.414
Mount Milligan	521164	MILL 1	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	332.887
Mount Milligan	521165	MILL 2	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	443.905
Mount Milligan	521177	MILL 3	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	444.089
Mount Milligan	521178	MILL 4	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	277.539
Mount Milligan	521179	MILL 5	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	462.756
Mount Milligan	521180	MILL 6	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	370.225
Mount Milligan	521181	MILL 7	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	351.719
Mount Milligan	521182	MILL 8	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	444.449
Mount Milligan	521183	MILL 9	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	370.374
Mount Milligan	521184	MILL10	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	296.301
Mount Milligan	521185	MILL 11	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	444.471
Mount Milligan	521186	MILL 12	TCM (100%)	Claim	2005-10-14	2028-01-13	GOOD	444.496
Mount Milligan	521187	MILL 13	TCM (100%)	Claim	2005-10-14	2028-01-13	GOOD	407.598
Mount Milligan	521189	MILL 14	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	370.632
Mount Milligan	521190	MILL 15	TCM (100%)	Claim	2005-10-14	2028-01-13	GOOD	463.037
Mount Milligan	521191	MILL 16	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	463.038
Mount Milligan	521192	MILL 17	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	370.431
Mount Milligan	521193	MILL 18	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	370.621
Mount Milligan	521194	MILL 19	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	463.276
Mount Milligan	521195	MILL 20	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	463.276
Mount Milligan	521196	MILL 21	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	444.632
Mount Milligan	521197	MILL 22	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	444.635
Mount Milligan	521198	MILL 23	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	463.375
Mount Milligan	521199	MILL 24	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	463.374



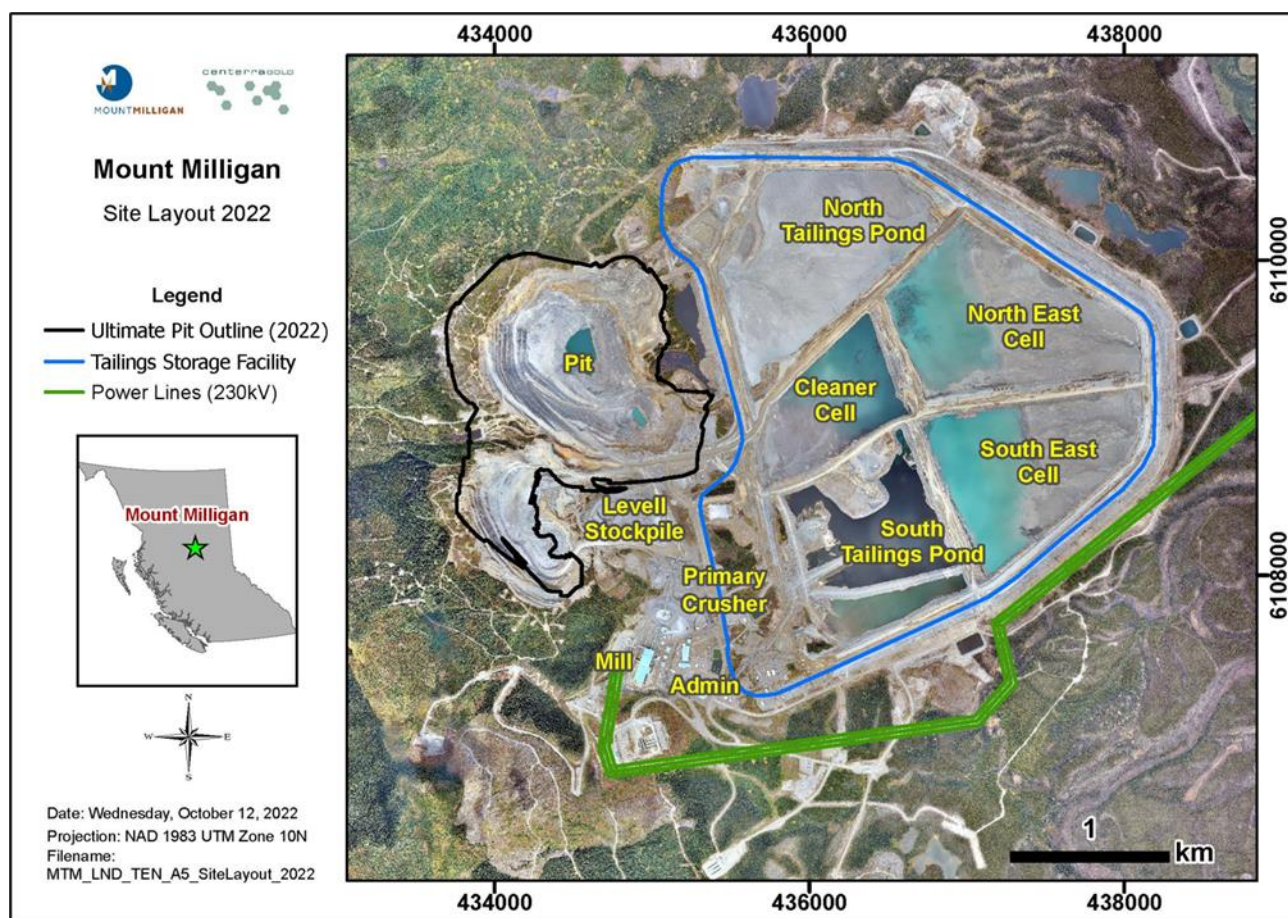
TECHNICAL REPORT ON THE MOUNT MILLIGAN MINE

Mount Milligan	521200	MILL 25	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	463.377
Mount Milligan	521201	MILL 26	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	185.351
Mount Milligan	521202	MILL 27	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	445.045
Mount Milligan	521203	MILL 28	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	445.047
Mount Milligan	521204	MILL 29	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	445.047
Mount Milligan	521205	MILL 30	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	445.049
Mount Milligan	521206	MILL 31	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	463.565
Mount Milligan	521207	MILL 32	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	370.852
Mount Milligan	521208	MILL 33	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	445.206
Mount Milligan	521209	MILL 34	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	445.207
Mount Milligan	521210	MILL 35	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	445.21
Mount Milligan	521212	MILL 36	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	333.905
Mount Milligan	521213	MILL 37	TCM (100%)	Claim	2005-10-14	2026-03-18	GOOD	166.952
Mount Milligan	524891	ARM	TCM (100%)	Claim	2006-01-08	2026-03-18	GOOD	463.039
Mount Milligan	524892	STRONG	TCM (100%)	Claim	2006-01-08	2026-03-18	GOOD	463.374
Mount Milligan	579598		TCM (100%)	Claim	2008-03-28	2026-03-18	GOOD	295.7519
Mount Milligan	579599		TCM (100%)	Claim	2008-03-28	2026-03-18	GOOD	295.6275
Mount Milligan	579600		TCM (100%)	Claim	2008-03-28	2026-03-18	GOOD	369.6889
Mount Milligan	579602		TCM (100%)	Claim	2008-03-28	2026-03-18	GOOD	369.5332
Mount Milligan	580741		TCM (100%)	Claim	2008-04-08	2026-03-18	GOOD	443.0304
Mount Milligan	580742		TCM (100%)	Claim	2008-04-08	2026-03-18	GOOD	443.0297
Mount Milligan	580743		TCM (100%)	Claim	2008-04-08	2026-03-18	GOOD	406.1485
Mount Milligan	580744		TCM (100%)	Claim	2008-04-08	2026-03-18	GOOD	461.7058
Mount Milligan	580745		TCM (100%)	Claim	2008-04-08	2026-03-18	GOOD	461.699
Mount Milligan	580746		TCM (100%)	Claim	2008-04-08	2026-03-18	GOOD	461.4626
Mount Milligan	580747		TCM (100%)	Claim	2008-04-08	2026-03-18	GOOD	461.6993
Mount Milligan	580748		TCM (100%)	Claim	2008-04-08	2026-03-18	GOOD	461.4618
Mount Milligan	580749		TCM (100%)	Claim	2008-04-08	2026-03-18	GOOD	461.4602
Mount Milligan	580750		TCM (100%)	Claim	2008-04-08	2026-03-18	GOOD	461.6977
Mount Milligan	595146		TCM (100%)	Claim	2008-12-01	2026-03-18	GOOD	443.6279
Mount Milligan	595163		TCM (100%)	Claim	2008-12-01	2026-03-18	GOOD	147.8759
Mount Milligan	677107	FURB	TCM (100%)	Claim	2009-12-01	2026-03-18	GOOD	462.4242
Mount Milligan	677785		TCM (100%)	Claim	2009-12-02	2026-03-18	GOOD	147.8006
Mount Milligan	678524		TCM (100%)	Claim	2009-12-03	2026-03-18	GOOD	464.0154
Mount Milligan	678527		TCM (100%)	Claim	2009-12-03	2026-03-18	GOOD	464.0028
Mount Milligan	678536		TCM (100%)	Claim	2009-12-03	2026-03-18	GOOD	389.7479
Mount Milligan	678564		TCM (100%)	Claim	2009-12-03	2026-03-18	GOOD	464.014
Mount Milligan	678583		TCM (100%)	Claim	2009-12-03	2026-03-18	GOOD	464.0256
Mount Milligan	678588		TCM (100%)	Claim	2009-12-03	2026-03-18	GOOD	464.2712
Mount Milligan	678603		TCM (100%)	Claim	2009-12-03	2026-03-18	GOOD	55.663
Mount Milligan	679483		TCM (100%)	Claim	2009-12-05	2026-03-18	GOOD	461.9455
Mount Milligan	679484		TCM (100%)	Claim	2009-12-05	2028-01-13	GOOD	221.7012
Mount Milligan	679485		TCM (100%)	Claim	2009-12-05	2026-03-18	GOOD	350.9391
Mount Milligan	679505		TCM (100%)	Claim	2009-12-05	2026-03-18	GOOD	369.2328
Mount Milligan	679506		TCM (100%)	Claim	2009-12-05	2026-03-18	GOOD	443.1255
Mount Milligan	679509		TCM (100%)	Claim	2009-12-05	2026-03-18	GOOD	462.1832
Mount Milligan	896789	MILL 9	TCM (100%)	Claim	2011-09-13	2026-03-18	GOOD	18.476
Mount Milligan	1030396	GD1	TCM (100%)	Claim	2014-08-19	2026-03-18	GOOD	369.1532
Mount Milligan	1030397	GD2	TCM (100%)	Claim	2014-08-19	2026-03-18	GOOD	664.135
Mount Milligan	1030398	GD3	TCM (100%)	Claim	2014-08-19	2026-03-18	GOOD	1106.8883
Mount Milligan	1036881	DB1	TCM (100%)	Claim	2015-06-23	2028-01-13	GOOD	277.0547
Mount Milligan	1036882	DB2	TCM (100%)	Claim	2015-06-23	2026-03-18	GOOD	110.7931
Mount Milligan	1050265		TCM (100%)	Claim	2017-02-24	2026-03-18	GOOD	1334.1725
Mount Milligan	1079602		TCM (100%)	Claim	2020-11-16	2023-11-04	GOOD	664.3425
Mount Milligan	1079603		TCM (100%)	Claim	2020-11-16	2023-11-04	GOOD	553.3019
Mount Milligan	1079604		TCM (100%)	Claim	2020-11-16	2023-11-04	GOOD	886.3434
Mount Milligan	1079605		TCM (100%)	Claim	2020-11-16	2023-11-04	GOOD	1034.1171
Mount Milligan	1079606		TCM (100%)	Claim	2020-11-16	2023-11-04	GOOD	885.7907
Mount Milligan	1079607		TCM (100%)	Claim	2020-11-16	2023-11-04	GOOD	737.7364
Mount Milligan	1079608		TCM (100%)	Claim	2020-11-16	2023-11-04	GOOD	885.7897
Mount Milligan	1079609		TCM (100%)	Claim	2020-11-16	2023-11-04	GOOD	737.7373

Mount Milligan	1079610		TCM (100%)	Claim	2020-11-16	2023-11-04	GOOD	738.3865
Mount Milligan	1079611		TCM (100%)	Claim	2020-11-16	2023-11-04	GOOD	645.5868
Mount Milligan	1093204		TCM (100%)	Claim	2022-02-10	2023-02-10	GOOD	1670.67
Mount Milligan	1093205		TCM (100%)	Claim	2022-02-10	2023-02-10	GOOD	1373.74
Max	1076910		TCM (100%)	Claim	2020-06-23	2026-03-18	GOOD	1786.3232
Max	1076911		TCM (100%)	Claim	2020-06-23	2026-03-18	GOOD	1857.9835
Max	1076912		TCM (100%)	Claim	2020-06-23	2026-03-18	GOOD	1860.6661
Max	1076913		TCM (100%)	Claim	2020-06-23	2026-03-18	GOOD	744.813
Max	1076914		TCM (100%)	Claim	2020-06-23	2026-03-18	GOOD	1860.4002
Max	1076915		TCM (100%)	Claim	2020-06-23	2026-03-18	GOOD	223.4167
Max	1076916		TCM (100%)	Claim	2020-06-23	2026-03-18	GOOD	725.9499
Max	1076917		TCM (100%)	Claim	2020-06-23	2026-03-18	GOOD	148.6954

The mine site comprises an open pit mine, TSF, mineralized stockpiles, a processing plant, workshop, warehouse, administration buildings, and camp. Figure 4-2 provides a plan view of the Project.

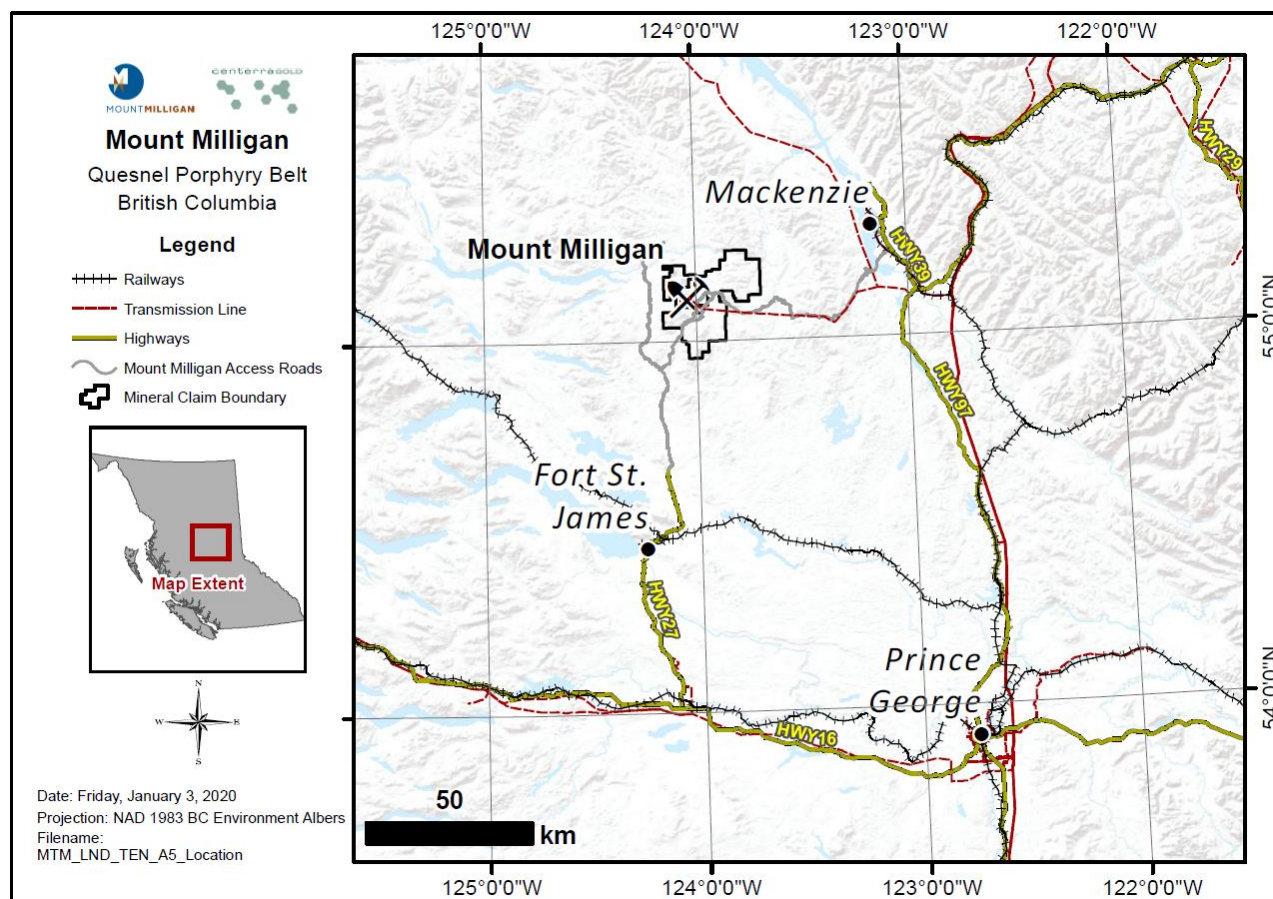
Figure 4-2: Plan View of Mount Milligan Mine Site



4.2. LOCATION

The Mount Milligan Mine is in north-central British Columbia, Canada, as shown in Figure 4-3.

Figure 4-3: Mount Milligan Mine Location



4.3. ROYALTIES AND STREAMING AGREEMENTS

H.R.S. Resources Royalty

In accordance with an option agreement dated July 16, 1986, as amended, between H.R.S. Resources Corp., successor in interest to Richard Haslinger, and TCM as the successor in interest to Goldcorp Canada Ltd., H.R.S. Resources Corp. is entitled to a royalty payment equivalent to a 2% NSR on production from four mineral claims collectively called the HEIDI claims. The HEIDI claims form a portion of the mining lease. In accordance with the terms of the royalty agreement, the royalty has been payable since June 25, 2016 (of the third year of commercial production). TCM has the right of first refusal on any proposed sale of the royalty by H.R.S. Resources Corp.

Stream Agreement with Royal Gold

Pursuant to an agreement, dated October 2010, as subsequently amended in December 2011, August 2012, December 2014, and October 20, 2016 (the Stream Agreement), with RGLD AG and Royal Gold, Inc. (collectively Royal Gold), TCM has agreed to sell to Royal Gold 35% of the gold produced and 18.75% of the copper production at the Mount Milligan Mine. Royal Gold pays \$435 per ounce of gold delivered and pays 15% of the spot price per metric tonne of copper delivered. When the Stream Agreement was originally entered into by TCM (prior to the Acquisition by Centerra), TCM received an upfront payment of \$781.5 million for the rights to receive future gold production. The Stream Agreement was amended by TCM as part of the Acquisition to reduce the percentage of gold production allocated to Royal Gold (from 52.25% to 35%) and to include 18.75% of the copper production from the Mount Milligan mine.

The Stream Agreement has an initial 50-year term, with automatic successive 10-year renewal periods.

TCM sells copper and gold concentrate from Mount Milligan Mine to customers and in connection with such sales, TCM purchases gold ounces and copper warrants in the market for delivery to Royal Gold in an amount based on a portion of the gold and copper content contained in the copper and gold concentrate sold to customers.

The Stream Agreement covers substantially the entire Property.

The Stream Agreement includes certain restrictions on assignment or transfer of the respective rights of both parties to the Stream Agreement.

BCGold Royalty

Pursuant to an agreement dated June 30, 2015, Terrane Metals Corp. agreed to pay BCGold Corp. a royalty equal to 2.5% of the NSR on all products regarding two mineral tenures, numbers 524891 and 524892, south of the Mount Milligan mining lease, historically referred to as BCGold's 'Rainbow Claims'. Terrane purchased the properties in exchange for cash (CA\$35,000) in addition to the NSR royalty and agreed to be responsible for any costs or acts required to keep the mineral titles in good standing. These tenures are outside of the current Mine Plan but within the greenfield exploration target areas.

5. ACCESSIBILITY, CLIMATE, LOCAL RESOURCES, INFRASTRUCTURE, AND PHYSIOGRAPHY

5.1. ACCESS

The Mount Milligan Mine is accessible from the east via Mackenzie on the Finlay Philip Forest Service Road and the North Philip Forest Service Road. These roads are maintained in good condition by the various user groups. A western route, completed in 2005, provides shorter access from Fort St. James via the North Germansen Road. This route includes 26.7 km of forest service roads, with the balance on public roads. Road travel to the site of the Mount Milligan Mine is 254 km from Prince George (population approximately 79,000).

5.2. CLIMATE

The area has short, cool summers and cold winters, which precipitation and temperatures are summarized on Table 5-1.

Table 5-1: Mount Milligan Climate Statistics

	Month												
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
Average Precipitation (mm)													
Rainfall	0.3	1.9	4.0	15.9	54.8	58.3	66.3	50.1	67.1	52.4	12.6	1.0	385.0
Snowfall	78.5	64.0	50.5	19.6	6.4	0.8	0.8	0.8	2.8	16.3	68.6	76.0	385.0
Total Precipitation	78.8	65.9	54.5	35.4	61.2	59.2	67.2	50.9	69.9	68.7	81.1	77.0	770.0
Lake Evaporation	-	-	-	42.2	69.2	81.5	84.7	68.4	37.0	7.1	-	-	390.0
Average Monthly Temperature (°C)													
Mean	-10.9	-8.1	-3.7	1.6	7.0	11.3	13.4	12.8	8.0	2.3	-4.7	-9.2	1.7

Source: 2009 technical report for the Property ("2009 Report"), verified Centerra, 2017

Mining and processing activities can occur throughout the year.

5.3. LOCAL RESOURCES

Labour and services are available from the surrounding towns of Prince George, Fort St. James, Mackenzie, Vanderhoof, Smithers and Fraser Lake.

5.4. INFRASTRUCTURE

Infrastructure available to the Project comprises the main forest service roads accessing the Property from the east and west. Electric power is accessed from the BC Hydro Kennedy Substation south of Mackenzie. Canadian National Railway service is available from Fort St. James and Mackenzie, which connects to the major western and eastern rail routes. Please see Item 18 for additional information regarding infrastructure.

5.5. PHYSIOGRAPHY

The Property lies near the northern boundary of the Southern Plateau and Mountain Region of the Canadian Cordilleran Interior System. More specifically, the Property is within the Nechako Plateau near the southern limits of the Swannell Range of the Omineca Mountains. The Property is dominated by a chain of peaks aligned in a north-south direction. Mount Milligan, which is 8 km north of the Project, is the highest of these peaks. It rises to the elevation of 1,508 m and is rounded and symmetrical in shape. The Project deposit is to the south of Mount Milligan on the eastern slopes of the chain, at an elevation of 1,100 m in an area of gentle relief.

The Nechako Plateau was covered by the Cordilleran ice cap, which moved eastward and northward from the Coast Ranges towards the Rocky Mountains near McLeod Lake, over-riding the mountains, coating the landscape with a blanket of glacial till, and altering the pre-glacial drainage patterns. Drumlins, flutings, eskers, and melt-water channels of various dimensions are noticeable features of the plateau surface. The Property is well-drained except for depressions where natural vegetation succession has filled in ponds to form bog-like fens. Drainage from the area is to the northeast via Nation River into Williston Lake, which forms part of the Peace-Mackenzie River basin.

6. HISTORY

6.1. PROPERTY OWNERSHIP

Table 6-1 summarizes historical ownership, mineral exploration and development activities of the area that has become the Mount Milligan mine and property in northcentral British Columbia.

Table 6-1: Mount Milligan Historical Ownership and Development Timeline

1929	Placer gold and platinum was discovered on Rainbow Creek by George Snell
1931	Whole length of Rainbow Creek (over 40 km) staked and worked by nearly 100 men, gold was flat, well-worn, and not of local origin (Galloway, 1931)
1937	Prospecting in the geographical Mount Milligan area, George Snell
1972	Pechiney Development Ltd. drilled 5 holes to test geophysical and geochemical anomalies
1984-1986	R. Haslinger staked mineral claims; optioned claims to Selco-BP (in 1986 signed option agreement with Lincoln Resources Inc.)
1987	MBX Main deposit discovery; Lincoln Resources Inc. #12 drill hole
1989	Southern Star deposit discovery; Lincoln Resources-Continental Gold Corp and JV with BP Resources
1990	Placer Dome purchased Mount Milligan
1991	Pre-Feasibility Study completed
1992	Intended to develop the mine in 1993 but project was determined to be sub-economic due to low metal prices
1993	Obtained permits required for commercial production at 60,000 tpd that expired in 2003
1996/1998	Re-evaluations completed
2004/2005	Project re-evaluation; drilling for metallurgical test work; updated resource block model; laboratory studies identified potential for improved metallurgy
2006	Barrick Gold Corp. purchased Placer Dome and sold Canadian assets to Goldcorp Inc.; Goldcorp sold Mount Milligan to Terrane Metals Corp. (Atlas Cromwell Ltd.); feasibility study and permitting process commenced
2008	Terrane completes Mount Milligan feasibility study and prepares to file Environmental Assessment (EA) application
2009	EA approved (March); <i>Mines Act</i> permit received (September)
2010	Thompson Creek Metals Company Inc. (TCM) acquired the development project through acquisition of Terrane; construction began mid-2010
2013	Mine commissioned (October)
2014	Commercial production achieved
2016	Completed ramp-up (January); secondary crusher construction and commissioning; TCM acquired by Centerra Gold Inc.

6.2. HISTORICAL EXPLORATION AND DEVELOPMENT ACTIVITIES (1937-2011)

1937-1972

In 1937, prospector George Snell found gold bearing float on the southwest flank of the geographical Mount Milligan (~8 km NNW of the Mount Milligan deposit). In 1945, Mr. Snell returned to the area and staked 10 two-post claims west of Mitzi Lake. Five pyritic andesite float samples returned assays ranging from trace to 150.2 g/t Au (4.38 ounces/ton; Nelson and Bellefontaine, 1996). The source of the float was not located, and no other gold-bearing mineralization was found in place.

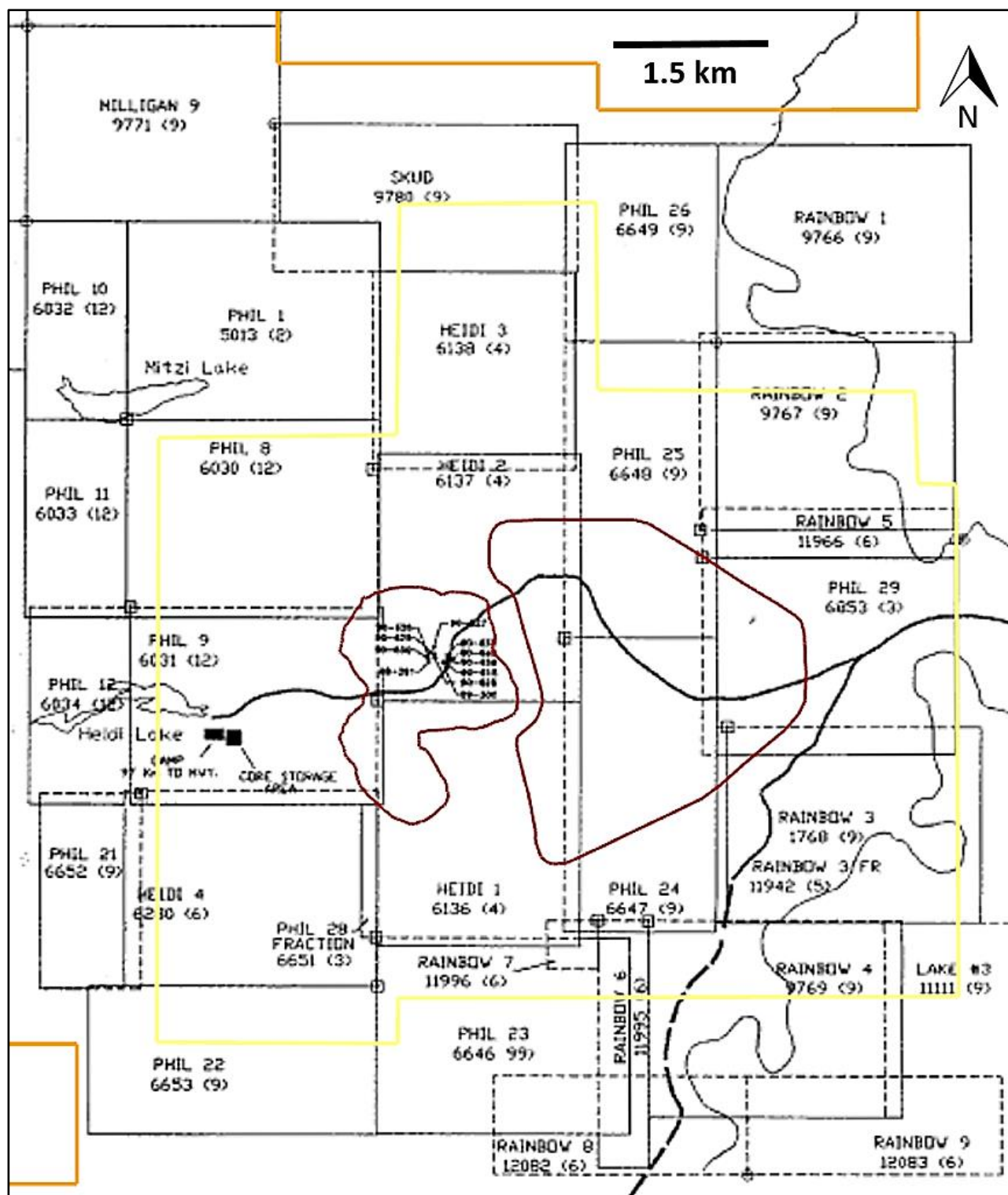
The first recorded claims in the Mount Milligan region were the Mosquito 1-10 two-post claims staked on August 4, 1972, by Pechiney Development Ltd. (Pechiney). Subsequent exploration work identified induced polarization (IP) and soil geochemical anomalies. Pechiney drilled five drill holes to evaluate the anomalies but identified no significant mineralization and allowed the claims to lapse.

1983-1989

In 1983, Selco Inc. ("Selco") staked the Phil 1 through 12 claims (Figure 6-1) in what is now the Brownfield-Greenfield exploration area west of the mine on the Mount Milligan property (also the ground covered by the original Mosquito claims). Preliminary surveys were completed. Additionally, during that summer, prospector Richard Haslinger discovered copper-gold mineralization in bedrock exposed in a creek in what is now called the Saddle zone (formerly Creek zone). In early 1984, Selco amalgamated with BP Resources Canada Ltd. ("BP Resources"). In April 1984, Richard Haslinger staked the Heidi claims to the east and adjacent with the Phil claims (Figure 6-1). Most of the current open pit is located on historical Heidi claims 1 and 2. BP Resources optioned the Heidi claims from Richard Haslinger in July 1984. In late 1984 and early 1985, BP Resources staked the Phil 21 through 29 claims south and east of the Heidi claims (Figure 6-1).

Systematic exploration near the Mount Milligan porphyry Au-Cu deposit area began in 1983. This region has sparse outcrop with most of the bedrock covered by a blanket of glacial till generally varying between <1 to 50 m in thickness with a maximum of about 100 m thickness; nonetheless, several geologic mapping programs in the project area were completed. Between 1983 and 1985, BP Resources, through their Selco Division, completed initial reconnaissance surface geochemical surveys and revealed an extensive area of anomalous copper and gold in the deposit area.

Figure 6-1: Historical Claims of the Mount Milligan Property



Note: (ARIS report 252299) with 2017 ultimate pit and TSF boundaries (dark red), current mine lease boundary (pale yellow) and property boundary (orange)

Follow up detailed mapping was completed in areas of anomalous gold-in-soil (Blanchflower, 1986). Around this time trenching, totalling 1,400 m, was completed by BP Resources focusing on the Creek, Boundary, and Esker zones, where pervasive sericite, epidote, and K-feldspar alteration was encountered with anomalous gold and copper values. The work identified polymetallic auriferous vein systems and weak Au-Cu porphyry mineralization near Heidi Lake. Geophysical programs completed by BP Resources over the deposit area included initial ground magnetic surveys in 1984 to 1985, very low frequency (VLF) electromagnetic surveys in 1985, and induced polarization (IP) surveys in 1985 and 1987. This work identified a broad coincident geophysical anomaly interpreted to be resulting from disseminated sulphides and associated alteration minerals.

On April 21, 1986, Lincoln Resources Inc. (Lincoln) entered into an agreement with BP Resources to continue exploration of the claims. The agreement allowed Lincoln to earn a 51% interest in the property, which was subsequently increased to 69.84% through the operation of dilution provisions. In July 1986, Lincoln entered into a new option agreement with Richard Haslinger on the Heidi claims. Further trenching was completed by Lincoln in 1986 and 1987, concentrating on the Esker, Creek, South Boundary, and North Slope zones.

In 1987, Lincoln commenced drilling (Figure 6-2), following up on targets identified during the BP Resources and Lincoln geophysical, geochemical, and trenching programs. Esker was the first zone drilled. This program's test results indicated moderate to high (1.33 to 42.70 g/t) gold grades over narrow widths. The Creek Zone similarly intersected low to moderate gold grades, generally in the range of 1 to 2 g/t over widths of 1 to 2 m.

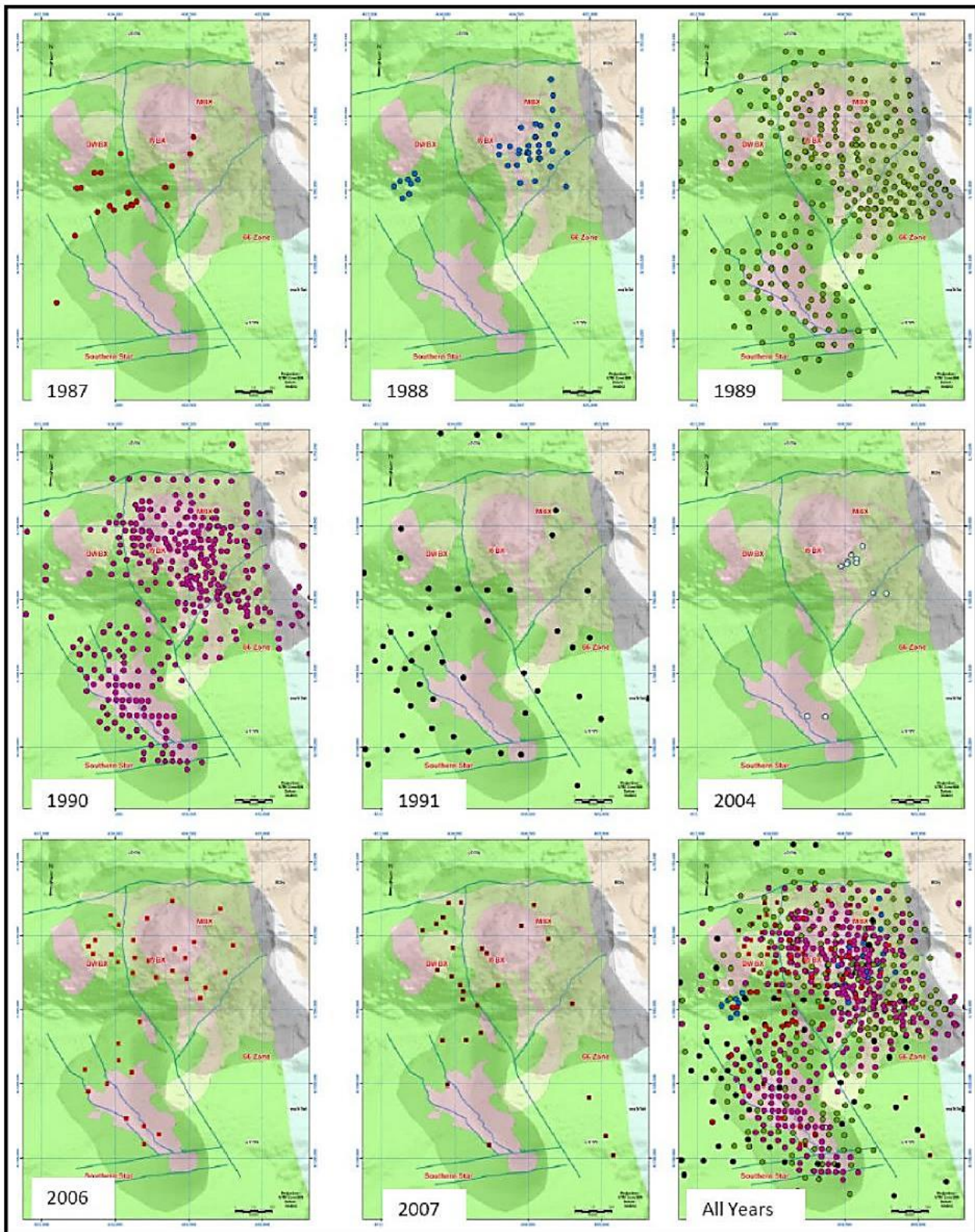
The MBX zone was initially targeted due to the presence of a magnetic anomaly located by reconnaissance ground magnetic surveys conducted by BP Resources and later by Lincoln (Rebagliati, 1987). The 1987 Lincoln survey was a detailed 31.1 km (50 m x 10 m) ground magnetic survey that outlined the MBX zone. There were also coincident IP and soil geochemical anomalies associated with the magnetic features. In September 1987, Lincoln undertook a drilling campaign to test targets identified by BP Resources. This work resulted in the first discovery of significant Au-Cu mineralization in the MBX zone (Rebagliati, 1988). Diamond drill hole 87-12 intersected intensely potassic altered monzonite dykes and volcanic rocks with disseminated pyrite and chalcopyrite, returning 50.51 metres grading 0.60 g/t gold and 0.24% copper. Hole 87-13 intersected 97 metres grading 0.62 g/t gold and 0.27% copper.

Exploration conducted by United Lincoln in 1988 included additional diamond drilling in the Esker and Creek zones but focused primarily on the porphyry Au-Cu style mineralization intersected in the MBX zone. On July 31, Lincoln reorganized to become United Lincoln Resources Inc. (United

Lincoln). In September, United Lincoln staked the Milligan, Rainbow 1 through 4, the SKUD mineral claims (Figure 6-1). In August 1988, Continental Gold Corp. ("Continental") acquired 64% of the shares of United Lincoln. On March 15, 1989, Continental and United Lincoln amalgamated and concurrently transferred the amalgamated undertaking to their subsidiary and successor company, DASS No. 39 Holdings Ltd., which changed its name to Continental Gold Corp. ("Continental") on the same date.

A major drilling program was launched in 1989 that included 87,662 m of diamond drilling in 336 holes. Hole 89-200 intersected mineralization of the Southern Star deposit to the south of the MBX Main deposit. The 1989 program outlined the known principal zones of the project, including the high-gold low-copper 66 zone (named after drill hole 88-66), the zone along the western margin of the MBX stock (WBX zone), and a deeper downthrown portion of the WBX, the DWBX zone, in the footwall block of the Harris fault. An airborne geophysical survey was also flown by Aerodat Limited in 1989, covering the area north and west of the deposit area; it consisted of magnetics and VLF-EM, flown on east-west oriented flight lines at 100-m spacing.

Figure 6-2: Drill Hole Location by Year (1987-2007)



Source: 2009 Technical Report for the Property, verified by Centerra, 2017

1990-1998

In 1990, Continental continued staking and acquiring claims in the region, including the RAINBOW 5 through 9, RAINBOW 3 Fraction, BEE and SEE mineral claims, MBX 14 through 29, and RAIN placer claims. It also acquired the BONANZA, MARTIN, and TRNAVA mineral claims. An infill diamond drilling program was carried out from January to September (386 drill holes totalling 82,924 m). Of these, 352 were drilled by Continental and the remaining 34 were drilled by Placer Dome who acquired control of the property in late 1990. In September, Placer Dome Inc. ("Placer Dome") had purchased BP Resources' share of the Phil and Heidi mineral claims. Placer Dome, and their wholly owned subsidiary PDI Subco, then acquired approximately 98% of the shares of Continental by takeover bid, and in November resumed exploration drilling.

In January 1991, PDI Subco acquired the balance of the outstanding Continental shares. With these acquisitions, Placer Dome became the primary proponent of the project and continued the process of seeking regulatory approval. In April 1991, a PFS was produced for the development of a 60,000-tpd open pit mine and flotation process plant. Placer Dome completed a 90-hole (17,969 m) drill program focused on the Southern Star deposit with additional exploration drilling to the west of Southern Star and north of Heidi Lake. Geotechnical, metallurgical, and condemnation holes were also completed during this period.

Other exploration activities in 1991 included a regional mapping program completed by consultant Atholl Sutherland-Brown and a trenching program conducted by Placer Dome geologists in the North Slope area, north of Heidi Lake. Geophysical work included IP surveys north of the BP Resources survey area and VLF surveys north of the deposit and in the Phillips Lake area. The IP surveys primarily outlined the pyritic haloes surrounding the intrusions and mineralized zones. Ground magnetic surveys were also completed in 1991 north of the deposit area and at Phillips Lake. Results showed both the MBX and Southern Star stocks are positive magnetic anomalies, although the MBX stock lies immediately south of a broad (5.5 km²) magnetic high anomaly that is interpreted to be related to a southern extension of the Mount Milligan Intrusive Complex, mapped about 5 km to the north of the Main deposit area (Nelson and Bellefontaine, 1996). Also, around this time, a small helicopter-borne DIGHEM survey was flown. The survey collected both magnetic and EM data; the EM data exhibits a pattern very similar to the IP ground surveys.

In 1992, it was determined the project was sub-economic and Placer Dome wrote-off the carried value of the property. Drilling that year was completed in the Phillips Lake area in the eastern part of the property to test some IP and magnetic targets. No mineralization was intersected.

In 1993, Placer Dome received provincial and federal approvals to develop the project (these approvals expired in 2003).

In 1996, Placer Dome re-evaluated the project and developed an updated geological model that included new domains and hard boundaries. Test pits were excavated into the bedrock surface to obtain additional geotechnical information.

In 1998, an economic re-evaluation was completed using the 1996 model for the MBX Main deposit and the 1991 model for Southern Star. A variety of alternate mining and processing scenarios were investigated.

2004-2011

Exploration resumed in 2004 when Placer Dome compiled and reprocessed historical data (geological, geochemical, geophysical) into a geographic information system (GIS) database. A three-dimensional (3D) geological model was developed, and enhanced interpretation studies were completed. Pulps from previous drilling programs were analysed with an ASD reflectance spectrometer to obtain alteration mineral spectra to aid in geological modelling (Lustig, 2007). A 14-hole (2,184 m) diamond drilling program was conducted for additional metallurgical testing.

In 2005, geochemical studies were carried out including a regional stream sediment sampling program to assess the downstream dispersion from the Mount Milligan deposit using several analytical and sampling techniques including Bulk Leach Extractable Gold (BLEG). Additionally, an M.Sc. study was initiated through UBC/MDRU to investigate alteration zoning and geochemical dispersion with the objective of developing a 3D alteration model.

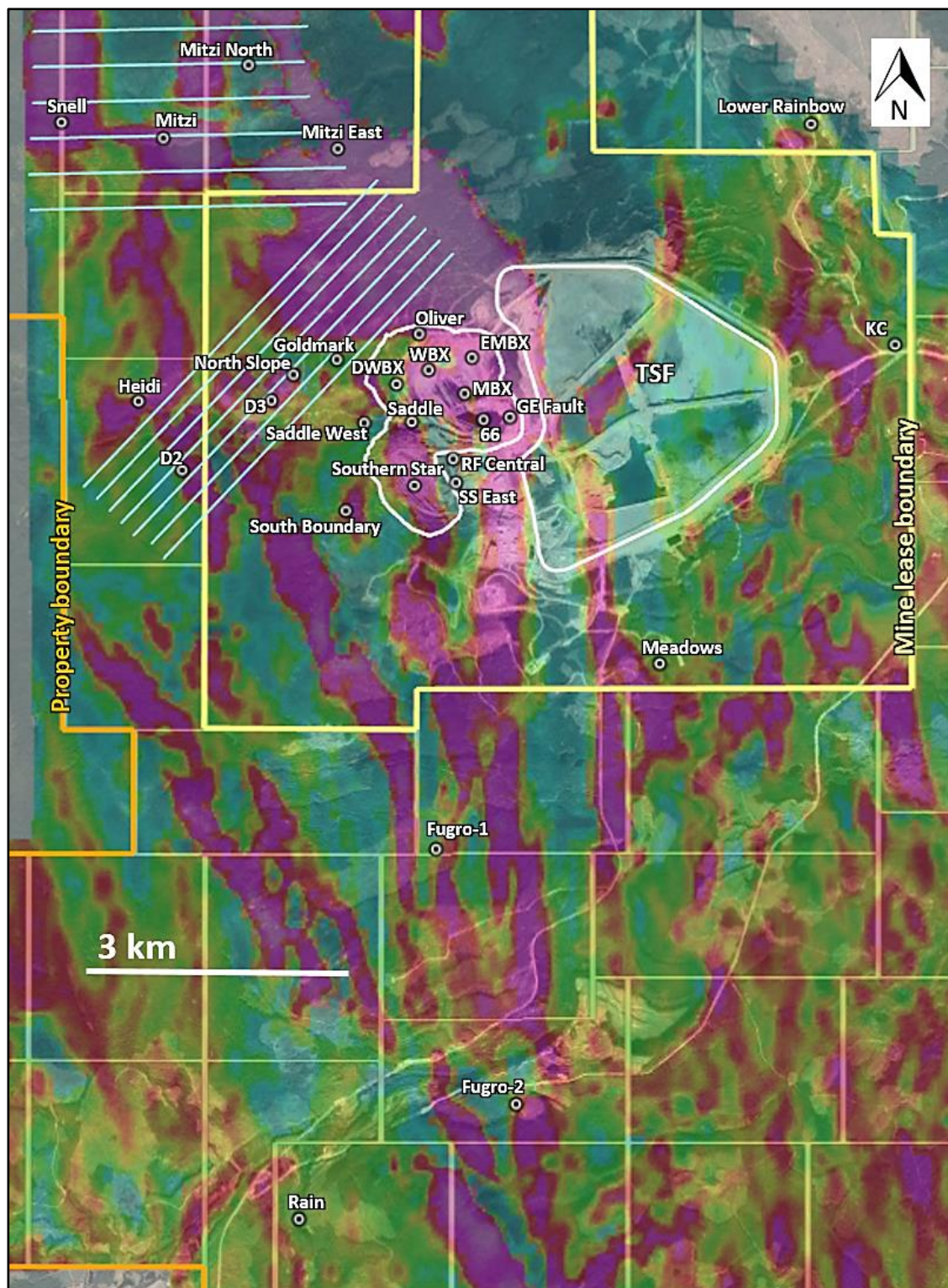
In February 2006, Placer Dome filed a NI 43-101 Technical Report that stated a Mineral Resource estimate (Placer Dome Inc. News Release dated February 20, 2006; Lustig, 2006). In May 2006, Barrick Gold Corporation purchased Placer Dome and sold Placer Dome's Canadian assets to Goldcorp Inc. (Goldcorp), including the Mount Milligan property. Goldcorp then sold certain assets (including Mount Milligan) to Atlas Cromwell Ltd. (Atlas Cromwell). In July 2006, Atlas Cromwell was renamed Terrane Metals Corporation (Terrane). In September, Terrane commenced a 36-hole (9,557 m) diamond drilling program to collect more material for metallurgical test work, to collect geological and geotechnical information along the margin of the MBX Main deposit, and infill drilling of target areas identified by previous operators.

In 2007, an additional 33-hole (10,515 m) diamond drilling program was completed for geotechnical purposes along the margin of the Southern Star deposit and to collect geological information in the planned mine infrastructure area.

In the summer seasons of 2008 and 2009, Terrane completed a mapping program, a regional geochemical stream survey and geophysical programs. In the mapping program, a total of 175 rock samples were collected from across the property, with 40 of these sent for thin section analysis and 42 sent for assays. Additionally, in May-June 2008, an airborne magnetics and HeliGEOTEM II electromagnetic survey consisting of 1,458 line-km was flown with coverage over 264 km² (66%) of the Mount Milligan property (Figure 6-3; Fugro Airborne Surveys Corp., 2008). A total of 101 traverse lines were flown with a line spacing of 200 m, and an additional 10 tie lines were flown with a line spacing of 2,000 m. This survey was designed to identify geophysical signatures characteristic of the project deposits in areas with thick overburden cover. The survey was successful in identifying numerous geophysical anomalies. In October 2008, Geotech Ltd. flew a ZTEM (Z-Tipper Axis Electromagnetic) survey over the Mount Milligan Test Block on behalf of Geoscience BC. The survey consisted of 25 approximately 8.0-km long, east-west oriented flight lines, totaling 200 line-km, that were obtained at nominal 250-m line spacings over an approximately 6- x 8-km area. The survey comprised airborne Tipper AFMAG (audio frequency electromagnetics) measurements, as well as aeromagnetism using a cesium magnetometer. ZTEM inversion results were produced in 2009 (Geotech Limited, 2009).

In 2009, ground-based IP surveys (53.6 line-km) were completed on two grids to investigate 12 of the HeliGEOTEM geophysical anomalies (Figure 6-3). The North Grid (20 line-km) comprised six east-west oriented lines at 400 m spacing; the South Grid (33.6 line-km) comprised seven SW-NE oriented lines at 200 m spacing. The 2D and 3D IP surveys (2DIP and 3DIP) were conducted by SJ Geophysics Ltd. correspondingly on the North and South grids (Hermiston, 2009). Survey results indicated that five of the HeliGEOTEM anomalies, including the Mitzi and Snell targets, had coincident aeromagnetic and IP chargeability signatures comparable to those associated with the MBX Main and Southern Star deposits. In October 2009, Terrane filed a NI 43-101 Technical Report with an updated Mineral Resource estimate (Wardrop, 2009).

Figure 6-3: Mount Milligan Exploration Targets



Google Earth (2019) image of Mount Milligan mine property, showing Resource Near-pit infill (NPI) and Exploration drilling target areas. Transparency overlay is airborne magnetic survey (1VD; Fugro Airborne Surveys, 2008). Pale blue grid lines are the North Grid (2DIP) and South Grid (3DIP) survey lines. The 2019 ultimate pit and TSF boundaries are shown in white lines. Property, mine lease and mineral claim boundaries are also shown.

In 2009 and 2010, Terrane completed a soil geochemical orientation survey over the Mount Milligan deposit, a soil geochemical survey over the North Grid to supplement the 2DIP survey, and a regional geochemical stream survey. The soil orientation survey comprised 97 sample sites, at which Soil Gas Hydrocarbon (SGH), Mobile Metal Ion (MMI) and conventional B-horizon soil samples were taken. A total of 46 sites were sampled in the stream sampling survey, with water, silt and BLEG samples collected at each site. A prominent copper-gold anomaly was outlined in the areas of the known mineralization, that also extended up-slope from the known deposit. The North Slope area is highly anomalous, but no significant near surface source was discovered to explain the full extent of the Cu-Au soil anomaly. A multi-element soil anomaly on Snell Hill, west of Mitzi Lake, exhibited a geochemical signature suggestive of a high-level intrusive source (Heberlein, 2010).

In 2010, Quantec Geoscience Ltd. completed a Titan-24 Direct Current resistivity (DC) and IP chargeability survey (9 lines; 35.8 line-km) and a 21.3 line-km Audio Magnetotelluric resistivity (MT) survey over the deposit area. Results identified 27 geophysical anomalies with potential for porphyry copper mineralization from near surface to >1,500 m depth, with nine high priority targets and seven secondary targets (Martinez Del Pino and Eadie, 2010). Nine diamond drill holes (4,944 m) were completed in the South Grid area (North Slope and D3 targets) during a September to November exploration drilling program. Narrow intervals of high-grade porphyry Au-Cu mineralization associated with potassic alteration (biotite-magnetite) and broader Au-rich, Cu-poor intersections with '66 zone' style alteration-mineralization (quartz-sericite-pyrite-carbonate; QSPC) were identified.

On October 20, 2010, TCM acquired the issued and outstanding equity of Terrane, including the Mount Milligan deposit and large land holdings. From November 2010 to March 2011, eight diamond drill holes (5,591 m) were completed to follow up previous drilling results in the footwall of the down-dropped MBX (DWBX) stock and other porphyry stocks (Goldmark and Unnamed stock). Results of the Titan-24 survey were used for targeting, and drilling tested for an extension of the DWBX zone to the west (Figure 6-3).

Table 6-2: Historical (1987-2011) Drilling Programs Summarized by Year

Year	Metres	Holes	Company
1987	2,304	23	Lincoln
1988	6,645	47	United Lincoln
1989	87,662	336	United Lincoln, Continental
1990	82,924	386	Continental, Placer Dome
1991	17,969	90	Placer Dome
1992	526	4	Placer Dome
2004	2,184	14	Placer Dome
2006	9,557	36	Terrane
2007	10,515	33	Terrane
2010-11	10,535	17	Terrane
Total	230,821	986	

The phased start-up of the mine commenced on August 15, 2013, and commercial production began on February 18, 2014. For details on exploration programs from 2014 onward, see the Exploration portion (Items 9 and Items 10) of this Technical Report.

On October 20, 2016, Centerra acquired the issued and outstanding common shares of Thompson Creek and, indirectly the Mount Milligan property.

6.3. HISTORICAL PRODUCTION

Mine waste stripping activities began in 2012, while mill commissioning began in Q3 2013. Commercial production was achieved in February 2014, defined as operation of the mill at 60% design capacity mill throughput for 30 consecutive days. Table 6-3 presents historical production tonnes, grade, recoveries and concentrate production for calendar years 2013 through 2021.

Table 6-3: Historical Production as of December 31, 2021

Years	Milled Ore Tonnage ('000 t)	Head Grade		Metal Recovery		Concentrate Production			Waste Tonnage ('000 t)
		Cu (%)	Au (g/t)	Cu Rec (%)	Au Rec (%)	Concentrate ('000 dmt)	Cu (M lb)	Au ('000 oz)	
2013	2,055	0.29	0.56	79.2%	54.3%	18.7	10.4	20.1	24,753
2014	14,290	0.27	0.63	80.4%	63.1%	125.4	68.0	184.0	11,224
2015	16,138	0.26	0.64	80.2%	68.6%	140.7	75.2	226.0	14,413
2016	19,277	0.19	0.58	74.7%	58.9%	125.6	61.6	212.0	20,363
2017	17,743	0.18	0.64	78.9%	62.4%	121.5	56.4	228.1	20,557
2018	13,556	0.20	0.71	81.4%	64.5%	106.0	49.6	199.5	19,764
2019	16,350	0.26	0.53	81.3%	67.4%	159.5	75.0	187.8	23,730
2020	20,067	0.26	0.41	77.40%	62.90%	185.3	87.3	166	22,043
2021	20,900	0.21	0.46	78.30%	65.80%	162.3	77	201.5	25,265
Total	99,410	0.23	0.62	79.5%	63.8%	797.3	396.2	1,257.4	134,804

All figures are shown on a 100% production basis without stream agreement deductions.

7. GEOLOGICAL SETTING AND MINERALIZATION

7.1. REGIONAL GEOLOGY

The Mount Milligan deposit is located within the Quesnel terrane of the North American Cordillera. The Quesnel terrane is part of the Intermontane Belt, a composite belt of volcanic arc and oceanic terranes that evolved outboard of the western margin of North America (Nelson et al., 2013). The terranes of the Intermontane Belt are interpreted to have accreted to North America during the Middle Jurassic (Mihalynuk et al., 2004).

The Quesnel terrane (Figure 7-1) represents a series of late Paleozoic to Jurassic island arcs. The oldest rocks assigned to the Quesnel terrane are Upper Paleozoic volcanic and sedimentary rocks of the Lay Range assemblage, overlain by the Middle to Upper Triassic Takla Group (Nelson and Bellefontaine, 1996). The Middle to Upper Triassic Takla Group is subdivided into a lower unit of basinal fine clastic sedimentary rocks, the Slate Creek Succession, and an upper unit of predominantly mafic to intermediate volcanic rocks, which is given a variety of names reflecting the presence of local volcanic centres and is referred to as the Witch Lake succession in the Mount Milligan deposit area (Nelson and Bellefontaine, 1996). The Takla Group in this area is overlain by intermediate volcanic rocks of the Lower Jurassic Chuchi Lake Succession. The Triassic and Early Jurassic volcanic packages have mildly alkaline, or shoshonitic, geochemical signatures (Barrie, 1993). The 180-km long Hogem intrusive complex represents the batholith core of the Quesnel terrane in northcentral BC. It generally evolved from more mafic peripheral to more felsic central phases, and from weakly alkaline to sub-alkaline compositions from the Late Triassic to Early Cretaceous; except for a more strongly alkaline Early Jurassic phase that includes the Chuchi syenite and Duckling Creek syenite complexes, known for being copper-gold prospective (Garnett, 1978; Devine et al., 2014). Mount Milligan is on trend with the Chuchi syenite complex at the southern end of the Hogem intrusive complex.

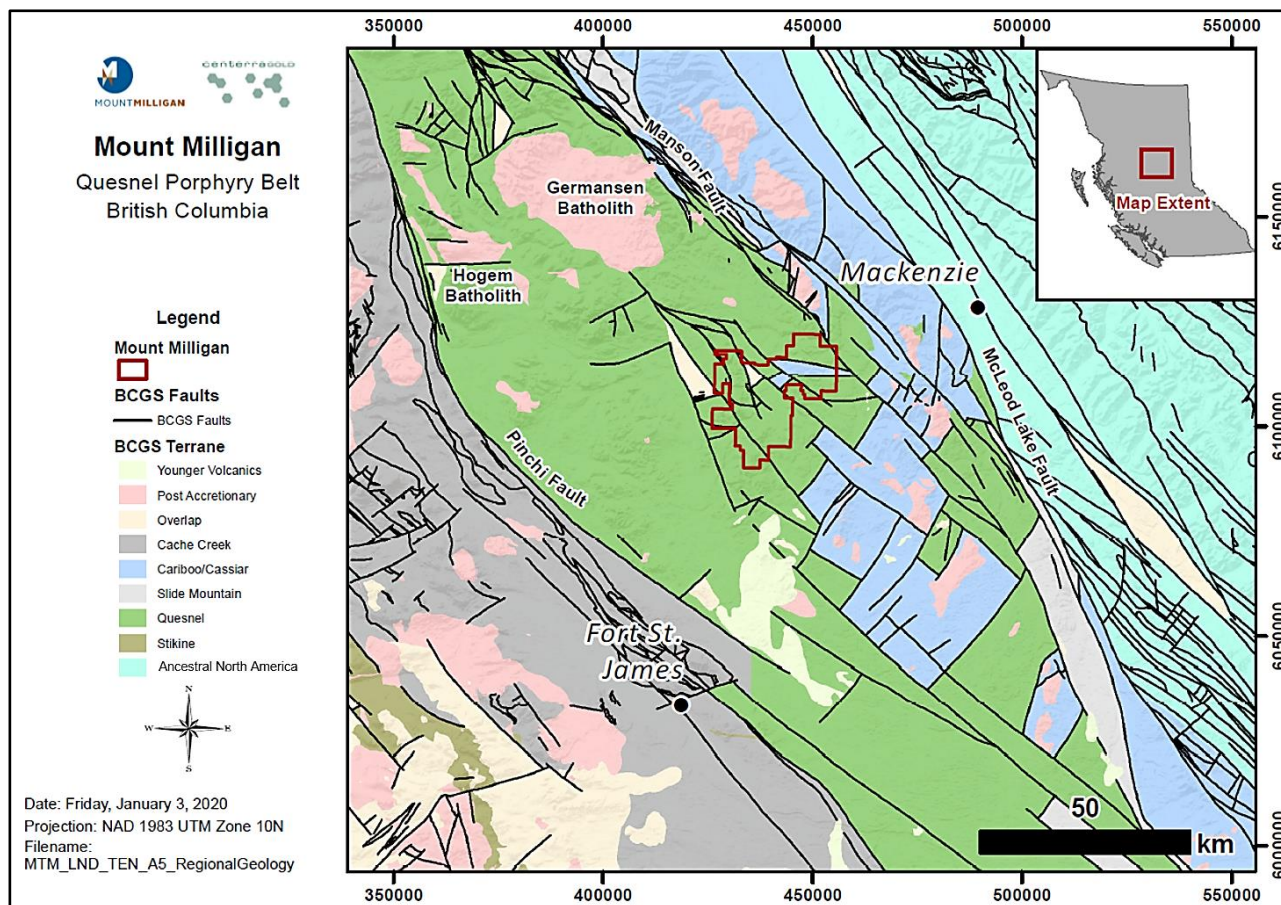
At the latitude of Mount Milligan, the Quesnel terrane is juxtaposed against the Cassiar platform and the Slide Mountain terrane to the east and the Cache Creek and Stikine terranes to the west (Figure 7-1). On the east, the parautochthonous Cassiar platform includes Proterozoic to Paleozoic carbonate and siliciclastic strata that formed along the passive margin of Ancestral North America (Laurentia). The Slide Mountain terrane is composed of Late Paleozoic oceanic rocks that formed

outboard of Ancestral North America. On the west, Stikine terrane is a Late Paleozoic to Mesozoic volcanic arc terrane with many similarities to Quesnel terrane. The Cache Creek terrane is a Late Paleozoic to Mesozoic accretionary complex that formed in an ocean basin outboard of the allochthonous Quesnel and Stikine terranes and became trapped between them during Mesozoic accretion and oroclinal closure (Mihalynuk et al., 1994).

On its western side, the Quesnel terrane is bounded by the Pinchi and Ingenika strike-slip faults and on its eastern side, it is bounded by northwest-trending thrust and strike-slip faults that include the Swannell fault, the Manson-McLeod fault system, and the Eureka thrust. The boundary between the Quesnel terrane and Cassiar platform is a complex structural zone that includes Early Jurassic east-directed thrust faults that juxtapose Quesnel terrane above the Cassiar platform. These faults and related folds are locally overprinted by somewhat younger west-directed structures that reverse this stacking order, as well as by dextral strike-slip and normal faults that were active in Cretaceous and Early Tertiary time (Schiavizza, 2005). The Quesnel terrane is juxtaposed against the obducted Slide Mountain terrane by a series of west-directed thrusts. Subsequent Cretaceous to Early Tertiary dextral faulting has dismembered the Slide Mountain terrane within fault zones including the Manson-McLeod system east of Mount Milligan.

Younger rocks commonly found in the region include Cretaceous granitic stocks and batholiths; Eocene volcanic and sedimentary rocks; flat-lying basalt of both Neogene and Quaternary age; and extensive blankets and veneers of Quaternary glacial till, glaciofluvial, and glaciolacustrine deposits.

Figure 7-1: Regional Geological Setting



7.2. LOCAL AND PROPERTY GEOLOGY

The Mount Milligan property is mostly underlain by Upper Triassic volcanic rocks of the Witch Lake succession (Figure 7-2). The Witch Lake succession is moderately-to-steeply east-northeast dipping and characterized by augite-phyric volcanoclastic and lesser coherent basaltic andesite to andesite, with subordinate epiclastic beds.

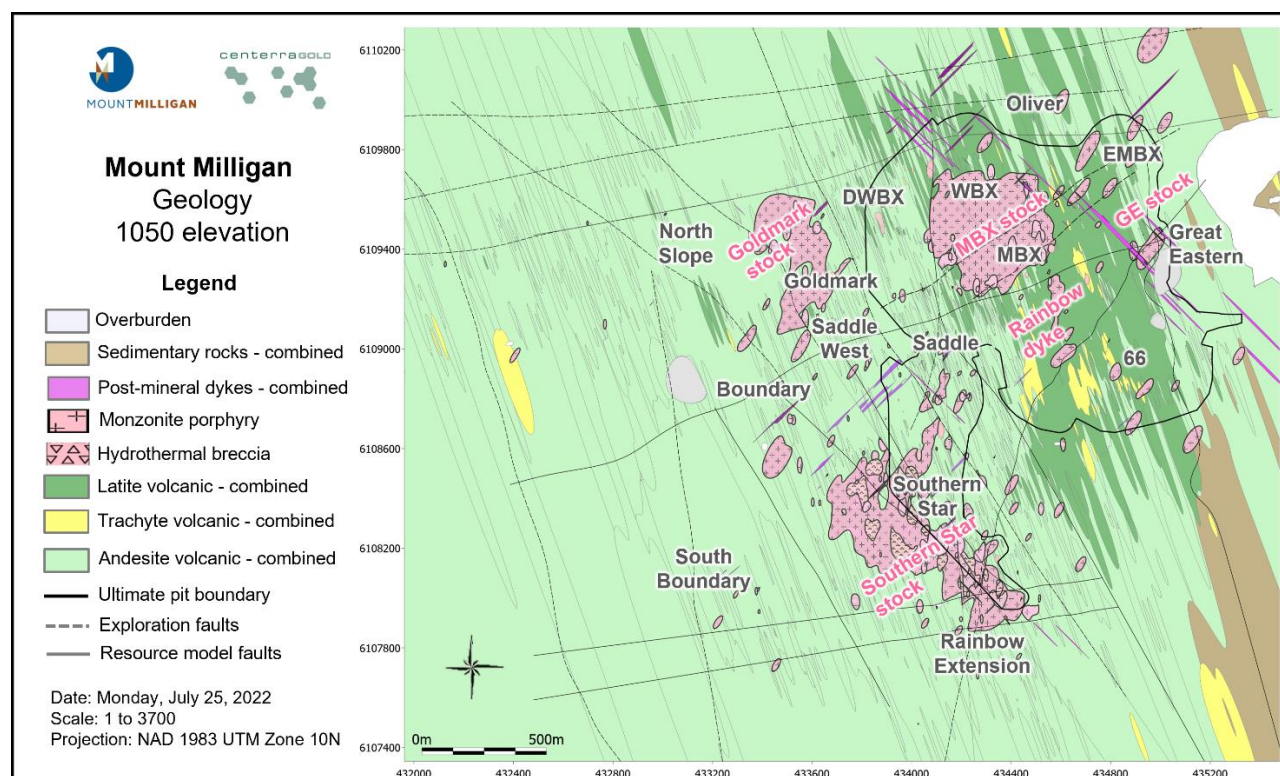
In the northwestern part of the Mount Milligan property, volcanic rocks are intruded by Early Jurassic to Cretaceous rocks of the Mount Milligan intrusive complex (Figure 7-2) located about five to nine kilometers north of the Mount Milligan porphyry deposit. The Early Jurassic component of the intrusive complex comprises monzonitic rocks with minor dioritic-monzodioritic and gabbroic-monzogabbroic rocks.

The Mount Milligan porphyry deposit, including the Magnetite breccia (MBX), Southern Star, Saddle, Goldmark, North Slope, and Heidi stocks and dyke complexes comprise the Heidi Lake stock cluster and are composed of monzonite porphyry rocks and various hydrothermal breccia phases. The Early

Jurassic intrusions are coeval with the Chuchi Lake succession, which is not currently recognized on the property but lies approximately 15 km to the west. The relationship of the Heidi Lake stock cluster to the Mount Milligan intrusive complex is unknown.

Younger intrusions in the area include Cretaceous granites of the Mount Milligan intrusive complex, and Wolverine Metamorphic Complex (WMC). The latter includes schistose to gneissic amphibolite-grade Windermere Group basement rocks of Ancestral North America that were rapidly exhumed in the Paleogene (Ferri et al., 1994; Staples, 2009). The WMC rocks are locally exposed in the northeastern corner of the property (Figure 7-1). Tertiary sedimentary rocks locally overlie older rocks in wedge-shaped half-graben features evident on the east side of the deposit across the Great Eastern Fault (Figure 7-2).

Figure 7-2: Geological Map of the Mount Milligan Deposits



7.3. LITHOLOGY

Witch Lake Succession

Volcanic and volcanoclastic rocks of the Late Triassic Witch Lake succession are host rocks of the Mount Milligan deposit (Figure 7-2). Monolithic fragmental varieties of andesite (augite lapilli tuff, crystal tuff, and tuff), form most of the unit. Minor augite porphyritic flows are present in the west side of the deposit. Small, discontinuous heterolithic debris flows and polymictic breccias are found scattered throughout the deposit and are interbedded with the monolithic fragmental rocks. Plagioclase and/or hornblende phenocrysts are locally present within flows, individual lapilli, and crystal tuff.

Much of the volcanic rocks to the east of the MBX and Southern Star stocks have historically been classified as latite (Figure 7-2). They were distinguished from andesitic volcanic rocks by a darker colour, a general absence of visible hornblende, and the presence of alteration minerals associated with the potassic assemblage including biotite, magnetite, and potassium feldspar. It is common to see only one or two of the potassic alteration minerals, and magnetite alteration is more common than biotite or potassium feldspar. These rocks are more appropriately interpreted as potassically altered andesite equivalents, but for consistency in drill sections and models, drilling campaigns have continued to use the historical latite lithology codes (Table 7-1).

Table 7-1: Rock Units of the Mount Milligan Deposits Used in Drill Programs Descriptions

Lithology		
Code	Name	Description
DRPD	Post-Mineral Dykes	Plag diorite porphyry
MZPD	Post-Mineral Dykes	Plag monzonite porphyry
TRD	Post-Mineral Dykes	Trachyte
HMZP	MBX and SS Stocks	Plag-Hbl monzonite porphyry
HYBX	MBX and SS Stocks	Hydrothermal breccia
MVHD	MBX and SS Stocks	Monzonite volcanic hybrid
MZPP	MBX and SS Stocks	Plag monzonite porphyry
XNMZ	MBX and SS Stocks	Xenolithic monzonite porphyry
TRBT	Trachytic Volcanics	Bedded trachytic flow
TRFW	Trachytic Volcanics	Trachytic flow
LAT	Latitic Volcanics	Latitic rocks - undifferentiated
LATF	Latitic Volcanics	Latitic Tuff
LNL	Latitic Volcanics	Px latitic lapilli tuff
LPFW	Latitic Volcanics	Px latitic porphyry flow
LPXT	Latitic Volcanics	Px latitic crystal tuff

Lithology		
Code	Name	Description
ANDS	Andesitic Volcanics	Andesitic rocks - undifferentiated
ANLT	Andesitic Volcanics	Px andesite lapilli tuff
ANTF	Andesitic Volcanics	Andesitic tuff
APFW	Andesitic Volcanics	Px andesite porphyry flow
APXT	Andesitic Volcanics	Px andesite crystal tuff
ARGL	Clastic Sedimentary Rocks	Argillite
SED	Undifferentiated Epiclastics	Sediments
HBHL	Undifferentiated Volcaniclastics	Lithic tuff breccia
HTDF	Undifferentiated Volcaniclastics	Heterolithic debris flow
LTTF	Undifferentiated Volcaniclastics	Lithic tuff
PBX	Undifferentiated Volcaniclastics	Polymictic volcaniclastic breccia
FALT	Fault	
GRDR	Felsic Intrusive Rocks	Granodiorite quartz porphyry
MNDR	Intermediate Intrusive Rocks	Plagioclase monzodiorite (+/-) porphyry
GABR	Mafic Intrusive Rocks	Gabbro
BSLT	Mafic Volcanic Rocks	Basalt
OVB	Soil Overburden	Undivided surficial materials
CASE	Casing	Everything above casing block

Rocks classified as trachyte are inter-bedded with 'latite' to the east and south of the MBX stock (Figure 7-2). They are characterized by high potassium feldspar content and a lack of mafic minerals. Minor fine-grained plagioclase is also present. Massive and bedded varieties of trachytic rocks are recognized in the volcanic stratigraphy. Bedded varieties are generally discontinuous and locally exhibit cross-bedding and graded bedding. Curvilinear pyrite-chlorite partings or bands of pyrite-chlorite are common along bedding planes. These rocks are now considered to be felsic volcanic and epiclastic units that have undergone potassic alteration. They are the most distinctive stratigraphic markers in the deposit area providing evidence that the geology is tilted to the east-northeast.

Nelson et al. (1991) reported that the abundance of potassium feldspar in the volcanic rocks led past authors to a field classification of augite-porphyritic latites and banded trachytes. Microscopic examination indicated that the potassium-rich nature of the rocks is due to infiltration of secondary potassium feldspar in veinlets and microfractures, as clumps with pyrite and epidote, as seams in plagioclase phenocrysts, and as fine-grained aggregates along bedding planes in the sediments.

Intrusive Rocks – Synmineral

The Main and Southern Star deposits are centred on two principal intrusive bodies, the MBX and Southern Star stocks. Several smaller intrusive bodies are recognized as being contemporaneous with the mineralizing events, including the Unnamed stock in the Saddle zone (formerly Creek zone) and the Goldmark stock (Figure 7-2).

The MBX stock is a composite stock approximately 400 m in diameter, that generally dips moderately to the west. In more detail, it has been recognized throughout the deposit that the multiple intrusive phases making up a composite stock may have different orientations, with both west-northwest and west-southwest dipping phases. Since 2020, the exploration model has used a west-northwest dip in modelling porphyry intrusions, shown on east-west section that compares the model to earlier years in Figure 9-3. Potassically altered hydrothermal breccia occurs along the MBX stock margins. In the southeastern portion of the Main deposit, the Rainbow Dyke (a large sheet intrusion up to 50 m wide) protrudes from the footwall of the MBX stock. The dyke changes morphology from a sill-like body near the stock to a shallowly east-dipping curvilinear dyke, 200 m east of the stock, suggestive of a cone sheet (high level igneous intrusion). The Southern Star stock is a moderately west-dipping composite monzonite porphyry complex, approximately 800 x 300 m in area. Its margins are more irregular and undulated than those of the MBX stock. Hydrothermal breccia is extensive throughout the Southern Star stock. Breccia bodies are typically graded from their margins inward by relatively unaltered monzonite to crackle brecciated monzonite, to well-developed mosaic breccia with strongly potassium feldspar altered matrix.

The MBX and Southern Star stocks generally contain up to 30% sub-parallel plagioclase feldspar phenocrysts, 1-10 mm in length. These phenocrysts occur within a fine-grained greyish pink groundmass of potassium feldspar with lesser plagioclase feldspar, and accessory quartz, hornblende, biotite and magnetite. The Southern Star stock has coarser plagioclase phenocrysts than the core of the MBX stock. Porphyritic monzonite units vary texturally and compositionally within the composite stocks. Syn-mineralization plagioclase hornblende monzonite porphyry dykes are common throughout the Southern Star stock.

Intrusive Rocks – Post-Mineral

Three major types of post-mineralization dykes cut the MBX Main and Southern Star deposits: trachytic, monzonitic, and dioritic varieties. These dykes are generally fresh-looking and lack sulphide mineralization.

Trachytic dykes are the earliest and most common in the southwestern portion of the MBX, northern portion of the Southern Star deposit, and throughout the Saddle zone (Figure 7-2). They are up to 15 m wide and dip moderately to the northwest. They are grey, aphanitic to fine-grained, and may contain accessory magnetite. They often have late carbonate veins (calcite/dolomite) with sericite selvages.

Monzonitic dykes are recognized throughout both deposits. They are up to 10 m wide and dip moderately to the northwest. The dykes are characterized by 15% - 35% fresh euhedral plagioclase phenocrysts up to 5 mm across in a dark aphanitic matrix and may contain augite phenocrysts up to 5 mm. Groundmass predominantly comprises fine-grained potassium feldspar as indicated by staining. Accessory magnetite is always present. Some monzonitic dykes are weakly propylitic altered.

Dioritic dykes are the youngest intrusive rocks. Although they are recognized in both deposits, they are most common in the northern (MBX Main) deposit area. They are up to 5 m wide and dip steeply to the northeast. The dioritic dykes are characterized by abundant plagioclase phenocrysts up to 10 mm (often zoned) in a fine-grained groundmass, and generally contain 2-4 mm long hornblende phenocrysts and minor quartz eyes up to 1 mm in diameter. Some dioritic dykes are weakly propylitic or carbonate altered.

The rock codes used in Mount Milligan drilling programs are summarized in Table 7-1. The historic rock names have been simplified due to the variability in naming by different geologists logging drill core over time and multiple programs.

Structure

The volcanic stratigraphy within and around the Mount Milligan deposits generally dips moderately to steeply to the east-northeast. This is congruent with the geometry of the MBX and Southern Star stocks as moderately westerly dipping bodies and suggests that the entire Mount Milligan hydrothermal system has been tilted to the east-northeast. Exactly how and when this tilting occurred remains unclear; however, there are at least two episodes (Early-Middle Jurassic and Cretaceous-Paleogene) of deformation recognized on the property. Based on volcanic bedding measurements in oriented core from recent drilling programs, the dip of the volcanic stratigraphy shallows moving from west to east across the deposit, with sub-vertically dipping units in the South Boundary zone (west end of the deposit) and sub-horizontally dipping units in the Great Eastern Fault zone (east end of the deposit). This geometry suggests the porphyry deposit may have formed along the hinge of a synform fold structure, which was subsequently tilted.

The structural style of the Mount Milligan area regionally, and at the scale of the deposit, is characterized by tilted fault blocks of differing stratigraphic levels bounded by steeply dipping east-northeast and northwest trending faults, as well as by north and northeast trending faults. At the regional scale the fault geometry has been interpreted to reflect a releasing bend between transcurrent northwest-trending dextral faults during Late Cretaceous to Eocene time (Nelson and Bellefontaine, 1996). A similar scenario is observed at the deposit scale where adjacent fault blocks exhibit alteration and mineralization characteristic of deeper (e.g., MBX) and shallower or peripheral (e.g., 66, DWBX) porphyry environments. The timing relationship between northeast- and northwest-trending faults remains ambiguous, likely due in part to multiple reactivations of structures.

A recent detailed structural investigation in the Mount Milligan open pit has recognized opposing shear sense indicators along northeast-trending structures, lending support for interpreted reactivation; these structures record an early dextral-reverse shear followed by later sinistral shear (Shaban and Barnett, 2019). Examples of faults in this orientation are the Alpine, North MBX, Saddle, Oliver and SS cross faults. Dextral reverse shear on these faults is interpreted to be Cretaceous in age, accommodating regional east-vergent shortening that was widespread in the Northern Cordillera at this time. The subsequent sinistral shear may have occurred during regional dextral transcurrent faulting along northwest-trending faults (Shaban and Barnett, 2019); the northeast-trending faults would be in the antithetic R' orientation with respect to the main northwest-trending dextral faults and therefore would have accommodated minor sinistral shear.

Shallow to moderately southeast- and east-dipping faults are documented to be the youngest structures in the immediate vicinity of the open pit (Shaban and Barnett, 2019). These include the Great Eastern fault which juxtaposes hydrothermally altered Witch Lake succession rocks in its footwall against Tertiary sedimentary and volcanic rocks in its hanging-wall to the east, and the Rainbow fault, which is a well-developed southeast-side-down normal fault. They are interpreted to have formed as part of the Late Cretaceous-Paleogene dilational jog in the regional dextral fault system.

Other northwest- and north-trending steeply-dipping faults occur outside of the pit and were not addressed in the recent structural investigation. These faults have been identified through drilling and geophysical interpretation and include the North Slope, King Richard, Goldmark, SS10, Triangle and Harris faults. They are interpreted as dextral-oblique normal faults predominantly, although some, such as the Harris fault, have apparent reverse sense of shear.

There is evidence that at least some of these structures are long-lived, with a component of deformation that predates or is synchronous with Early Jurassic mineralization and accretion of the

Quesnel terrane onto Ancestral North America. Several of the recognized east-northeast- and northwest-trending structures have been successfully targeted for high gold-low copper mineralization during recent exploration programs and exhibit enhanced phyllic alteration with late-stage pyrite-dominant veins. Similarly, at a regional scale, Nelson and Bellefontaine (1996) note that preferential emplacement of Early Jurassic intrusions along northeast and northwest trends suggests localization along pre-existing crustal scale structures. These would correspond to arc-parallel and arc-transverse structures globally recognized to localize magmatism and porphyry mineralization (Richards et al., 2001).

7.4. ALTERATION

Alteration of host rocks at Mount Milligan is well developed. The alteration assemblages include potassic, sodic-calcic, inner- and outer-propylitic, and carbonate-phyllic (Jago et al., 2014). Overprint of the assemblages is common and can be locally additive or destructive of mineralization grade (Jago et al., 2014). Porphyry gold-copper mineralization broadly coincides with zones of initial potassic alteration (DeLong, 1996), and high-gold low-copper mineralization with structurally controlled carbonate-phyllic (quartz-sericite-pyrite-carbonate; QSPC) alteration. In the following descriptions, vein classifications follow the convention of Jago et al., (2014), which was also adapted for geological logging.

Potassic Alteration

Regional mapping and petrographic studies in the Mount Milligan area have demonstrated that Witch Lake volcanic rocks and derived epiclastic sedimentary rocks are affected by strong potassic alteration as far as 4 km from the deposits (Nelson et al., 1991). Potassic alteration is best developed along the contact margins of the MBX and Southern Star stocks as well as the Rainbow Dyke and decreases in intensity both towards the core of the stocks, and outward into the host volcanic rocks. The alteration assemblage within the stocks is characterized by potassium feldspar, biotite, and magnetite with minor quartz and apatite (the potassic assemblage of Jago, 2008). The alteration assemblage in the surrounding volcanic rocks is characterized by the same assemblage with the addition of actinolite as well as minor albite, calcite, and chlorite (the calc-potassic assemblage of Jago, 2008). Potassium feldspar is the predominant alteration mineral within the intrusions, whereas biotite is more prevalent within the host andesites (DeLong, 1996; Jago and Tosdal, 2009; Jago et al., 2014). Secondary biotite forms up to 30% of wall rocks near intrusive contacts, replacing the andesite protolith but also occurs as envelopes to potassium feldspar veinlets (DeLong et al., 1991). Associated vein types include all early-stage veins, including potassium feldspar (E1), quartz ±

potassium feldspar (E2), chalcopyrite (E3), chalcopyrite-pyrite (E4), and magnetite (E5) veins as well as transitional pyrite-magnetite (T1) veins.

Sodic-Calcic Alteration

Sodic-calcic alteration is spatially restricted to the immediate periphery of the preserved potassic alteration zone within volcanic rocks. It comprises albite, actinolite, chlorite and pyrite with minor epidote, calcite, magnetite, apatite, titanite, quartz and possibly riebeckite (Jago et al., 2014). Albite typically replaces potassium feldspar in groundmass along with uralitization of augite phenocrysts. Sodic-calcic alteration represents an intermediate alteration between the high temperature potassic alteration and the relatively low temperature propylitic alteration (Jago et al., 2014). This spatial association of sodic-calcic alteration is recognized in other BC alkalic porphyry systems including Copper Mountain (Lang et al. 1995) and Galore Creek (Micko et al., 2014). Associated vein types include chalcopyrite-pyrite (E4), actinolite-chlorite (T2), and pyrite (T3) veins. It is also well-developed in the Heidi stock (Heidi zone), a copper target located outside the Mount Milligan mine lease, 3.5 km west of the open pit.

Inner Propylitic Alteration

The inner propylitic alteration assemblage occurs outboard of the sodic-calcic shell and it overprints earlier potassic alteration (Jago et al., 2014). It contains epidote, albite, calcite, actinolite, and chlorite with minor potassium feldspar, biotite, apatite, titanite, and quartz (Jago et al., 2014). This assemblage is relatively narrow (~20 m wide) and discontinuous in the MBX zone, marking the spatial transition between sodic-calcic and outer propylitic alteration assemblages. Associated veins include transitional pyrite (T3) and peripheral epidote-albite-pyrite (P1) veins.

Outer Propylitic Alteration

The outer propylitic alteration assemblage is widespread and peripheral to all other alteration zones but also crosscuts older alteration assemblages along minor faults (DeLong et al., 1991; Jago et al., 2014). Epidote is the most common mineral and is associated with chlorite, calcite, actinolite, pyrite and hematite with minor quartz, titanite, apatite, and montmorillonite (Jago et al., 2014). It is associated with peripheral epidote-albite-pyrite (P1) and pyrite-epidote (P2) veins as well as late-stage pyrite-calcite (L3/L4) veins.

Carbonate-phyllitic Alteration

Carbonate-phyllitic QSPC alteration is well-developed along stratigraphic horizons in the MBX and 66 zones, and along major east-northeast and north-northwest trending faults. It comprises dolomite, ankerite, potassium feldspar, adularia, albite, muscovite/sericite, chlorite, pyrite, and hematite with minor illite, quartz, epidote, riebeckite, titanite, and rutile. In the 66 Zone, the best gold grades (4-5 g/t Au) occur at the interface between potassic and carbonate-phyllitic alteration, where a quartz-sericite-pyrite assemblage predominates (Jago et al., 2014). This zone of carbonate-phyllitic alteration is interpreted as representing a shallower level of the Mount Milligan porphyry system (such as the phyllic alteration shell in typical calc-alkalic porphyry systems) that has been downthrown approximately 100 m by faulting and juxtaposed on the MBX zone (Jago et al., 2014). HGLC mineralization along major east-northeast and north-northwest trending faults is associated with a sericite-carbonate-chlorite-hematite assemblage hosting late-stage vein types including dolomite-sericite-quartz-pyrite (L1), dolomite-ankerite (L2), pyrite-calcite (L3/L4), chlorite-calcite (L5), and hematite (L6) veins.

7.5. MINERALIZATION

Mineralization at the Mount Milligan deposit comprises two styles, early-stage porphyry Au-Cu and late-stage HGLC. The early-stage porphyry Au-Cu mineralization comprises mainly chalcopyrite and pyrite, occurs with potassic alteration and early-stage vein types, and is spatially associated with composite monzonite porphyry stocks (especially at their hanging-wall and footwall margins), hydrothermal breccia, and narrow dyke and breccia complexes. Late-stage, structurally controlled pyritic HGLC style mineralization is associated with carbonate-phyllitic alteration and intermediate- to late-stage vein types, and is spatially associated with faults, fault breccias and faulted lithological contacts (i.e., faulted monzonite porphyry dyke margins). It crosscuts and overprints the earlier stage porphyry Au-Cu mineralization.

Porphyry style Au-Cu mineralization occurs in the hanging-wall and footwall zones of the MBX, Saddle, Southern Star, Great Eastern, and Goldmark stocks (Figure 7-2). Disseminated and vein/veinlet-hosted mineralization is associated with the composite monzonite stocks, their brecciated margins and variably altered volcanic host rocks. Core zones of auriferous chalcopyrite-pyrite mineralization with magnetite rich potassic alteration transition laterally and vertically to pyrite rich HGLC zones within the inner propylitic (albitic) and carbonate-phyllitic alteration shells; the latter appear to be late stage and exhibit strong structural control.

Brownfield target areas west of the mine have mainly peripheral alteration and mineralization signatures; these include near surface HGLC zones proximal to structures. Narrow intrusive units, dykes or “pencil” porphyries, are common and are locally Au-Cu mineralized. The Heidi target farther west and outside the mine lease is a shallow copper prospect with low gold.

Hypogene Mineralization

The bulk of the mineralization comprises disseminated and vein-hosted chalcopyrite (in potassic and sodic-calcic alteration assemblages) and disseminated and vein-hosted pyrite (in all alteration assemblages). Late stage polymetallic (sub-epithermal) veins are rare, but most pronounced in the DWBX zone, and along the Saddle fault. These veins contain sphalerite, galena, and sulfosalt in addition to chalcopyrite and pyrite.

Chalcopyrite

Chalcopyrite is associated with potassic alteration at the contact margin between volcanic and intrusive rocks. It occurs most commonly as fine-grained disseminations and fracture fillings, and less commonly as veinlets and in veinlet selvages. Adjacent to the MBX stock, chalcopyrite may be accompanied by pyrite to form coarse sulphide aggregates. Chalcopyrite-bearing veins contain pyrite and magnetite in a gangue of potassium feldspar, quartz, and calcite. In massive trachytic rocks, chalcopyrite occurs with pyrite along curvilinear partings and as disseminations. Chalcopyrite also occurs in gold-bearing early quartz veins in the Southern Star, WBX, DWBX, and Great Eastern zones.

Pyrite

Pyrite content increases with distance from the MBX and Southern Star stocks and is most abundant in propylitically and intensely QSPC altered rocks. Pyrite occurs as disseminations, veinlets, large clots, patches, and as replacements of mafic minerals. Gold mineralization in the 66 zone is associated with 10-20% pyrite. Cross-cutting vein relationships indicate several generations of pyrite mineralization.

Gold

Gold occurs as grains from 1 to 100 µm in size, as observed in process samples. Grains occur as microfracture fillings and are attached to pyrite, chalcopyrite, or bornite (Ditson, 1997). Gold also forms inclusions within pyrite, chalcopyrite, and magnetite grains. SEM work indicates electrum throughout the deposit with varying gold to silver ratios.

Petrographic work completed in 2019 on 22 drill core samples from the 66, Saddle and DWBX zones shows the settings of electrum (~3-20 µm) can be summarized as; a), isolated inclusions in cores of pyrite crystals; b), associated with microclusters of chalcopyrite and arsenopyrite in larger pyrite crystals; c), in or near microfractures filled with chalcopyrite cutting pyrite; d), at the margins of sulphide minerals such as between tetrahedrite and pyrite, or between pyrite and chalcopyrite; and e) along microfractures through pyrite. The mean Au/Ag ratio of the entire suite of 74 electrum particles was 67.0% Au, or roughly 2:1 Au/Ag.

Silver

Silver is a minor metal found in the Mount Milligan deposit. Polymetallic gold-silver bearing sulphide and sulfosalt rich veins are present in volcanic rocks peripheral to the MBX and Southern Star stocks. Tetrahedrite is the most widespread Ag-bearing sulfosalt mineral at Mount Milligan, hosted in polymetallic veins. Silver is also known to occur throughout the deposits in conjunction with copper and gold mineralization, from micron-scale electrum grains and inclusions.

Polymetallic Veins

Gold- and silver-bearing sulphide, sulfosalt and carbonate-rich veins are present in andesitic volcanic rocks peripheral to the MBX and Southern Star stocks on the western margin of the deposit, in the DWBX and Saddle zones. They comprise pyrite with lesser chalcopyrite, sphalerite, galena, molybdenite, arsenopyrite, tetrahedrite-tennantite, and minor quartz and carbonate. Alteration envelopes are not always present but are recognized to consist of chlorite-sericite-carbonate in propylitically altered andesitic volcanic rocks. Polymetallic veins intersected in the DWBX zone and along the Saddle fault may exhibit subepithermal textures (Sillitoe, 2010).

Supergene Mineralization

Supergene enrichment is poorly developed at Mount Milligan. It is recognized in the MBX, WBX, and Southern Star zones, and is deeper and more extensive in the MBX and WBX zones than in Southern Star (Placer Dome, 1991). Supergene enrichment is restricted to the sporadic occurrence of secondary copper minerals including sulphides (covellite, chalcocite, and djurleite), oxides (cuprite and tenorite), carbonates (malachite and azurite) and native copper. The secondary sulphides occur as rims around chalcopyrite. Oxides, especially cuprite, occur as surface coatings on native copper. Secondary copper minerals commonly occur with iron oxides (goethite, magnetite, and hematite) and iron carbonate (siderite), particularly where malachite and azurite are present. Hydrous iron oxides, which are generally referred to as limonite and include goethite, commonly replace

chalcopyrite and pyrite. Limonite either completely replaces sulphide minerals or occurs as coatings on surfaces of fractures and hairline cracks. Limonitic coatings commonly occur at depths greater than those at which pyrite or chalcopyrite are completely replaced. Supergene enrichment is best represented in the MBX and WBX zones between 6109150N to 6109800N, and between 434000E to 434850E, where it is partially developed over a 20 m thickness. Locally, the supergene mineralization reaches 50-60 m in thickness, particularly along the north and eastern margins of the MBX stock.

Length Width Continuity

Mount Milligan is a roughly tabular, near-surface silica-saturated alkalic gold-copper porphyry deposit that measures approximately 2,500 m north-south, 1,500 m east-west, and extends to a vertical depth greater than 500 m. There is 1 to 100 m depth of till overburden in the deposit area. Within this system, the overall shapes of the mineralized bodies are irregular and gradational with off-set along major structures. The two principal deposits, MBX Main and Southern Star, are centred on composite monzonite porphyry stocks. The MBX Main deposit has been subdivided into zones based on structural boundaries and/or changes in alteration-mineralization. These zones include the MBX, WBX, DWBX, and 66. The Southern Star deposit has not been subdivided into zones, but may be considered to include the Saddle zone at its northern end. The limits of these mineralized zones can vary greatly in size and shape depending on which grade cut-offs are used. The known deposit extents are limited by the lack of drill hole information and could potentially be extended down-dip of monzonite porphyry stocks to the west, down-dip of stratigraphy to the east, and to the north and south along the linear trend of the stock cluster.

8. DEPOSIT TYPES

The Mount Milligan deposits are categorized as silica-saturated alkalic Cu-Au porphyry deposits (Lang et al., 1995; Panteleyev, 1995) associated with alkaline monzodioritic-to-syenitic igneous rocks and are recognized in only a few mineral provinces worldwide (Deyell and Tosdal, 2004). Alkalic Cu-Au porphyry deposits form one endmember of a continuum of porphyry deposits, the other endmember being calc-alkalic Cu-Mo deposits. Alkalic Cu-Au porphyry deposits are thought to form towards the back arc when associated with convergent plate boundaries and are syn-collisional (Westra and Keith, 1981; Logan and Mihalynuk, 2014).

Porphyry copper $\pm$ gold deposits commonly consist of vein stockworks, vein sets, veinlets, and disseminations of pyrite, chalcopyrite $\pm$ bornite that occur in large zones of economic bulk-mineable mineralization within porphyritic igneous intrusions, their contact margins, and adjoining host rocks. The mineralization is spatially, temporally, and genetically associated with hydrothermal alteration of the intrusive bodies and host rocks.

While alkalic and calc-alkalic porphyry deposits share many similarities, there are significant differences that distinguish them. As the name suggests, mineralization in these systems is temporally and spatially associated with variably alkaline intrusions. Due to the lower overall silica content relative to alkalis, alkalic deposits typically have no (silica-undersaturated), or few (silica-saturated) quartz veins. Instead, mineralization occurs as sulphide veinlets and sulphide replacement of iron (Fe^{2+}) bearing minerals. Alkalic porphyry deposits are commonly smaller than the calc-alkaline type and occur in clusters with variable intensity and character of mineralized zones within the cluster. Mineralization can be centred on narrow (<200 m diameter), vertically elongated pipes (pencil porphyries), dyke complexes or may be hosted in larger composite plutonic complexes (Holliday and Cooke, 2007). Metal ratios in alkalic systems can have considerably higher gold tenor than calc-alkalic systems.

Alkalic systems also have distinct alteration assemblages, which typically have a narrower footprint than in calc-alkaline deposits (Bissig et al., 2010). Commonly, there is a magnetite-rich potassic alteration assemblage at the core of the deposit (that may contain hydrothermal breccia), which laterally transitions through calc-potassic alteration comprising calcium-bearing minerals and/or an albite rich sodic-calcic assemblage, and then an outer shell of propylitic alteration that may be reddened by hematite dusting. Intervening phyllic or argillic assemblages, as recognized in calc-alkalic systems, are not well-developed. Instead, high-level stratabound alteration zones characterized by albite-sericite alteration assemblages may be the alkalic equivalent of lithocap root

zones (Holliday and Cooke, 2007). The presence of abundant late carbonate alteration is also common to alkalic systems.

Examples of alkalic Cu-Au porphyry deposits in British Columbia include Galore Creek, Mount Polley, Copper Mountain, New Afton, Mount Milligan and Lorraine. British Columbia deposits occur in both the Quesnel and Stikine island arc terranes and range in age from Late Triassic to Early Jurassic (Logan and Mihalynuk, 2014). Global examples include Ok Tedi in Papua New Guinea as well as Northparkes and Cadia in Australia.

9. EXPLORATION

This Item focuses on components of the 2014 to 2021 exploration programs, other than drilling, specifically the compilation of historical reports and data, reinterpretation of historic data, development of a three-dimensional (3D) exploration model, surface geological mapping, geochemistry and geophysics programs. Detailed information on exploration drilling programs from 2014 to 2021 is included in Item 10 (Drilling). Information on exploration programs prior to 2014 is covered in Item 6 (History).

9.1. RECENT EXPLORATION ACTIVITIES (2014-2021)

Exploration, 2014-2016

In 2014, TCM conducted a two-phase exploration program from August-October on the northeastern side of the property, an area which was previously under explored. TCM engaged Equity Exploration Consultants Ltd. (Equity) to conduct the program. Phase-1 comprised geological mapping, prospecting and soil sampling. A total of 147 line-km were mapped and 1,527 samples assayed. Ten samples were sent to a specialized laboratory for thin section studies. Phase-2 included ground geophysical surveys (IP and magnetics) and a more detailed mapping, prospecting and soil sampling program that focused on target areas. This work identified a new target 17 km northeast of the deposit, Prospect 26, with strong propylitic alteration and anomalous Cu, Zn, and Sb in soil values. The target is associated with a resistivity low ringed by a zone of high chargeability (Hughes and Perk, 2014). Several additional zones with anomalous As, Sb, Mo, Pb, and Zn values in soil were identified but remain unexplained.

Exploration, 2017-2021

In January 2017, Centerra-TCM conducted a thorough review and compilation of historic exploration reports, documents and databases relating to the initial prospecting, discovery and development of the Mount Milligan property and surrounding land held by the TCM land tenure. Sources were located from TCM archive servers, from Centerra-TCM's lead consultants and contractors, and from government publications. The first compilation report was delivered by Equity on February 15, 2017, summarizing approximately 227,000 files comprising 404 GB of data.

In July 2017, Centerra-TCM identified exploration targets and additional high-quality resources on the property, to design and safely conduct systematic exploration programs to test these, to use

enhanced and improved exploration methodologies, to develop a comprehensive 3D exploration model and well-managed database, and to support improved resource block modelling and mine planning. Spatially, the Mount Milligan exploration strategy was divided into three principal domains that define exploration programs and roughly coincide with three scales of measurement within a clustered system of mineralized and potentially mineralized monzonite porphyry stocks. Boundaries between the three domains in this context may have some overlap:

- Near-pit/Within-pit (~1.5-km diameter area) – porphyry core scale, known centre of porphyry system; originally defined as within the 2017 ultimate pit shell and boundary (as defined by Centerra Gold Inc., March 22, 2017, NI 43-101 Technical Report on Mount Milligan Mine authored by Andrews, Berthelsen, and Lipiec).
- Brownfield (5-8 km diameter area) – porphyry cluster scale, linear trends of monzonite porphyry stocks related to a single cluster (Heidi Lake stock cluster); outside the 2017 ultimate pit shell but within the mine lease boundary.
- Greenfield (up to 30 km diameter area) – stand-alone deposit scale, additional deposit that is unrelated to the Mount Milligan deposit and is not within the Heidi Lake stock cluster; outside the mine lease boundary but within the Mount Milligan property mineral tenure block.

The 2017 Centerra-TCM exploration program included a ground-based geophysical survey conducted from August to November by Peter E. Walcott & Associates Limited (“Walcott”). A geophysical program totalling 67.6 line-km (23 lines) of IP survey was completed over two target areas, Fugro-Rain (south of the mine lease in the greenfield area) and KC (east of the mine lease overlapping both brownfield and greenfield areas; Figure 6-3, Figure 9-1). A ground-based magnetic survey was also completed and comprised 309 line-km at 100 m line spacing over the same areas. Results include a 1.3 x 1.7 km sub-circular magnetic high anomaly with coincident chargeability high at the Fugro-Lip target, one km south of the mine lease; a west-northwest trending 1.9 x 0.8 km chargeability high with coincident magnetic low anomaly at the Fugro-2 target, 3.5 km south of the mine lease; and an arcuate 1.2 x 0.3 km magnetic high anomaly at the KC target, 500 m east of the TSF within the mine lease where two surface grab samples collected in August 2017 by Centerra-TCM geologists returned anomalous values of 0.13% Cu, 0.16 g/t Au and 0.15% Cu, 0.12 g/t Au.

Also completed in 2017, were two memos provided in June and October by an external geophysicist (M. Nosyrev), which reinterpreted historical Mount Milligan geophysical data including the 2010 Titan-24 DCIP geophysical survey (Quantec Geoscience Ltd.) and the 2009 South Grid 3DIP survey (SJ Geophysics Ltd.). The memos attempted to identify the geophysical features of known mineralization to help guide exploration drill targeting. Target geophysical signatures included the marginal gradient zones of steep or shallow dipping chargeability high anomalies, and marginal gradient zones of west-dipping moderate chargeability and resistivity high features interpreted to be tilted porphyry intrusions. The first memo also provided a map of faults across the Mount Milligan

deposit based on geophysical signatures; several of these were unmapped in historic geological models for the deposit and were added to the exploration model. Both memos were used to develop drill targets in all three exploration domains: in-pit, brownfield and greenfield.

In 2018, a 15.5 line-km IP geophysical survey (six lines at 100 m spacing; five east-west lines and one north-south line) was completed in May-June by Walcott in the KC target area east of the TSF (Figure 6-3, Figure 9-1). Several anomalous ring-shaped features identified in the 2017 ground magnetic survey and a northeast trending airborne resistivity high feature had lines run across them. One 400 m wide moderate chargeability and moderate-high resistivity target was identified near Rainbow Creek, one km southeast of the TSF near the Meadows groundwater target area within the mine lease.

In 2019, geophysics programs completed by Walcott included those listed below. The detailed airborne magnetic surveys were conducted using a stinger mounted airborne magnetic system utilizing an Astar B2 helicopter provided by Silver King Helicopters Inc. of Smithers, B.C.

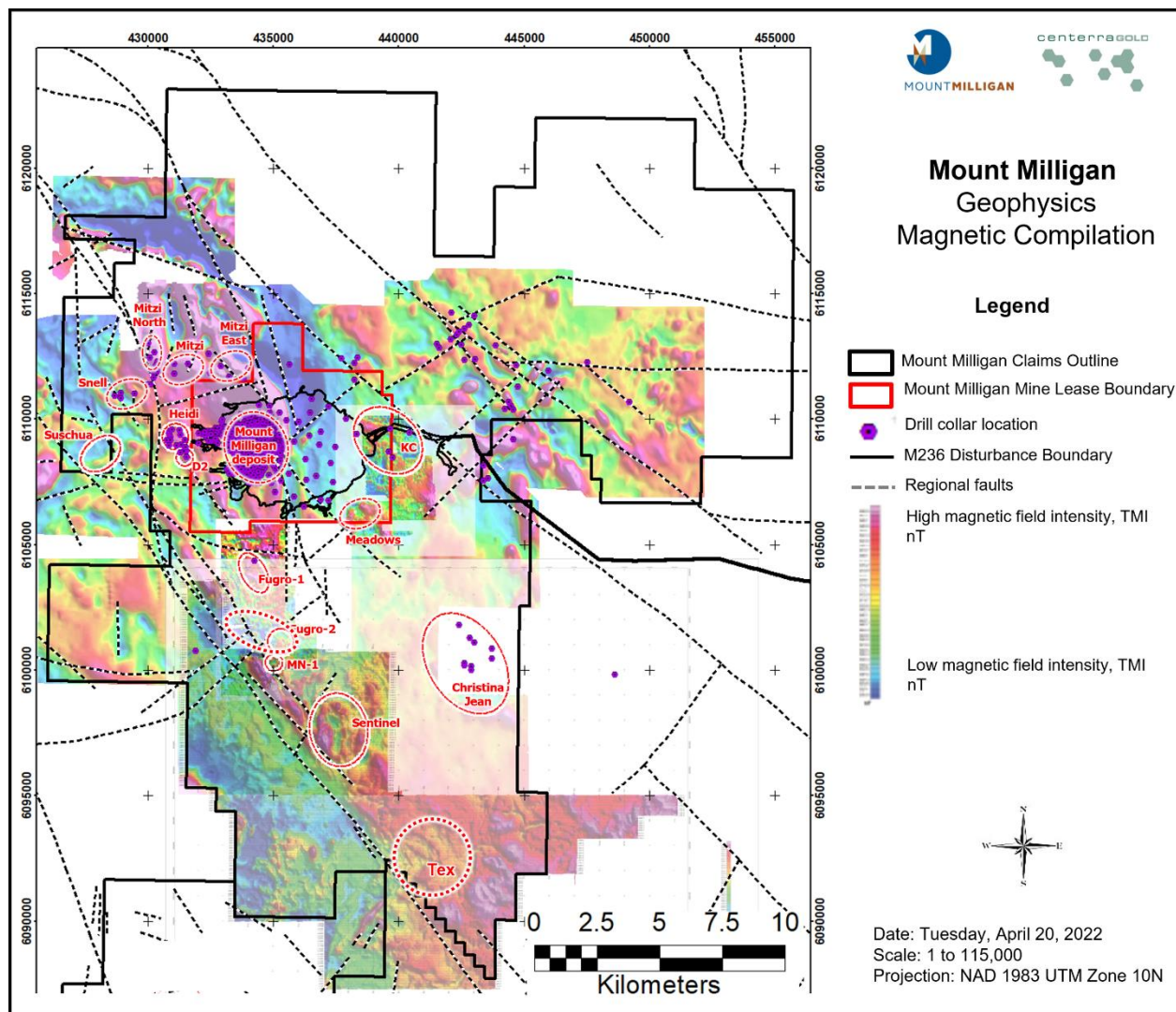
A 631 line-km (38.9 km² area) low altitude aeromagnetic survey at 75 m line spacing flown on May 31st over the western side of the Mount Milligan property in both brownfield (49.1%) and greenfield (50.9%) exploration areas.

A 1,437 line-km (129.0 km² area) low altitude aeromagnetic survey at 100 m line spacing covering the groundwater exploration area and surrounding areas north and east of the mine and TSF. Within lease (brownfield) coverage was 19.4 km² (15.0%) and the rest was greenfield coverage (within property claim block) and beyond (Figure 9-1).

First phase of an IP survey (12 lines, 16.7 line-km, 200 m line spacing) was conducted from September 18th to October 17th across the North Slope, Goldmark and Oliver zones across a series of east-northeast trending faults.

Joint ZTEM-MT inversion was completed using the ground MT (magnetotelluric resistivity) data collected in the 2010 Titan-24 DCIP geophysical survey to help guide the 2009 3D ZTEM inversion (Geotech Limited, 2009). The purpose of the joint inversion was to start developing a deeper view of the Mount Milligan porphyry system.

Figure 9-1: Geophysics Magnetic Survey Compilation of Mount Milligan



Plan map of the Mount Milligan claims block with compilation of ground-based (Fugro and KC) magnetic surveys completed in 2017 and low altitude aeromagnetic surveys completed in 2019-2021 by Centerra-TCM and Peter E. Walcott & Associates Limited. The maps show contours of total magnetic field intensity (TMI; nT). Scales of nT values vary between maps.

In 2020, a total of 25.6 line-km of IP geophysical surveys were completed in October-November by Walcott, to develop Brownfield exploration targets in two areas north and south of the 2017 ultimate pit boundary. The first grid was completed in the South Boundary zone, a HGLC mineralization target with auriferous veins west-southwest of the 2017 ultimate pit boundary. The survey consisted of 14 north-easterly orientated lines ~1.4 km long at 200-m spacing, with two orthogonal tie lines ~1.5 km long at 400 m spacing. A single north-northwesterly orientated line 1.8 km long was completed from the northeastern edge of the ultimate pit boundary in the Oliver zone, to assess the potential for significant mineralization to the north of the deposit. Results show a deepening of the main moderate to high chargeability anomaly associated with the deposit to the north-northwest, suggesting a northerly dip component to the Mount Milligan porphyry system.

In 2021, greenfield exploration consisted of geophysical surveys, soil sampling, geological mapping, and prospecting. A total of approximately 1,640 line-km of airborne magnetic geophysical surveys were completed in March by Walcott to develop greenfield exploration targets south of the mining lease. The surveys were flown on east-west orientated lines flight lines with a nominal line spacing of 100 m, with orthogonal tie lines with a nominal line spacing of 1000 m. The survey identified a nested ring feature centred on a magnetic high with destructive halo and outer magnetic high rim, which was named the Sentinel target (Figure 9-1). A follow-up soil sampling grid at approximately 400 by 200 m spacing in eleven east-west lines approximately four km long, was completed by CGS Exploration from June-September 2021, totalling 133 soil samples. Between October 22nd to November 23rd, a total of 33.4 line-km of brushing was completed over nine east-west lines spaced 400 m apart at Sentinel. A total of 10.6 km of ground IP survey was completed by Walcott. Desktop studies included surficial geology mapping of the Milligan claims block that was used to derive till sampling suitability and drift thickness maps, completed by Palmer Environmental Consulting from Vancouver, B.C.

In 2021, brownfield exploration target development at the South Boundary zone included mechanical excavation of a 200-m long trench oriented approximately east-west, across an area where auriferous veins were modelled to project to surface. The trench was mapped for geology including oxidation, lithology, alteration, mineralization, and sampled in July 2021, totalling 75 samples.

9.2. SUMMARY AND INTERPRETATION OF EXPLORATION ACTIVITIES

Since mid-2017, several conceptual exploration targets have been generated through the development of a 3D exploration model and complete drilling database, reinterpretation of historical

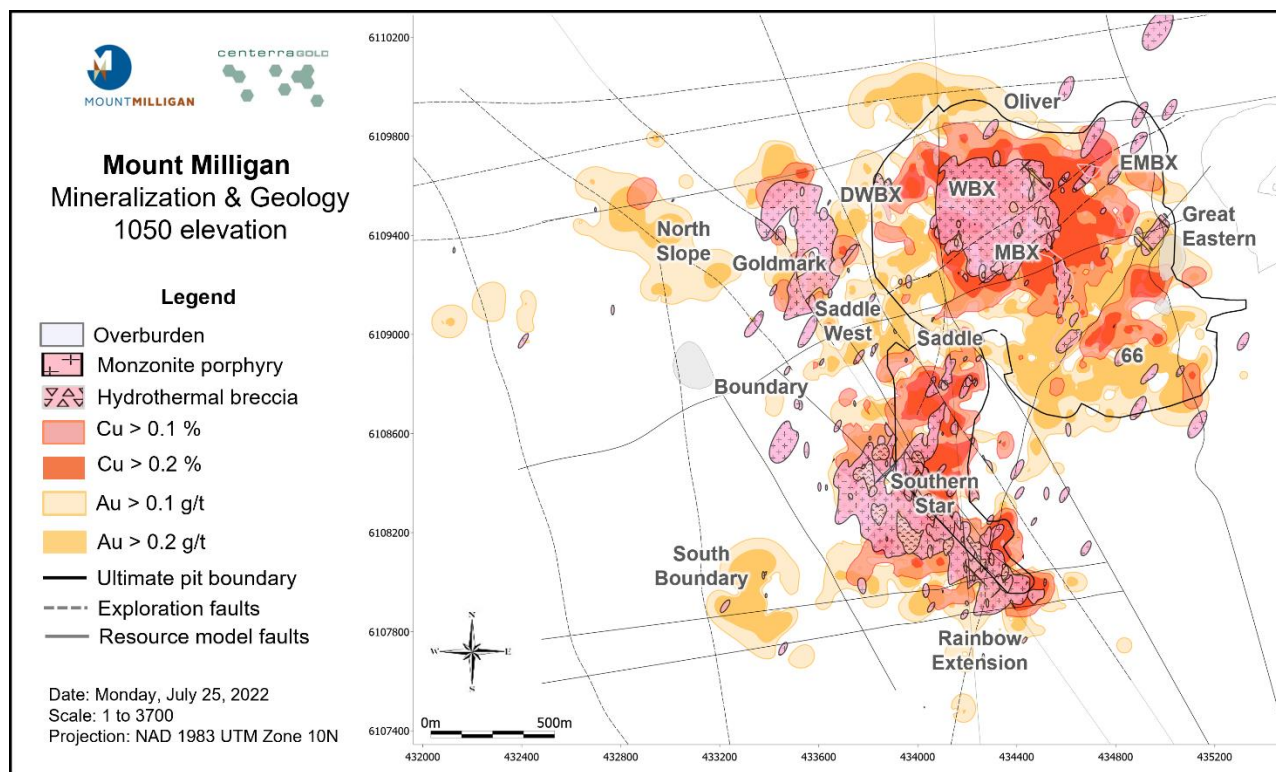
geophysical data, enhancement of structural model for the deposit, ground-based and airborne geophysics programs, and exploration diamond drilling (Item 10). These activities have resulted in the identification of two distinct styles of mineralization at the Mount Milligan deposit.

Early-stage porphyry Au-Cu mineralization (and early-stage vein types) associated with composite monzonite porphyry stocks and related hydrothermal breccia, and narrower dyke and breccia complexes. Early-stage porphyry Au-Cu mineralization typically has a one-to-one gold to copper ratio.

Late-stage, structurally controlled HGLC mineralization (and intermediate- to late-stage vein types) associated with faults and fault breccias, that crosscuts/overprints the earlier stage porphyry mineralization and is more spatially widespread. HGLC mineralization locally increases the gold to copper ratio.

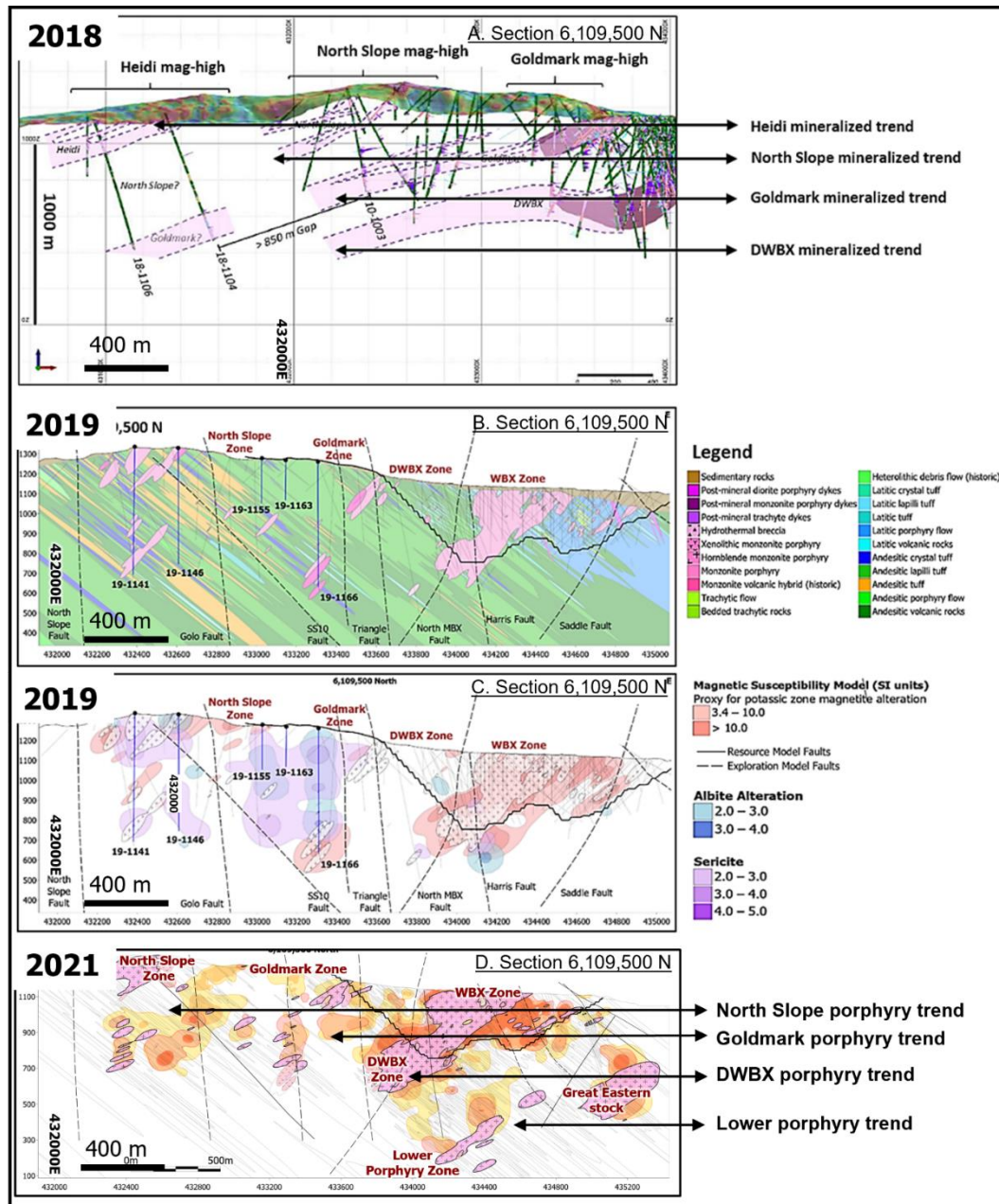
These two mineralization styles are zonal, overlapping, have different predominant sulfide species (chalcopyrite-pyrite versus pyrite only), and require different strategies in exploration, mining and metallurgy. Combined, these may expand economically viable mineralization in several directions including to the NNE-SSW along the axis of the main porphyry trend (the geometry of which may include a series of offsets on subparallel cross-faults; Figure 9-2); and west of existing resources following the ENE-WSW structural fabric and down-dip orientation of the porphyry stocks and dyke/breccia zones. There is an apparent shallow dip of mineralization to the west of the existing resource which is interpreted to be in part due to the moderate eastward tilt (westward dip) of the porphyry system as well as by repeated east-side-down offsets on northwest-trending subparallel normal, or listric, faults (Figure 9-3).

Figure 9-2: Mineralization of Mount Milligan Porphyries



Schematic zonal and overprinting distribution of porphyry Au-Cu and high-gold low-copper (HGLC) mineralization. Porphyry Au-Cu mineralization shown in red, occurs typically as halos surrounding stock centres shown in pink. The distribution of high-gold low-copper (HGLC) mineralization shown in yellow, is more variable and may be associated with fault zones or volcanic stratigraphy rather than with stock centres. Open pit boundary is the 2021 pit shell boundary at 1050 m elevation above mean sea level.

Figure 9-3: Sectional Views of Mount Milligan Mineralization



View is to the north on all sections, near section 6,109,500 N. A: Schematic interpretation provided by Equity of a series of shallowly west-dipping mineralized trends associated with the DWBX, Goldmark, North Slope and Heidi zones from east to west (2018). Magnetics (1VD) draped over topography. B: Interpreted moderate dip of monzonite intrusive units in the Centerra-TCM exploration model. The difference in attitude with the trends shown in top image may be due to a series of steeply dipping east-side-down normal faults and related offsets (2019). C: Alteration model showing magnetic susceptibility shells representing potassic alteration, and albite and sericite shells which are a proxy for the extent of the mineralizing paleo-hydrothermal system. Open pit is the 2017 pit shell boundary (2019). D: Interpreted geology model showing the new mineralization trend found below the 2021 ultimate pit shell boundary (Lower Porphyry and Great Eastern) and the extension of the DWBX stock to the west (2021). Copper grade shells > 0.1% are shown in red, gold shells > 0.1 g/t shown in yellow, intrusions are shown in pink, exploration faults are shown as dashed lines.

10. DRILLING

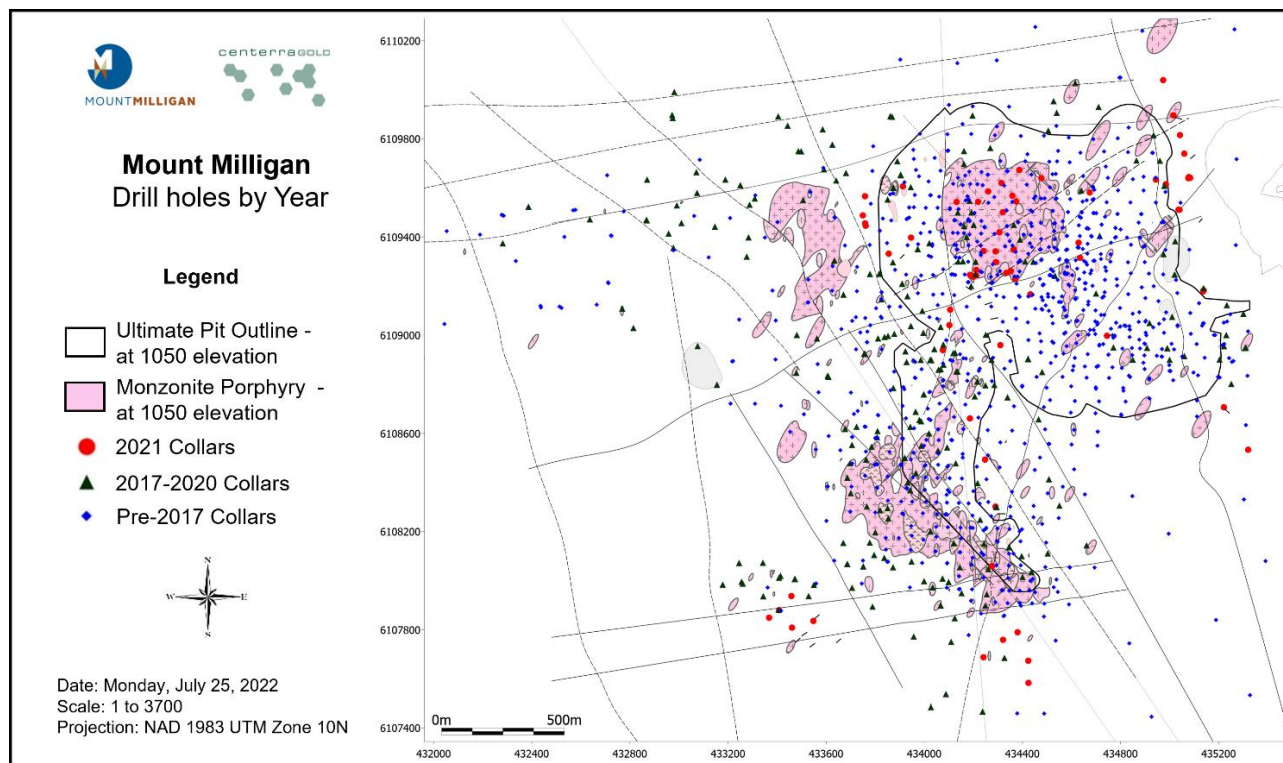
Diamond drilling at the Mount Milligan property was designed to test and delineate mineralized material, to obtain metallurgical samples, to condemn areas planned for infrastructure and to gather geotechnical and environmental information. A total of 235,349 m of core from 1,005 drill holes were drilled by Lincoln, United Lincoln, Continental Gold, Placer Dome, Terrane and TCM (pre-Centerra Gold acquisition) between 1987 to 2016. Most of the drilling samples were collected from NQ (47.6 mm core diameter) diamond drill core. Since 2017, Centerra-TCM has completed a total of 152,825 m of drilling in 346 drill holes in drilling programs with various purposes including; metallurgical, NPI and Resource expansion, brownfield (within mine lease) exploration, and greenfield (within property outside mine lease) exploration (Figure 10-1, Table 10-1).

The complete historical drilling campaigns are summarized in Table 10-1. Only values for mineral-focused exploration, Resource infill/expansion and metallurgical drilling are included (excludes groundwater exploration drilling and near surface till drilling for construction material). Figure 10-1 shows all drill hole collar locations for the property, including historical and recent drilling programs.

Table 10-1: Drilling Programs Summarized by Year

Year	Metres	Holes	Company
1987	2,304	23	Lincoln
1988	6,645	47	United Lincoln
1989	87,662	336	United Lincoln, Continental
1990	82,924	386	Continental, Placer Dome
1991	17,969	90	Placer Dome
1992	604	4	Placer Dome
2004	2,184	14	Placer Dome
2006	9,557	36	Terrane
2007	10,515	33	Terrane
2010	5,767	10	TCM
2011	5,637	17	TCM
2015	1,786	5	TCM
2016	1,795	4	TCM
2017	7,692	21	Centerra-TCM
2018	30,942	76	Centerra-TCM
2019	42,821	107	Centerra-TCM
2020	31,865	74	Centerra-TCM
2021	39,505	68	Centerra-TCM
Total	388,174	1,351	

Figure 10-1: Drill Hole Collar Map (all drilling)



10.1. DATA COLLECTION

Pre-2004 Drilling

Geotechnical information was collected for all drill holes after drill hole 87-70, and geological information was collected for all drilling conducted at the property. The information was collected on drill log forms that varied depending on the operator conducting the work and the type of deposit being targeted. Some drill holes were assayed for gold and not for copper, others for copper and not for gold. Lincoln, United Lincoln, and Continental Gold used a written form, with a header and descriptive format. The header included information on drill hole number, location, orientation, drilling company, geologist and dates. The descriptive section was broken down to intervals based on primary lithology, with descriptive reference to alteration, structure, and mineralization.

Placer Dome used a coding system known as GEOLOG, with specific information captured on coded geologic logs, which were then entered into a computer and verified using GEOLOG software. The header section included drilling metadata similar to that recorded by previous operators. Geological information included lithology, alteration, and mineralization including both intensity/amount and mode of occurrence. GEOLOG forms were supplemented by graphical logs as well as descriptive remarks.

Geotechnical information was recorded on forms and captured in spreadsheets. Recorded information included core recovery, RQD, hardness or compressive strength (CS), degree of breakage, degree of weathering or oxidation, fracture and joint frequency, and specific gravity (SG).

Geochemical analyses of silver were routinely conducted in early drilling with 14,896 samples analyzed from holes 88-61 to 89-212. Samples from later drilling were not analyzed for silver.

2004 Drilling (Placer Dome)

Placer Dome's 2004 drill program was conducted to obtain core for metallurgical test work, with drill holes located within the MBX, 66, and Southern Star zones (Figure 9-2, Figure 10-1). Holes were twinned with or drilled near existing holes, with a total of 2,184 m drilled in 14 holes. Aggressive Diamond Drilling Ltd. of Kelowna, British Columbia conducted the drilling using a Boyles 56 drill rig.

Drilling information was captured directly to an acQuire database. All samples from the program were also analyzed for silver.

2006-2007 Drilling (Terrane)

In early September 2006, Terrane commenced a four-phase diamond drilling program. The first phase was designed to acquire samples for metallurgical test work. Initiated in late 2006, the second phase targeted mineralization on the west side of the MBX stock. The third phase commenced in March 2007 and was designed to gather geotechnical information across the MBX zone. The fourth phase commenced in June 2007 and was designed to gather geological and geotechnical information on the margins of the Southern Star zone, and in areas of proposed infrastructure.

Cyr Drilling International Ltd. (Cyr Drilling) of Manitoba conducted all four phases using a skid-mounted machine and HQ (63.5 mm) diameter drill rods. An ACE core orientation tool was used to provide oriented core for subsequent measurements.

Geological information was compiled on Excel spreadsheets for all drill holes completed by Terrane. The geological spreadsheets included drill hole numbers, locations, orientation, drilling dates and geologist name. The detailed geological information included lithology, alteration, and mineralization, including both intensity/amount and mode of occurrence. Magnetic susceptibility information was collected at two metre intervals. Geotechnical information was recorded on a separate set of Excel spreadsheets and captured core recovery, RQD, and fracture intensity. SG was also measured. As part of the sampling protocol developed by Terrane for the 2006-2007 drill program, every second sample was analyzed by a multi-element ICP package which included silver.

More detailed description of the four-phase diamond drilling program is as follows.

Phase I (32 drill holes totalling 7,690 m) was designed to acquire fresh material for metallurgical test work. The program tested a range of lithologies and alteration types across the mineralized domains of the MBX and Southern Star deposits. The holes were collared in fences across the strike length of deposits, and targeted material that had been well-defined from historical drilling.

Phase II (23 drill holes of HQ diameter core totaling 9,292 m) was designed to target areas of mineralization that had been less densely drilled by previous operators. Drilling targeted the DWBX which was intersected as part of the Phase I drill program. In addition, several holes were located between the Southern Star and MBX Main deposits.

Phase III (5 drill holes of HQ diameter core totaling 1,525 m) was designed to gather further geological and geotechnical information along the perimeters of the MBX Main deposit, including on the western limits of the MBX stock through the Harris fault, on the northern margin of the MBX stock into altered host andesite volcanic rocks, and to the SSE from the MBX zone into the 66 zone. Twelve samples (20 cm each; 2.4 m total) were collected for unconfined compressive strength (UCS) testing; and 130 samples (20 cm each; 26 m total) for point load (PL) testing.

Phase IV (9 drill holes of HQ diameter core totaling 1,565 m) included three geotechnical drill holes on the perimeter of the Southern Star deposit, and six condemnation drill holes designed to gather subsurface information in areas of the proposed infrastructure. The program followed completion of Resource modeling for the 2008 report and results were included in Resource modelling for the 2009 report. Condemnation drilling was located outside of the Resource model limit and did not encounter significant mineralization.

2010-2016 Drilling (TCM)

In 2010-2011, 17 drill holes (10-1003 to 10-1012 and 11-1013 to 11-1019), totalling 10,535 m, were drilled for exploration potential. The program targeted an extension of the DWBX zone west of the Main deposit and north of the King Richard Creek.

In 2011, an additional 10 holes were drilled for metallurgical characterization of material to be mined within the Phase-3 pit envelope defined in 2009. These holes were named MET 11-01 to MET 11-10. A total of 867.5 m was drilled, logged, sampled and assayed.

Apex Diamond Drilling Ltd. (Apex) of Smithers, B.C. completed the 2010-2011 drilling using a skid-mounted machine and HQ and NQ diameter drill rods. No oriented core data was collected. Detailed core logging and sampling was done by Equity Exploration Consultants Ltd. ("Equity") of Vancouver, B.C. and included lithology, alteration and mineralization. Geotechnical information captured

included core recovery, RQD and fracture intensity. Data was captured using Geospark software, an Access-based relational database system that stores drill hole and geological data in individual data tables. Upon completion of logging, data was validated and exported as Excel files for each data table.

Exploration drilling resumed in 2015. TCM contracted Equity to manage a five-hole (1,786 m) exploration drilling program focused on the Mitzi and Snell target areas in the northwest part of the property claim package, five km northwest of the open pit (Figure 6-3, Figure 9-1). The program ran from October to November and one of the five drill holes was abandoned at 41.5 m depth. The aim of the program was to test the Mitzi and Snell coincident magnetic, chargeability and geochemical anomalies for porphyry-style Au-Cu mineralization. Previous work in the Snell target area included 10 drill holes completed in 1990 that failed to identify significant mineralization; these were drilled to relatively shallow depths below surface (45 m to 188 m). In 2008, the area was included in a property-wide HeliGEOTEM II airborne geophysical survey; and in 2009 was covered by the North Grid 2DIP survey (Figure 6-3) and grid soil sampling in 2009-10. This work rekindled interest in the Snell and Mitzi target areas as prospective for porphyry Au-Cu deposits. Drilling at Mitzi (588 m) tested the southwest edge of a ring-shaped chargeability high anomaly and intersected albite and phyllic (sericitic) alteration and anomalous copper values (301 ppm Cu over 9.52 m; DDH 15-1024). Drilling results at Snell (1,198 m) suggested the high chargeability and multi-element geochemical anomalies are related to epiclastic argillaceous rocks in a narrow northwest trending fault-related basin (Branson and Voordouw, 2016).

In 2016, TCM completed a four-hole (1,795 m) exploration drilling program with two holes at Mitzi target (1,065 m) and two holes at Snell target (730 m) (Figure 9-1). The first two drill holes tested the Mitzi aeromagnetic and surrounding ring-shaped chargeability high anomalies. At the Snell target, two drill holes were drilled to the north of the 2015 drill holes in the coincident aeromagnetic and chargeability high anomaly with anomalous multi-element geochemistry. Intervals of biotite-chlorite and sericite-chlorite alteration were intersected, and local weak pyrite and pyrrhotite mineralization with some chalcopyrite noted. Assays returned narrow intervals anomalous to weak Au-Cu mineralization.

L.D.S Diamond Drilling Ltd. of Kamloops, B.C. completed the 2015-2016 drilling programs using a skid-mounted machine and NQ diameter drill rods. No oriented core data was collected. Detailed core logging and sampling was done by Equity. Data was captured directly in a Geospark access database and included lithology, alteration, mineralization and structure. Magnetic susceptibility was measured every 3 m using a Terraplus KT-10 magnetic susceptibility metre. Geotechnical

information captured included core recovery and RQD. At completion of logging, data was validated and exported as a Microsoft Access database file (*.mdb).

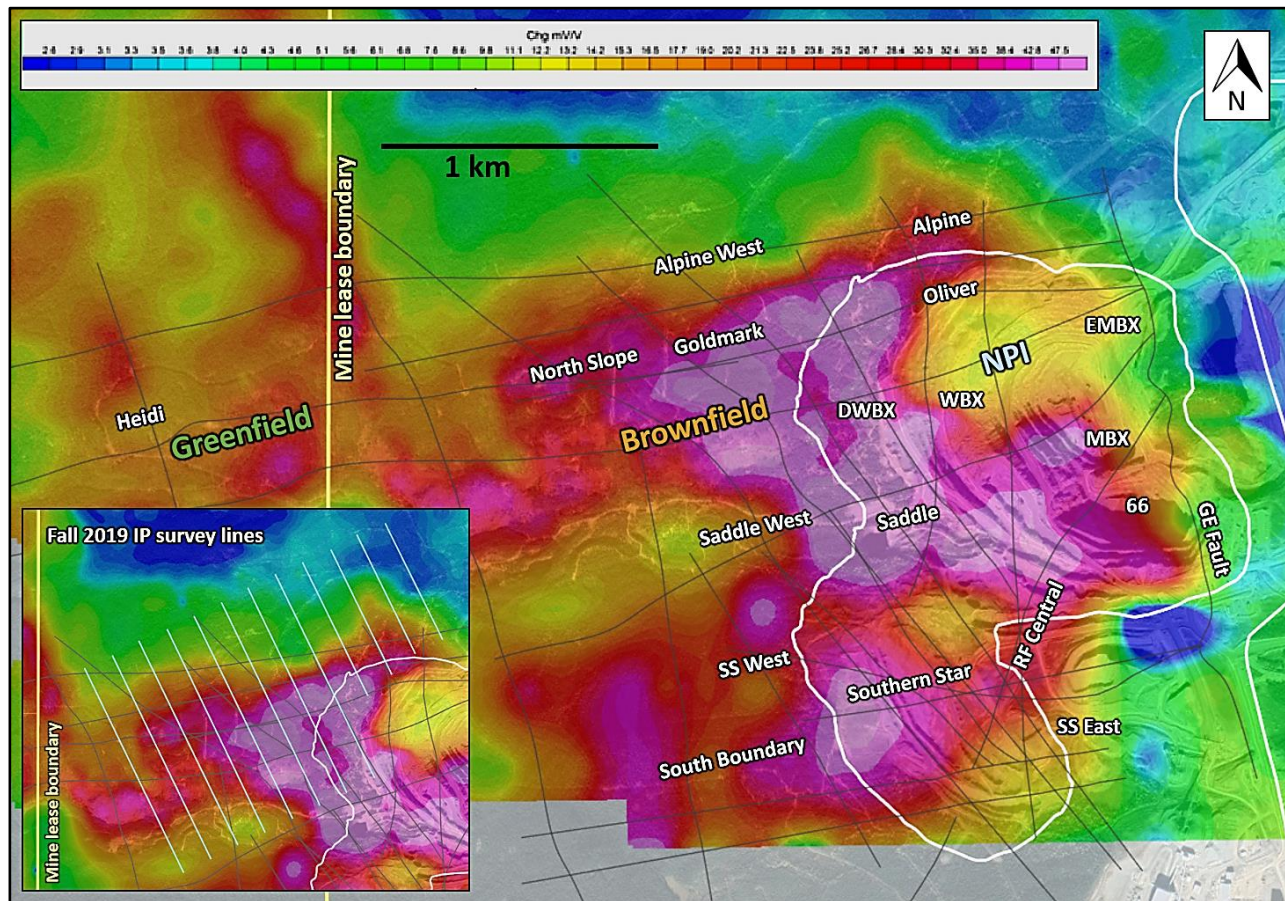
2017-2021 Drilling (Centerra-TCM)

In 2017, an eight-hole (1,082 m) geometallurgical drilling program was completed in May in the MBX and 66 zones. This was followed in August to December by a 13-hole (6,611 m) near-pit infill and expansion drilling program in the WBX, DWBX, MBX and Saddle zones within the 2017 ultimate pit boundary (as defined by the Centerra Gold Inc., March 22, 2017, NI 43-101 Technical Report on Mount Milligan Mine authored by Andrews, Berthelsen, and Lipiec). All eight of the geometallurgical drill holes (66 zone) and four of the 13 exploration drill holes were included in the 2017 year-end Mineral Resource update. Only four of the exploration holes were included because of delays with commercial labs and required QA/QC re-assays. The remaining nine holes were used in the 2018 year-end Mineral Resource update.

In addition, a total of 32 till samples were collected for gold grain counts and indicator mineral grain analyses during a till drilling program in the second quarter from the Phase 5 and Phase 8 mining areas. Annual till drilling programs by TCM were designed to test for construction material for dam building purposes. Samples (~20 kg wet each with mean sample length of 3.3 m) were collected from eight shallow drill holes over a 915 m east-west trend at the northern margin of the Southern Star zone (Phase-8), and in 10 drill holes over an 890 m trend at the north-northeast corner of the MBX pit, outside the EMBX zone (Phase-6; Figure 9-2). Mean drilling depth was 30 m. For the first batch of 15 samples analyzed by Overburden Drilling Management Limited (ODM) of Ottawa, Ontario, the total calculated visible gold grain count in heavy mineral concentrate (HMC) ranged from 43 - 1,653 ppb Au, mean 511 ppb Au (0.51 g/t Au). A second batch of 17 samples ranged between 35 - 62,755 ppb Au, mean 4,667 ppb Au (4.67 g/t Au). Excluding the two highest grade samples (62,755 ppb and 9,344 ppb) the mean of the remaining 15 samples was 483 ppb Au (0.48 g/t Au).

In 2018, Centerra-TCM completed a 33-hole (12,167 m) Phase-1 NPI and expansion drilling program that ran from February to July in multiple zones within the 2017 ultimate pit boundary, including the DWBX, WBX, MBX, Saddle, Southern Star, EMBX, and 66 zones (Figure 9-2, Figure 10-2). An 18-hole (6,490 m) Phase-2 NPI program ran from September to November in the Saddle and Southern Star zones. Results continued to demonstrate continuity of significant mineralization both within and below the 2017 ultimate pit shell.

Figure 10-2: Mount Milligan Exploration Zones



Note: Exploration zones for NPI, Brownfield and Greenfield areas of the Mt. Milligan deposit and Mount Milligan property. Grey lines include both modelled and conceptual faults. White lines are 2017 ultimate pit boundary and TSF. Background is IP chargeability and LiDAR. The chargeability scale ranges from 0-47.5 mV/V. Inset map shows the September-October 2019 ground-based IP survey lines (18 line-km).

Centerra-TCM exploration programs in 2018 included a ground-based 12-hole (6,669 m) brownfield (within mine lease) drilling program that ran from June to August in the Goldmark, North Slope and Saddle West (King Richard) zones west of the 2017 ultimate pit boundary (Figure 9-2, Figure 10-2). Results indicated two domains of shallow HGLC mineralization in the combined Goldmark-North Slope zone and Saddle West zones, as well as indications of deep porphyry Au-Cu mineralization, particularly in the North Slope zone.

A total of 46 drill holes from the NPI and brownfield programs were included in the year-end Mineral Resource update, expanding the 2018 Measured and Indicated Mineral Resource shell (at CAD \$8.12 NSR cut-off) to the west and at depth.

A helicopter-supported 13-hole (5,616 m) greenfield (outside mine lease) drilling program ran from June to August in the Mitzi East, Heidi, D2 and Fugro zones located north, west and south of the mine lease (Figure 6-3, Figure 9-1). The drilling at Mitzi East completed a 4,000 x 500 m wide fence with 2015-2016 Mitzi drilling outside the northwest corner of the mine lease and was used to

delineate the margins and far-field signature of the porphyry system to the north. There were no significant assay results. A shallow copper target was identified in the Heidi zone, 800 m west of the mine lease (Figure 9-1, Figure 10-2).

In April 2018, gold grain analysis results from HMC were received for an additional 20 till samples taken from two drill holes in the 2018 till drilling program (S18-073 and S18-076). Material (mean 7.7 wet kg per sample) was collected at 1.5 m to 3.0 m intervals through the complete till column of two interpreted east-west trending paleo-channels at the north and south end of the Southern Star zone, the channels separated by 900 m. Calculated total visible gold in HMC ranged from 0-893 ppb, with median value of 290 ppb. The median value for all 52 till samples collected in the near-pit area since 2017 is 296 ppb Au.

In December 2018, Centerra-TCM completed a five-hole (567 m) Phase-1 scout drilling program for groundwater exploration east and north of the TSF. This program tested results of Nuclear Magnetic Resonance (NMR) and Transient Electromagnetic (TEM) geophysical surveys completed between August and November and was designed to support water well targeting efforts for process plant operations. Three of the five drill holes in two target areas (Alpine Lake and Lower Rainbow) produced low flow water and three holes reportedly reached top of bedrock below the surficial geology. No significant assay values were returned from bedrock samples.

In 2019, Centerra-TCM completed a 53-hole (19,832 m) Phase-1 NPI drilling program that ran from February 22 to September 14 in multiple zones within the 2017 ultimate pit boundary or marginal to it, including Southern Star (and SS Satellite Pit appendage), Saddle, WBX, GE Fault, and Oliver zones (Figure 10-2). A 19-hole (7,455 m) Phase-2 NPI program ran from September 15 to November 30 in the WBX, Oliver, GE Fault, Rainbow Fault Central and Southern Star East zones. Significant mineralization was intersected in each zone (less so for RF Central). Assay results continue to demonstrate significant intervals of additional porphyry Au-Cu mineralization both within, below and outside the margins of the 2019 ultimate pit shell.

Exploration drilling programs in 2019 included a ground-based 19-hole (8,672 m) brownfield program that ran from April 30 to August 26 in the North Slope, Goldmark, Saddle West (King Richard), SS West and South Boundary zones west of the 2017 ultimate pit boundary. A 12-hole (5,634 m) Phase-2 brownfield drilling program ran from August 26 to December 16 in the same zones. Results further defined shallow HGLC mineralization as being spatially associated with parallel east-northeast-west-southwest (and possibly north-northwest-south-southeast) trending fault structures and related breccias; verified low-grade porphyry Au-Cu mineralization in the North Slope Zone between ~340-620-m depth related to a dyke and breccia complex that remains open in several directions; and

intersected low-grade porphyry Au-Cu mineralization at comparable depth in the Southern Star West (~560-670 m) and Saddle West zones (~400-430 m). In addition, a ground-based four-hole (1,228 m) greenfield (outside mine lease) drilling program ran from July 5 to 28 in the Heidi Zone that further defined the extent and style of shallow low-grade copper mineralization.

In February 2019, Centerra-TCM completed a five-hole (611 m) Phase-2 scout drilling program for groundwater exploration east and northeast of the TSF. This program continued following up results of NMR and TEM geophysical surveys completed in 2018 and was designed to support water well targeting efforts for process plant operations. One hole in the Upper Rainbow (19-WE08) target area showed indications of groundwater and one hole in the Lower Rainbow (19-WE09) produced flowing artesian conditions. No holes reached bedrock.

In 2020, Centerra-TCM completed a 34-hole (15,582 m) NPI and Resource expansion drilling program that ran from February 20 to December 3 in multiple zones within, or marginal, to the 2019 ultimate pit boundary (as defined by the Centerra Technical Report 2019), including the Southern Star, Saddle, DWBX, and Great Eastern Fault zones (Figure 9-2, Figure 10-2).

Brownfield exploration drilling in 2020 consisted of a 28-hole (14,928 m) program completed from February 12th to December 1st. The program included two deep exploration drill holes totalling 2,760 m, which tested a geophysical target generated from reprocessed historical airborne ZTEM data. The ZTEM images resistivity anomalies at depths of >1 km below the 2019 ultimate pit boundary. Zones targeted from north to south include Zone-4, Goldmark, North Slope, King Richard, Saddle West, South Boundary, and Rainbow Extension/Southern Star South. During the 2020 NPI and Brownfield drilling programs (totalling 30,510 m), three drill holes were abandoned before target depth due to technical difficulties.

Centerra-TCM also completed a 12-hole (1,355 m) metallurgical drilling program in the Southern Star Zone from September 3 to September 18, 2020.

No Mineral Resource model update was completed at year-end 2020. The 2020 drilling (30,510 m) was included in the 2021 year-end Mineral Resource model update.

In 2021, Centerra-TCM completed a 41-hole (25,591 m) Resource NPI drilling program that ran from February 23 to December 11 along the western ultimate pit margins (DWBX zone), below the 2019 ultimate pit boundary including in-pit exploration (MBX and WBX Deep, Saddle zones), to the southwest of the ultimate pit margins (Southern Star zone) and along the eastern ultimate pit margins (Great Eastern Fault Zone) (Figure 10-1, Figure 10-2).

Brownfield exploration drilling in 2021 comprised a 27-hole (13,914 m) program extending from February 12 to December 1. Primary exploration targets include the newly discovered Great Eastern stock below the eastern margin of the pit, the shallow HGLC South Boundary zone, as well as the potential for Resource expansion south of the 2019 ultimate pit boundary in the Rainbow Extension zone.

Results continued to demonstrate continuity of significant mineralization both within, along the margins, and below the 2019 ultimate pit shell, expanding the 2021 Measured and Indicated Mineral Resource shell to the west and at depth. A total of 60 drill holes from NPI and Brownfield exploration drilling (from 21-1297 to 21-1356, excluding abandoned drill hole 21-1347) were used in the 2021 year-end Mineral Resource model update. For five drill holes, partial batches were pending re-assay QA/QC checks by the lab (from drill holes 21-1338, 21-1339, 21-1345, 21-1346, and 21-1352). The internal assay cut-off date for the 2021 year-end Resource model was February 8, 2022. As of the cut-off, results were still pending for eight drill holes from NPI and Brownfield exploration drilling (from drill holes 21-1357 to 21-1364) with partial results pending from one drill hole (21-1358), and these results were not included in the 2021 Mineral Resource model update.

Project management of mineral-focused drilling programs in 2017-2018 was conducted jointly by Centerra-TCM geologists and Equity consultants, and by Centerra-TCM geologists for 2019-2021 drill programs. L.D.S Diamond Drilling Ltd. of Kamloops, B.C. completed the 2017-2021 drilling using Longyear 38 skid-mounted core drills with mainly NQ and lesser HQ diameter drill rods. Apex Diamond Drilling Ltd. of Smithers, B.C. completed the helicopter supported greenfield exploration program in 2018 using two Hydracore 2000 helicopter-portable core drills and NQ diameter drill rods. Omineca Diamond Drilling Ltd. of Burns Lake, B.C. completed drill programs during the 2020-2021 drill campaigns using both skid-mounted and helicopter-portable core drills with mainly NQ and lesser HQ diameter drill rods.

Oriented core data was collected from 2018-2021 using a Boart Longyear Trucore tool, marked on the bottom of each run. The azimuth of the drill was aligned using a hand-held magnetic compass or the Reflex TN14 north-seeking gyrocompass aided by foresights and back sights laid out by the mine survey team using a Trimble differential GPS (DGPS) unit. All drill core was logged, photographed, sampled, and assayed.

At the core logging facility, geotechnicians would reassemble the core and draw a blue orientation line along the bottom of the core. The angular difference between runs, the quality of core reassembly, and the quality of the orientation mark were recorded as measures of orientation QA/QC. Alpha and Beta angles of structural features were recorded by a structural geologist using

a Reflex IQ-logger tool and plotting software. The IQ-logger can capture structural measurements and plot them in real time on a stereonet. Structural data measurements are automatically organized by depth and corrected to true-north azimuth and dip.

Detailed core logging and sampling was done by Equity in 2017, both Equity and Centerra-TCM in 2018-2019, and by Centerra-TCM from 2020-2021. Data was captured directly in a Geospark access database and included lithology, alteration, mineralization, detailed vein data using a modified vein classification scheme after Jago et al. 2014, and structure. Starting with drill hole 19-1215 onward, data was captured directly into an acQuire database.

Geotechnical information collected included core recovery, RQD, SG, and PL testing. Magnetic susceptibility was measured every metre using a Terraplus KT-10 magnetic susceptibility metre. SG measurements were collected every 10 m on all Resource and Brownfield exploration drill holes. On average, samples selected for SG measurements were 10 cm long. SG values were calculated by dividing the dry mass of a sample by its wet mass. The wet mass was measured by measuring the mass of the rock sample suspended in water. PL measurements were conducted by geotechnicians every 10 m on all Resource drill holes using a Roctest PIL-7 point-load tester.

10.2. SUMMARY AND INTERPRETATION OF 2017-2021 DRILLING (CENTERRA-TCM)

This subsection summarizes the current understanding of exploration target zones resulting from drilling programs in 2017 to 2021. The summary of zones goes from north to south generally (Figure 10-2), and each zone is indicated as being within the scope of NPI, Brownfield, or Greenfield drilling programs. For each zone, selected significant composite assay intervals are provided. To be considered significant for exploration results, composite assay intervals must be longer than 2.0 m, have grade greater than 0.1 g/t Au or 0.1% Cu and include maximum internal waste of 4.0 m where it exists. Intervals less than 2.0 m but with grade above 1.0 g/t Au are also reported.

Recent drilling has continued to define targets below the ultimate pit boundary (e.g., MBX Deep zone), as well as along the ultimate pit margins to the west (e.g., Goldmark, DWBX, North Slope zones), to the east (eg. Great Eastern zones), to the north (e.g., Oliver zone), and to the south (e.g., South Boundary and Rainbow Extension zones).

North Slope Zone and Zone-4 – Brownfield

Porphyry Au-Cu mineralization in the North Slope zone is associated with a variably faulted, monzonite porphyry dyke and breccia/fracture zone complex. The dyke and breccia complex controls the apparent shallowly west-dipping 'North Slope' mineralized trend which is suprajacent to the deeper western extension of the 'Goldmark' trend (Figure 9-3). Significant porphyry Au-Cu style mineralization ranges from about 300 m to over 600 m depth in the west part of the zone within a NNW trending moderate chargeability feature. Farther east towards the Goldmark zone, porphyry style mineralization appears to shallow slightly and ranges from about 250-400 m depth. Overprinting HGLC mineralization becomes more prevalent in the east part of the zone towards the chargeability high associated with the Oliver fault (Figure 10-2). This can also be seen in the alteration model where there is a higher intensity of sericitic $\pm$ albitic alteration between the North Slope and Goldmark zones (Figure 9-3).

Selected North Slope zone intercepts from 2018-2019 drilling include:

- 18-1097: 99.00 m @ 0.346 g/t Au, 0.175% Cu from 283.50 m;
- *including* 2.86 m @ 2.580 g/t Au, 0.682% Cu from 305.08 m;
- 18-1107: 18.34 m @ 1.408 g/t Au, 0.076% Cu from 66.00 m;
- *including* 1.34 m @ 16.200 g/t Au, 0.327% Cu from 83.00 m;
- 19-1141: 68.47 m @ 0.172 g/t Au, 0.183% Cu from 436.53 m;
- 19-1212: 71.00 m @ 0.205 g/t Au, 0.207% Cu from 220.00 m.

There is potential for additional discovery on the northern part of the North Slope zone, north of the Oliver fault, where it adjoins a geophysics target called Zone-4. This area was drill tested in 2020 drilling proximal to a modelled fault intersection, associated with a steep chargeability gradient anomaly which intersects west-dipping coincident moderate resistivity-chargeability anomalies and shallow magnetic highs.

Selected Zone-4 intercepts from 2020 drilling include:

- 20-1257: 26.00 m @ 0.793 g/t Au, 0.019% Cu from 514.00 m;
- *including* 1.92 m @ 3.915 g/t Au, 0.030% Cu from 519.08 m;
- *and* 3.89 m @ 2.393 g/t Au, 0.018% Cu from 529.00 m.

Goldmark Zone and Goldmark-Oliver Trend – Brownfield

The Goldmark zone is centered on the N-S trending, west-dipping Goldmark monzonite porphyry stock and dyke complex. It is situated in the center of the chargeability halo that surrounds the MBX Main zone and interpreted to be fault bound by the Oliver fault to the north. The Goldmark mineralized trend is suprajacent to the western extension of the composite DWBX stock and associated mineralization (Figure 9-3). Shallow low-grade porphyry Au-Cu style mineralization is concentrated at the hangingwall/footwall margins of dykes and at the footwall margin of the Goldmark stock; and at greater depth (~450 m and deeper) at the margins of the underlying DWBX stock. The Goldmark zone is dominated by inner to outer propylitic alteration, with localized areas of sodic-calcic alteration. Potassic alteration is weakly developed in both the Goldmark intrusions and surrounding volcanics compared to the larger stock centres (MBX, Southern Star, Great Eastern). HGLC style mineralization occurs throughout the zone and at shallow levels. It is spatially associated with minor fault and fracture zones, overprinting quartz rich sericitic (QSPC) alteration, transitional to late-stage pyrite $\pm$ calcite vein(lets) and with lithological contacts including the faulted margins of the Goldmark stock and some hornblende monzonite porphyry dykes. The HGLC mineralization trend also appears to follow the ENE fault trend and associated high chargeability zone (Figure 10-2).

Selected Goldmark zone intercepts from 2018-2019 drilling include:

- 18-1086: 23.12 m @ 6.210 g/t Au, 0.043% Cu from 20.19 m;
- *including* 3.67 m @ 38.052 g/t Au, 0.081% Cu from 38.33 m;
- 18-1090: 39.00 m @ 0.231 g/t Au, 0.189% Cu from 139.00 m;
- 19-1166: 12.00 m @ 2.498 g/t Au, 0.169% Cu from 250.00 m;
- *including* 0.93 m @ 30.000 g/t Au, 0.254% Cu from 252.07 m;
- 19-1234: 13.31 m @ 5.828 g/t Au, 0.032% Cu from 152.20 m;
- *including* 6.80 m @ 11.070 g/t Au, 0.043% Cu from 152.20 m;
- 19-1198: 57.00 m @ 0.396 g/t Au, 0.383% Cu from 101.00 m.

There is potential for adding near surface HGLC Mineral Resources close to the northern margin of the ultimate pit boundary following the ENE trending Oliver fault if continuity of mineralization can be demonstrated. The 2019 Goldmark-Oliver IP survey was completed across this trend from the North Slope zone in the west to Oliver zone in the east and will guide further exploration for shallow HGLC mineralization close to the pit (Figure 10-2). Drill targeting in this area generally tests a shallow east dipping chargeability high feature and gradient zone. There is also potential for both shallow and deep Au-Cu porphyry Resource addition west of the ultimate pit boundary, in both the Goldmark porphyry zone and the underlying DWBX porphyry zone (Figure 9-2, Figure 9-3). Until 2022,

exploration drilling in this area was kept to historical exploration trail disturbance that were re-opened in recent years' drilling campaigns (since 2018), and the fault block adjacent to the western margin of the pit boundary had only been drilled at wide spacing, generally >150 m. However, with recent west wall pushback mine development, this area has become more accessible for detailed drilling.

Oliver and EMBX Zones – NPI and Brownfield

The Oliver zone covers the northern margin of the ultimate pit boundary, north of the Oliver fault. This is an area where potassic altered andesite has been identified historically, suggesting the potassic alteration shell of the MBX Main deposit is open to the north where drilling is sparse. The target area also includes the northern portion of the axial trend of porphyry stocks, the eastern portion of the Goldmark-Oliver high gold trend, and their intersection (Figure 9-2). Drill targeting in this area generally tests the same shallow east dipping chargeability feature as in the Goldmark zone, but the feature is deeper at Oliver.

Significant mineralized intercepts from 2018-2019 drilling are spatially related to faults and faulted margins of monzonite porphyry dykes with more of an HGLC signature. These include:

- 19-1221: 32.00 m @ 0.372 g/t Au, 0.049% Cu from 362.00 m;
- *including* 1.45 m @ 2.295 g/t Au, 0.048% Cu from 367.00 m;
- 19-1225: 24.70 m @ 0.624 g/t Au, 0.011% Cu from 245.80 m;
- *including* 8.00 m @ 1.564 g/t Au, 0.007% Cu from 256.50 m.

The EMBX zone (East MBX zone) also lies within the northern extent of the potassic alteration shell associated with the MBX Main deposit but is within and below the 2021 ultimate pit boundary. Significant intercepts from the 2018-2019 drilling are spatially related to faults and faulted lithological contacts within the host volcanic rocks and two stages of alteration (potassic and an albite bearing propylitic overprint).

- 18-1078: 19.15 m @ 1.574 g/t Au, 0.076% Cu from 98.05 m.
- *including* 5.20 m @ 3.431 g/t Au, 0.093% Cu from 112.00 m.

MBX, WBX, and DWBX Zones – NPI

The MBX, WBX, and DWBX zones are mostly within the 2021 ultimate pit shell except at depth. Drill holes from 2018-2021 programs are dual purpose as they are collared in the pit for infill drilling (Mineral Resource classification upgrade) and extended below the ultimate pit shell for Resource

expansion. Drilling has shown continuity of significant mineralization both within and below the ultimate pit shell. Programs in the MBX and WBX zones have targeted the footwall of the MBX stock, provided more pierce points through the Harris Fault, tested the extent of an intrusive sill feature subjacent to the Rainbow Dyke, and infilled low density drilling where needed to support Mineral Resource classification. Deep geophysical targets include a steeply dipping chargeability gradient zone that may be related to a north-south trending fault that is not recognized in surface mapping. Both porphyry Au-Cu and HGLC style mineralization have been identified in drilling.

Significant intercepts in the MBX and WBX zones from 2018-2019 drilling include:

- 18-1069 : 100.75 m @ 0.417 g/t Au, 0.364% Cu from 23.00 m ;
- 18-1070 : 72.34 m @ 0.519 g/t Au, 0.232% Cu from 3.66 m ;
- *including* 3.75 m @ 3.495 g/t Au, 0.456% Cu from 37.23 m ;
- *and* 3.40 m @ 1.129 g/t Au, 0.384% Cu from 57.00 m ;
- 19-1207 : 120.00 m @ 0.250 g/t Au, 0.222% Cu from 164.00 m.

The DWBX zone lies on the western side of Harris Fault in a downthrown fault block relative to the adjacent block that hosts the WBX and MBX zones. Because of this, the DWBX composite stock (an interpreted offset root of the MBX stock) is at greater depth than the MBX stock and there is more of a peripheral alteration-mineralization signature near surface (Figure 9-3). Recent drilling has been designed to infill the west wall of the ultimate pit boundary and extend mineralization related to the DWBX stock westward and deeper. Distal to the stock are narrow sections (<2 m wide) of gold mineralization in late-stage carbonate + pyrite ± quartz ± chalcopyrite ± galena ± tetrahedrite veins often with sericite halos. Porphyry-style mineralization is associated with potassic alteration and early quartz veins within the DWBX composite stock and at the stock margins, with the highest grades typically in the stock hangingwall. At least two phases of monzonite porphyry intrusions are interpreted to make up the DWBX composite stock, with different inferred dip directions and different intensities of potassic alteration and chalcopyrite mineralization. The DWBX stock and associated breadth of mineralization appears to taper towards the north, however a moderately west-dipping hydrothermal breccia body associated with Au-Cu mineralization has been identified at shallow depth in host volcanic rocks in the northern part of the DWBX zone. This feature is believed to be associated with emplacement of porphyry stocks and remains to be tested down-dip to the west and into the Goldmark stock footwall volcanic rocks. Recent results from the DWBX zone show the potential for Resource expansion to the west of the current ultimate pit boundary.

Significant mineralized intercepts in the DWBX zone from 2017-2021 drilling include:

- 17-1049 : 15.00 m @ 13.488 g/t Au, 0.031% Cu from 290.00 m ;
- *including* 7.12 m @ 28.068 g/t Au, 0.026% Cu from 296.58 m;
- 17-1049 : 57.34 m @ 1.074 g/t Au, 0.076% Cu from 328.16 m ;
- *including* 13.45 m @ 3.908 g/t Au, 0.068% Cu from 351.90 m;
- 17-1051 : 45.20 m @ 1.780 g/t Au, 0.044% Cu from 60.20 m ;
- *including* 7.50 m @ 5.510 g/t Au, 0.059% Cu from 60.20 m;
- *and* 20.60 m @ 1.569 g/t Au, 0.058% Cu from 75.30 m;
- 18-1071 : 102.00 m @ 0.304 g/t Au, 0.247% Cu from 184.00 m ;
- 21-1336 : 196.50 m @ 0.329 g/t Au, 0.347% Cu from 84.50 m ;
- 21-1340 : 183.00 m @ 0.265 g/t Au, 0.371% Cu from 198.00 m ;
- 21-1350 : 181.65 m @ 0.915 g/t Au, 0.267% Cu from 333.00 m ;
- *including* 0.60 m @ 116.100 g/t Au, 1.880% Cu from 334.56 m;
- *and* 0.74 m @ 38.000 g/t Au, 0.240% Cu from 342.76 m.

MBX Deep, WBX Deep and Lower Porphyry Zones – NPI and In-Pit Exploration

The MBX Deep, WBX Deep and Lower Porphyry zones were drilled in 2020-2021 from the bottom of the existing pit surface, dominantly in the Phase-4 area, for both infill drilling and Resource expansion below the 2019 ultimate pit boundary. Assay results returned from the MBX Deep zone drilled in 2020-2021 show wide intercepts of potentially significant mineralization below the ultimate open-pit boundary. Porphyry-style mineralization is associated with potassic altered latite-andesite units in the footwall of the MBX/WBX monzonite porphyry stock, often overprinted by quartz-sericite-pyrite-carbonate alteration. All lithologies are variably crosscut by early, transitional, and late-stage veins. Late-stage pyrite-calcite $\pm$ polymetallic veins locally enhance gold, often associated with silver, zinc, and lead anomalies. To date, the tested dimensions of the MBX Deep zone are approximately 200 m (north-northeast to south-southwest) by 140 m (east to west) with a mean width of 110 m for significant intervals, and potential to expand to the south and to the east. The zone is interpreted to be bound to the west by the Harris fault and to the south by the Saddle fault.

Significant intercepts in the MBX Deep and WBX Deep zones from 2020-2021 drilling include:

- 20-1269 : 212.66 m @ 0.686 g/t Au, 0.183% Cu from 142.30 m ;
- *including* 31.06 m @ 2.155 g/t Au, 0.224% Cu from 302.94 m;
- 21-1305 : 111.61 m @ 0.392 g/t Au, 0.215% Cu from 22.20 m ;
- *including* 4.00 m @ 2.254 g/t Au, 0.248% Cu from 43.50 m;

- *and* 5.05 m @ 1.613 g/t Au, 0.276% Cu from 75.00 m ;
- 21-1313 : 302.34 m @ 0.402 g/t Au, 0.268% Cu from 12.66 m ;
- *including* 11.46 m @ 1.021 g/t Au, 0.375% Cu from 275.54 m;
- 21-1351 : 148.00 m @ 0.466 g/t Au, 0.172% Cu from 211.00 m ;
- *including* 14.00 m @ 1.457 g/t Au, 0.168% Cu from 304.00 m.

A second series of fault-bound monzonite porphyry intrusions with potassic alteration $\pm$ mineralized early veins named the Lower Porphyry zone, have been drilled at depth underlying the ultimate pit boundary and the MBX stock by approximately 400 m to 600 m. This Lower Porphyry trend is interpreted to be associated with the Great Eastern stock, which lies shallower in an adjacent fault block to the east (Figure 9-3). The Lower Porphyry zone hosts wide runs of low-grade Au-Cu mineralization associated with monzonite porphyry intrusions and potassically altered volcanics in their margins. Grades are enhanced in drill hole 21-1354 associated with a bedded trachyte unit, interpreted to be a favorable volcanic horizon for mineralization in multiple zones including the 66 zone.

Significant intercepts in the Lower Porphyry zone from 2021 drilling include:

- 21-1354 : 112.20 m @ 0.714 g/t Au, 0.340% Cu from 424.25 m ;
- 21-1344 : 69.57 m @ 0.257 g/t Au, 0.179% Cu from 518.00 m ;
- 21-1348 : 117.70 m @ 0.126 g/t Au, 0.161% Cu from 488.00 m.

Great Eastern Fault and Great Eastern Stock Zones – NPI and Brownfield

Drilling in the Great Eastern zones throughout 2019-2021 continued to define two targets, a shallow target associated with the Great Eastern fault (GEF), and an underlying Au-Cu porphyry target associated with the recently discovered Great Eastern composite stock.

The shallow GEF zone lies east of the 66 zone on the east side of the GEF within its hangingwall. Historically, the Great Eastern Fault was interpreted to be steeply dipping and truncate the deposit on the east side. However, recent TEM geophysical modelling and drilling suggests the fault is moderately east dipping and may represent a faulted disconformity within the volcanic stratigraphy. As such, there is potential for extension of 66 zone mineralization to the east in the footwall block of the Great Eastern Fault and beyond the limits of the ultimate pit. Drilling in 2019-2021 has shown that significant mineralization is spatially associated with faults and fault breccia with monzonite porphyry clasts and early-stage vein fragments, and QSPC alteration overprinting early-stage

potassic. Drilling has defined a tabular body along hangingwall of the GEF, with dimensions of approximately 400 m (north-northwest to south-southeast along strike) by 230 m (east to west) with an average width of 25 m for significant intervals of variable Au-Cu grade. The zone may be interpreted as preserved portions of the MBX deposit, milled in the damage zone of the GEF.

Significant intercepts in the GEF zone from 2019-2021 drilling include:

- 19-1169 : 42.88 m @ 0.215 g/t Au, 0.226% Cu from 111.00 m ;
- 19-1213 : 33.40 m @ 0.662 g/t Au, 0.056% Cu from 152.00 m ;
- *including 3.07 m @ 4.700 g/t Au, 0.180% Cu from 160.93 m;*
- 20-1250 : 18.28 m @ 0.493 g/t Au, 0.729% Cu from 54.86 m ;
- 21-1316 : 78.00 m @ 0.450 g/t Au, 0.311% Cu from 24.00 m ;
- 21-1318 : 40.00 m @ 0.303 g/t Au, 0.426% Cu from 31.00 m.

Drilling east of the ultimate pit boundary first intersected the Great Eastern stock zone at the end of 2020 drill programs which was successfully intersected up-dip to the east-southeast in subsequent exploration drilling in 2021. Assay results returned from the Great Eastern stock zone show wide runs of significant mineralization in the hangingwall and footwall margins of the potassic altered monzonite porphyry composite stock and surrounding potassic altered volcanic-volcaniclastic units with variable QSPC overprint. The highest-grade intervals are associated with zones of quartz stockwork within the stock and in the footwall margins, particularly bedded trachytic volcanic host units. Three different phases of intrusions have been logged in detail within the Great Eastern composite stock, which may also affect the grade distribution. To date, the dimensions of the Great Eastern stock defined by drilling are approximately 300 m east-west by 250 m north-south with room to extend to the southeast in the interpreted up-dip direction.

Significant intercepts in the Great Eastern Stock zone from 2020-2021 drilling include:

- 20-1274 : 76.12 m @ 0.439 g/t Au, 0.254% Cu from 390.70 m ;
- 21-1310 : 42.00 m @ 0.313 g/t Au, 0.382% Cu from 341.00 m ;
- 21-1314 : 107.00 m @ 0.297 g/t Au, 0.223% Cu from 262.00 m ;
- 21-1314 : 59.89 m @ 0.228 g/t Au, 0.135% Cu from 375.11 m ;
- 21-1314 : 119.00 m @ 0.462 g/t Au, 0.369% Cu from 490.00 m.

66 Zone – NPI

The 66 zone lies on the southeast side of Rainbow Fault, predominantly in the hanging wall block. It is interpreted as a downthrown fault block (~100 m of throw) and a preserved portion of the sericitic (QSPC) alteration shell overprinting potassic alteration, from a higher level of the porphyry deposit, likely the upper part of the MBX Main zone. It is the foremost setting for HGLC style mineralization and associated vein types at the Mount Milligan deposit. This mineralization is more widespread in the 66 zone than elsewhere in the deposit where it is more narrowly confined to veins, faults, and breccias. Recent work has highlighted the importance of narrow porphyry dykes to mineralization in the volcanic stratigraphy. Higher gold grade appears to be localized in the hanging wall of a north-dipping monzonite porphyry unit, possibly an offset southeastern extension of the Rainbow Dyke. If an extension of the Rainbow Dyke, it is conceivable that this mineralized intrusion represents a cone sheet that formed above the MBX stock during emplacement and spans across the MBX and 66 zones. Bedded/banded and massive trachytic rocks in the volcanic stratigraphy are favourable hosts to mineralization when proximal to, or intersecting with, porphyry dykes. A deeper porphyry unit, historically called the “Lower Monzonite”, appears to be un-mineralized but is not well tested by drilling.

Recent drilling programs have followed up HGLC intersections from the 2017 metallurgical drilling program and tested for continuity and controlling factors of mineralization. The southeast-dipping Rainbow Fault appears to cut off mineralization at depth but there is potential for additional HGLC mineralization down-dip of the fault within its hanging wall below the 2019 ultimate pit shell.

Significant intercepts in the 66 zone from 2018 drilling include:

- 18-1080 : 32.99 m @ 1.100 g/t Au, 0.003% Cu from 3.96 m ;
- 18-1080 : 125.00 m @ 0.914 g/t Au, 0.011% Cu from 47.00 m ;
- *including* 7.00 m @ 7.408 g/t Au, 0.022% Cu from 72.00 m ;
- 18-1084 : 55.95 m @ 1.990 g/t Au, 0.044% Cu from 3.05 m ;
- *including* 7.00 m @ 1.560 g/t Au, 0.042% Cu from 25.00 m ;
- *and* 12.10 m @ 6.589 g/t Au, 0.035% Cu from 42.00 m.

Saddle and Saddle West Zones – NPI and Brownfield

The Saddle zone (formerly Creek zone) is where Mr. Haslinger made his creek bed discovery. This area was less densely drilled historically due to accessibility issues related to King Richard Creek.

The creek was dammed as part of mine construction and drilling resumed with Centerra-TCM in 2017. The zone is centred on a north-south trending, moderately west-dipping monzonite porphyry stock called the Unnamed stock and subjacent sheeted dyke complex, situated midway (~400 m) between the MBX stock to the north-northeast and the Southern Star stock complex to the south-southwest (Figure 9-2). The Saddle zone comprises shallow HGLC style mineralization related to transitional and late-stage veins near faults in the Unnamed stock hanging wall; a subjacent magnetite hydrothermal breccia and early-stage vein zone with porphyry Au-Cu style mineralization in the footwall of the Unnamed stock; a subjacent wedge of well mineralized andesite, and a sheeted porphyry dyke complex of which the coarse-crystalline crowded plagioclase monzonite units appear to be the best Au-Cu mineralized. In geophysics, the Saddle zone is represented by a west dipping moderate chargeability and resistivity feature that is flanked by domains of high chargeability and low resistivity.

Drilling in 2018-2019 tested for up- and down-dip extension of mineralization in the footwall of the stock; for mineralization at depth below the ultimate pit boundary; for a high-grade shoot of porphyry Au-Cu mineralization projecting southward towards the Southern Star zone; and infill drilling for the Resource model.

Selected intercepts in the Saddle zone from 2018-2019 drilling include:

- 18-1056 : 57.20 m @ 0.630 g/t Au, 0.274% Cu from 212.50 m ;
- *including* 5.40 m @ 2.340 g/t Au, 0.986% Cu from 230.60 m;
- *and* 7.70 m @ 0.959 g/t Au, 0.477% Cu from 240.00 m;
- 18-1057 : 149.45 m @ 0.456 g/t Au, 0.435% Cu from 178.40 m ;
- *including* 9.96 m @ 1.418 g/t Au, 1.048% Cu from 219.18 m;
- *and* 2.00 m @ 1.114 g/t Au, 0.788% Cu from 258.52 m;
- 18-1058 : 89.85 m @ 0.445 g/t Au, 0.466% Cu from 160.92 m ;
- *including* 2.64 m @ 1.181 g/t Au, 0.517% Cu from 203.38 m;
- 19-1158 : 76.75 m @ 0.318 g/t Au, 0.375% Cu from 64.00 m.
- *including* 2.00 m @ 1.127 g/t Au, 1.150% Cu from 128.00 m.

The Saddle West (or King Richard zone) zone is centered on the east-northeast trending Saddle Fault, south of the Goldmark zone within Heidi Lake valley. Till overburden in the valley ranges from 10-100 m. Drilling in this area has targeted a down-dip extension of the Saddle zone. Results suggest the area is prospective for shallow HGLC style mineralization in a parallel east-northeast trend to the Goldmark-Oliver trend (~700 m to the north). Significant intercepts show mineralization is associated

with faulted/fractured margins of monzonite porphyry dykes, and with transitional to late-stage veins with sericitic alteration overprinting early-stage potassic alteration.

Significant intercepts in the Saddle West zone from 2018-2019 drilling include:

- 18-1093 : 23.04 m @ 0.982 g/t Au, 0.073% Cu from 242.00 m ;
- *including* 2.89 m @ 6.537 g/t Au, 0.115% Cu from 262.15 m;
- 18-1108 : 7.89 m @ 4.687 g/t Au, 0.034% Cu from 112.11 m ;
- *including* 3.89 m @ 9.357 g/t Au, 0.024% Cu from 112.11 m;
- 19-1185 : 14.92 m @ 2.135 g/t Au, 0.263% Cu from 97.08 m ;
- *including* 7.07 m @ 4.293 g/t Au, 0.449% Cu from 99.00 m.

Southern Star and South Boundary Zones – NPI and Brownfield

The west-dipping Southern Star composite monzonite porphyry stock, dyke and breccia complex is crossed by a set northwest trending faults interpreted as east-side-up reverse faults at roughly 100 m spacing; and east-northeast-trending cross faults near the southern part of the complex which bound a tapering southern monzonite lobe (Figure 9-2). Porphyry Au-Cu style mineralization in the Southern Star deposit is hosted in monzonite porphyries and volcanic host rocks crosscut by abundant variably mineralized hydrothermal breccias and xenolithic monzonite bodies. Breccias occur at the hangingwall and footwall margins of stocks; some are gradational from crackled to milled, and they can be dilutive where un-mineralized. The source of mineralized porphyry clasts in some breccias may be farther west and at greater depth than current drilling. Geophysical targets include deep southwest-dipping coincident chargeability-resistivity gradient zones with high magnetism.

Recent drilling has focused on infill for Resource estimation in support of Phase-8 mine planning, and tested for the extension of mineralization below the ultimate pit boundary and along dip in the hangingwall and footwall of the intrusive complex. Significant assay results from 2018-2019 drilling suggest there is a porphyry Au-Cu mineralized shell in the north and west part of the Southern Star deposit which transitions to a HGLC shell in the southern part of the deposit. Ratios of Au:Cu, used as a proxy for the evolving state of a mineralizing fluid, are lowest (~0.60-0.95) in the north and west part of the deposit, increase to (~1.1-1.6) in a middle overlap zone, and are highest (~1.6-5.7) in the south part, ignoring highest outliers. This suggests that the top of the Southern Star porphyry deposit is to the south, where the high gold shell remains open. This is a similar relationship as seen in the MBX zone (porphyry Au-Cu) and 66 zone (HGLC) across the Rainbow Fault; and a southwest

extension of the Rainbow Fault may be related to the juxtaposition of these mineralization styles in the Southern Star zone as well (Figure 9-2). A north-south trending fault (M1 Fault) appears to cut off significant HGLC grades to the east of Southern Star.

Selected results in the Southern Star zone from 2018-2019 drilling include:

- 18-1123 : 239.27 m @ 0.480 g/t Au, 0.192% Cu from 12.19 m ;
- *including* 8.00 m @ 1.208 g/t Au, 0.404% Cu from 46.00 m;
- *and* 4.00 m @ 1.387 g/t Au, 0.539% Cu from 70.00 m;
- *and* 8.00 m @ 0.979 g/t Au, 0.330% Cu from 84.00 m;
- *and* 11.25 m @ 1.282 g/t Au, 0.452% Cu from 178.00 m;
- *and* 6.00 m @ 1.124 g/t Au, 0.402% Cu from 195.00 m;
- 19-1132 : 148.75 m @ 0.484 g/t Au, 0.258% Cu from 14.63 m ;
- *including* 3.45 m @ 1.258 g/t Au, 0.682% Cu from 81.55 m;
- *and* 11.40 m @ 1.556 g/t Au, 0.516% Cu from 91.00 m.

The South Boundary zone is the southernmost of the three subparallel east-northeast trending structurally-controlled HGLC targets (Goldmark-Oliver, Saddle West, South Boundary) west of the main porphyry stock trend (Figure 9-2). South Boundary lies west of the Southern Star zone in a similar relationship as the Saddle West to the Saddle zone and Goldmark to the DWBX-WBX zones. The area was trenched by BP Resources in the 1980's and wide-space drilled by United Lincoln/Continental Gold in 1989 and 1991. This exploration work identified a high gold 'polymetallic vein' style target (Sketchley, Rebagliati and Delong, 1995). Drilling resumed with Centerra-TCM in 2019 and was designed to validate historical HGLC drilling results, test for expansion and continuity of mineralization, test for a monzonite porphyry stock or dyke complex, and more generally test the hangingwall block above the westward dipping Southern Star stock. Geophysical targets include a shallow-to-moderately west dipping chargeability-resistivity gradient feature with high chargeability and low resistivity in its hangingwall (shallow) and footwall (deep), and a coincident magnetic high feature. This is representative of the Southern Star stock and its westward down-dip extension. In the core of the South Boundary zone is another magnetic high feature that has a ~200 m diameter ring-shape which was also targeted.

Results of the 2019-2021 drilling indicate that mineralization is spatially related to narrow faults, breccia and fracture zones in volcanic rocks with QSPC alteration and associated transitional-to-late stage veins including pyrite stringers and semi-massive pyrite $\pm$ magnetite veins; and to faulted and

brecciated monzonite porphyry dykes with early stage alteration-mineralization and associated veins that have been overprinted by the QSPC assemblage and veins. Some chalcopyrite is noted in late-stage veins suggesting remobilization of earlier stage mineralization. To date, the tested dimensions of the South Boundary zone are approximately 400 m (northwest to southeast) by 150 m (east to west) with an average width of 30 m for significant intervals. The main auriferous vein orientation dips moderately west-northwest, measured in oriented core and ground-truthed by an exploration trench completed in 2021. The South Boundary zone may represent the gold rich cap of an underlying porphyry deposit, similar to the geometry of the 66 zone and the MBX stock. This implies an untested Au-Cu porphyry target to the northwest towards the historical Boundary zone, covered during a ground IP survey in 2020, that shows a near surface high chargeability feature.

Selected results in the South Boundary zone from the 2019-2021 drilling include:

- 19-1200 : 49.30 m @ 1.214 g/t Au, 0.053% Cu from 354.00 m ;
- *including* 1.68 m @ 5.883 g/t Au, 0.166% Cu from 375.32 m;
- *and* 1.10 m @ 1.100 g/t Au, 0.021% Cu from 388.00 m;
- *and* 4.30 m @ 8.796 g/t Au, 0.306% Cu from 399.00 m;
- 19-1211 : 34.00 m @ 0.882 g/t Au, 0.111% Cu from 149.00 m ;
- *including* 2.00 m @ 11.400 g/t Au, 0.506% Cu from 150.96 m;
- 20-1249 : 22.50 m @ 1.542 g/t Au, 0.017% Cu from 67.50 m ;
- 20-1255 : 38.00 m @ 0.535 g/t Au, 0.036% Cu from 139.00 m ;
- 21-1327 : 46.00 m @ 1.246 g/t Au, 0.034% Cu from 107.00 m ;
- 21-1330 : 56.50 m @ 0.429 g/t Au, 0.009% Cu from 121.50 m.

Rainbow Extension Zone – Brownfield

The Rainbow Extension zone was drill tested in 2020-2021, exploring for Resource expansion potential south of the ultimate pit boundary where the interpreted extension of the moderately southeast dipping Rainbow Fault is projected to continue south of the Southern Star deposit (Figure 9-2). The target was covered during the 2020 ground IP survey showing a broad moderate chargeability feature in the footwall of the modelled Rainbow Extension fault. Drilling intersected low-grade Au-Cu porphyry mineralization associated with narrow hornblende monzonite porphyry dykes and transitional pyrite-chalcopyrite stringers. The geology is dominantly andesitic lapilli tuffs with inner to outer propylitic alteration and a variable QSPC overprint.

Significant intercepts in the Rainbow Extension zone from 2021 drilling include:

- 20-1235 : 96.87 m @ at 0.291 g/t Au, 0.131% Cu from 232.00 m ;
- 21-1320 : 121.56 m @ at 0.123 g/t Au, 0.107% Cu from 185.73 m ;
- 21-1326 : 63.00 m @ at 0.303 g/t Au, 0.122% Cu from 212.00 m.

Heidi Zone – Greenfield

The Heidi zone is located approximately 2.7 km west of the ultimate pit boundary and 900 m outside the mine lease boundary (Figure 9-1, Figure 9-3, Figure 10-2). The area was explored and wide-spaced drilled (<300 m depth) by Placer Dome in 1991. A monzonite porphyry stock (Heidi stock) was identified and mapped, and drilling results indicated shallow low-grade copper mineralization. Follow-up geochronological analysis gave a U-Pb date of 189 ± 3.3 Ma (Mortensen et al., 1995) which is the oldest measured monzonite in the Heidi Lake porphyry stock cluster, which averages ~184.8 Ma (Jago, 2008). Centerra-TCM resumed drilling the target in 2018 towards the end of a helicopter supported greenfield drilling program. Geophysically, the target area comprises a 1.6 km long east-west linear magnetic high anomaly that continues eastward into the North Slope zone, as well as a 350 m diameter ring-shaped magnetic high anomaly in the core of the Heidi zone; also a coincident southwest-northeast trending moderate chargeability and high resistivity feature that may be indicative of the Heidi stock.

Results of drilling in 2018-2019 showed that significant mineralized intervals are related to narrow monzonite porphyry dykes in andesitic rocks. Alteration includes a distinct mix of early-stage potassic and sodic-calcic (albite-actinolite-chlorite) assemblages overprinted by propylitic and late-stage QSPC assemblages. Early to transitional vein types include pyrrhotite-bearing polymetallic veins that can be massive and disseminated pyrrhotite appears to be part of the sodic-calcic alteration assemblage. Higher gold grades occur at depth in 18-1104 which may be a distal western extension of the shallow-dipping Goldmark mineralization trend (Figure 9-3), and is spatially associated with a faulted hanging wall of a monzonite porphyry dyke.

Selected results in the Heidi zone from 2018-2019 drilling include:

- 18-1104 : 20.00 m @ 0.206 g/t Au, 0.132% Cu from 568.00 m ;
- 18-1106 : 33.60 m @ 0.074 g/t Au, 0.205% Cu from 17.00 m ;
- *including* 3.52 m @ 0.176 g/t Au, 0.416% Cu from 18.48 m;
- *and* 2.00 m @ 0.116 g/t Au, 0.254% Cu from 33.00 m;

- 19-1179 : 83.73 m @ 0.048 g/t Au, 0.154% Cu from 24.00 m ;
- *including* 5.51 m @ 0.302 g/t Au, 0.268% Cu from 24.00 m;
- *and* 8.00 m @ 0.087 g/t Au, 0.182% Cu from 56.00 m.

10.3. SURVEY CONTROL

Pre-2006 Drilling

The first grids on the property were established in 1984 and 1985 when BP Resources conducted soil geochemical surveys along flagged and tagged lines. The lines were put in using chain and compass and were not corrected for the local magnetic deviation apparently caused by the Mount Milligan Intrusive Complex to the northwest.

Between 1986 and 1988, Lincoln and United Lincoln established a new grid for geophysical surveys. These lines were cut and marked by aluminum tags and pickets. The first grid lines were put in using a compass and were not corrected for the local magnetic deviation. Later lines were turned off a baseline by a transit and electronic distance measurement (EDM) and were put in using a backsight and foresight. This grid was used to establish the locations of diamond drill holes 87-1 to 89-120.

In November 1988, McElhanney Surveying Ltd. (McElhanney) established a mine grid whose relative position was set by labeling a point adjacent to the legal corner post of the PHIL 9 and PHIL 12 mineral claims as being 10,000 N and 10,000 E. The northern and eastern boundaries of the PHIL 9 mineral claim were surveyed to tie the mine grid to the legal corner post, and to establish control points for surveying in drill holes.

Drill holes 87-1 to 88-70 were surveyed to the mine grid as time permitted during the 1988 diamond drill program. The other diamond drill holes were surveyed as they were drilled.

The mine grid and legal survey grids are tied at a point called PCON, which is on a small hill approximately 500 m northwest of the MBX deposit. PCON is tied to a regional Datum (NAD27 UTM Zone 10) through regional geodetic points on Knob Hill northeast of the MBX zone and to a point on the 124th meridian in the area south of the deposit.

For 2004 Placer Dome drilling, McElhanney of Prince George were contracted to survey the drill sites.

All pre-2006 drill hole collars were surveyed to the mine grid and elevations have been corrected to the 2008 LiDAR surface.

2006-2007 (Terrane)

Planned drill holes were initially spotted by the Terrane field geologist using a handheld GPS, and markers were placed for drill collar location and field sites.

In April and May of 2007, AllNorth Consultants Ltd. of Prince George was contracted to survey in the 2006–2007 program drill holes. This survey was conducted using a LEICA RTK GNSS base station and rover combination rated for sub-decimeter accuracy. The latest survey was completed on September 11, 2007 and the field data was subsequently post-processed into NAD83 UTM Zone 10 coordinates.

2010-2016 Drilling (TCM)

In 2010-2011, 27 drill holes (10-1003 to 10-1012, 11-1013 to 11-1019, and MET 11-01 to MET 11-10), were surveyed using a Trimble DGPS capable of sub-decimeter accuracy. The elevation of these collars is based on the 2008 Lidar survey.

During June 2013, 20 holes were drilled to define the thickness of the overburden in the east part of the Main and 66 zones. Collars of vertical holes S13-01 to S13-20 were surveyed by mine personnel using a Trimble DGPS. Elevations were verified using the 2008 Lidar survey.

2015-2016 drill collars were surveyed using a handheld Garmin GPS which was adequate for the exploratory nature of the drill holes. These holes were not used for mineral estimation.

2017-2021 Drilling (Centerra-TCM)

For 2017-2021, all final drill hole locations within the 2019 ultimate pit boundary (Resource and near-pit exploration) were surveyed using mine survey Trimble DGPS equipment (models Trimble TSC7 controller paired to a SPS882 GNSS antenna or a Trimble TSC3 controller paired to a SPS986 GNSS antenna). For greenfield and brownfield exploration, collars were surveyed using Trimble TSC3 capable of sub-decimeter accuracy, a Trimble GeoExp6000 capable of sub-metre accuracy, or a handheld GPS (Garmin 62), depending on availability and radio link signal strength of the rover GNSS antenna away from the base station.

10.4. DOWNHOLE SURVEYS

Pre-2004 Drilling

Downhole dips for diamond drill holes 87-1 to 90-758 (pre-Placer Dome) were determined by acid tests. Downhole azimuths were not determined because it was believed that the results obtained by conventional downhole survey methods utilizing a compass would have been inaccurate due to the concentrations of magnetite. Nine holes were surveyed with a Sperry Sun gyro instrument. Data from this study was analyzed and a weighted average azimuth change of +0.02597 degrees per metre was calculated and applied to all holes with bottom of hole dips of $\leq 85^\circ$.

The holes drilled under the management of Placer Dome employed a Sperry Sun magnetic survey instrument, with the location of survey points controlled by magnetic susceptibility readings of the drill core. Downhole surveys during the 2004 drill program were completed with an Icefield Instruments inclinometer tool.

2006–2007 Drilling (Terrane)

Terrane employed a Reflex EZ-Shot magnetic survey instrument to measure azimuth, dip, roll angle, temperature, and magnetic field strength. Magnetic field strength measurements were used to identify measurements that may have been influenced by magnetic interference. Anomalous readings were flagged and excluded from the data used in Mineral Resource estimation.

2010-2016 Drilling (TCM)

For 2010-2011 and 2015 drill programs, downhole surveys were completed using a Reflex EZ-Shot downhole tool to take azimuth and dip readings approximately every 50 m. Magnetic field strength measurements were used to identify measurements that may have been influenced by magnetic interference. Anomalous readings were flagged and excluded from the data used in Mineral Resource estimation.

No downhole surveys were carried out on the 20 holes drilled in 2013 to define the thickness of overburden.

In 2016 a Reflex TN-14 Gyrocompass was used for drill hole alignment. A Reflex Gyro was used for downhole surveys with tool with azimuth and dip readings taken approximately every 50 m downhole. The Reflex Gyro tool was used in 2016 to reduce anticipated influence from highly magnetic rock.

The 2015 and 2016 drilling programs were exploratory in nature, located 5 km northwest of the Mount Milligan deposit, and are not included in any Mineral Resource estimate.

2017-2021 Drilling (Centerra-TCM)

Down hole surveys for 2017 to 2021 drill holes were completed using both a Reflex EZ-trac downhole tool (greenfield exploration drilling) and Reflex EZ-Gyro (brownfield exploration and Resource infill drilling).

The EZ-trac survey tool was used to capture azimuth and dip readings approximately every 50 m downhole. A magnetic declination value of 17.8° was used in for the 2017-2019. Magnetic field strength measurements were used to help identify measurements that may have been influenced by magnetic interference. Anomalous readings were flagged and excluded from the data used in Mineral Resource estimation.

The EZ-Gyro is a north-seeking, non-magnetic compass unaffected by magnetic interference. At completion of drilling, a single measurement was conducted every 50 feet (~15 m) intervals, and 3 consecutive measurements at a single depth were conducted every 500 feet (EZ-Gyro 'Optimized' mode) for quality control purposes.

While drilling vertical holes, EZ-Gyro surveys were used to provide real time indication that the drill hole was within acceptable limits of deviation.

10.5. CORE RECOVERY

All the drill holes are diamond drill core holes with predominantly NQ size core diameter. Some HQ size has also been drilled since 2006, mainly during the 2006-2007 feasibility drilling period and for metallurgical test work in 2020. A few six-inch holes were drilled for metallurgical samples. Geotechnical information was routinely recorded for all diamond drilling programs including core recovery, RQD, hardness or CS, degree of breakage, degree of weathering or oxidation, fracture and joint frequency, and SG. Core recovery routinely exceeds 90% and with a mean of 96%. Based on the high core recovery there is no known material impact on the reliability of results.

11. SAMPLE PREPARATION, ANALYSES, AND SECURITY

11.1. PRE-2004 SAMPLES

Sampling Method and Approach

Most of the diamond drill hole samples were collected from NQ diameter (47.6 millimetre [mm]) drill core. Core was marked with intervals and split lines for sampling by the geologist on nominal 1 m (Lincoln and United Lincoln), or 2 m (Continental Gold) intervals except where veins, alteration or lithologic contacts were encountered. Core was split in half using a core splitter, with one half shipped to the laboratory for analysis and the other half retained on site.

Core was stored in outside core racks or dead-stacked on the ground. All racks were vandalized in 2005, leaving only the dead-stacked core intact. Check assay programs were run by both Continental and Placer Dome and the data passed QA/QC, prior to the core vandalism in 2004. Some of this core was re-organized during the 2004, 2005, 2006, and 2007 field seasons.

Figure 11-1: Heidi Lake Core Storage Yard in 2005 after Occurrence of Vandalism



Photo by Darren O'Brien.

Sample Preparation and Laboratory

Samples for drill holes 87-1 to 88-60 were prepared and assayed by Acme Analytical Laboratories Ltd. (Acme); drill holes 88-61 to 90-758 by Mineral Environments Laboratories Ltd. (Min-En); and drill holes 90-759 to 91-862 by the Placer Dome Research Centre (PDRC). All laboratories were in Vancouver, British Columbia and independent of the Project owner.

Lab certification details for Acme, Min-En, and PDRC are unknown during the time the analyses were performed.

Samples were prepared as follows:

- Samples were first dried at 95°C.
- Samples were jaw crushed to nominal 6 mm.
- Samples were then roll crushed to nominal 0.3 mm.
- Samples were riffle split until a 300 to 400 gram (g) sub-sample was generated.
- Sub-samples were then pulverized to 95% passing a 120-mesh screen.

Assaying

Gold (Au) was assayed by the laboratory at which they were prepared (Acme, Min-En, and PDRC) by standard fire assay with an atomic absorption (AA) finish on a 30 g pulp sample. Copper (Cu) was assayed by digesting 2 g of sample in aqua regia and determining the assay value by atomic absorption spectrometry (AAS). Au assay batches consisted of 24 samples and Cu assay batches of 70 samples. Bondar Clegg and Chemex, both of Vancouver, British Columbia, performed check assays on selected sample pulps using the same protocol. Both Bondar Clegg and Chemex were independent of the Property owner at the time and independent of Centerra.

Min-En performed metallic screen fire assays for Au by weight-averaging the entire +120 mesh fraction with the average of two assays of the -120 mesh fraction.

Trace element geochemical analysis of silver (Ag) by AAS (Min-En) was routinely conducted on samples from drill holes 88-61 to 89-212.

The assay procedures were performed by methods conforming to industry standard practices for Au-Cu porphyry deposits.

Quality Control

Standards, blanks, and duplicates were not routinely inserted into the sample stream during the historic drill programs from 1987 to 1992. However, there were external check assay programs in place, both by Continental Gold and Placer Dome, and one of these check programs did employ standards as a check on accuracy. Additional check assay programs have been initiated over the years of the pre-2004 assay samples and are described in detail in Thompson Creek Metals Company Inc.'s (TCM) 2015 Technical Report (refer to TCM January 21, 2015, NI 43-101 Technical Report check assay programs). The report concluded from the check assay programs that the data was acceptable and free of significant bias.

It is the QP's opinion that the quality control program for the pre-2004 Mount Milligan drill program met generally accepted industry standards at the time and confirm the accuracy, precision, and lack of contamination.

Security

No specific sample security measures were in place during the pre-2004 drill programs.

11.2. 2004 PLACER DOME SAMPLES

Sampling Method and Approach

Diamond drill hole samples were collected from NQ-2 diameter (50.5 mm) drill core. Core was marked with intervals and split lines for sampling by the geologist on 2 m intervals except where pyrite veins or lithologic contacts were encountered. Core was split in half using a core splitter, with one half shipped to the laboratory for analysis and the other half retained on site.

The 2004 drill core was subject to vandalism in 2005 and is no longer available (Figure 11-1).

Sample Preparation and Laboratory

Samples for drill holes 04-920 to 04-933 (2004) were prepared and assayed by Eco-Tech Laboratories Ltd. (Eco-Tech) of Kamloops, British Columbia which maintained an ISO 9001:2000 standard for the provision of assay and geochemical analytical services. Eco-Tech is independent of the Property owner at the time and independent of Centerra. Samples were prepared as follows:

- Samples were jaw crushed to nominal 5 mesh.

- Samples were crushed to nominal 10 mesh (equipment used is not documented).
- Samples were then riffle split until a nominal 1 kilogram (kg) sub-sample was generated.
- Sub-samples were then pulverized to nominal 140 mesh.

Assaying

Au was assayed by Eco-Tech by standard fire assay with an AA finish on a 30 g pulp sample. Cu assays utilized an aqua regia sample decomposition with analysis by inductively coupled plasma atomic emission spectroscopy (ICP-AES). Cu analyses greater than 7,000 parts per million (ppm) were rerun using AAS.

The method codes used by Eco-Tech are not documented on the assay certificate, but the lab is accredited as described above.

Quality Control

Placer Dome introduced a QA/QC program through the systematic insertion of one blank, one pulp duplicate, and one company standard inserted every 37 core samples sent for analysis to Eco-Tech. Eco-Tech received a flask labeled “Standard A”, which was used as the company standard. “Standard A” consisted of CGS-1 from CDN Resource Laboratories Ltd. of Delta, BC. CGS-1 was prepared from ore supplied by B.C. Metals Corporation from the Red Chris Porphyry deposit in British Columbia. Assay results of the CGS-1 standard were consistently higher than the recommended standard value but within the acceptable ranges defined on the certificate. Analyses of pulp duplicates yielded generally excellent correlation for Cu and fair to poor correlation for Au. All blanks produced null to low Au and Cu values. No external lab check assays were performed for the 2004 samples.

It is the QP’s opinion that the quality control program for the 2004 Mount Milligan drill program met generally accepted industry standards at the time and confirm the accuracy, precision, and lack of contamination and the results are suitable for resource estimation.

Security

No specific sample security measures were documented for the 2004 drill program.

11.3. 2006–2007 TERRANE SAMPLES

Sampling Method and Approach

Terrane collected samples from HQ diameter (63.5 mm) diamond drill core. Intervals were nominally 2 m of core length but were shortened at the discretion of the geologist at lithological, structural or major alteration contacts. Prior to marking the sample intervals, technicians photographed and geotechnically logged the core. Technicians then marked the sample intervals and assigned sample numbers. After the sample intervals were marked, the geologist logged the core in detail and the core was sent for sampling.

Drill core was split in half using a hydraulic core splitter, with one-half placed in a sample bag for shipping to the sample preparation laboratory and the other half placed back in the core box for future reference. Core boxes were cross-stacked on pallets and remain at the project site.

Sample Preparation and Laboratory

The initial splitting of drill core at the project site was the only aspect of sample preparation performed by Terrane employees. The half-core samples were then shipped to ALS Chemex (now ALS Minerals (ALS)) in Vancouver, British Columbia for sample preparation and analysis. ALS laboratories in North America are registered to ISO 9001:2000 for the provision of assay and geochemical analytical services by QM Quality Registrars. In addition, ALS' main North American laboratory in North Vancouver, British Columbia is accredited by the Standards Council of Canada (SCC) for specific tests listed in the Scope of Accreditation No. 579, which is available at <https://www.scc.ca>. This accreditation is based on international standards (ISO 17025) and involves extensive site audits and ongoing performance evaluations. ALS was independent of the Property owner at the time and independent of Centerra.

Samples were prepared using Method Code PREP-31 as follows:

- Samples were logged into the tracking system and barcodes applied.
- Samples were dried and weighed.
- Samples were fine crushed >70% passing 2 mm.
- Samples were split to 250 g and pulverized to >85% passing 75 µm.

Assaying

Drill core samples were analyzed for Au content using ALS method Au-AA25. Au assays utilized a fire assay fusion sample decomposition of a 30 g pulp with an AAS finish. Cu analysis was completed using ALS method Cu-OG46. Cu assays utilized aqua regia sample decomposition with analysis by ICP-AES.

Every second sample was also analyzed for multiple elements using ALS method ME-MS41. This method analyzed trace levels for 50 elements by aqua regia digestion and a combination of ICP-AES and inductively coupled plasma mass spectrometry (ICP-MS). Ag was analyzed as part of this multi-element package.

Assay methods used by Terrane (Au-AA25, Cu-OG46, and ME-MS41) at ALS are listed on their ISO 17025 accreditation.

Quality Control

In addition to an internal laboratory quality control program utilized by ALS, Terrane maintained an additional QA/QC program through the systematic use of standards, blanks, and duplicates. For every 20 samples, one standard and one blank were inserted into the sample stream by core sampling personnel at the project site. Two different Cu-Au standards were purchased from CDN Resource Laboratories Ltd. in Delta, British Columbia. Standards were alternated for each batch of 20 samples. In addition, for every twentieth sample, the sample preparation laboratory created a duplicate pulp for a comparative analysis.

The results of the QA/QC program were reviewed, and corrective action was taken on sample batches with QA/QC samples exceeding acceptable limits. A slight high bias of the Cu analyses of 2-5% relative to the accepted value of the certified reference material used was noted. A similar bias was noted in the laboratory's internal QC analyses. This bias was within the range of some of the labs participating in the round robin analyses of the standards. There was no appreciable bias of the Au analyses relative to the standards.

External pulp duplicate check analyses at a separate laboratory (Acme) indicated no significant relative bias in the Au analyses and a consistent high bias of the primary analyses of ~5% for Cu.

Assessment of precision based on routine analyses of preparation duplicates (split after initial crushing) indicated the precision of Au analyses is slightly less than ideal, and Cu is slightly greater.

Routine analyses of blank material indicated no systematic contamination during sample preparation. A few isolated higher analyses resulted in batch re-assays with acceptable results.

It is the QP's opinion that the quality control program for the 2006-2007 Mount Milligan drill programs met generally accepted industry standards at the time and confirm the accuracy, precision, and lack of contamination.

Security

Samples were sealed in large rice sacks and stored in the core sampling shed to improve the security of the samples while at the project site, and to ensure the validity and integrity of the samples taken. Twice weekly, the sacks were shipped from the project site directly to the ALS preparation laboratory via Russell Transfer, a bonded independent expeditor based in Fort St. James, British Columbia.

11.4. 2010-2011 TCM SAMPLES

Sampling Method and Approach

TCM collected samples from predominately NQ diameter diamond drill core. Intervals were nominally 2 m of core length but were shortened, at the discretion of the geologist, at lithological, structural, or major alteration contacts. The drill core was marked with the sample intervals, assigned sample numbers and photographed before the core was sent for cutting.

Drill core was cut in half using a diamond drill core saw by TCM staff, with one half placed in a sample bag for shipping to the sample preparation laboratory and the other half placed back in the core box for future reference. Core boxes were cross-stacked on pallets and remain at the project site.

Sample Preparation and Laboratory

For the 2010-2011 drilling campaigns TCM continued with the preparation method and analytical protocol utilised by Terrane for the 2006-2007 program. Drill core was cut at the project site and the half-core samples were then shipped to sent to the ALS sample preparation facility in Terrace BC for sample preparation. Splits of the pulps were subsequently shipped to the ALS laboratory in Vancouver for analysis. ALS was independent of the Property owner at the time and independent of Centerra.

Assaying

The 2010-2011 drill core samples were analyzed for Au content using ALS method Au-AA25. Au assays utilized a fire assay fusion sample decomposition of a 30 g pulp with an AAS finish. Cu analysis was completed using ALS's method Cu-OG46. Cu assays utilized aqua regia sample decomposition with analysis by ICP-AES.

Every second sample was also analyzed for multiple elements using ALS's method ME-MS41. This method analyzed trace levels for 50 elements by aqua regia digestion and a combination of ICP-AES and ICP-MS. Ag was analyzed as part of this multi-element package.

Assay methods used by TCM (Au-AA25, Cu-OG46, and ME-MS41) at ALS are listed on their ISO 17025 accreditation.

Quality Control

All campaigns utilized the same QA/QC protocol initiated during the 2004 drill program, with the insertion of standard reference material (SRM), duplicates and blanks into the sample stream to monitor precision, accuracy, and contamination of the sampling and analytical process. Results of these analyses were continuously and independently monitored, with a 'failure table' documenting the quality control samples that exceeded acceptable limits and tracking the corrective measures taken. Where necessary, analytical batches were re-assayed to achieve final analyses that met industry standards of quality.

Precision was measured through duplicate samples taken at various stages of sample size reduction; quarter core field duplicates, preparation duplicates taken after coarse crushing and pulp duplicates taken routinely as part of the ALS and BV internal quality control.

Coarse barren limestone or quartzite was routinely added to each batch to monitor possible contamination during the sample preparation stage. No significant contamination was indicated during the assay programs.

In addition to the QA/QC samples submitted to the primary laboratory, approximately 5% of the sample pulps were submitted to Acme Labs in Vancouver for the 2010-2011 campaigns as an independent check against analytical bias and accuracy.

Through continuous monitoring of the QA/QC results, significant issues affecting the results were identified and resolved, and it is the QP's opinion that the quality control and check assays completed

confirmed that the 2010-2011 Mount Milligan assay data were accurate, precise, and free of contamination to industry standards and is of sufficient quality to be used in a resource estimation.

It is the QP's opinion that the quality control program for the 2010-2011 Mount Milligan drill programs met generally accepted industry standards and confirm the accuracy, precision, and lack of contamination and the results are suitable for resource estimation.

Security

Samples were sealed in large rice sacks and stored in the core sampling shed to improve the security of the samples while at the project site, and to ensure the validity and integrity of the samples taken. Twice weekly, the sacks were shipped from the project site directly to the ALS preparation laboratory via Russell Transfer, a bonded independent expeditor based in Fort St. James, British Columbia.

11.5. 2015-2016 TCM SAMPLES

Sampling Method and Approach

TCM collected samples from predominately NQ diameter diamond drill core. Intervals were nominally two metres of core length but were shortened, at the discretion of the geologist, at lithological, structural, or major alteration contacts. The drill core was marked with the sample intervals, assigned sample numbers and photographed before the core was sent for cutting.

Drill core was cut in half using a diamond drill core saw by TCM staff, with one half placed in a sample bag for shipping to the sample preparation laboratory and the other half placed back in the core box for future reference. Core boxes were cross-stacked on pallets and remain at the project site.

Sample Preparation and Laboratory

Bureau Veritas Laboratory (BV) in Vancouver, British Columbia was utilized for sample preparation and analysis for the 2015 and 2016 drilling campaigns located approximately 5 km northwest of the current open pit. BV's laboratories in North America are registered to ISO 9001 for the provision of assay and geochemical analytical services. The BV analytical laboratory in Vancouver, British Columbia is accredited by the SCC for specific tests listed in the Scope of Accreditation No. 720, which is available <https://www.scc.ca>. This accreditation is based on international standards (ISO 17025) and involves extensive site audits and ongoing performance evaluations. BV was independent of TCM at the time and independent of Centerra.

Drill core was cut at the project site and half-core samples sent to BV in Vancouver, British Columbia for sample preparation and analysis.

Samples were prepared as follows (code PRP80-250):

- Samples were dried and weighed.
- 1 kg sub-samples were crushed to $\geq 80\%$ passing ~ 2 mm.
- Crushed material was then riffle split and a 250 g sample pulverized to $\geq 85\%$ passing $75\ \mu\text{m}$.

Assaying

Drill core samples from 2015-2016 were analyzed utilizing BV's 37 multi-element AQ251 method. The method utilizes ICP-MS analysis of a 15 g sample after modified aqua regia digestion for low to ultra-low determination. The analytical flow sheet called for fire assay of samples returning >0.20 g/t Au by ICP but the 2015 program returned no samples exceeding that grade, with the highest value running 0.15 g/t Au.

Quality Control

The campaigns utilized the same quality control protocol initiated during the 2004 drill program with the insertion of SRM, duplicates and blanks into the sample stream to monitor precision, accuracy and contamination of the sampling and analytical process. Results of these analyses were continuously and independently monitored, with a 'failure table' documenting the QA/QC samples that exceeded acceptable limits and tracking the corrective measures taken. Where necessary, analytical batches were re-assayed to achieve final analyses that met industry standards of quality.

For QA/QC, duplicate samples were taken at various stages of sample size reduction. Quarter core field duplicates, preparation duplicates taken after coarse crushing, as well as pulp duplicates were all taken routinely as part of the BV quality control.

Coarse barren limestone or quartzite was routinely added to each batch to monitor possible contamination during the sample preparation stage. No significant contamination was indicated during the assay programs.

The results of the QA/QC program for the 2015 and 2016 drill programs indicated that Cu and molybdenum (Mo) analyses were accurate, precise and free of contamination. Au assays were also uncontaminated but significantly less accurate and precise. The low accuracy in the Au values can

be attributed to the imprecision of the analytical method used. The aqua regia digestion with ICP-MS finish procedure is known to be less accurate and precise than fire assay due to the smaller test weight and limited Au solubility in refractory minerals.

The 2015 and 2016 drilling programs were exploratory in nature, located 5 km northwest of the Mount Milligan deposit. The QA/QC analysis shows that the data are sufficiently accurate and precise for exploration purposes and were not used in resource modelling.

It is the QP's opinion that the quality control program for the 2015-2016 Mount Milligan drill programs met generally accepted industry standards and confirm the accuracy, precision, and lack of contamination and the results are suitable for resource estimation.

Security

Samples were sealed in large rice sacks sealed with individually numbered security straps. Samples were shipped from Mount Milligan Mine to BV's preparation facility in Smithers, British Columbia in 2015 and to BV's analytical facility in Vancouver, British Columbia in 2016. BV reported all bags were received in good condition, with security tags intact and with no evidence of tampering.

11.6. 2017-2021 CENTERRA-TCM SAMPLES

Sampling Method and Approach

Diamond drill core samples were collected from predominately NQ diameter diamond drill core. HQ diameter (63.5 mm) drill core was also sampled when logged including metallurgical drill holes from 2017 and 2020. Sample intervals were nominally of 2 m core length but were shortened at the discretion of the geologist at lithological, structural, or major alteration contacts. Prior to detailed logging and marking of the sample intervals, technicians logged the core for geotechnical characteristics. After the geologist logged the core in detail, the drill core was marked with the sample intervals, cut lines, and assigned sample numbers. The wet and dry drill core was photographed prior to being sent for cutting.

Most of the NQ and HQ drill core was cut in half using a diamond drill core saw, with one half placed in a sample bag for shipping to the laboratory and the other half placed back in the core box for future reference. For metallurgical test work in 2017, selected core intervals had their remnant HQ drill core halves cut an additional time to create a quarter core sample to be sent to the lab. In 2019, NQ drill

core from holes 19-1132 to 19-1134 was cut an additional time so three quarters of the core could be sent to BV for analysis and one quarter was placed back in the box.

Sample Preparation and Laboratory

From 2017 to present, drill samples were cut at the project site and half-core samples were sent for sample preparation and pulverization to BV in Vancouver, British Columbia.

A subset of approximately 1,166 samples from 3 drill holes (17-1049, 17-1050, 17-1051) was shipped to Activation Laboratories (ActLabs) in Kamloops, British Columbia. The ActLabs laboratory in Kamloops, British Columbia is accredited by the SCC for specific tests listed in the Scope of Accreditation No. 790, which is available at <https://www.scc.ca>. This accreditation is based on international standards (ISO 17025) and involves extensive site audits and ongoing performance evaluations.

A total of 281 samples from one drill hole (20-1280) were sent to SGS Laboratories (SGS) in Vancouver, British Columbia, as well as a subset of 5% of samples from 2018-2021 drill core samples as an independent check for analytical bias and accuracy. The SGS laboratory in Vancouver, British Columbia is accredited by the SCC for specific tests listed in the Scope of Accreditation No. 744, which is available at <https://www.scc.ca>. This accreditation is based on international standards (ISO 17025) and involves extensive site audits and ongoing performance evaluations.

BV, ActLabs and SGS are all independent of Centerra.

At BV, samples were prepared as follows (code PRP80-250):

- Samples were dried and weighed.
- Entire sample crushed to $\geq 80\%$ passing ~ 2 mm.
- Crushed material was then riffle split and a 250 g sample pulverized to $\geq 85\%$ passing 75 μm .

At ActLabs, samples were prepared as follows (code RX1):

- Samples were dried and weighed.
- Entire sample crushed to $\geq 80\%$ passing 2 mm.
- Crushed material was then riffle split and a 250 g sample pulverized (mild steel) to $\geq 95\%$ passing 105 μm .

At SGS, samples were prepared as follows (code PRP89):

- Samples were dried and weighed.
- Entire sample crushed to $\geq 75\%$ passing 2 mm.
- Crushed material was then riffle split and a 250 g sample pulverized (mild steel) to $\geq 85\%$ passing 75 μm .

Assaying

At BV, samples from 2017-2021 were analyzed for precious and base metals, as well as multi-elements. Au was assayed using a 30 g fire assay with AAS finish (BV lab code FA430). Au results over the upper detection limit of the method (≥ 10 ppm) triggered 30 g fire assay with gravimetric finish (FA530). All samples were also analyzed for a 45-element package, including Cu and base metals, using a 4-acid digestion and inductively coupled plasma mass spectrometry/emission spectroscopy (ICP-MS/ES) on a 0.25 g aliquot (BV lab code MA200). Cu results $\geq 1\%$ triggered analysis using ICP-MS with AAS finish (MA404) on a 0.50 g aliquot. Ag results ≥ 100 ppm triggered 30 g fire assay with gravimetric finish (FA530). Sulphur $>10\%$ triggered Leco analysis (TC000). Mercury was analyzed, using an aqua regia digest with cold vapor AAS finish (CV400).

Assay methods used by Centerra-TCM for Au (FA430, FA530), 45 element package (MA200), and over-limit Cu and sulfur (MA404, TC000) are listed on BV's ISO17025:2017 accreditation.

At ActLabs, samples from 2017 were analyzed for similar precious and base metals as BV. Au was assayed using a 30 g fire assay with AAS finish (ActLabs code 1A2). Au results over the upper detection limit of the method (≥ 10 ppm) triggered 30 g fire assay with gravimetric finish (code 1A3). Samples were also analyzed for a 47-element package, including Cu and base metals, using a 4-acid digest and ICP-MS on a 0.25 g aliquot (Actlabs code UT-4M). Cu, lead and zinc were further analyzed by using a 4-acid digestion with inductively coupled plasma optical emission spectroscopy ICP-OES finish. Sulphur $>10\%$ triggered Leco analysis (Actlabs code 4F).

Assay methods used by Centerra-TCM for Au, Ag and Cu are listed under Act Labs ISO17025:2017 accreditation.

At SGS, samples were analyzed for similar precious and base metals as completed by both BV and ActLabs. Au was assayed using a 30 g fire assay with atomic emission spectrometry (AES) finish (SGS code GE_FAI31V5). Au over the upper detection limit of the method (≥ 10 ppm) triggered 30 g fire assay with gravimetric finish (SGS code GO_FAG30V). Samples were also analyzed for a 49-element package, including Cu and base metals, using a 4-acid digest and (ICP-MS on a 0.2 g aliquot (SGS code GE_ICM40Q12). Cu, lead and zinc were further analyzed by using a 4-acid

digestion with ICP-AES finish. Sulphur >10% triggered Leco analysis (SGS code GE_CSA06V). Mercury was analyzed using an aqua regia digest and cold vapour with AAS finish (SGS code GE_CVA37A25).

Assay methods used by Centerra-TCM for Au, Ag and Cu are listed under SGS ISO17025:2017 accreditation.

Quality Control

For 2017-2021 diamond drilling, certified reference materials (CRM), blanks, and duplicates were used to monitor QA/QC of the core sampling, processing, and assaying processes. All samples were marked with a unique sample ID number and sample tag in the core box by a logging geologist and cut using an electric core saw. While sampling drill core, the logging geologists inserted CRMs and coarse blank samples alternately into the sample sequence every 10 samples. Clean coarse marble landscape rock weighed in ~1 kg samples were used for blank material. Two different Cu-Au CRM's were purchased from CDN Resource Laboratories Ltd. in Delta, British Columbia, and two different Cu-Au and multi-element certified CRM's from OREAS North America Inc. in Sudbury, Ontario. The CRM's were selected to match low, medium and high-grade mineralization ranges and are dominantly sourced from Cu-Au bearing porphyry intrusive rocks.

In addition to the inserted reference materials, field and coarse reject duplicates were inserted alternately into the sample sequence every 20 samples. Field duplicates were prepared by quartering one half of the core, with one quarter sent for analysis with a unique sample ID, and the other remaining in the core box. Coarse reject duplicates were prepared at BV labs prior to sample pulverization. The QA/QC insertion rates are acceptable according to current CIM best practice standards, with QA/QC samples accounting for ~15% of the 2017-2021 assay database.

After the assay results were received from the lab, Au and Cu assays were checked by a Centerra database manager using control charts for the CRMs, blanks and duplicates. Any quality control failures (samples bracketing CRMs with assay values $\pm$ three standard deviations of the expected value) were documented and relevant batches of samples were requested for re-assay by BV labs using the primary pulp. After completion of the programs approximately 5% of the sample pulps from BV were submitted to the SGS laboratory in Burnaby, British Columbia as an independent check for analytical bias and accuracy.

QA/QC of drill core from 2017 to 2021 demonstrates that assay data are accurate, precise and free of contamination to industry standards and in the QP's opinion, of sufficient quality to be used in a resource estimation.

Security

After drill hole samples were cut and bagged in poly-plastic sample bags, they were prepared for sample shipment by a geotechnician. Individual sample bags were collected in rice bags and sealed with zip ties and individually numbered security tags. A copy of the sample submission form with a list of samples and security tags for each batch was included in the last rice bag of samples and emailed to the exploration office and BV labs.

Production Blast Hole Samples

Blast hole samples are collected daily by the Mine Geology team and delivered to the Mount Milligan (MTM) Assay Laboratory for analysis. Sample results are used for daily grade control purposes including the determination of ore/waste boundaries, grade differentiation between ore blocks, and ore blending strategies.

Blast hole sample results are not used in annual Mineral Resource estimate updates, but are reconciled monthly with the Geology block model, and actual mill results.

Sample Delivery, Intake, and Preparation

Blast hole samples are collected twice a day by the Ore Control Geologist/Technician and delivered to the on-site Mount Milligan Assay Laboratory for analysis.

Samples are delivered in individual heavy-duty plastic bags with unique barcode ID tags.

Upon delivery at the assay laboratory, a Geology Sample Submission form is submitted to the laboratory stating the date, submitter and the number of samples along with the sample identifications and the analysis required (Cu, Au, Ag, S, Fe, ABA).

Each sample is transferred from the bag to individual drying pans and is scanned into a batch created in the Laboratory Information Management System (LIMS) software.

Samples are dried, crushed, and riffle split down to ~300 g.

The split material is then pulverized to -200 mesh (75 microns), transferred to a barcoded envelope and staged for analysis by wet chemistry and fire assay.

Wet Chemistry

Metal analysis is completed via acid digestion and AA analysis for each sample. Cu, Fe and Ag values are read on a PerkinElmer PinAAcle atomic absorption spectrometer using internal calibration with QC standards to include blanks, drift check standards, duplicates and CRM standards.

Acid Base Accounting (ABA) analysis comprises volumetric titration to determine neutralization potential and sulphur analysis to determine acid potential. Sulphur analysis is completed on a LECO CS230SH to include quality control blanks and certified reference material standards.

Fire Assay

Flux preparation, fusion, cupellation, parting and atomic absorption analysis are completed for each sample, and Au values are read on a PerkinElmer PinAAcle atomic absorption spectrometer using internal calibration with quality control standards to include blanks, drift check standards, duplicates and CRM standards.

Quality Control and Remedial Action Procedure

All MTM Assay Laboratory procedures are accompanied by appropriate, industry standard instrument calibration and QA/QC measures, including quarterly third-party analysis checks. Ore and acid-base accounting analyses standard operating procedures (SOP) includes steps to confirm on-site laboratory method accuracy, precision, contamination control, sample tracking, and recordkeeping. The assay laboratory also receives blind duplicate samples from the Ore Control Geologist/Technician which are compared against daily sample analysis. This is managed as part of the MTM Assay Laboratory Quality Management System.

Adequacy of Sample Preparation, Analysis and Security

In the opinion of the QP for this Item of the Technical Report, sample preparation, security, and analytical procedures utilized during drilling programs were adequate and conducted according to CIM Estimation of Mineral Resources and Mineral Reserves best practice guidelines.

12. DATA VERIFICATION

The QP completed several data verification checks in support of the Mount Milligan Mineral Resource estimate. The verification process included a two-day site visit to the Project site to review geological procedures, chain of custody of drill core samples, drill collar inspections and the collection of independent samples for metal verification. Other data verification included spot check comparisons of Au and Cu assays from the drill hole database against original assay records (lab certificates) and a review of QA/QC performance for drilling completed between 2004-2021, with attention given to the Centerra specific drilling from 2017-2021.

12.1. SITE VISIT

A site visit to the Project site was carried out by Jennifer Simper, P.Geo., from July 28, 2021, to July 29, 2021. The site visit included the following activities:

- Confirmation core logging and independent assay verification on selected drill core samples
- Inspection of drill hole collar locations
- Review of procedures and collection of data

Details of the site visit and data verification are summarized in the following sub-Items.

12.2. INDEPENDENT LOGGING AND SAMPLE VERIFICATION

The QP selected intervals from six drill holes from both historical (2006 and 2007) and Centerra (2018 and 2019) drill programs for independent logging and sample analysis. Figure 12-1 and Figure 12-2 provide examples of verification intervals from drill holes 06-957 and 19-1132, showing examples of Au and Cu mineralization hosted in andesite porphyry flow and plagioclase monzonite porphyry, respectively. The Centerra drill logs were found to match the observed core reasonably well and no material issues were identified.

Twelve random verification samples were obtained from the logged core sample intervals, corresponding to the original intervals. The QP did not use the exact same analytical methods as Centerra, and the samples were analysed at an independent laboratory. Samples were sent to the ALS Laboratory in Kamloops, BC for preparation and were analysed at the ALS Geochemistry laboratory in North Vancouver, BC. The QP chose fire assay analysis (30 g pulp) with AA finish (AU-AA25) for Au and aqua regia digestion and ICP finish (0.4 g pulp) for Cu (Cu-OG46). Centerra typically analyses for Au by 30 g fire assay with AAS finish and Cu by 4-acid digestion and ICP-MS.

Figure 12-1: Drill Hole 06-957, Andesite Porphyry Flow**Figure 12-2: Drill Hole 19-1132, Plagioclase Monzonite Porphyry**

Table 12-1 summarizes the core intervals sampled and compares the verification results to the Centerra assay values.

Table 12-1: Independent Sample Verification Intervals

Drill Hole ID	From (m)	To (m)	Thick (m)	Centerra Au (g/t)	Centerra Cu (wt. %)	QP Au (g/t)	QP Cu (wt. %)
18-1074	131	133	2	0.599	0.992	0.700	1.065
18-1074	123	125	2	1.049	0.562	1.180	0.635
19-1132	83	85	2	1.190	0.670	1.360	0.816
19-1132	99	101	2	2.027	0.644	1.770	0.701
18-1111	109.11	111	1.89	2.258	1.550	2.440	1.420
18-1111	117	119	2	1.198	1.700	1.140	1.670
06-957	186	188	2	0.730	0.740	0.620	0.953
06-957	194	196	2	0.770	0.750	0.380	0.715
06-967	368	370	2	0.180	0.570	0.190	0.561
06-967	380	382	2	0.360	0.710	0.180	0.432
07-988	112	114	2	0.320	0.600	0.310	0.569
07-988	128	130	2	0.440	0.660	0.150	0.271

Figure 12-3 and Figure 12-4 provide a graphical comparison of the QP verification and Centerra assays for Au and Cu, respectively. The Au samples compare well, while there is more variation in the Cu samples, however, this is likely due to the differences in analysis method. The QP did not identify any material bias in the Centerra sample data, and the comparison results were found to be reasonable given the nature of mineralization in the deposit.

Figure 12-3: XY Scatterplot Comparison of Verification Sample Results for Au

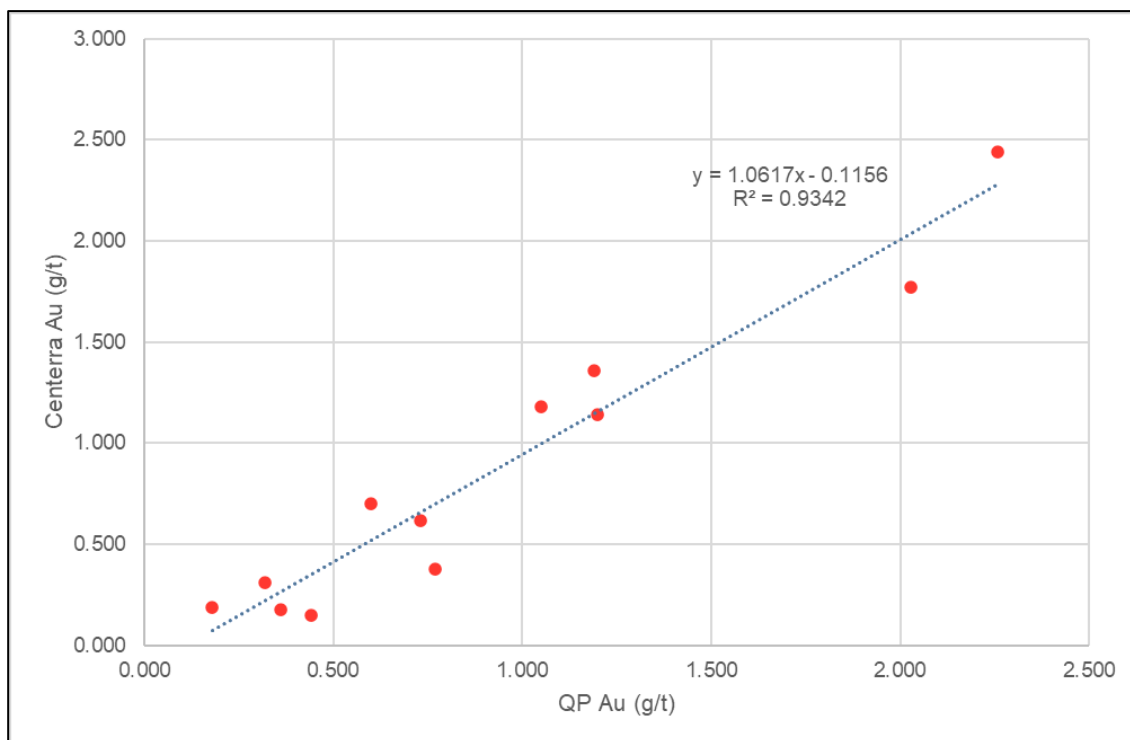
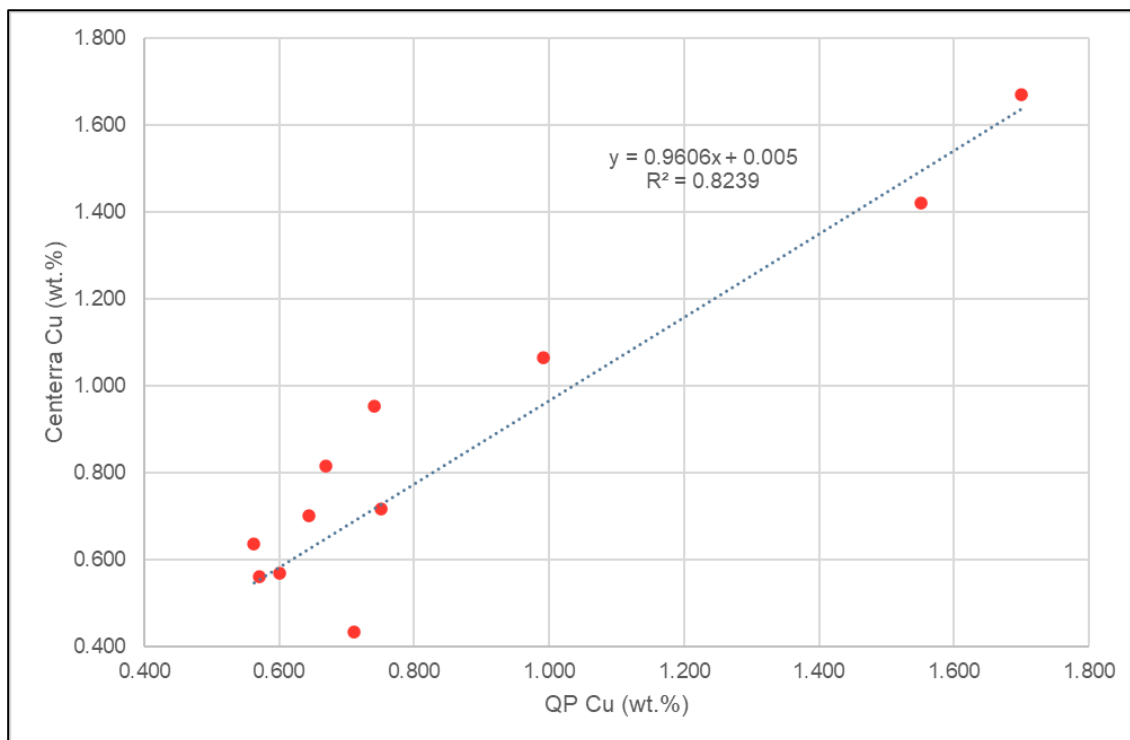


Figure 12-4: XY Scatterplot Comparison of Verification Sample Results for Cu



12.3. DRILL COLLAR INSPECTION

Six drill collar locations were visited and surveyed using a handheld GPS to confirm the collar survey data provided by Centerra. All collar locations were found to be within the accuracy of the GPS as summarized in Table 12-2. Figure 12-5 represents the collar location for drill hole 19-1194. One drill was active at the time of the site inspection, and the QP visited the drill rig at 21-1328, as shown below in Figure 12-6.

Historical (pre-2017) collar locations could not be confirmed due to the ongoing mining at Mount Milligan, and the fact that the vast majority of these holes sit within the pit boundary or were reclaimed. All pre-2006 drilling was in mine grid coordinates, which were transformed to UTM NAD 83 by AllNorth in 2007.

In 2019 Centerra completed several validations of the historical collar and survey as it was being migrated into an acQuire database. These included:

- Review of the 2007 All North transformation to confirm pre-2007 drill holes originally surveyed in the local mine grid were transformed to NAD83 UTM Zone 10 consistently.
- Verification of downhole survey data from raw data files where available for 2004 to 2019 drill holes.

Table 12-2: Comparison of Drill Hole Collar Coordinates

Hole ID	Easting QP (UTM)	Easting Centerra (UTM)	Difference Easting (m)	Northing QP (UTM)	Northing Centerra (UTM)	Difference Northing (m)
19-1194	433,528.6	433,528.5	-0.1	6,107,937.1	6,107,937.0	-0.1
20-1258	433,544.2	433,549.8	5.6	6,107,981.9	6,107,982.5	0.6
20-1259	434,084.8	434,088.1	3.3	6,107,540.3	6,107,537.8	-2.5
21-1325	434,943.7	434,943.8	0.1	6,109,633.5	6,109,634.0	0.5
21-1326	434,240.0	434,241.3	1.4	6,107,688.7	6,107,687.8	-0.9
21-1327	433,364.8	433,367.3	2.5	6,107,846.8	6,107,848.7	2.0

Figure 12-5: Drill Hole Collar Location of Hole 19-1194**Figure 12-6: Active Drill Hole 21-1328**

12.4. QA/QC REVIEW

The QP independently reviewed the QA/QC data provided by Centerra for drilling completed at Mount Milligan between 2017 to present. Centerra employs an onsite QA/QC process including the insertion of standards, blanks and duplicate samples at regular intervals in the sample stream. Standards were inserted every 10 even samples, blanks every 10 odd samples, and duplicates every 30 odd samples. Visual logging of the core determines whether a high, mid or low-grade standard was inserted into the sample stream. Table 12-3 summarizes the four main Certified Reference Material (CRM) standards used at Mount Milligan in 2021.

Table 12-3: Centerra CRM's 2021

Standard	Type	Au (g/t)	Cu (wt. %)
CDN-CM-28	High Grade	1.38	1.36
CDN-CM-43	Low to mid grade	0.309	0.233
OREAS 501C	Low grade	0.221	0.276
OREAS 503D	Mid grade	0.666	0.524

The QA/QC data consisted of assay values for CRMs, blanks, and duplicates. The assay results were analysed for the most commonly used standards, which consisted of Oreas 501C, Oreas 503D, CDN- CM-28 and CDN-CM-43, as well as all blanks (decorative white stone). Out of 2,719 standards, 207 issues were found where the assayed values were considered to be a “hard failure,” outside of plus or minus 3 standard deviations of the certified mean value. Of the 3,826 blank samples analysed, 19 were considered a “hard failure”. On review of these failures, it was determined that Centerra had followed up on these issues and appropriate actions had been taken to ensure the quality of the data.

The 2020 NI 43-101 Technical Report summarized the QA/QC processes as follows:

- **Pre-2004:** standards, blanks and duplicates were not routinely inserted into the sample stream prior to 1992. Some external check assay programs did use standards. Limited records exist, but previous reports indicated that the data was free of significant bias.
- **2004 Placer Dome:** one blank, one pulp duplicate, and one company standard were inserted every 37 core samples. The standards were routinely higher than the mean values on the certificate (CDN Resource Laboratories Ltd, CGS-1), but within the tolerance limits. Blanks were null to low for Au and Cu. Duplicates had excellent correlation for Cu, but fair to poor for Au.

- **2006-2007 Terrane:** one standard and one blank were inserted into the sample stream every 20 samples. Two different Cu-Au standards were purchased from CDN Resource Laboratories Ltd and were alternated every batch. Additionally, for every 20th sample the preparation lab prepared a pulp duplicate. There was a slight high bias for the Cu, (2-5%) relative to the certificate value, and no bias for Au. Blanks returned no systematic contamination. The pulp duplicates returned a slightly less than ideal bias for Au and higher for Cu.
- **2010-2011 TCM:** standards, blanks and duplicates were routinely inserted, though no intervals were reported. The Technical Report indicated that no significant bias was detected, and the assays were accurate and precise and free of contamination.
- **2015-2016 TCM:** standards, blanks and duplicates were routinely inserted, though no intervals were reported. Cu and Mo analyses were accurate, precise and free of contamination. Au assays were also uncontaminated but significantly less accurate and precise.

The QP reviewed the behaviour of the pre-2017 QA/QC data for the standards and blanks. The QP review found that there was good overall performance by the various standards utilized since 2004, with only a few instances of samples returning values outside the expected range (± 3 standard deviations (SD) from the mean) for Au and Cu. Review of the blanks found more variation, with some years and laboratories having consistently better results than others, however, the results returned were determined to be reasonable.

12.5. ASSAY DATABASE VERIFICATION

A total of 1,995 samples (approximately 1% of total database) were chosen for database verification, to be compared to original laboratory certificates. These samples were randomly selected from the Centerra assay database and represent all samples from across the deposit and sample dates ranging from historic through to 2021 and cover all areas of the deposit. No errors were found during the review.

Centerra provided the QP with their Geotechnical Data Collection and Procedures document, which detailed the methods for collecting SG data at site. Wet and dry masses for one sample per every 10 m are recorded by the logging geologist and then entered in the acQuire database where the SG is then calculated by the program from the sample masses and the interval length. The QP reviewed the provided documentation, and independently calculated the SG from the data for each sample and found no issues with the process.

12.6. QP STATEMENT ON DATA VERIFICATION

On completion of the data verification process for Mount Milligan, it is the opinion of the QP that the geological data collection, analytical methods, and QA/QC procedures used by Centerra are consistent with the 2018 CIM Mineral Exploration Best Practice Guidelines.

The QP conducted verification sampling and the results indicated no significant bias to be present in the data. The historical drill hole database has been reviewed and verified by Centerra and significant efforts have been made to ensure the drill collar location and downhole survey information is consistent with current drilling.

Centerra actively monitors the QA/QC samples through the sampling process and ensures that any batches that require re-assaying are flagged and followed up with the laboratory in a timely fashion. The QP recommends that Centerra continue with their current QA/QC practices, including umpire laboratory testing, to ensure a robust and bias free database.

It is the QP's opinion that the exploration data and observations from the various drilling programs completed have been appropriately verified for the purpose of completing a geological model, estimating Mineral Resources, and preparing an NI 43-101 compliant Mineral Resource estimate.

13. MINERAL PROCESSING AND METALLURGICAL TESTING

13.1. INTRODUCTION

Mount Milligan (MtM) is a copper-gold porphyry deposit, consisting of two principal zones, the Main Zone and the Southern Star (SS) Zone. The Main Zone includes four contiguous sub-zones: MBX, WBX, DWBX and 66 (low-copper and high-gold grades, southeast of the MBX sub-zone). These geologic zones are the basis for the metallurgical test work.

The Mount Milligan Mine deposit is being mined using conventional open-pit equipment, with the ore being processed through a gyratory crusher, secondary crushing and a SAG-ball mill-pebble crusher (SABC) combination together with a rougher and cleaner flotation plant, producing a marketable gold-rich copper concentrate.

13.2. SUMMARY

Metallurgical investigations conducted by various research laboratories prior to commencement of operations conclusively showed that froth flotation is the optimum process for the recovery of a copper concentrate containing gold and silver. These investigations were the basis of the performance models used in previous resource modelling. Up until 2020, recovery models in use were produced primarily using plant data from 2017.

In 2020 and 2021, since the previous update of the technical report, additional improvement in the operation led to higher average throughput rates. Subsequently, the flotation recoveries of copper and gold negatively impacted metal recoveries. A cleaner circuit expansion project was completed and commissioned in May 2022 to deliver additional metal recoveries in the flotation circuit. Using updated performance models, the LOM average recoveries are estimated at 80.2% for copper and 68.0% for gold, targeting a concentrate grade with a LOM average of 21.5% copper. Concentrate assays indicated that impurity elements were occasionally higher than specification for some select smelters; however,

controls via blending and composite sampling of the load out trucks have and will continue to ensure impurities will be kept to minimum levels.

13.3. UPDATED METALLURGICAL INTERPRETATION

Background

In the second half of 2020 and 2021, improved SAG mill liner geometry and increased secondary crusher utilization resulted in the highest throughput levels experienced at Mount Milligan. As a result, flotation recoveries for copper and gold decreased slightly due to reduced flotation retention time and overall circuit capacity with the higher throughput rates. This warranted a review and update to the recovery curves used at Mount Milligan. In the following update, a throughput term was added to both metal recovery curves to compensate for the effect described above.

13.4. COPPER AND GOLD RECOVERY CURVES

The current recovery curve for copper was produced in 2017 from data that was largely based on average throughput rates of 2,300 tonnes per operating hour (tpoh). The form of the recovery curve was a combination of rougher and cleaner recovery regressions (linear) whereby total recovery was calculated by multiplying rougher recovery and cleaner recovery.

$$\text{Rougher/Scavenger Cu Recovery} = 24.382 \times \text{Cu head grade} + 0.8012$$

$$\text{Rougher/Scavenger Cu Concentrate Grade} = 6.3425 \times \text{Cu head grade} + 0.0016$$

$$\text{Cleaner Cu Recovery} = -0.0022 \times \frac{\text{Cleaner Concentrate Grade}}{\text{Rougher Concentrate Grade}} + 0.9754$$

At the time, the intent was to produce an extra equation whereby concentrate grade/upgrade ratio could be manipulated to produce different recovery projections. The name it was given was the Direction Regression (DR) model.

In the case of gold, the recovery curve in use was created in August 2019 and carried the name "DR1908". The form of the equation is more traditional compared to that of copper and shown below:

$$\text{Au Recovery} = 68.68 * \text{Au Head Grade}^{0.0896}$$

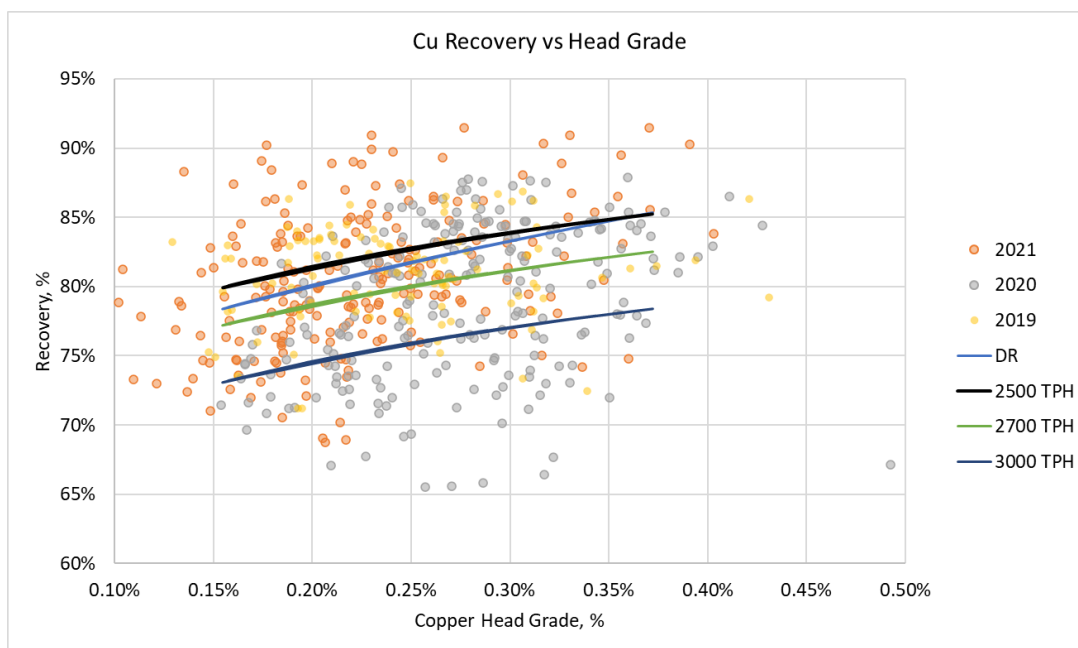
Updated Data Set and Fitted Model

Operational strategies remained unchanged in the flotation circuit since 2019 and hence, the data set was selected as the basis to predict future copper recoveries. The plant operating data from 2020 to September 2021 was filtered to exclude throughput rates below 2,500 tpoH to better reflect more recent process conditions. The fitted model for predicting copper recovery is shown below:

$$\text{Cu Recovery} = -0.000136731 \times \text{TPOH} + 1.59926 \times (\text{Cu head grade})^{0.0522036}$$

This model was compared to the actual plant performance to determine the model accuracies. The standard deviation of the residuals for the model was 4.04%. The actual plant copper recoveries were compared to the predicted copper recoveries using the new recovery model and were seen to be within 6.7% with 90% confidence. The resulting curve can be seen plotted at different throughput levels alongside the abovementioned DR curve in Figure 13-1.

Figure 13-1: Copper Recovery as a Function of Head Grade

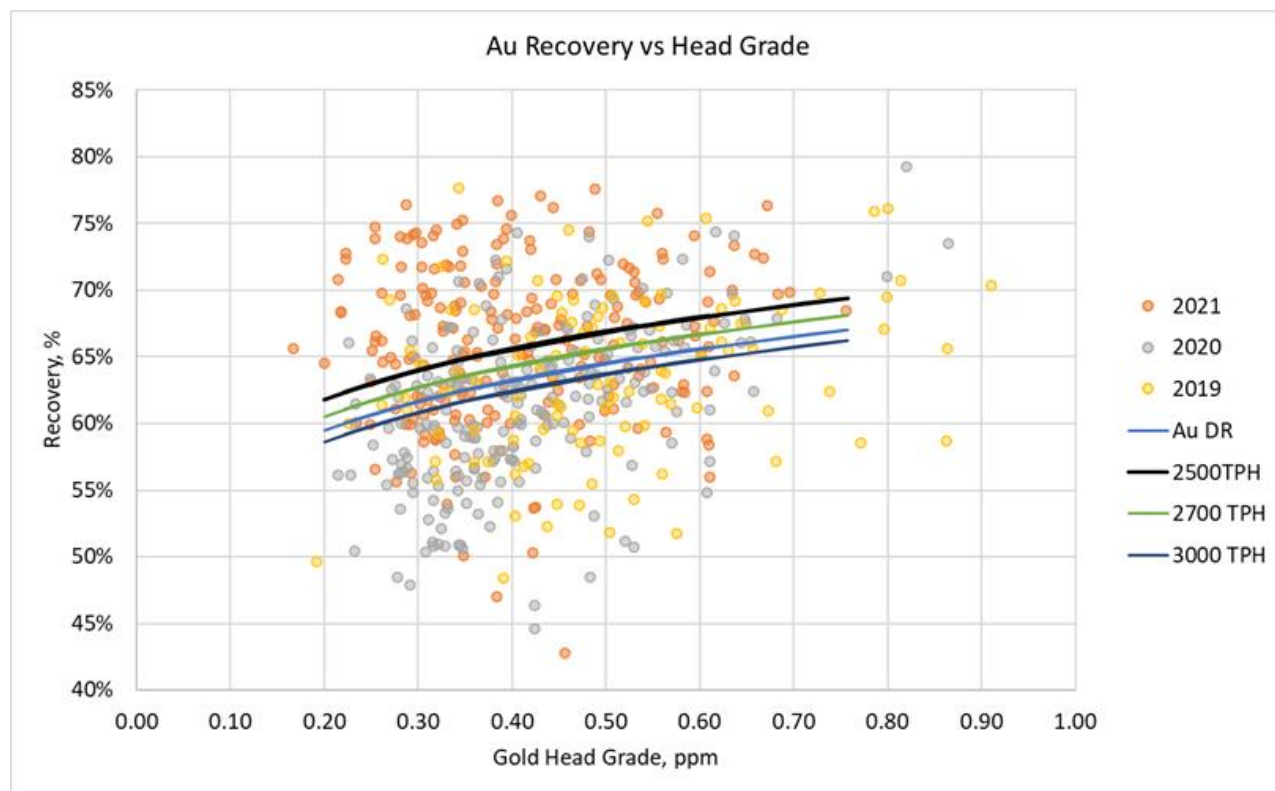


In the case for gold, the throughput effect was much less. With the same generic equation form, the fitted curve is shown below:

$$Au\ Recovery = -0.0000633099 \times TPOH + 0.868817 \times (Au\ head\ grade)^{0.0700520}$$

The resulting curve can be seen plotted at different throughput levels alongside the abovementioned DR1908 curve in Figure 13-2.

Figure 13-2: Gold Recovery as a Function of Head Grade



The new recovery curves are similar in shape when compared to previous ones in use. Evidently, recoveries for copper and gold decrease with reduced head grades as well as increased throughput. Efforts will continue to strive for improved recoveries while maintaining throughput rates.

13.5. LIFE-OF-MINE RECOVERY CURVE DISCUSSION

The previous sections describe the development and use of the copper and gold recovery curves for budgeting and forecasting purposes. Efforts are also undertaken to blend the process plant feed ore, maintaining the Py/Cpy ratio below 12 to achieve predicted

concentrate grades & recoveries. This is supported by previous plant data that mineral recoveries could not be maintained at the targeted concentrate grade with the heavy pyrite rejection required in the cleaner flotation circuit. Alternatively, to achieve target copper concentrate grade, the recoveries of copper and gold were below expectation for ore with higher than 12 Py:Cpy ratio.

Staged Flotation Reactors

Since 2017, numerous test programs were completed to assess the economic benefits of installing SFRs in the cleaner flotation circuit. In late 2020, upon conclusion of two pilot plant campaigns and in conjunction with previous test programs and available plant data, a feasibility study was completed on the installation of three (3) SFR units upstream of the existing cleaner circuit, commonly referred to as a cleaner scalper circuit.

The feasibility study concluded that expected overall gold and copper recoveries would increase by 4% and 2% respectively. The project would require capital investment of \$7.6M CAD with the expected payback period of approximately 6 months after commissioning.

Detailed engineering commenced in February of 2021. The installation of the cleaner scalper circuit resulted in the updated set of LOM recovery curves shown below:

$$Cu\ Recovery = -0.000136731 \times TPOH + 1.59926 \times (Cu\ head\ grade)^{0.0522036} + 0.02$$

$$Au\ Recovery = -0.0000633099 \times TPOH + 0.868817 \times (Au\ head\ grade)^{0.0700520} + 0.04$$

At the time of writing this Technical Report, the preliminary, realized recovery gains were 3.9% for gold and 2.7% for copper. Some small deficiencies during commissioning were investigated and remedial measures are underway. This will enable the SFR circuit to be optimized.

13.6. CONCENTRATE QUALITY AND MARKETING

The concentrate from MtM is marketed as gold rich copper concentrate with a target grade of 21.5% copper by weight. While there is a list of penalty elements applicable to the smelter terms, mercury content is the only element of importance. Occasionally, elevated mercury content is observed in the ore feed, which reports to the copper concentrate at undesirable

concentrations. However, the MtM facilities offer a total of approximately 16,000WMT of blending space between the site storage shed and the load out facility. Controls via assaying with the available storage space allow for blending activities to achieve minimum mercury concentrations in the copper concentrate and avoid penalties. Additional information related to marketing is described in Item 19.

13.7. SCAVENGER TAILINGS PUMP INSTALLATION

Previously, the tailings from the scavenger flotation circuit (majority of the tailings) flowed to the TSF via gravity. Over time, as tailings are deposited into the TSF, the elevation of the dam would eventually rise to the point where gravity would no longer be able to keep the tailings flowing. This was expected to occur in May 2022. For this reason, slurry pumps were installed to allow mill operations to continue to discharge tailings into the TSF. The tailings pumping station has been installed and commissioned.

13.8. CONTINUOUS IMPROVEMENT PROJECTS

Secondary Crushing Circuit Improvements

The secondary crushing plant project was completed in late 2016. Since then, winter conditions caused frequent plug-ups of transfer chutes in conjunction with various deficiencies in the design of the material handling system. In 2020, MtM initiated improvement plans to address the numerous deficiencies of the secondary crushing system by way of modified chute designs and conveyor speeds adjustments. Upon completion, the secondary crushing circuit run times has significantly improved to more than 65% compared to slightly over 50%..

In 2022, additional engineering is underway to further enhance the run time of the screening operation in the secondary crushing circuit during winter months.

Reclaim Feeder Improvement

Frozen fine ore has been problematic during winter season since the beginning of the mill operation. In late 2020, a lump breaker was installed in one of the four (4) reclaim feeders to break the lumps of frozen ore before they could plug the feeder chute. This has led to an

increase in average feeder uptime from 88% to 95% in the winter seasons of 2019 and 2020. Additional lump breakers were installed in 2021 for all reclaim feeders and resulted in a 4% average increase in feeder runtime in the winter months.

Grinding Advanced Process Control (APC)

Continued efforts are put into improving process control for every unit operation within the processing plant. Additional preventive maintenance measures for various instruments were implemented in 2020 to ensure the measurement system is functioning accurately and is sustained. This is the basis to ensure good regulatory and supervisory controls are maintained. The original APC program is a fuzzy-logic based system. This was used for the comminution, flotation and dewatering circuits. In 2020, a novel grinding APC technology was implemented in mid 2021 with the goal of improving the mill average feed rate. This technology adopts multiple control methodologies, which solve control problems individually using separate techniques. The controllers covered all key pieces of equipment in the SABC circuit. Upon implementation, results indicate the Grinding APC increased throughput by an average of 4.44%.

Pebble Crushing Circuit Upgrade

With improvements to the secondary crushing circuit, the proportion of pre-crushed ore feed to the SABC circuit increased. This was beneficial in increasing the throughput rates of the plant. Due to the nature of the pre-crushed feed containing high amounts of pebbles, the SABC circuit became increasingly constrained around the pebble crushers. Instead of reducing the grate aperture in the SAG mill to help alleviate the workload of the pebble crushers, with corresponding reduced throughput, an upgrade to the existing pebble crushers was pursued, notably because crushing is inherently more energy efficient than grinding. In September 2022, a capacity increase program (with a target of 25%) was completed by converting the crushers from MP1000 to MP1250. Performance was being evaluated at time of writing this Technical Report. Ultimately, the objective is to maximize comminution energy utilization on the crushing component of the SABC circuit, subsequently increasing the SAG mill power draw from the current 20.5 MW-21.0 MW to 23.5 MW installed by increasing ball mill load and total mill load. It has previously been demonstrated that the SAG mill can draw the full 23.5 MW at higher mill charge levels than current.

Flotation Advanced Process Control

The original flotation APC utilized fuzzy-logic controls and was not achieving the desired stability of circuit operation. At the time of writing, a project is underway to implement a novel flotation APC with the goal of achieving more consistent operation and improved outputs. The new APC utilizes cell pull rates and active manipulation of all final control elements to increase operating efficiencies. The project implementation is scheduled for Q3 of 2022 and the anticipated recovery increase is 0.5% for both copper and gold.

13.9. RECOMMENDED IMPROVEMENTS

Improved Liner Design

Since the Technical Report update in 2020, the MtM processing plant continued to operate at its nameplate capacity of 62,500 tonnes per operating day. As a result of the increased throughput since the second half of 2021, the wear on the liner packages in the grinding mills accelerated resulting in shorter liner life, as measured on calendar time. Previously, SAG mill liners were replaced every 20-24 weeks to ensure sufficient protection and satisfactory performance of the mill. Accelerated liner wear has reduced liner life to 19 weeks due to increased throughput. The accelerated wear decreases mill availability and increases liner cost.

Design reviews are being progressed to achieve longer service life for the SAG mill liners which is the constraining factor in plant availability. Additionally, in pursuit of higher equipment utilization, SAG discharge grate design was reviewed. The current grate aperture is 70mm with the circuit bottleneck being the pebble crushing circuit. In the future, smaller aperture grates will be installed to reduce the top size feeding the pebble crushing circuit and transfer some workload to the SAG mill which is not drawing all the available motor power of 23.5 MW.

Pre-Feasibility Study 2022

In the first half of 2022, a pre-feasibility study was completed to evaluate several debottlenecking initiatives with positive economic outlook. During the study, efforts were made towards increasing the availability of the secondary crushing circuit, particularly around winter months. Additionally,

by having more consistent feed to the SABC circuit, the operating target of the grinding circuit particle size P80 should be increased to approximately 200 microns as compared to approximately 160 microns to further enhance throughput rates.

With the enhanced throughput rates, additional flotation cells and regrind mills were also investigated to increase the flotation capacity of the plant. The added flotation cells would enhance the retention of the flotation circuit as well as the mass pull while the regrind mill would satisfy the added liberation requirement with increased rougher flotation concentrate at increased particle size P80. The conclusion of the study resulted in the following recommendations for further evaluation:

1. To enclose the banana screen in the secondary crushing circuit.
2. To install two high shear flotation cells in front of the rougher flotation circuit.
3. To install an additional horizontally stirred regrind mill.
4. To install two additional plates in the existing dewatering filter press.
5. To upgrade ball mill motors' capacity.

14. MINERAL RESOURCE ESTIMATES

14.1. INTRODUCTION

Centerra prepared a Mineral Resource estimate for the Project using data from both historic drilling and drilling from recent years undertaken by TCM. A database was compiled consisting of data from 1,511 core holes, with collar and down-hole survey, geological and assay information. It is important to note that 49% of the holes in the current drill hole database do not have surveyed down-hole azimuth values due to magnetite content which would cause errors in the magnetic readings used prior to 1990 drilling programs. Consequently, those drill holes are considered to create a degree of risk in the mineralized domain interpretation and grade distribution interpretation. Spot checks (1.1% of total sample database) were conducted on the database (.csv files) for errors in the collar, survey, assay, and geology files. The checks didn't reveal any significant errors that would have an impact on the estimates.

The mineral resource block model discussed herein considers assay data available as of February 08, 2022. Table 14-1 summarizes the quantity of data available for the updated resource estimation.

Table 14-1: Summary of Geological Database

Number of DDH	Length (m)
1,511	397,191

In assembling the database for the Mineral Resource estimate, a systematic approach was used to select assay data from the core drill samples. Intervals noted in the drill logs as being subject to any discrepancy with the assay certificates were investigated and corrected, or removed, from the sample database.

The Mineral Resource estimate was derived from a computerized resource block model using Datamine Studio RM version 1.10.200.0. The construction of the model started with drill hole data, which served as the basis for the definition of 3D mineralized domain envelopes with resources limited to the material inside those envelopes, with the exception of the surrounding low-grade material defined as zone 100. The next step was the selection of drill hole data within the mineralized envelopes in the form of fixed length composites. The composite data was constrained within the individual domains and used to interpolate the grade of blocks on a regular grid and populate the mineralized envelopes with estimated grades from the composite samples within the same envelopes. All the interpolated blocks below the overburden and within the conceptual resource pit surface comprise the Mineral Resources that were then classified based on their proximity to composites and the corresponding precision/confidence level. Centerra, based on information provided from the exploration database, have re-interpreted the mineralized domains and completed the Mineral Resource estimation.

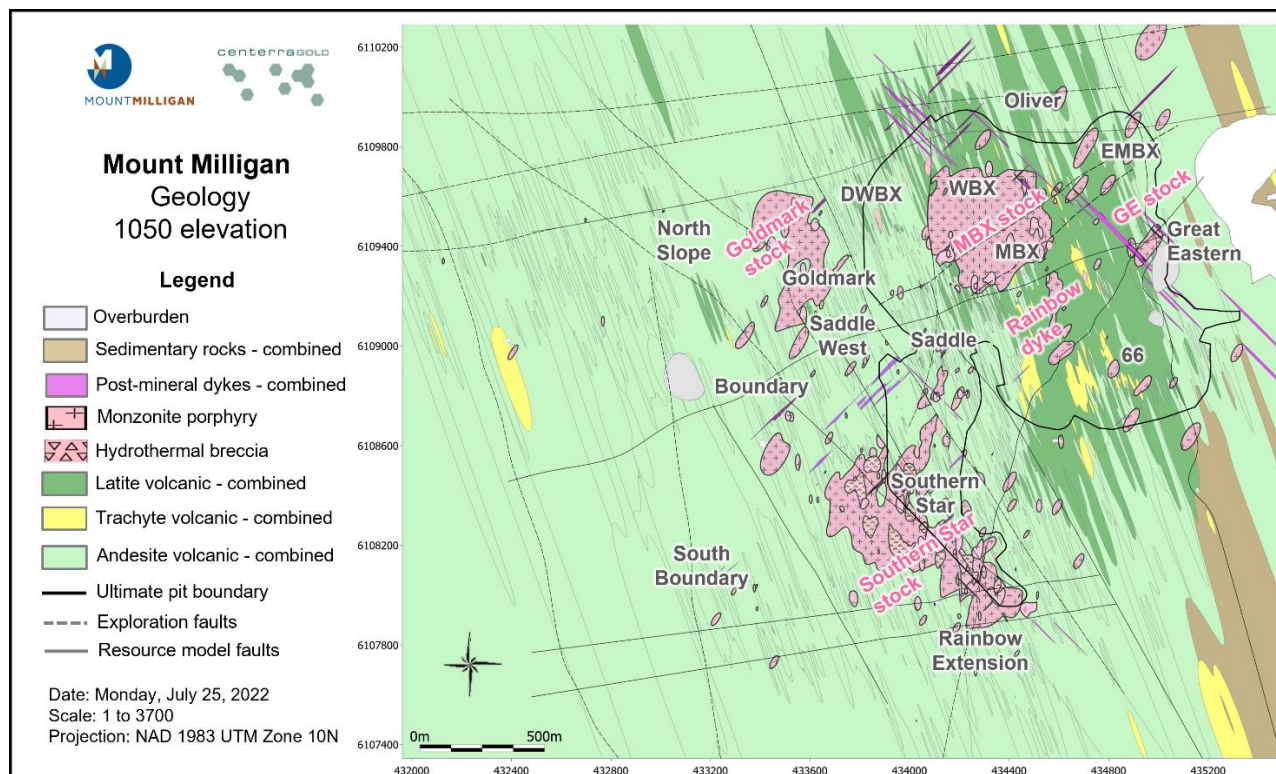
14.2. GEOLOGY AND MINERALIZATION MODEL

Geology Model

The geological model is based on the most recent interpretation of geology, alteration, and structure.

A site-wide plan view of the Mount Milligan intrusive complex is included as Figure 14-1.

Figure 14-1: Geological Map of the Mount Milligan Deposit



Mineralization Model

Centerra updated the mineralized domain models to constrain the Mineral Resource estimate (MRE). Updated Domains resulted in a volume increase of 107 million m³. The domain models were based on a 0.1% CuEq cut-off (see Item 14.8 for CuEq Cutoff formula), and were created by digitizing polylines on east-west sections at approximately 40 metre intervals throughout the deposit, incorporating moderate length of grades below 0.1% CuEq (Figure 14-2) to account for waste inclusions and allow for continuity of mineralization. In some instances, the section intervals were greater, up to 60m, relative to the drilling density. The interpretation of the mineral domain models was correlated with the litho-structural interpretation completed by Centerra's exploration and mine geology groups. Polyline interpretation was verified in north looking sections and horizontal view for consistency. These polylines were linked and triangulated to create three-dimensional wireframe

solids. Figure 14-3 to Figure 14-8 display the interpreted mineral domain models. Table 14-2 shows the characteristics of the various domains.

Mineralized domains were characterized by:

- Delineation of waste inclusions (1000, 3002 sub domains)
- Strong characterization of the anticlinal signature with the variability in geometry (dip and plunge).
- Distinctive gold distribution generally along the eastern flank of the deposit, (66, Saddle and SS zones) but having relatively high variability.
- High-grade subdomains modelled (1001,2000, 3000 subdomains).
- The mineralization occurring outside of the modelled domains was captured within the pit shell and interpreted accordingly. An isometric view of lithology and several cross sections of colour-coded mineral domains are provided in Figure 14-2 and Figure 14-3 to **Error! Reference source not found..**

Figure 14-2: 3D Litho-Structural Interpretation Isometric View-Looking N-NE (Not to Scale)

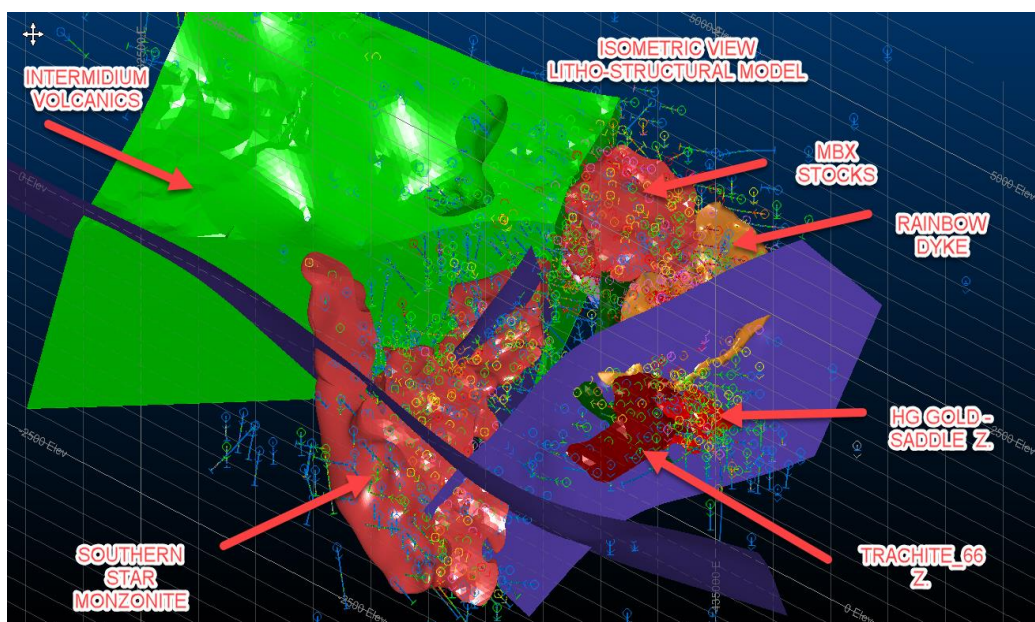


Figure 14-3: 3D Mineralized Domains Isometric View Looking N-NE (Not to Scale)

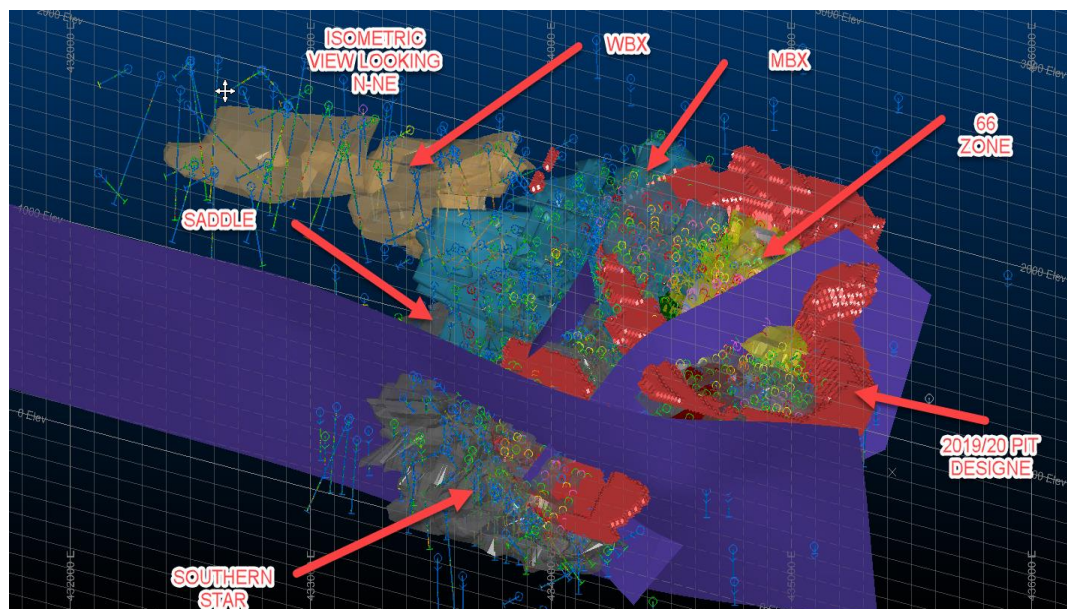


Figure 14-4: Drill Holes and Limits of Domains in Cross Section 6,108,900N, Looking North (50-m grid spacing, Au top, Cu middle, CuEq bottom)

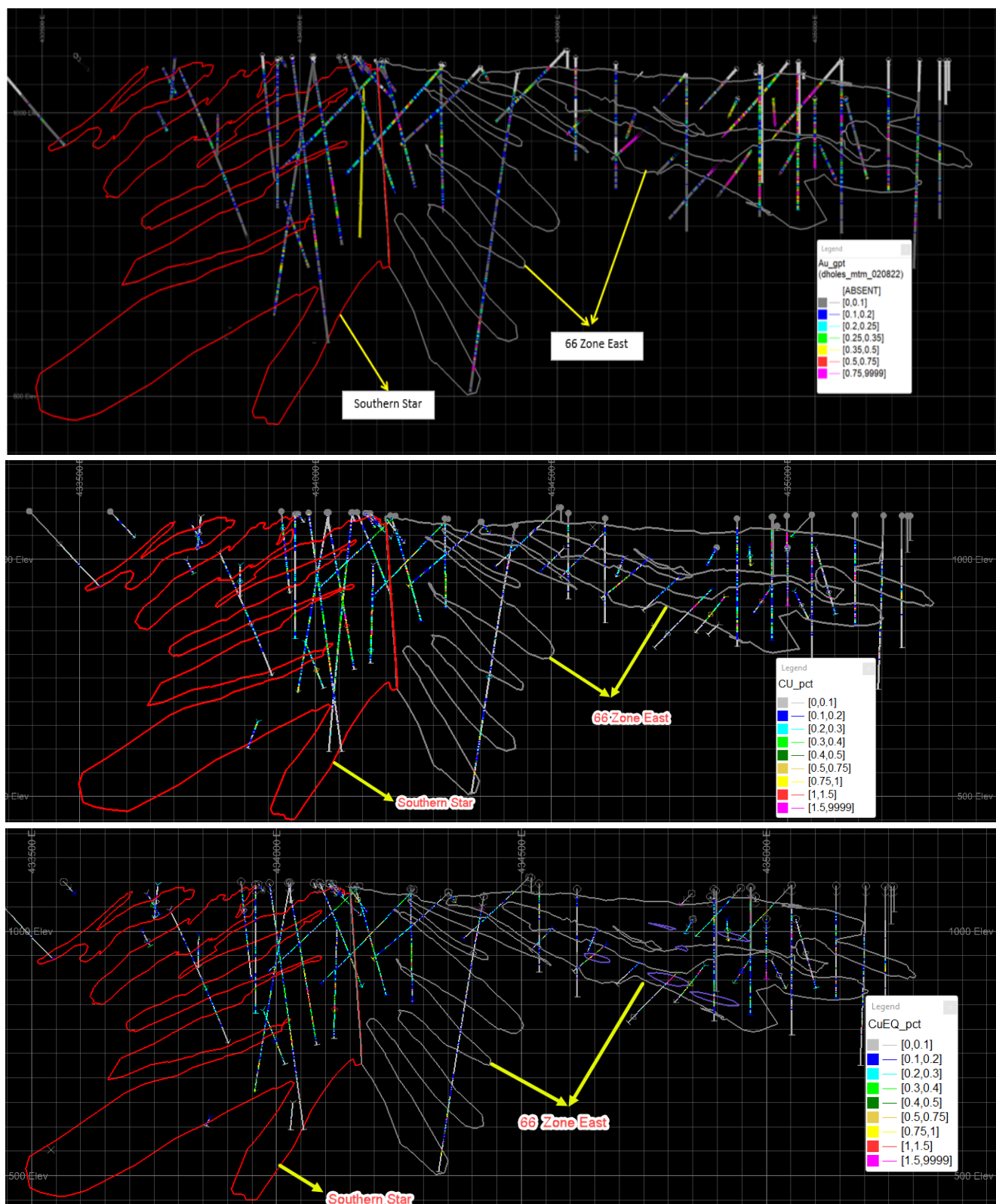
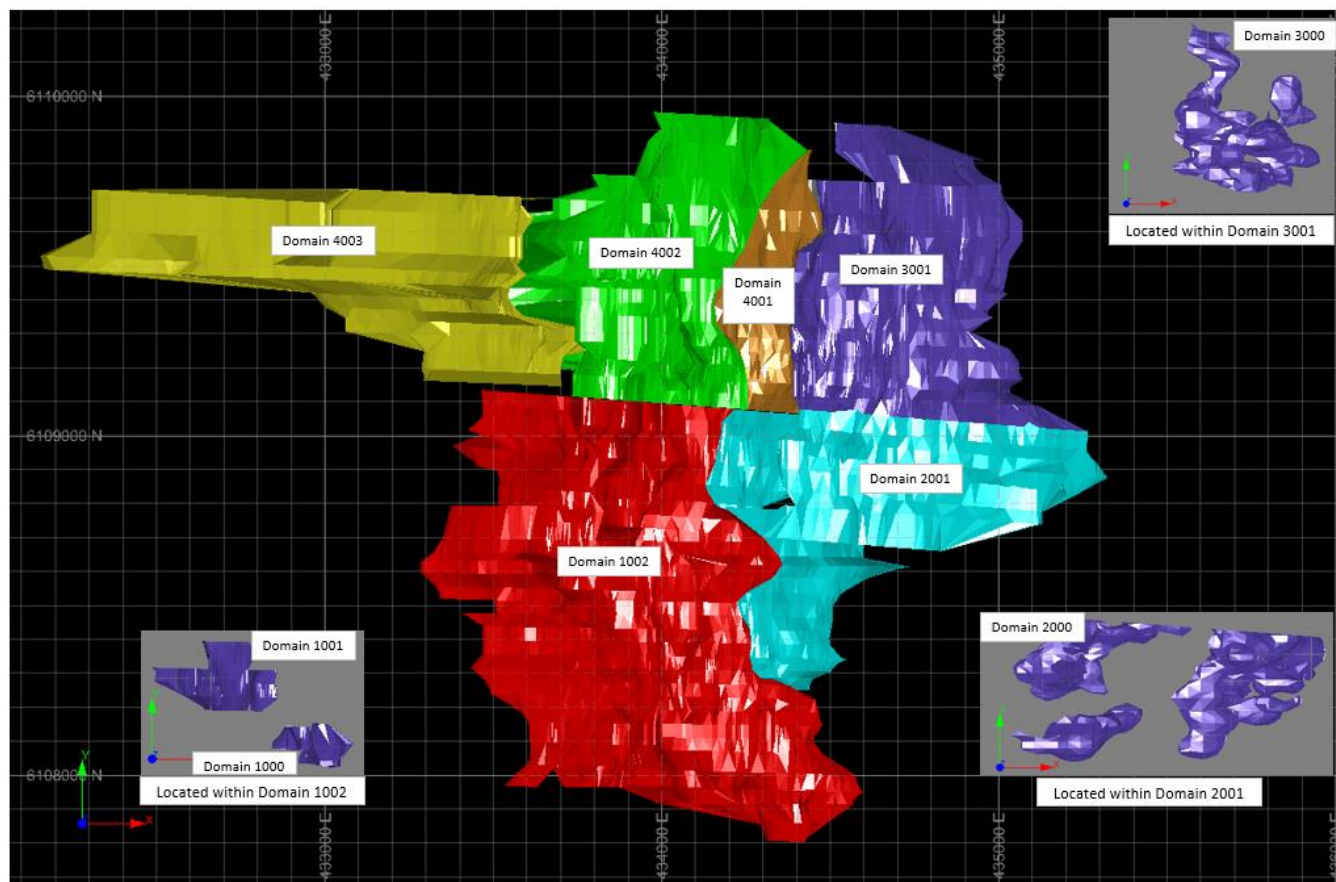


Table 14-2: Mineralized Domain Coding

Domain Name	Domain Code	Description
Southern Star (SS)	1002	Southern most situated copper-gold zone; Intrusive controlled characterized with high grade gold breccia bodies
	1000	Low Grade subdomain within 1002
	1001	High Grade subdomain within 1002
Saddle	2001	Central part, south from MBX, north from Southern Star Intrusive and Domain
	2000	High Grade subdomain within 2001
MBX Volcanic East	3001	Gold Hosted zone, east of the MBX, adjacent to the Rainbow Fault
	3000	High Grade subdomain within 3001
	3002	Low Grade subdomain within 3001
MBX Cont. East	4001	Central deposit, good Cu-Au correlation; distributed within the MBX stock and volcanogenic unit to the east
MBX Cont. West	4002	Central deposit, good Cu-Au correlation; distributed within the MBX stock and volcanogenic unit to the west
WBX Domain	4003	West from MBX, relatively narrow, flat lying zone
Shell	100	Area with Samples outside of named domain wireframes

Figure 14-5 is the level plan at 1040 EL with a +/-50m window, showing interpretation of mineralized domains relative to the main lithological units.

Figure 14-5: Plan View Schematic of Mineralized Domains

14.3. EXPLORATORY DATA ANALYSIS (EDA)

EDA was completed on raw assay and composite samples within the interpreted mineralized zones as discussed in the following sections.

Assay

There are 179,983 assay intervals in the database used for the current Mineral Resource estimate.

Table 14-3 summarizes the range of gold and copper values from the utilized assay data and Table 14-4 summarizes the number of assays per mineral domain.

Table 14-3: Summary of Gold and Copper Assay Data Available for Resource Estimation

Item	Sample Length Mean (m)	Number of Records	Au g/t and Cu %		
			Minimum	Maximum	Mean
Gold (g/t)	1.87	179,983	0.001	236.8	0.23
Copper (%)	1.87	179,983	0.0001	7.2	0.11

Most of the surface drill holes are inclined between -45 and -70 degrees to the east or west on E-W cross sections at 40-60 m spacing and vertical pierce points spaced at 40-60 m vertically on the same section.

Table 14-4: Detailed Database Summary Table

Location	Domain Code	No. of Assays
Southern Star	1000	594
	1001	537
	1002	34,158
66 Zone East	2000	701
	2001	9,829
MBX Volcanic East	3000	749
	3001	20,119
	3002	1,225
MBX Cont. East & West	4001	8,697
	4002	20,540
WBX	4003	2,117
Total Samples		99,266

14.4. CAPPING

The exploratory data analysis revealed some outlier sample data with extremely high copper and gold values that needed to be controlled prior to grade estimation to limit grade bias in the block model. Each individual domain was analyzed, and an appropriate cap grade was determined and applied to the raw assay data.

A standard approach to capping high-grade outlier values comprises developing probability plots and examining the distribution to search for any natural gap, or change, in orientation in those distributions.

Cumulative probability plots are shown in Figure 14-6 and Figure 14-7 for gold and copper, respectively. In almost all cases, the log scale cumulative frequency plots for high grades show obvious abrupt change of slope, which correspond to a natural gap in the grade population data, as indicated by the green lines.

Table 14-5 displays the capping limits applied and global statistics of gold and copper values. Table 14-6 compares the summary statistics for raw, capped, and composited samples.

Table 14-5: Applied Capping Limits Raw Data

Domain	Gold Cap (g/t)	Copper Cap (%)
Domain 1000	1.50	1.00
Domain 1001	3.00	1.40
Domain 1002	6.00	1.80
Domain 2000	15.00	0.50
Domain 2001	16.50	1.30
Domain 3000	22.00	1.40
Domain 3001	14.50	2.50
Domain 3002	4.00	1.00
Domain 4001	8.00	1.00
Domain 4002	15.00	1.50
Domain 4003	4.50	1.00
Domain 100	3.00	2.00

Figure 14-6: Example Cumulative Probability and Histogram Plots for Gold in the 3001 Domain

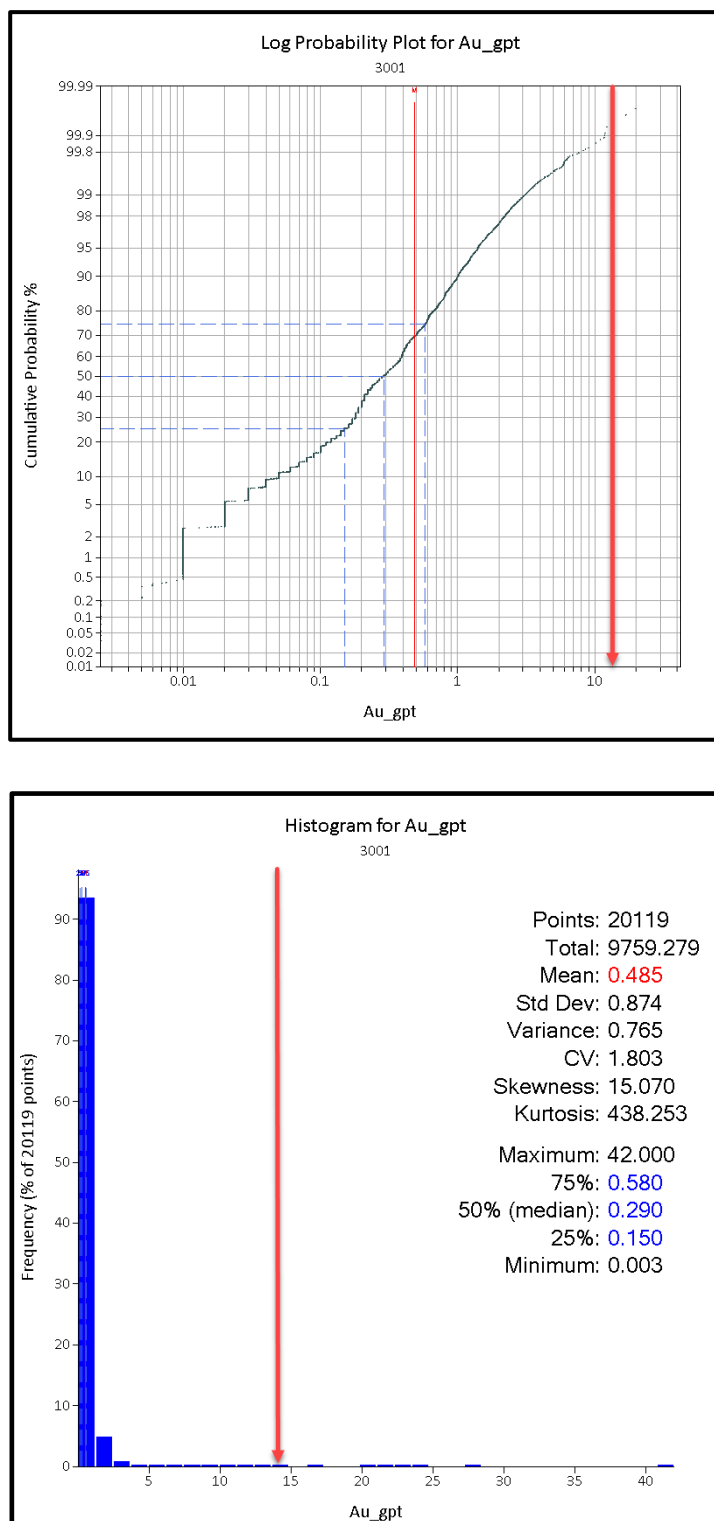
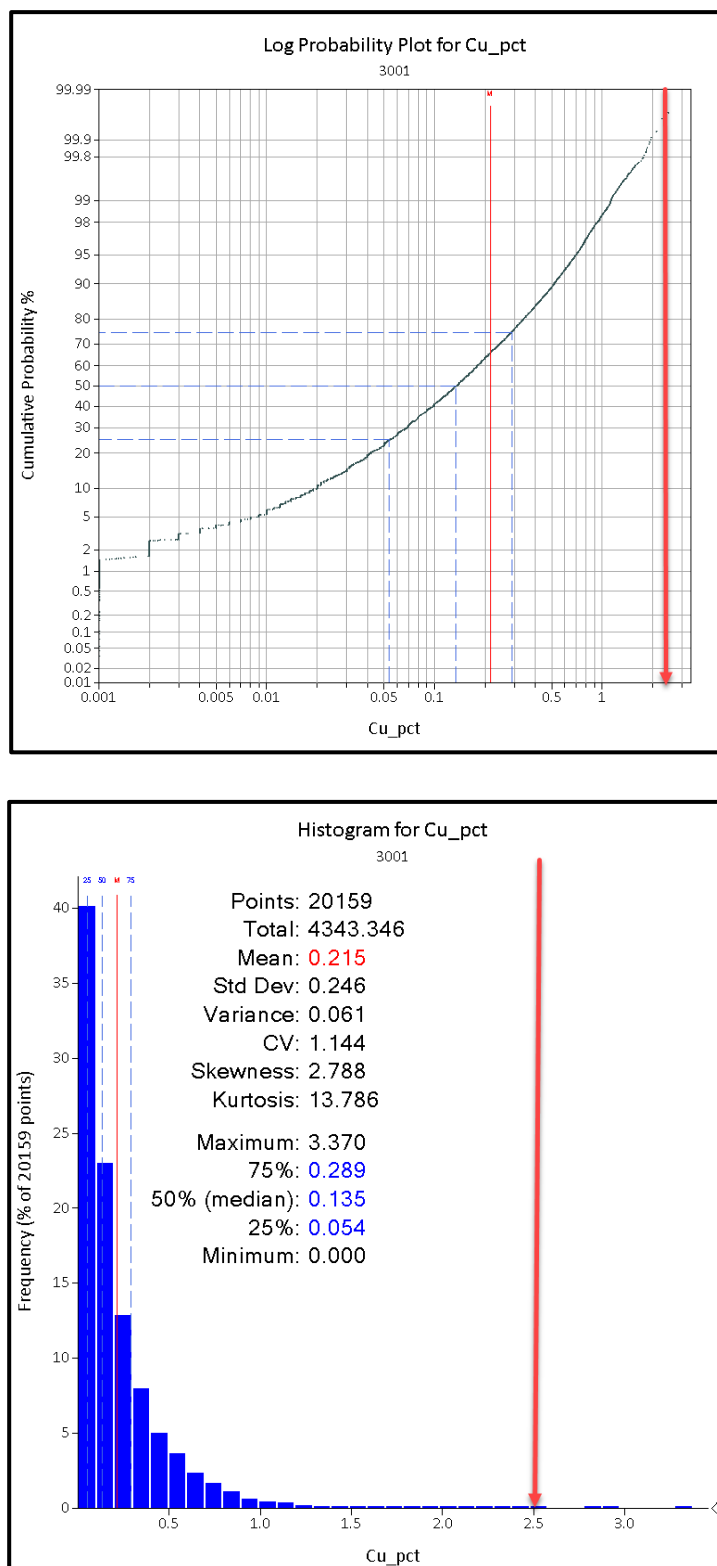


Figure 14-7: Example Cumulative Probability and Histogram Plots for Copper in the 3001 Domain



14.5. COMPOSITES

The analysis of the spatial continuity of gold and copper grade in each mineralized domain was assessed based on 4 m down-hole composites. This composite length was selected as a multiple of the 2 m modal sample length and provided sufficient sample support for the 15x15x15m block size. The residual length after compositing was added to the previous interval.

Table 14-6 and Table 14-7 list descriptive statistics of the raw/uncapped data and 4 m capped composites. Compositing started at the overburden-bedrock contact and composites were broken within the mineralised domains.

Table 14-6 shows the summary statistics for capped raw gold and copper assays with a comparison of the coefficient of variance (CV) for capped and un-capped raw assay values (Gold).

Table 14-6: Summary Statistics for Raw, Capped, and Capped Composite Data, by Domain (Gold, g/t)

Zone	Type	Count	Minimum	Maximum	Mean	Standard Deviation	CV	Variance
Southern Star - 1002	Raw	35,290	0.00	17.10	0.23	0.39	1.72	0.15
	Capped Raw	35,290	0.00	6.00	0.23	0.34	1.51	0.12
	Capped Composite	16,125	0.00	5.21	0.22	0.26	1.21	0.07
66 Zone East - 2001	Raw	10,533	0.00	51.50	0.51	1.51	2.95	2.27
	Capped Raw	10,533	0.00	16.50	0.49	1.13	2.30	1.28
	Capped Composite	4,981	0.00	13.09	0.48	0.91	1.89	0.83
MBX Volc. East	Raw	22,098	0.00	236.85	0.57	2.46	4.29	6.05
	Capped Raw	22,098	0.00	22.00	0.57	0.98	1.79	0.96
	Capped Composite	9,837	0.00	13.44	0.53	0.79	1.51	0.93
MBX Cont. East	Raw	8,696	0.00	17.90	0.25	0.58	2.23	0.34
	Capped Raw	8,696	0.00	8.00	0.24	0.50	2.03	0.25
	Capped Composite	4,055	0.00	6.14	0.23	0.35	1.51	0.12
MBX West	Raw	20,542	0.00	50.60	0.29	0.92	3.16	0.84
	Capped Raw	20,542	0.00	15.00	0.28	0.75	2.63	0.56
	Capped Composite	9,107	0.00	13.90	0.27	0.49	1.82	0.24
WBX	Raw	2,117	0.00	30.00	0.23	0.96	4.14	0.93
	Capped Raw	2,117	0.00	4.50	0.20	0.47	2.31	0.22

Zone	Type	Count	Minimum	Maximum	Mean	Standard Deviation	CV	Variance
	Capped Composite	964	0.00	4.08	0.19	0.35	1.83	0.13

14.6. 2022 DENSITY MODEL

Density Data Validation

A composite 4m length-weighted dataset of 33,416 samples was utilized for modelling of rock density. Values of the composites ranged up to 10.070 tonnes per cubic metre (t/m^3) with a mean value of 2.786 t/m^3 . It was determined that rare high values can be attributed to narrow intervals of near solid chalcopyrite. To prevent smearing of the high values, histograms and probability plots were used to obtain a cut-off grade with which to constrain the high values.

Estimation Methodology

The 2022 density model was estimated in Hexagon MinePlan 3D software, version 15.70. The model was flagged with the 10 domain solids which were developed and vetted in Datamine Studio RM Pro software. The domain solids were grouped in the estimation for an East component consisting of domains 2001, 3001, 4001, and a West component consisting of domains 1002, 4002. The high-grade domains of 1000, 2000, and 3000, located within the bounds of other domains, were not used. The domains were combined for density estimation after analysis demonstrated little variation between the 10 separate domains.

Domain 100, the assignment given to all material outside of the 10 mineralized domains, was run as a separate pass. Each estimation pass was completed using Inverse Distance Weighting raised to the third power (IDW^3). Density values greater than 3.769 t/m^3 were restricted to a maximum 30-m search distance. The value of 3.769 t/m^3 was determined from probability plotting by domain. Post estimation run; any blocks not previously estimated along the model margins were backfilled with the mean value of 2.786 t/m^3 . Table 14-7 details the estimation parameters used in the density model. The model was validated manually by comparing assays (from samples listed in the diamond drill database plotted on cross sections) to the model output, and by minor comparisons to the previous work on the model.

Table 14-7: Specific Gravity Estimation Parameters

Estimation Pass	Search Radius (m)	Estimation Method	Minimum Samples	Maximum Samples	Max Samples per Hole
East Pass 1	250	IDW ³	3	8	3
East Pass 2	500	IDW ³	1	8	3
West Pass 1	250	IDW ³	3	8	3
West Pass 2	500	IDW ³	1	8	3
Domain 100	500	IDW ³	2	8	4

Overburden Density

An internal study was completed to determine the density of overburden materials determined from proctor compaction tests and truck weigh scales. The overburden was assigned an average in-situ density of 2.363 t/m³ for glacial till material suitable for core dam construction and 2.370 t/m³ for common glacial fluvial.

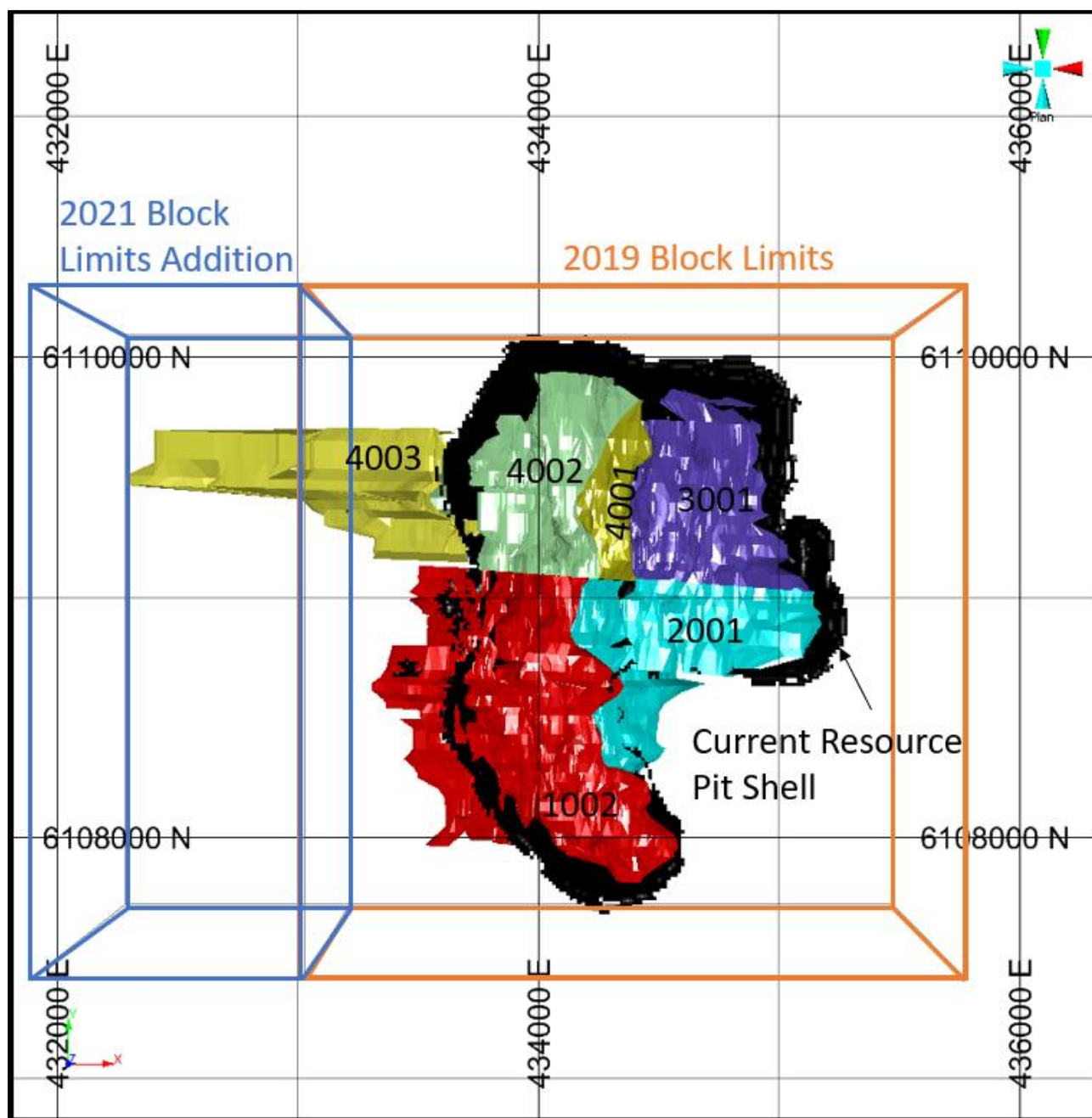
14.7. MINERAL RESOURCE ESTIMATION

This Mineral Resource estimate was completed by the Centerra. Updates in this model include:

- The addition of 156 new holes (85 exploration, 59 resource infill support, and 12 metallurgical of new drill core data
- Three high-grade gold zones were statistically identified within the Southern Star, 66 zone and the Saddle zone. They were interpreted and modelled using hard boundaries, having positive impact on the interpretation of the gold distribution and overall validation of the model. A new low-grade zone was modelled within the Southern Star. This low-grade zone was generated as a result of grade control drilling and poor production reconciliation of copper grades.
- The extents of the block model were increased to the western side to incorporate the WBX domain which wasn't part of the previous 2019 estimation (Figure 14-8).
- A previous gap between wireframes in the saddle zone was eliminated in the updated domain models.
- Approximately 107 million m³ of wireframe volume was added since 2019.
- Variogram orientations were updated, and search volume multipliers were increased based on reinterpretation of the domain variography.
- The assumed, long-term gold price was increased from \$1,500/oz to \$1,550/oz.
- Recovery curves for gold and copper were used in the calculation of CuEq, replacing the static values used in 2019.
- The exchange rate was increased from CA\$1.25 to CA\$1.30 to US\$.

- Secondary smoothing of the resource categories was implemented to eliminate the “Spotted Dog” effect.

Figure 14-8: Plan View of the Changes in the Block Model Extents between 2019 and 2021 Models



Block Model Limits

Table 14-8 summarizes the block model limits. The 2021 block model was extended to the west as compared to the 2019 model to accommodate the 4003 domain.

Table 14-8: Block Model Parameters

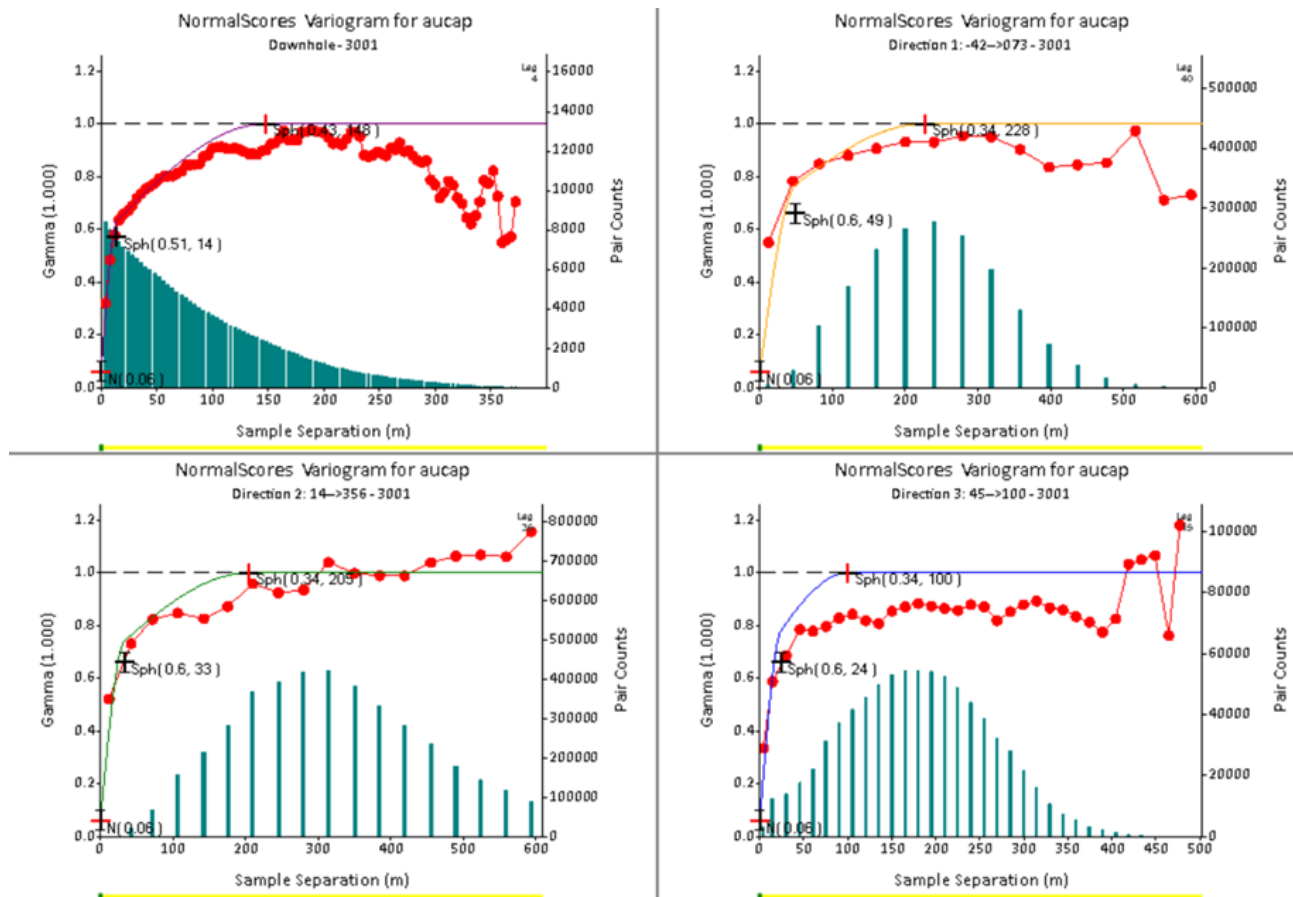
Parameter	X	Y	Z
Minimum	432000	6107500	305
Maximum	435690	6110245	1445
Number of Blocks	246	183	76
Block Size (m)	15	15	15

The block size selected was 15 m x 15 m x 15 m as this reflected an appropriate Selective Mining Unit (SMU) relative to the bench height employed at the mine. The total number of blocks in the model is 3,421,368.

Spatial Analysis

The spatial continuity of the composite grades, in each mineralized domain, was assessed using Snowden Supervisor. An example for Au in the 3001 domain is shown in Table 14-10. Variograms for the Mineral Resource estimate were defined in Datamine and Snowden Supervisor software based on capped composite samples. Nugget values were derived from downhole variograms, and 3D directional variography was determined from variogram maps and sweeps in the major and semi-major directions, based on geological understanding of mineralization controls. All variograms were modelled as double-structured spherical models, from experimental variograms (Table 14-9).

Figure 14-9: Gold in the 3001 Domain



Experimental variograms and variogram models are on file at Mount Milligan Mine. Directions investigated were the N-S average strike, the vertical average dip, and the E-W across average strike and dip. In each case, in addition to the principal directions, an average variogram along drill holes was calculated to better assess the magnitude of any nugget effect.

In most domains, the spatial continuity of the grade of composites is characterized by: (1) low relative nugget effects from 0.5% to 10% of the sill, and (2) generally poorly defined anisotropies with best continuity (lowest curve) along the average N-S strike or the E-W dip direction.

Copper and gold mineralization exhibit strong local trends, which are a function of the mineralizing system. In the SS domain, mineralization is strongest along the footwall contact of the Monzonite-Volcanic contact, with decreasing intensity away from this contact. Continuity is greatest in the plane of the contact.

Copper mineralization in the Main deposit is most pronounced along the MBX intrusive and Volcanic boundary, forming a concentric pattern around the MBX stock. Copper grade diminishes away from the stock towards the 66 Zone. Gold mineralization also forms a concentric pattern around the MBX stock, but is more dispersed, exhibiting continuity in the north-south direction, parallel to the stratigraphic strike of the footwall rocks.

In most domains, the spatial continuity of both copper and gold grade of composites were characterized by:

- Relatively low nugget effect.
- Generally, poorly defined anisotropy but rather having an isotropic distribution in the strike-dip plane.

Table 14-9: Variogram Parameters based on Domains

Domain			Rotation Angles			Modeled Range Distance (m)		
	Nugget	Sill	VANGLE1	VANGLE2	VANGLE3	X	Y	Z
100	0.123	0.446	-90	40	180	28	23	28
		0.43	-90	40	180	504	478	191
1002	0.136	0.688	-95	60	50	38	38	23
		0.176	-95	60	50	156	309	107
2001	0.114	0.625	165	90	-135	53	24	7
		0.261	165	90	-135	270	263	103
3001	0.107	0.687	100	45	-110	40	33	24
		0.206	100	45	-110	228	205	100
3002	0.088	0.624	90	70	180	75	29	37
		0.288	90	70	180	158	111	56
4001	0.125	0.74	100	90	150	50	36	30
		0.135	100	90	150	178	145	61
4002	0.134	0.726	-90	60	35	73	26	16
		0.14	-90	60	35	219	226	77

Resource Block Modeling – Gold and Copper Grade

Interpolation

Drill hole spacing across the deposit is generally on the order of 40-80 metres in the north-south and east-west directions. A block size of 15 x 15 x 15 metres, with no rotation, was selected to accommodate the drill hole spacing, width of the mineralization, and the mining bench height (15m).

To preserve trends, multiple estimation passes (Ordinary Kriging and Inverse Distance Squared) were used in the estimation of copper and gold. The first pass (SVOL1) established the search at approximately 0.7x the variogram range, the second pass (SVOL2) at 2x the 1st SVOL range, and the third pass (SVOL3) at the 3x variogram range. The 100 (shell) domain used a 0.7x variogram range, 1.5x and 2x respectively for SVOL 2 and SVOL3. A separate block model was created to run a Nearest Neighbour (NN) estimate for validation purposes, where the data for the NN estimate was composited at the length of the block size (15 m).

The search parameters for copper and gold are summarized in Table 14-10.

Table 14-10: Search Parameters used in Grade Estimation

Domain	Pass	Factor	Range X (m)	Range Y (m)	Range Z (m)	Datamine Rotation			Datamine Rotation Axis			Min. Comp	Max. Comp	Max. Comp/Hole
1000 & 1001 & 1002	1	1	45	40	30				3	1	3	10	18	4
	2	2	90	80	60	-95	60	50	3	1	3	6	20	4
	3	3	135	120	90				3	1	3	6	22	4
2001	1	1	45	35	30				3	1	3	10	18	4
	2	2	90	70	60	165	90	-135	3	1	3	6	20	4
	3	3	135	105	90				3	1	3	6	22	4
3001 & 3000	1	1	50	40	30				3	1	3	10	18	4
	2	2	100	80	60	100	45	-110	3	1	3	6	20	4
	3	3	150	120	90				3	1	3	6	22	4
3002	1	1	60	40	35				3	1	3	10	18	4
	2	2	120	80	70	90	70	180	3	1	3	6	20	4
	3	3	180	120	105				3	1	3	6	22	4
4001	1	1	45	35	30				3	1	3	10	18	4
	2	2	90	70	60	100	90	150	3	1	3	6	20	4
	3	3	135	105	90				3	1	3	6	22	4
4002 & 4003	1	1	50	35	30				3	1	3	10	18	4
	2	2	100	70	60	-90	60	35	3	1	3	6	20	4
	3	3	150	105	90				3	1	3	6	22	4
Outside the Wireframes (100))	1	1	50	40	35				3	1	3	10	18	4
	2	2	100	80	70	90	60	190	3	1	3	8	20	4
	3	3	150	120	105				3	1	3	3	18	4

Search directions were established as per variogram orientation. The estimation strategy is summarized in Table 14-11. For all blocks, a minimum of three or two drill holes were required per each pass to make an estimate, with generally a minimum of 6-10 and a maximum of 18-22 composites.

Table 14-11: Grade Estimation Parameters

Interpolation Parameters	1st Pass	2nd Pass	3rd Pass
Primary Interpolation Method	Ordinary Kriging	Ordinary Kriging	Ordinary Kriging
Minimum number of Composites	10	6	6
Maximum number of Composites	18	22	22
Maximum number of Composite per hole	4	4	4
Discretization	3X3X3	3X3X3	3X3X3

14.8. COPPER EQUIVALENT CALCULATIONS

Copper equivalent (CuEq) grade was calculated for individual blocks, using estimated in situ copper and gold grades and the following parameters:

- Gold Price (AuPrice): \$1,550/oz
- Copper Price (CuPrice): \$3.50/lb
- Copper Recovery (CURE): Copper Recovery Curve%
- Gold Recovery (AURE): Gold Recovery Curve%
- Conversion of gold price to \$/lbs: 14.5833

Formula used for CuEq calculation:

$$CuEq = CU_{\%} + \left(\frac{AURE * AuPrice * 14.5833}{CURE * CuPrice} \right) * \frac{Au_{gpt}}{10000}$$

A typical bench plan and cross section of the copper and gold block estimate distribution is shown in Figure 14-10 to Figure 14-13.

Figure 14-10: 1010 M EL Bench – Copper Block Estimate Distribution

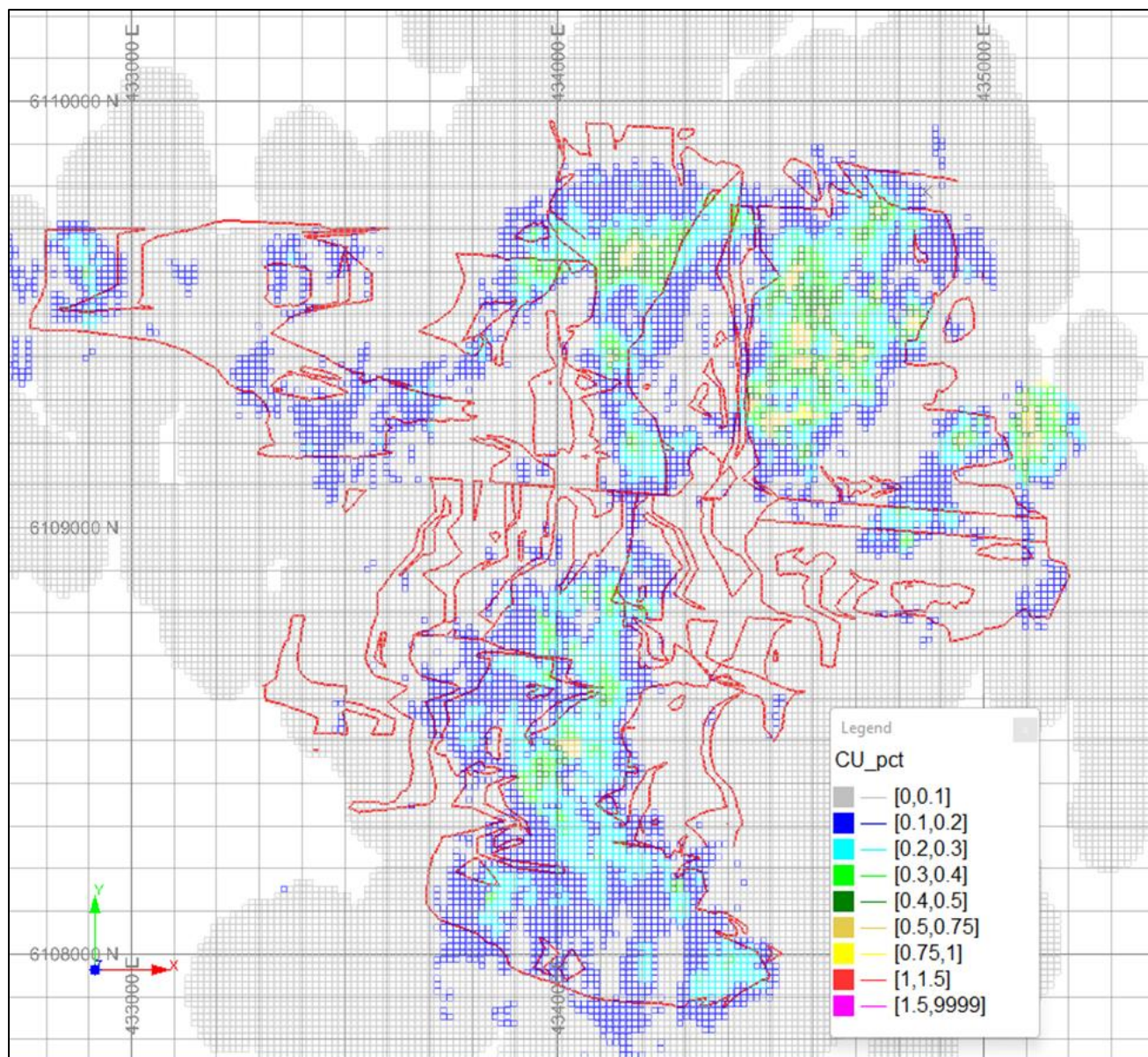


Figure 14-11: 1010 M EL Bench – Gold Block Estimate Distribution

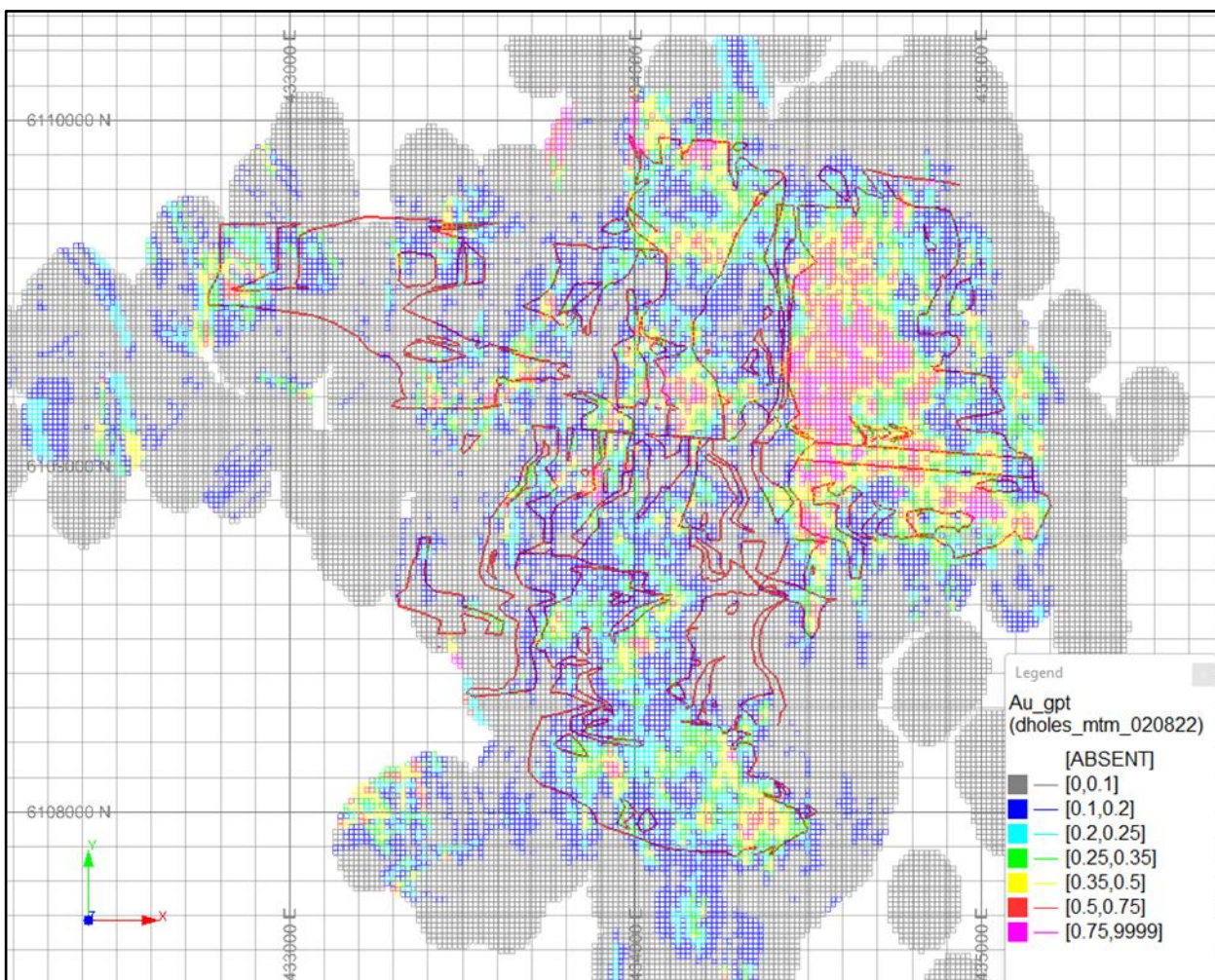
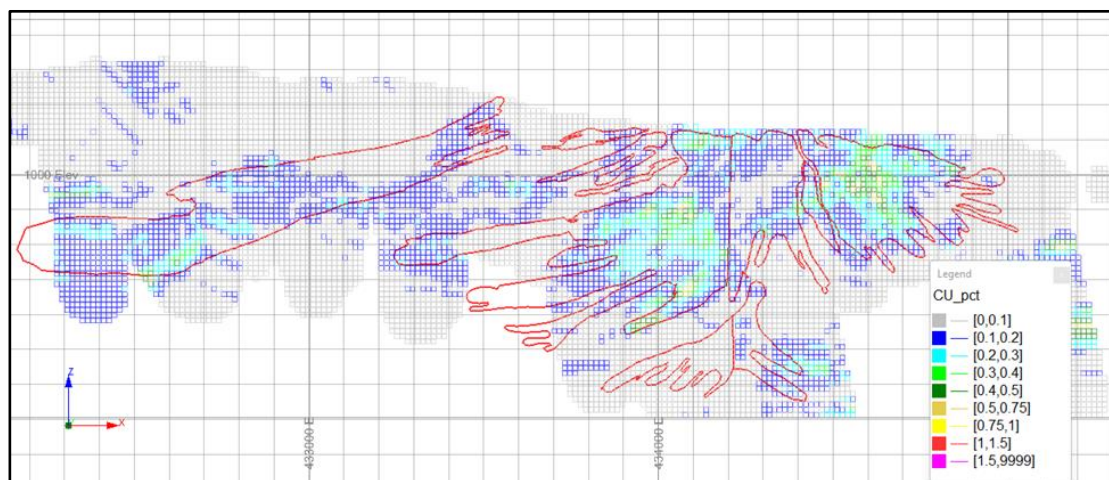


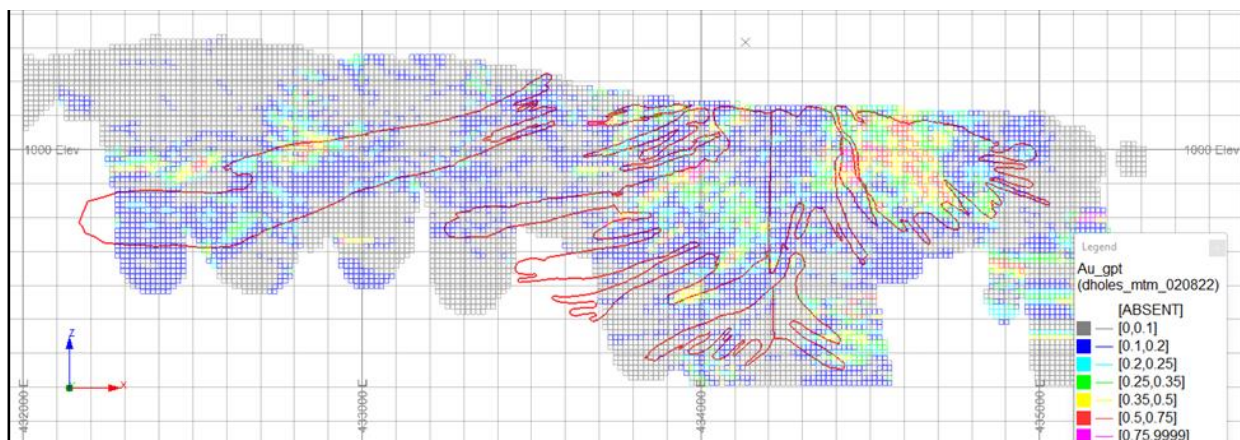
Figure 14-12: Section 6109500N – Copper Block Estimate Distribution – Looking North



Source: Centerra, 2021. Gridlines are 100m.

Effective Date: December 31, 2021

Page 14-22

Figure 14-13: Section 6109500N – Gold Block Estimate Distribution - Looking North

Source: Centerra, 2021. Gridlines are 100m.

Model Validation

A visual comparison of the block model grades and composite data was conducted in multiple cross sections and plan view. The estimated block grades were found to be consistent with the composite grades, with no major discrepancies observed. Example cross section comparisons for gold are shown in Figure 14-14 and Figure 14-15.

Centerra conducted a reconciliation of the current 2022 resource estimate model vs actual mill recovery and Ore Grade blast hole models for the 2016-2021 productions years. The 2022 resource estimate model performed superior to the 2019 resource estimate model obtaining results within ~ 1% that of mill recovery.

Figure 14-14: Cross Section at 6,108,480N Showing Mineralized Blocks

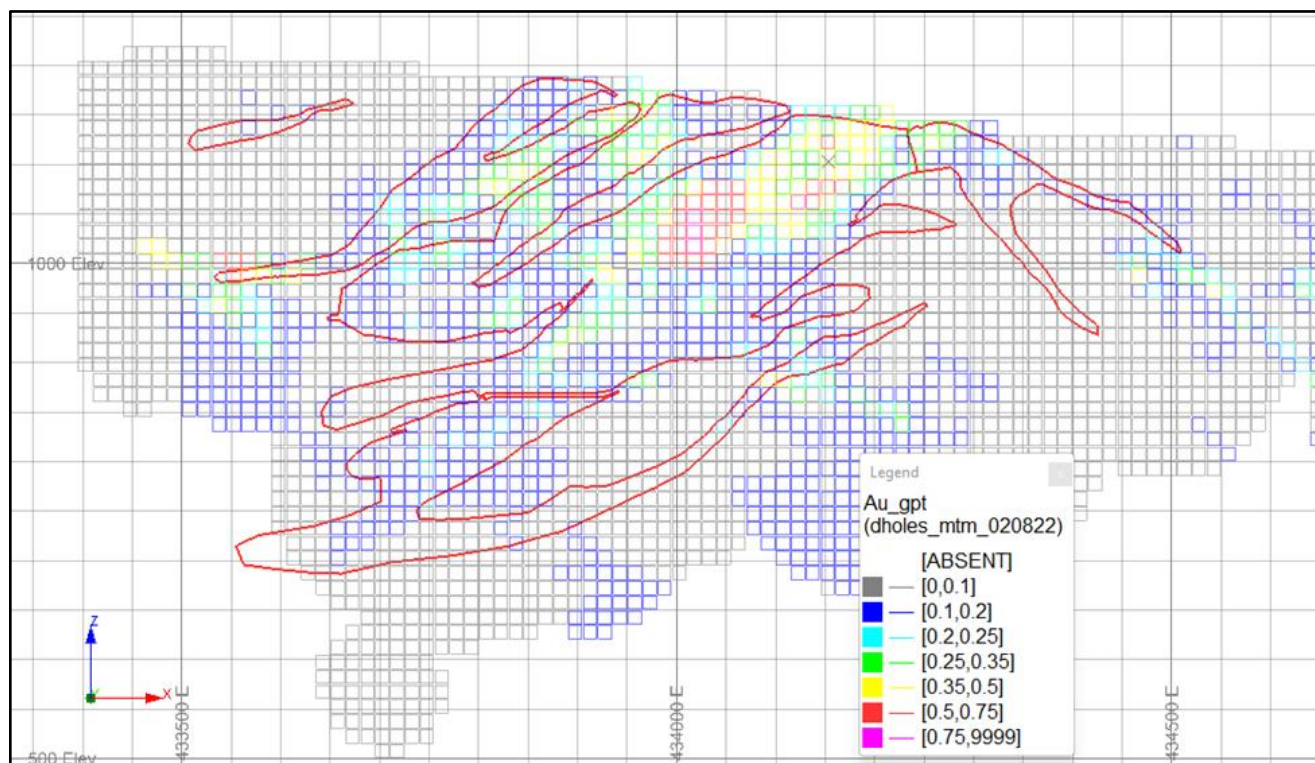
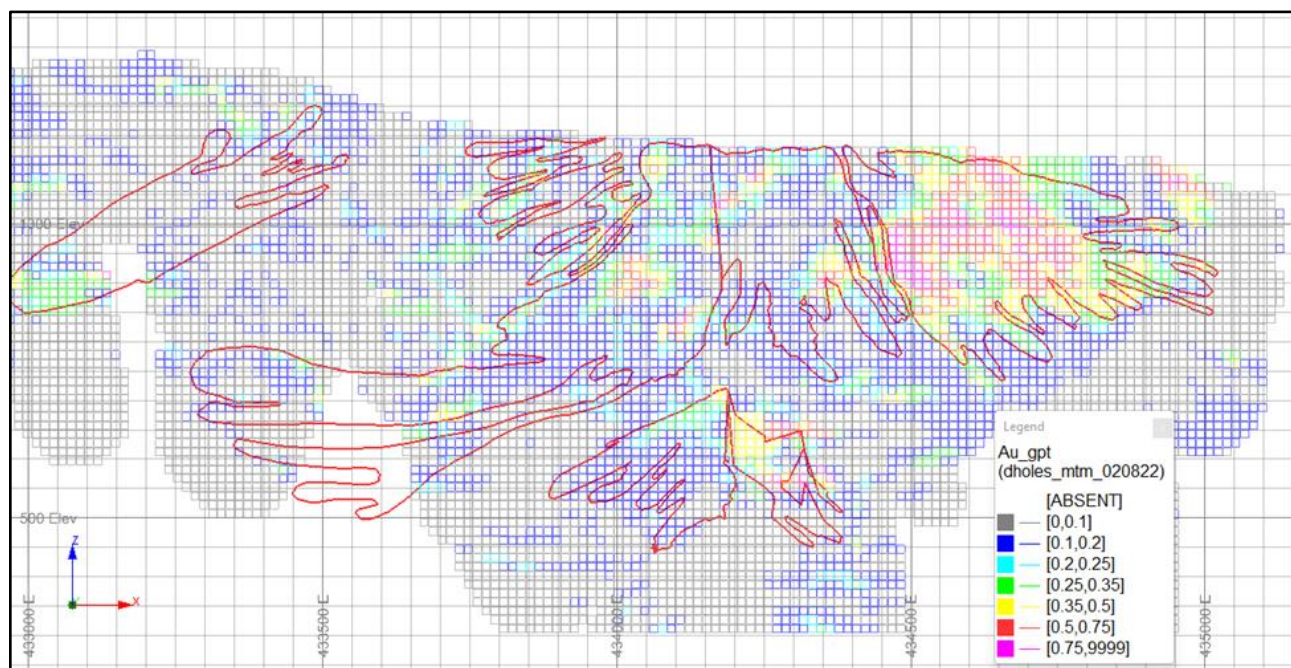


Figure 14-15: Cross Section at 6,109,420N Showing Mineralized Blocks



In addition, global statistical validation was conducted among the two modelling methodologies used during the estimation process.

Based on improved statistical parameters and SWATH plots correlation, Ordinary Kriging (OK) was the final modelling methodology selected for reporting the Mineral Resource estimate (Figure 14-16 to Figure 14-18).

This deposit is characterized by relatively high variability in the gold grade distribution which is reflected in the grade estimation and block model validation.

Figure 14-16: Swath Plot for Cu % in X Direction within Resource Pit

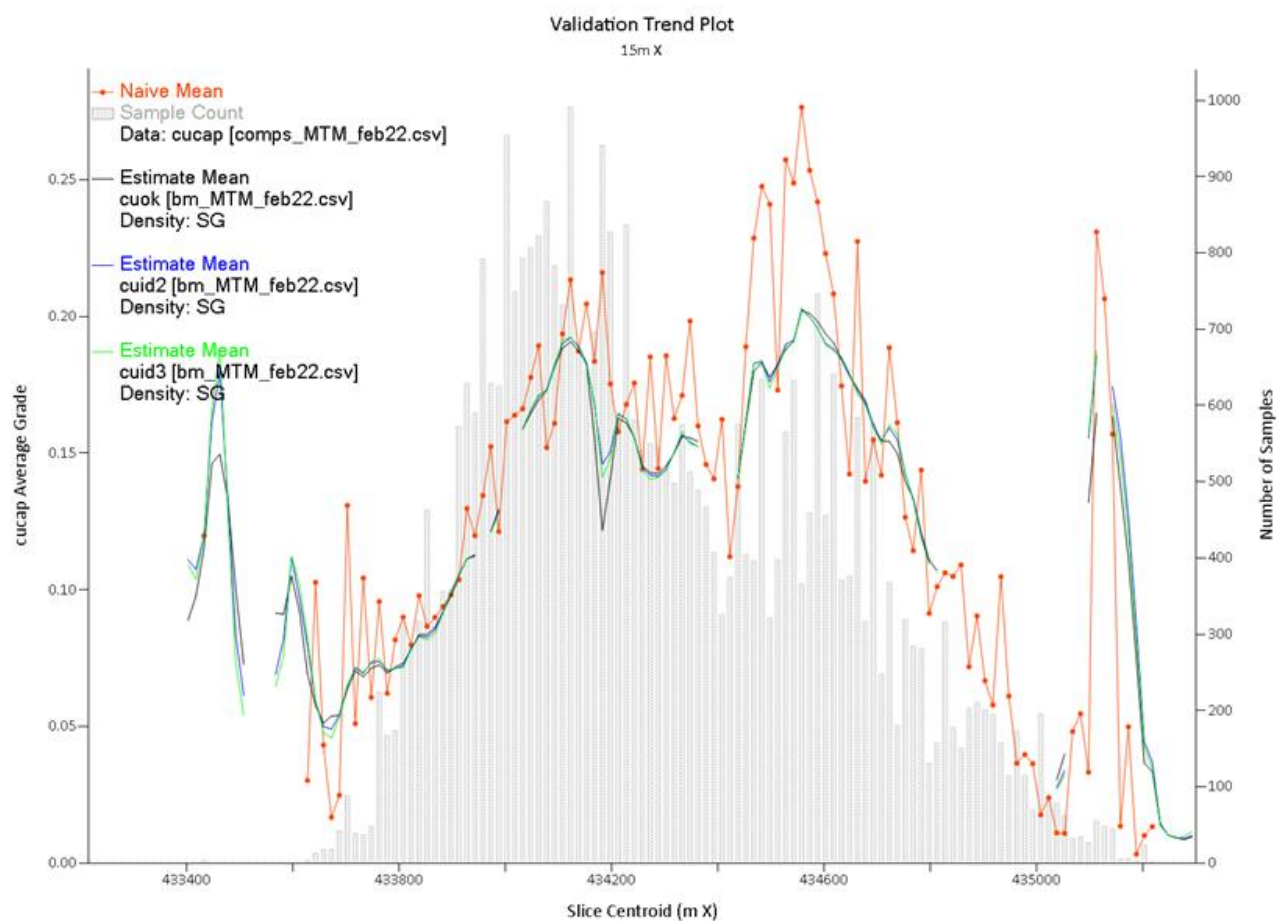


Figure 14-17: Swath Plot for Au g/t in X Direction within Resource Pit

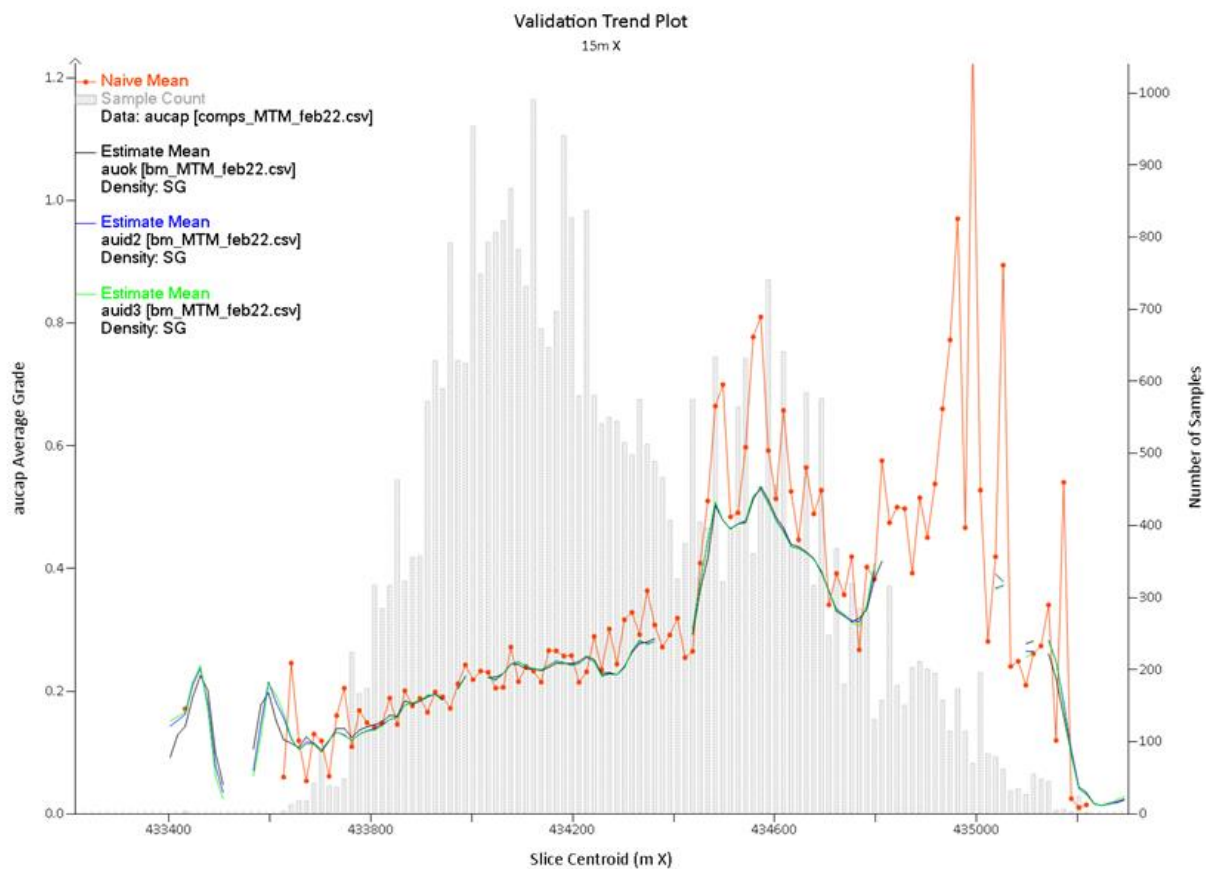


Table 14-12 presents global mean grade comparison of the different estimation types used.

Table 14-12: Global Statistical Validation of OK vs ID2 vs NN

	Gold				Copper			
	Southern Star - Domain 1002							
	OK Est	ID2 Est	ID3 Est	NN Est	OK Est	ID2 Est	ID3 Est	NN Est
Mean (g/t)	0.178	0.179	0.178	0.177	0.14	0.141	0.141	0.139
	66 Zone East - Domain 2001							
	OK Est	ID2 Est	ID3 Est	NN Est	OK Est	ID2 Est	ID3 Est	NN Est
Mean (g/t)	0.38	0.384	0.384	0.371	0.072	0.073	0.073	0.07
	MBX Volcanic East - Domain 3001							
	OK Est	ID2 Est	ID3 Est	NN Est	OK Est	ID2 Est	ID3 Est	NN Est
Mean (g/t)	0.416	0.418	0.419	0.402	0.179	0.179	0.179	0.176
	MBX Cont. East - Domain 4001							
	OK Est	ID2 Est	ID3 Est	NN Est	OK Est	ID2 Est	ID3 Est	NN Est
Mean (g/t)	0.212	0.213	0.214	0.211	0.132	0.133	0.133	0.13
	MBX West - Domain 4002							
	OK Est	ID2 Est	ID3 Est	NN Est	OK Est	ID2 Est	ID3 Est	NN Est
Mean (g/t)	0.219	0.219	0.219	0.222	0.142	0.143	0.143	0.144
	WBX							
	OK Est	ID2 Est	ID3 Est	NN Est	OK Est	ID2 Est	ID3 Est	NN Est
Mean (g/t)	0.15	0.154	0.155	0.147	0.111	0.114	0.114	0.12

Note: Additional block model validation Swath Plots are on file at Mount Milligan Mine.

Validation between the Ordinary Kriging, ID2 and ID3, and NN stands as acceptable with reasonable discrepancies as summarized in Table 14-12.

14.9. RESOURCE CLASSIFICATION

Block model quantities and grade estimates were classified according to the CIM Definition Standards for Mineral Resources and Mineral Reserves (2019) and adopted by NI 43-101. Mineral resource classification is typically a subjective concept, and industry best practices suggest that resource classification should consider both the confidence in the geological continuity of the mineralized structures, the quality and quantity of exploration data supporting the estimates, and the geostatistical confidence in the tonnage and grade estimates. Appropriate classification criteria should aim at integrating these concepts to delineate regular areas at similar resource classification.

This classification was based upon a review of data distribution, data density, and the spatial continuity of both gold and copper. Plots of kriging variance, average distance to composites, and the minimum and maximum number of composites used were generated and reviewed.

The following criteria were considered in the classification of the Mineral Resource estimate:

- Distance as defined by kriging pass, established from variogram models.
- Kriging variance
- Number of drill holes used to make an estimate
- Number of samples used to make an estimate

The classification of Mineral Resources for the MBX, WBX, 66 Zone, SS and Saddle Zones was based primarily on copper equivalent grades. All blocks were estimated using composites from at least two drill holes. All blocks were trimmed by a constraining open-pit shell (or cone) before reporting.

Measured Mineral Resource

Blocks classified as Measured Mineral Resource were estimated in the first estimation pass, thus lie within 0.7x the variogram range (Pass 1). In addition, three or more holes were used to make an estimate and were estimated by a minimum of eight composites with kriging variance less than or equal to 0.33.

Indicated Mineral Resource

Blocks classified as Indicated Mineral Resource were estimated in the second estimation pass, thus lie within 2 x the value of Pass 1 (variogram range). In addition, these blocks have a kriging variance of less than 0.66, used three or more holes to make an estimate and were estimated by a minimum of six composites.

Inferred Mineral Resources

Inferred Mineral Resource blocks were classified from 3 x the value of Pass1 (variogram range) and were estimated by a minimum of six and maximum of 24 composites and could possibly be estimated using a single drill hole.

A secondary smoothing pass was manually completed in cross section to improve the continuity of the classification in areas where there were inconsistencies.

14.10. MINERAL RESOURCE STATEMENT

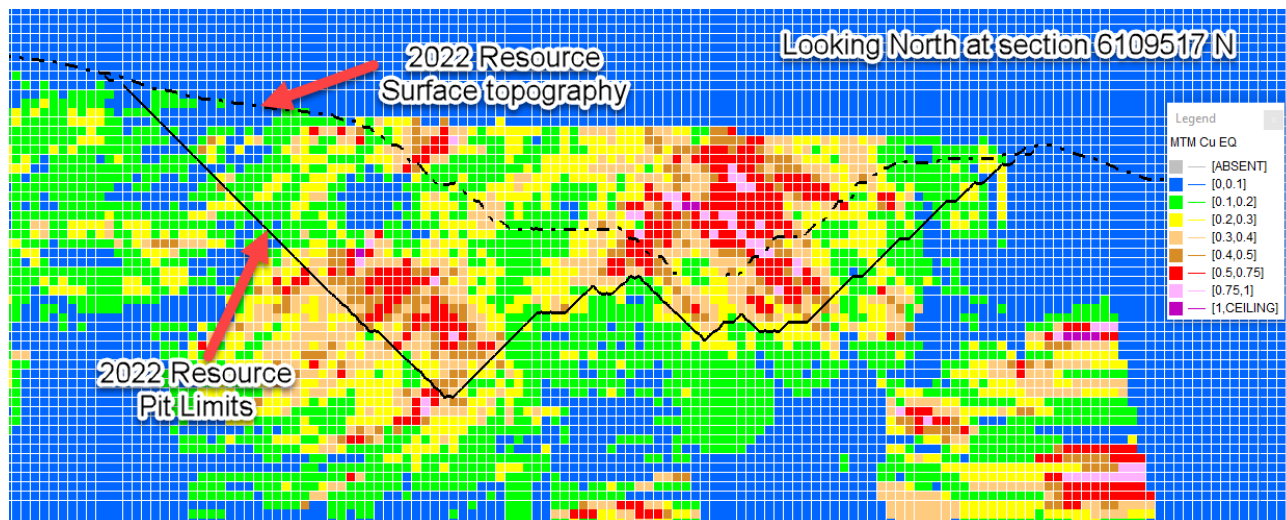
To determine the quantities of material offering “reasonable prospects for economic extraction” (as defined in CIM Definition Standards for Mineral Resources & Mineral Reserves, 2019) for an open pit operation, Centerra used Datamine NPV Scheduler software to run a pit optimization exercise. The applied optimization parameters are presented in Table 14-13. The reader is cautioned that the results from this pit optimization are used solely for the purpose of testing the “reasonable prospects for economic extraction” based on an open pit mining scenario and do not represent Mineral Reserves.

Table 14-13: Pit Optimization Parameters for Mineral Resource Estimation

Parameters	Unit	Value
Gold Price	\$ per ounce	1,550
Copper Price	\$ per pound	3.50
Exchange Rate	CA\$:US\$	1.30:1.00
Mining Cost (including sustaining capital)	\$ per tonne mined (average LOM cost)	3.00
Processing Cost (including sustaining capital)	\$ per tonne of milled	6.69
G&A Cost (including sustaining capital)	\$ per tonne of milled	2.85
NSR Cut-Off Value	CA\$/tonne milled	9.62
Mining Dilution	percent	Inherent to re-blocking process
Mining Loss	percent	Inherent to re-blocking process
Pit Wall Angles	degrees	Variable – see section 16
Milling Rate	Mt per year	21.9
Copper Recovery	percent	Variable – see section 13
Gold Recovery	percent	Variable – see section 13

After reviewing operating costs relevant to NSR cut off value calculations and block CuEq calculations, it was found that 0.20% CuEq was an appropriate cut-off grade to report the Mineral Resource estimate for the Mount Milligan Mine (please refer to cross section below (Figure 14-18). That is, the Mineral Resources at 0.20% CuEq correspond well with the CA\$9.62/tonne NSR value. The QP considers that the blocks above the 0.20% CuEq cut-off grade and located within the conceptual pit envelope show “reasonable prospects for economic extraction”.

Figure 14-18: Cross Section 6,109,740N (MBX and 66 Zone)



Caution to readers: In this Item, all estimates and descriptions related to mineral resource estimates are forward-looking information. There are many material factors that could cause actual results to differ materially from the conclusions, forecasts or projections set out in this item. Some of the material factors include differences from the assumptions regarding the following: estimates of Cut-off Grade (COG) and geological continuity at the selected cut-off, metallurgical recovery, commodity prices or product value, mining and processing methods, and G&A costs. The material factors or assumptions that were applied in drawing the conclusions, forecasts and projections set forth in this Item are summarized in other Items of this report.

Mineral resources have been estimated in conformity with CIM Estimation of Mineral Resource and Mineral Reserves Best Practices Guidelines (Nov 2019) and are reported in accordance with the Canadian Securities Administrators' National Instrument 43-101.

The Mineral Resource estimates are reported exclusive of Mineral Reserves. The effective date of the Mineral Resource Statement is December 31, 2021 (Table 14-14).

Table 14-14: Mineral Resource Statement, Effective Date December 31, 2021 (exclusive of Mineral Reserves)

Category	Cut-off CuEq (%)	Tonnes (kt)	Copper Grade (%)	Gold Grade (g/t)	Contained Copper (Mlb)	Contained Gold (koz)
Measured (M)	0.2	36,529	0.21	0.26	169	305
Indicated (I)	0.2	152,796	0.17	0.31	573	1,523
Total M+I	0.2	189,325	0.18	0.30	742	1,828
Inferred	0.2	4,638	0.07	0.47	7	70

Notes:

- (1) CIM definitions were followed for the classification of Mineral Resources.
- (2) Mineral Resources are reported within a conceptual pit shell at a 0.2% CuEq cut-off value using assumed metal prices of \$3.50 per pound copper and \$1,550 per ounce gold, and a US\$/CA\$ exchange rate of \$1.00/CA\$1.30. Refer to Item 14.8 for CuEq Formula and Item 14.10 for conceptual pit shell parameters.
- (3) All figures have been rounded to reflect the relative accuracy of the estimates.
- (4) Mineral Resources that are not Mineral Reserves do not have a demonstrated economic viability.
- (5) Mineral Resources are reported exclusive of Mineral Reserves within a conceptual Mineral Resource Pit and considering mining progress as of December 31, 2021.
- (6) The QP for the Mineral Resource estimate is Mr. Brian Thomas, P.Geo., an independent QP as defined under NI43-101 and an employee of Golder Associates Ltd.

The Mount Milligan Au-Cu porphyry deposit contains a combined Measured and Indicated Mineral Resource, exclusive of Mineral Reserves, of an estimated 189.3 Mt at 0.18% Cu and 0.30g/t Au containing an estimated 742 million pounds of copper and 1.8 million ounces of gold and an Inferred Mineral Resource of 4.6 Mt at 0.07% copper and 0.47g/t gold.

Comparison to Previous Mineral Resource Statement

The December 31, 2021, Mineral Resource estimate (see Table 14-14) saw a net increase of 63,955 kt of material resulting in an increase of 225 Mlb Cu and 420 koz Au of Measured and Indicated category above the previous resource estimate published in 2019 (non-current) (see Table 14-15). Inferred category saw a net loss of 3 Mlbs Cu and an increase of 15 koz Au.

Table 14-15: 2019 Non-Current Mineral Resource Statement (exclusive of Mineral Reserves)

Category	Cut-off CuEq (%)	Tonnes (kt)	Copper Grade (%)	Gold Grade (g/t)	Contained Copper (Mlb)	Contained Gold (koz)
Measured (M)	0.2	50,582	0.16	0.44	182	713
Indicated (I)	0.2	74,788	0.20	0.29	335	695
Total M+I	0.2	125,370	0.19	0.35	517	1,408
Inferred	0.2	3,736	0.12	0.46	10	55

Notes:

- (1) CIM definitions were followed for the classification of Mineral Resources.
- (2) Mineral Resources are reported at a 0.2% CuEq cut-off value using assumed metal prices of \$3.50 per pound copper and \$1,500 per ounce gold, and a US\$/CA\$ exchange rate of \$1.00/CA\$1.25.
- (3) All figures have been rounded to reflect the relative accuracy of the estimates.
- (4) Mineral Resources that are not Mineral Reserves do not have a demonstrated economic viability.
- (5) Mineral Resources are reported exclusive of Mineral Reserves.

These differences can be attributed mainly to increased drilling, changes in modelling methodology, and changes in economic assumptions as described in 14.7.

14.11. MINERAL RESOURCE RISKS AND RECOMMENDATIONS

The Mount Milligan Mineral Resource estimate has inherent risks associated with it and the QP has summarized the potential risks in this section. Recommendations are also suggested and are meant to mitigate current and future risks to estimates.

- A single variogram was chosen to represent both copper and gold distribution for each domain (similar to 2019 Resource Estimate). The correlation of gold and copper at Mount Milligan is unique to each domain but some have poor to no correlation as in the case of the 2001 Domain. This can lead to the improper distribution of grade in the estimate. The QP recommends that variograms representing gold and copper be modelled separately in the future. In terms of mitigating the current risk in Domain 2001, additional drilling to confirm extents of the 2001 high gold zone should be conducted prior to production.
- Estimation is constrained within each grade shell domain except for the 100 or “shell” domain which represents everything (peripherally, below or interlobate) not contained within a domain wireframe. Estimation within the 100 Domain uses the entire drillhole database in an unconstrained manner which suggests a potential risk of grade overestimation within the 100 Domain. The one caveat to this risk is that the reconciliation of the current model provided superior results to that of the 2019 model when compared against several past years of mill production. The QP recommends that Centerra Gold eliminate the use of the 100 Domain in future model estimations and incorporate those areas into existing or reinterpreted wireframes.
- Estimation search volumes, specifically the 2nd and 3rd are somewhat higher than that used in the 2019 model which has the overall effect of potentially spreading grades further but also increasing the volume of material falling within the Indicated category based on the Centerra classification methodology. This, coupled with the previously indicated unconstrained estimation of the 100 Domain, could lead to the potential miscategorization of blocks around the periphery of the constrained domain wireframes. A detailed review of block categorization outside the periphery of the domain wireframes is recommended in future models.
- The reinterpretation of the domain wireframes should be considered with more focus on incorporating structural trends as part of the grade distribution. Such an approach would allow localized trends to be realized at the mine planning stage.
- Domain wireframes should be reviewed for construction errors. Several have crossovers and gaps which can cause volume issues depending on the modeling software used.
- An analysis of the distribution of potentially deleterious minerals should be completed.

The QP is unaware of any known environmental, permitting, legal, title, taxation, socio-economic, marketing, political, or other relevant factors that could materially affect the Mineral Resource estimate.

15. MINERAL RESERVE ESTIMATES

15.1. SUMMARY

The mineral reserves for the Mount Milligan Mine are based on the conversion of the Measured and Indicated Resources within the current mine plan. Measured Resources are converted to Proven Reserves and Indicated Resources are converted directly to Probable Reserves.

The QP has not identified any known legal, political, environmental, or other risks that would materially affect the potential development of the Mineral Reserves.

15.2. MINING METHOD AND MINING COSTS

The Mount Milligan Mine is an open pit operation using conventional mining equipment. The mining process is owner operated. Contractor equipment is hired periodically to augment the owner's fleet during the construction season of the tailings dam which also doubles as the waste dump.

All work is based on current mine operating plans generated by AGP and verified by Mount Milligan Mine personnel.

Cost estimates are based on actual operating costs and proposed budgets for the remaining mine life. They were derived using first principle cost models.

The current resource model dated December 31, 2021, is used for all mine design work. Only Measured and Indicated Resources were used in the determination of reserves for the Mount Milligan Mine.

Geotechnical Considerations

Highwall slope angle criteria vary by geotechnical sectors and pit areas. The designs are based on recommendations of Knight Piésold (KP) provided in 2021. A geotechnical review of the life of mine open pit design was also completed by KP and confirmed in their memo dated July 13, 2022, entitled "Mount Milligan Mine – Geotechnical Review of Revised Life of Mine Open Pit Design for Updated NI 43-101 Technical Report". The confirmed designs were used to generate reserves in this Technical Report.

In general, the inter ramp angles vary from 30 to 47 degrees depending on pit area and wall orientation. This is mainly dependent on the geological domain and structural system in which the walls are located. Specific wall angles for each domain, as determined by the geotechnical consultants, are illustrated in Table 15-1.

Heights between safety catch benches vary by material type. In the overburden (glacial till), the bench height is 15 m; and in the waste rock, it is 30 m. Berm widths vary from 14 m to 17 m depending on the zone and pit. In overburden and in both pits a safety catch bench of 17 m is used in the design; in the Main (MBX) Pit, a catch bench of 16 m is employed. In the MBX pit on the North, South, and West walls as well as in the Southern Star (SS) Pit and all walls in the intrusives and volcanics, a safety catch bench of 14 m is used.

A larger catch or haul ramp is incorporated into the pit walls wherever the slope height exceeds 150 m.

The various criteria have been loaded into the geologic model by lithological unit and geological domain for use in mine planning. The geotechnical criteria are applied to the pit optimization and pit design work.

Economic Pit Shell Development

The final pit designs are based on pit shells using the Lerch-Grossman algorithm and procedure in MinePlan. The parameters for the pit shells are shown in Table 15-1.

Table 15-1: Pit Optimization Parameters

Parameter	Units	Mount Milligan Mine Values
Metal Price (Varying metal Prices used)		
Gold Price	\$/oz	1,350
Copper Price	\$/lb	3.25
Geotechnical Inter-Ramp Angles		
MBX Pit and SS Pit all walls overburden	degrees	30
MBX Pit East and Southeast Walls Intrusives, Volcanics	degrees	42
MBX Pit North, South, West Walls Intrusives, Volcanics	degrees	47
SS Pit All wall intrusives, volcanics	degrees	47
Process Recovery		
Process Tonnes per Operating Hour	tpoh	2,700
Recovery – Gold	%	Variable
	Formula	$-0.000136731 \times \text{tpoh} * + 1.59926 \times (\text{Cu grade})^{0.0522036} + 0.02$
Recovery – Copper	%	Variable
	Formula	$-0.0000633099 \times \text{tpoh} + 0.868817 \times (\text{Au grade})^{0.0700520} + 0.04$
Costs		
Ore Mining	CAD/t mined	2.83
Waste Mining	CAD/t mined	2.95
Overburden Mining	CAD/t mined	2.45
Incremental cost per bench above/below 1085m elevation	CAD/t all materials mined	0.002
Processing	CAD/t milled	6.74
Admin	CAD/t milled	2.88
Processing and Admin	CAD/t milled	9.62
Blocks used	Resource Classification	M+I

* tpoh – tonnes per operating hour

Pits were generated using various combinations of revenue factors of \$1,350/oz gold price and \$3.25 copper price.

Cut-off Grade

The mill feed cut-off grade employed for the LOM plan was determined as follows; in-situ material with an NSR value per tonne (VLPT) of greater than CA\$ 9.62/t (processing and administration costs) and with a copper grade of 0.14% copper or higher is designated as standard ore. Material with an NSR value greater than CA\$ 9.62/t but copper grades lower than 0.14% is designated as HGLC (high-gold low-copper) ore. The cut-off values used for the statement of reserves are shown in Table 15-2.

Table 15-2: Mount Milligan Mine Reserve Cut-off Grades

Ore Type	VLPT (\$)	Copper Grade (%)
Standard Ore	9.62	≥ 0.14
HGLC	9.62	< 0.14

Dilution

The resource model is a homogenous model created by re-blocking a subblocked model. Grades for gold and copper are estimated using subblocks along the contacts of defined mineralized domains. When scaling the subblocks to the reserve block size, mineral grade estimates from subblocks for the percentage of material inside the domain are diluted against the percentage of material outside of domain.

Pit Design

The detailed pit phase designs at Mount Milligan Mine are based on the optimized pit shells generated with the current resource model and the geotechnical parameters outlined in Table 15-2. Pit designs were developed for the MBX and SS pit areas. The initial phases were designed for the purpose of access and blending material within the revenue factor (RF) equal to 0.95 pit shell. The final phase, Phase 11, used the RF=0.98 pit shell.

Design sizing for ramps and working benches is based on the use of a CAT 7495 rope shovel and CAT 793 rigid frame dump trucks which are the largest units in the pit. The operating width of the truck is 8.3 m. This means double lane widths are 34 m (three times the operating width plus berm and ditch widths). Ramp uphill gradients are maximum 10% in the pit and maximum 10% uphill on the dump access roads. Working benches were designed for 60 m minimum on pushbacks, although some pushbacks did work in a retreat manner to facilitate access.

The reserves mine schedule is based on the end of December 31, 2021, mining position. The production schedule delivers 246.2 Mt of Proven and Probable ore grading 0.18% copper and 0.37 g/t gold to the process plant over a design life of 11 years. The ore tonnage is made up of 76.5 Mt of Proven Reserves and 169.7 Mt of Probable reserves, including Proven ore in the stockpile, as of the end of day December 31, 2021.

Mined waste tonnage totals 211.7 Mt, which will be placed in the various waste rock management facilities. The overall strip ratio is 0.92:1 (waste:ore) tonnes mined.

Mineral Reserves Statement

The reserves for the Mount Milligan Mine are based on the conversion of Measured and Indicated Resources within the current mine plan. Measured Resources are converted to Proven Reserves and Indicated Resources are converted directly to Probable Reserves. The Mineral Reserve estimates were prepared under the supervision of Gordon Zurowski, P.Eng., a QP as defined under NI 43-101.

Cut-off grades for the reserves were based on the block model file VALPT CAD 9.62.

This estimate is as of December 31, 2021, end of day. The total reserves for the Mount Milligan Mine are shown in Table 15-3.

Table 15-3: Proven and Probable Reserves – Summary for Mount Milligan Mine

Mineral Category	Reserve	Tonnes (kt)	Gold (g/t)	Copper(%)	Contained Gold (koz)	Contained Copper (Mlbs)
Proven		76,477	0.37	0.20	914	337
Probable		169,681	0.37	0.18	2,011	659
Total		246,158	0.37	0.18	2,925	996

Notes:

1. CIM definitions were followed for Mineral Reserves estimation.
2. The Mineral Reserve estimate, prepared by AGP Mining Consultants, is as of Dec. 31, 2021, and is based on the Mineral Resource estimate dated Dec. 31, 2021, for Mount Milligan Mine.
3. The Mineral Reserve calculation and estimate was completed under the supervision of Gordon Zurowski, P.Eng. of AGP, who is a Qualified Person as defined under NI 43-101.
4. Mineral Reserves are estimated at \$7.40/t (CA\$9.62/t) NSR cut-off value using metal prices of \$3.25 per pound copper and \$1,350 per ounce gold, and a US\$/CA\$ exchange rate of \$1.00/CA\$1.30.
5. Figures may not total exactly due to rounding.
6. As of June 30, 2022, Au Contained and Cu Contained have been depleted from the above figures by approximately 82 koz and 38 Mlbs from the above December 31, 2021, Mineral Reserve balances, respectively.
7. Production at Mount Milligan is subject to a streaming agreement with RGLD Gold AG and Royal Gold, Inc. (collectively, "Royal Gold") which entitles Royal Gold to 35% of gold sales from the Mount Milligan Mine. Under the stream arrangement, Royal Gold will pay \$435/oz of gold delivered. Mineral Reserves and Resources for the Mount Milligan property are presented on a 100% basis.

16. MINING METHODS

16.1. OVERVIEW

Open pit mining is the mining method in use at Mount Milligan based on the size of the resource, grade tenor, grade distribution and proximity to topography. With current metal pricing levels, knowledge of the mineralization and previous mining activities, open pit mining remains the most reasonable approach for development.

The mine production drill fleet comprises two electric rotary blast hole drills (311-mm diameter), one diesel blast hole drill (222-mm diameter), and a smaller diesel blast hole drill (152 – 222-mm diameter) preparing blast holes on 15 m benches. Loading of blasted rock is completed with two 41 m³ electric rope shovels, one 22 m³ hydraulic excavator, and two 19 m³ front end loaders. The haulage fleet comprises thirteen haul trucks of 229-metric-tonne (t) capacity and two 181-t capacity trucks. A typical fleet of support and ancillary equipment is employed, including track and rubber-tire dozers, graders, and a fleet of service vehicles. A fleet of articulated dump trucks are used in dam construction and project activities.

The LOM plan is based on Measured and Indicated Mineral Resources and **is developed from the estimated end of 2022 pit topography onward for the years 2023 to 2033**. The mine plan is developed using a series of phases or pushbacks in the Main (MBX) and Southern Star (SS) mining areas. The mill feed cut-off grade employed for the LOM plan was a value of greater than CA\$ 9.62/t (processing and administration costs) and with a copper grade of 0.14% copper or higher. This is termed standard ore. Material with a value greater than CA\$ 9.62/t but copper grades lower than 0.14% is designated as HGLC (high-gold low-copper) ore.

The result of the mine plan and conversion of resources to reserves means the LOM plan will provide 218.2 Mt of Proven and Probable in-situ material at average grades of 0.37 g/t gold and 0.18% copper to the process plant. Waste tonnages totalling 199.7 Mt will be mined in that period for an average strip ratio of 0.92:1.

16.2. GEOTECHNICAL PARAMETERS

Pit slope angles vary by sector and pit area within the open pit and are based on recommendations by Knight Piésold (KP) provided in 2021 and confirmed in a memorandum, dated July 13, 2022, titled “Mount Milligan Mine – Geotechnical Review of Revised Life of Mine Open Pit Design for Updated NI43-101 Technical Report”.

Previous geotechnical evaluations of the Mount Milligan mine had outlined five major geological domains:

- Overburden (Glacial Till)
- Intrusives (Monzonite)
- Volcanics (Andesite and Latite)
- Trachyte
- Rainbow Dyke

The lithological and structural setting at the Mount Milligan deposit were simplified into the following three major geological systems:

- Early easterly dipping volcanic formation, including the Andesite, Latite, Trachyte flow units, and lithology contacts and major faults (e.g., the Harris Fault in the MBX Pit, and the North-South Faults in the SS Pit).
- Sub vertical to slightly westerly dipping Monzonite Intrusives, including the MBX intrusion and SS intrusion.
- Late northeast to east trending extensional faults, including the Oliver Fault, Saddle Zone Fault, and Rainbow Fault in the MBX Pit and the East-West Fault in the SS Pit.

With this interpretation, potential structural features are likely to be exposed along the easterly and southerly facing West to North walls of the open pit designs.

Pit slope designs are based on double-benching 15 m high benches and consider haulage ramp positioning, safety berms and other geotechnical features required to maintain safe inter-ramp angles (IRA) and overall slope angles. The respective pit slope design parameters that have been used are illustrated in Table 16-1. The geotechnical areas are shown in Figure 16-1.

Table 16-1: Pit Slope Design Parameters

Pit/Sector/Geology	Inter-Ramp Angle (degrees)	Face Angle (degrees)	Berm Spacing (m)	Berm Width (m)
MBX & SS Pit / All Walls / Overburden	30	60	15	17
MBX Pit / East & Southeast Walls / Intrusives, Volcanics	42	60	30	16
MBX Pit / North, South & West Walls / Intrusives, Volcanics	47	65	30	14
SS Pit / All Walls / Intrusives, Volcanics	47	65	30	14

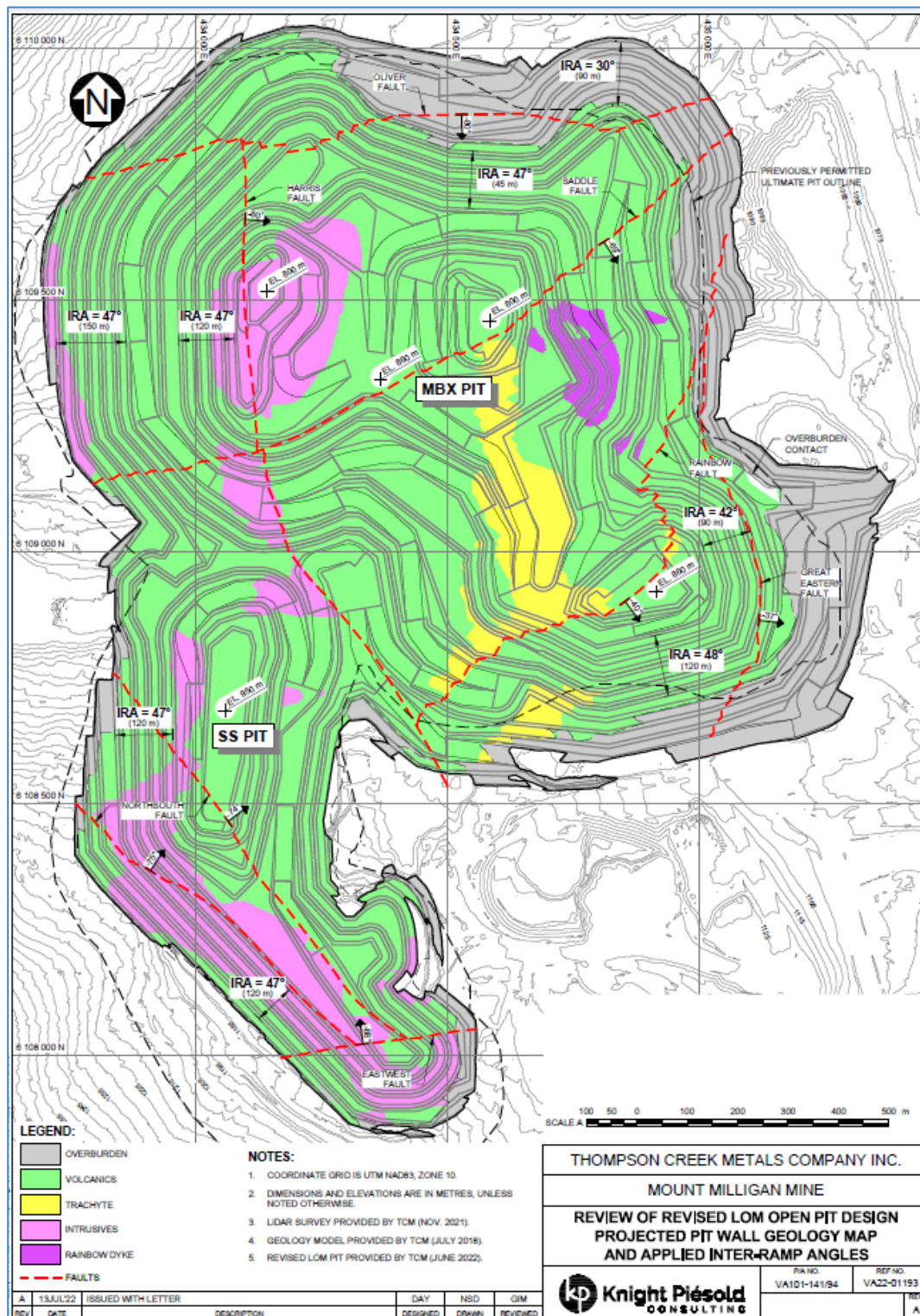
Slopes over a height of 150 metres are to include a geotechnical berm. The geotechnical berm is to be a wider bench (step-out) or a haul road ramp incorporated into the pit wall.

The LOM design was reviewed by KP, and the results of its review are shown in Table 16-2. All slope design areas complied with the design criteria in the review.

Table 16-2: Applied Inter-ramp Angles in the LOM Open Pit Design

Pit/Sector/Geology	Recommended IRA (degrees)	Applied IRA (degrees)	Max. Stack Height (m)	Potential Failure Mode and Major Faults
MBX & SS Pit / All Walls / Overburden	30	60	75	Bench ravelling due to surficial degradation and erosion
MBX Pit / East & Southeast Walls / Intrusives, Volcanics	42	60	120	Bench wedge, Rainbow fault broken zone rockfall
MBX Pit / North, South & West Walls / Intrusives, Volcanics	47	65	150	North to West Wall bench to multi-bench wedge, sub vertical fault zones
SS Pit / All Walls / Intrusives, Volcanics	47	65	120	West Wall bench to multi-bench wedge, sub vertical fault zones

Figure 16-1: Geotechnical Domains



Source: Knight Piésold 2022

As part of the KP review of the LOM plan, the following recommendations were made:

- Implement best practice, controlled wall blasting for both the interim and final pit walls
 - Including full double-bench height (30 m) pre-shearing, and
 - Smaller diameter blast holes, closer spacing and reduced powder factor
- Ongoing program of proper bench scaling prior to advancing to lower benches
- Construct perimeter diversion ditches along the West crests of the SS and MBX pits to effectively manage surface water flows
- Install horizontal drain holes in the middle to lower west walls of both pits for aggressive slope depressurization
- Implement a real time slope deformation monitoring system for targeted critical walls (e.g. north to west walls)

Further steepening of the final pit wall IRAs may be possible with the implementation of the above measures. Continued diligence in ongoing structural mapping, rock mass characterization, wet spot identification and seepage monitoring are recommended by KP to increase the confidence in rock mass behaviour for further slope stability verifications, design optimization and ongoing pit operations.

16.3. GEOLOGICAL MODEL IMPORTATION

The 2021 resource model was created by Centerra Gold personnel in Datamine format, later converted to MinePlan, and included the entire MinePlan folder with topography and other infrastructure items. The model was created with the grades in the whole block.

Framework details of the open pit block model are provided in Table 16-3. The final open pit mining model item descriptions are shown in Table 16-4. MinePlan was used for all mine planning activities using the Lerchs Grossmann (LG) algorithm for shell generation, pit design and mine scheduling tools.

Table 16-3: Open Pit Model Framework

Framework Description	Open Pit Model (Value)
MinePlan file 10 (control file)	Dm2210.dat
MinePlan file 15 (model file)	Dm2215.dat
X origin (m)	432000
Y origin (m)	6107500
Z origin (m) (max)	305
Rotation (degrees clockwise)	0
Number of blocks in X direction	246
Number of blocks in Y direction	183
Number of blocks in Z direction	76
X block size (m)	15
Y block size (m)	15
Z block size (m)	15

Table 16-4: Open Pit Model Item Descriptions

Field Name	Min	Max	Precision	Units	Comments
ALTAB	0	60	0.0001		New Alteration assemblage code from LF solids
AP	0	1000	0.0001	g/t	Acid potential
AS	0	1000	0.0001	ppm	Arsenic (ppm) estimation in MS
AU	0	10	0.0001	ppm	Gold (ppm) estimation in DM
AURE	0	1	0.0001	%	Gold recovery by script
BVAL	-500000	1000000	1	\$	Engineering request
C%	0	1	0.0001	%	C-Till material by solids coding & scripting
CLASS	0	4	1		Resource classification by script in DM
CPY	0	10000	0.0001	ppm	Chalcopyrite by script
CU	0	10	0.0001	%	Copper (pct) estimation in DM
CUEQ	0	10	0.001	%	Copper equivalent by script
CURE	0	1	0.0001	%	Copper recovery by script
DOM	0	4100	0.01	-	Domains coded from new DM solids
FANG	0	100	1	degrees	Face angle coded from script and solids for Geotech
FE	0	10000	0.0001	%	Chalcopyrite by script estimation in MS
GEOLO	0	91000	0.1	-	Legacy geology code from LF solids
HG	0	1000	0.0001	%	Mercury (ppm) estimation in MS
K-ALT	0	10	1	CA\$/t	Legacy alteration code from LF solids
LITCD	0	94	0.0001		Legacy lithology code from LF solids
MTOPO	0	1	0.0001	%	Mined topographical surface as of Dec 21, 2021
NPFIN	0	1000	0.0001	-	Neutralization potential final estimation in MS
NPR	0	5000	0.0001	-	Neutralizing Potential Ratio
NSCU	0	22	0.0001	-	DM variable undefined
OB%	0	1	0.0001	%	Percentage in Overburden solids
OBCD	0	15	0.0001		Overburden coding
OTOPO	0	1	0.0001	%	Original topography (LiDAR survey)

Field Name	Min	Max	Precision	Units	Comments
OX%	0	1	0.0001	%	Percentage in Oxide solids
OXIDE	0	1	0.001	%	Percentage in Oxide (rock) solids
P-ALT	0	10	1		Legacy alteration code from LF solids
PIT19	0	3	1		Pit shell coding for model comparison
PITAR	0	3	0.0001		Legacy area code from MS solids
PTYPE	0	100	0.0001		Production material type from script in MS
PY	0	10000	0.0001	ppm	Pyrite by script estimation in MS
PYCPY	0	10000	0.0001	-	Ratio
RCODE	0	1	0.0001		DM variable undefined
RECV	0	100	0.0001	t/m3	Total core recovery by estimation in MS
RESOM	1002	4100	0.01	%	DM variable undefined
ROCK%	0	1	0.0001	%	Percentage of rock at rock-overburden interface
RQD	0	1000	0.0001	-	Rock quality determination by estimation in MS
S	0	1000	0.0001	-	Sulphur estimation in MS
S%	0	1	0.0001	%	Percentage of S-till by solids coding & scripting
SB	0	1000	0.0001	ppm	Antimony (ppm) by estimation in MS
SG	0	5	0.0001	-	Specific gravity by estimation in MS
SLOPE	0	50	1	degrees	Slope angle coded from script and solids for geotechnical
VALOM	1002	4100	0.01	CA\$	DM variable undefined
VALPR	0	500	0.0001	CA\$/t	Value per tonne resource by NSR script
VALPT	0	500	0.0001	CA\$/t	Value per tonne reserve by NSR script
VTPO	0	1	0.0001	%	Rock topographical surface
ZONE	0	63	0.0001	-	Legacy zone coding from solids
CUUPR	0	9999	0.0001	-	Copper Upgrade Ratio
HGCON	0	9999	0.0001	ppm	Mercury in concentrate
HGUPR	0	9999	0.0001	-	Mercury Upgrade Ratio
HGTYP	0	9999	0.0001	-	Mercury Type

16.4. DILUTION

Mining dilution was determined by Mount Milligan personnel based on operating experience and provided in the supplied resource model.

The grades for gold and copper are estimated using subblocks to model the contact in the geology resource model. When the mining model is generated into a whole block model, the percentage of the material not within the mineralized domain is added into the block and the grades reduced accordingly in the overall block. Blocks that are fully contained within the mineralized domain, and not subblocked to model the contact, have no additional dilution added. In this manner the dilution represents contact dilution expected to be encountered by the mining equipment.

The whole blocks are assumed to be fully diluted for use in the pit shell and mine plan development.

16.5. PIT SHELL DEVELOPMENT

The open pit ultimate size and phasing requirements were determined with various input parameters including estimates of the expected mining, processing, and G&A costs as well as metallurgical recoveries, pit slopes, and reasonable long-term metal price assumptions. The parameters were discussed between AGP and Mount Milligan personnel to select appropriate operating cost parameters. Mining costs are based on current costs at site. Process and G&A costs were provided by the appropriate Mount Milligan team members.

Pit shells were run using the diluted grades in the model.

The parameters used are shown in Table 16-5. The net value calculations are in Canadian dollars (CA\$) unless otherwise noted. Costs and revenues are converted to Canadian dollars for use in pit shell determination where necessary. The mining cost estimates are based on the current equipment fleet with the estimated incremental hauls based on the current long term TSF design.

Table 16-5: Pit Shell Parameter Assumptions

Description	Units	Value
Resource Model		
Block classification used	Class	Measured + Indicated (M+I)
Block Model height	m	15
Mining Bench height	m	15
Metal Prices and Diesel		
Gold Price	\$/oz	1350
Copper Price	\$/lb	3.25
Diesel fuel cost to site	CA\$/L	1.31
Metallurgical Information		
Process Tonnes per Operating Hour	tpoh	2,700
Recovery - Gold	%	Variable
	Formula	$-0.000136731 \times \text{tpoh} + 1.59926 \times (\text{Cu grade})^{0.0522036} + 0.02$
Recovery - Copper	%	Variable
	Formula	$-0.0000633099 \times \text{tpoh} + 0.868817 \times (\text{Au grade})^{0.0700520} + 0.04$
Mining Cost		
Overburden Mining	CA\$/t	2.45
Waste base rate - 1085 elevation	CA\$/t	2.95
Incremental rate - above	CA\$/t/15m bench	0.02
Incremental rate - below	CA\$/t/15m bench	0.02
Mill feed base rate - 1085 elevation	CA\$/t	2.83
Incremental rate - above	CA\$/t/15m bench	0.02
Incremental rate - below	CA\$/t/15m bench	0.02
Processing and G&A Cost		
Processing cost	CA\$/t mill feed	6.74
General and Administrative Cost (G&A)	CA\$/t mill feed	<u>2.88</u>
Total Process + G&A	CA\$/t mill feed	9.62

Wall slopes for pit optimization were based on the KP design criteria, as provided and discussed in Item 16.2. Solids were used to code the model SLP item, then overall slopes were applied by area as shown in Table 16-6. These include the overall angle used with consideration for ramps.

Table 16-6: Pit Shell Slopes

Areas	Bench Geometry				
	Design Height	Design Angle	Design Width	Overall Angle	Inter-Ramp Angle
	(m)	(°)	(m)	(°)	(°)
Overburden	15	60	17	28	30
MBX (East-South) - Intrusive -Volcanics	30	60	16	39	42
Remaining MBX, SS - Intrusive, Volcanics	30	65	14	42	47
Remaining MBX, SS - Intrusive, Volcanics with extra Geotech berm	30	65	14	42	47

Nested LG pit shells were generated to examine sensitivity to gold and copper prices with a target of \$1,350/oz gold and \$3.25/lb copper. This was to gain an understanding of the deposit and highlight potential opportunities in the design process to follow. Diluted Measured and Indicated resources were used in the analysis. The metal prices were varied by applying revenue factors of 0.10 to 1.10 at 0.05 increments, thus generating a set of nested LG shells with a separation of 0.01 between 0.95 and 1.00. The chosen set of revenue factors result in an equivalent gold price varying from \$135/oz up to \$1,485/oz. All other parameters were fixed. The resulting nested pit shells assist in visualizing natural breakpoints in the deposit and selecting shells to act as design guidance for phase design. The net profit before capital for each pit was calculated on an undiscounted basis for each pit shell using \$1,350/oz gold and \$3.25/lb copper. Mill feed material/waste tonnages and potential net profit were plotted against gold price and are displayed in Figure 16-2.

The histogram shown in Figure 16-2 contained several break points in the pit shells. These were used as a guide for sequencing pit phase designs. With each incremental increase in the waste tonnage, and to a lesser degree the mill tonnage, the undiscounted net profit also increased. In the case of the first break point shown at \$1,283/oz gold and \$3.09/lb copper (RF=0.95), the cumulative waste tonnage is 106 Mt, with a corresponding mill feed tonnage of 179 Mt or a strip ratio of 0.59:1. The net profit also increased beyond this point, showing that there was still value to be obtained by going with a higher metal price or an additional phase. This break point represented 97% of the net value of a \$1,350/oz gold pit but with only 48% of the waste of the larger pit shell. This shell was used to guide the internal phase designs.

The second and final pit shell selected represented the ultimate pit at \$1,323/oz gold, and \$3.19/lb copper (RF=0.98). This resulted in a jump in the waste tonnage by 80 Mt from the first break point with a gain of 66.7 Mt of mill feed material for an incremental strip ratio of 1.2:1. The net profit continues to increase beyond this break point, although at a flatter rate than before. The cumulative value of the two break points was 100% of the \$1,350/oz gold and \$3.25/lb copper pit shell with 84%

of the waste movement of the larger pit required. Limited potential pit value was available beyond this pit shell and did not justify another phase. This final pit shell was used to define the ultimate pit, for what would become Phase 11.

Figure 16-2: Potential Profit vs. Revenue Factor

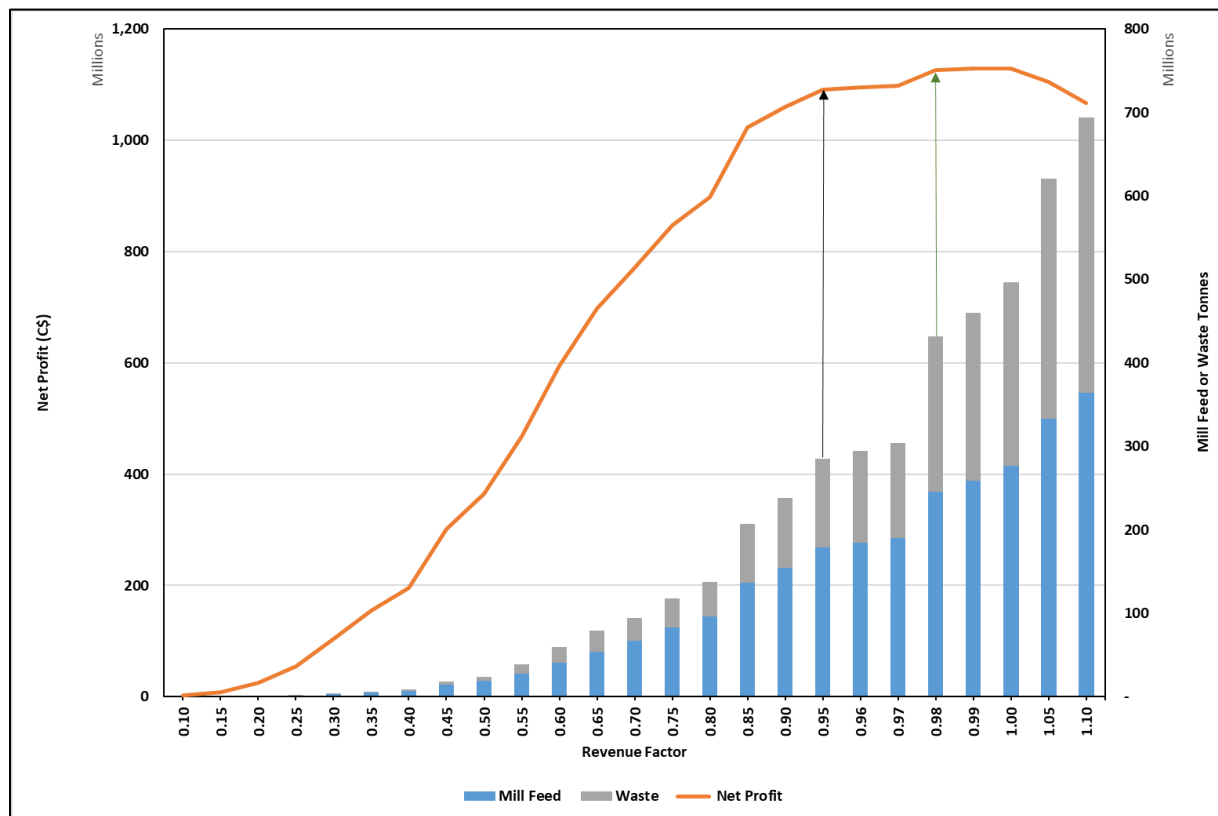


Figure prepared by AGP, 2022

16.6. PIT DESIGNS

Pit designs were developed for the MBX and SS pit areas. The initial phases were designed for the purpose of access and blending material within the revenue factor (RF) equal to the 0.95 pit shell. The final phase, Phase 11, used the RF=0.98 pit shell.

The phase tonnages and grades of the final phases are shown in Table 16-7.

Geotechnical parameters outlined in Table 16-1 and Table 16-2 were applied to pit designs.

Equipment sizing for ramps and working benches is based on the use of 229 t rigid-frame haul trucks. The operating width used for the truck is 8.3 m. This means that double lane widths are 34 m (three times the operating width plus berm and ditch). Ramp gradients are 10% in the pit.

Table 16-7: Final Design – Phases, Tonnages, and Grades

Phase	Mill Feed (Mt)	Au (g/t)	Cu (%)	VLPT (\$/t)	Au Cont. (M Oz)	Cu Cont. (M lbs)	Waste (Mt)	Total (Mt)	Strip Ratio
Phase 5	42.45	0.36	0.20	19.19	0.336	149.96	33.26	75.71	0.78
Phase 6	30.25	0.31	0.22	19.08	0.208	120.76	21.70	52.01	0.72
Phase 7A	15.56	0.72	0.06	24.00	0.265	14.36	17.64	33.10	1.14
Phase 7B	22.43	0.55	0.07	19.29	0.287	27.60	35.00	57.43	1.56
Phase 9	13.80	0.34	0.19	18.18	0.104	46.21	6.80	20.60	0.49
Phase 10	42.03	0.25	0.22	17.41	0.227	170.82	24.50	66.53	0.58
Phase 11	51.75	0.31	0.19	17.72	0.355	180.38	60.72	112.47	1.17
Total	218.18	0.371	0.18	18.77	1.782	710.09	199.72	417.89	0.92

The phase designs are described in further detail in the following subsections.

Phase 5

The Phase 5 design is shown in Figure 16-3. While designated as Phase 5, the mining of this phase starts when Phase 7A is almost complete and ties into the ramp system present from Phase 7A and Phase 7B. This phase bottoms at the 800-m above sea level (masl) bench.

Figure 16-3: Phase 5

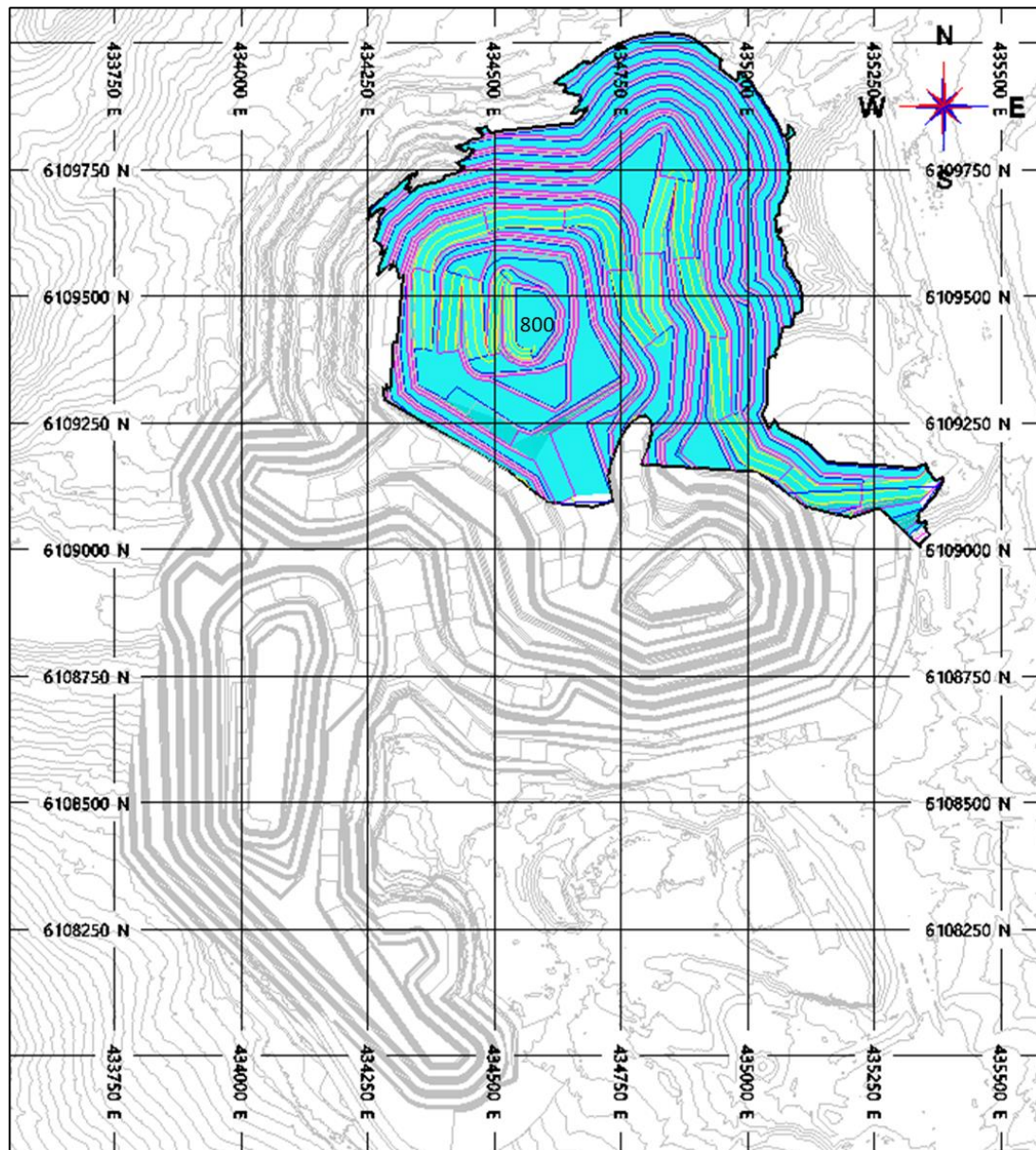


Figure prepared by AGP, 2022

Phase 6

Phase 6, like Phase 5, is concurrent or after Phases 7A and 7B. They both take advantage of the ramp systems at depth developed from those phases. Where Phase 5 mines the NE corner of the pit to the final pit limits, Phase 6 is not quite to the ultimate pit dimension. Access is started on the north side and pushes the pit to the NW and mines to the 860 level. The Phase 6 design is shown in Figure 16-4.

Figure 16-4: Phase 6

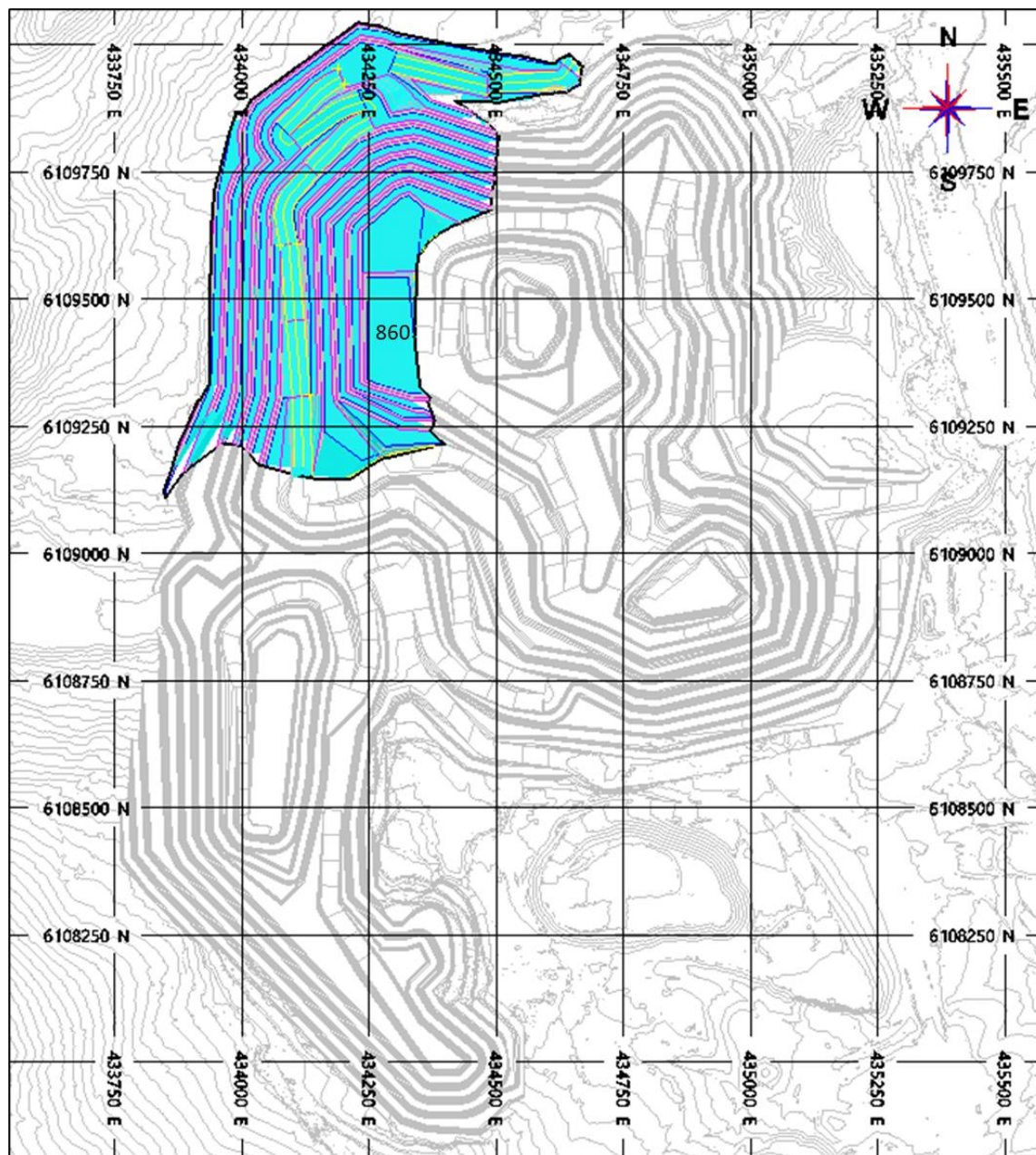


Figure prepared by AGP, 2022

Phase 7A

Phase 7A is already in progress and mines in the HGLC area of the pit. This phase helps tie in some access concerns that exist at present and expected in the future. This phase will bottom at the 920-masl elevation. The Phase 7A design is shown in Figure 16-5.

Figure 16-5: Phase 7A

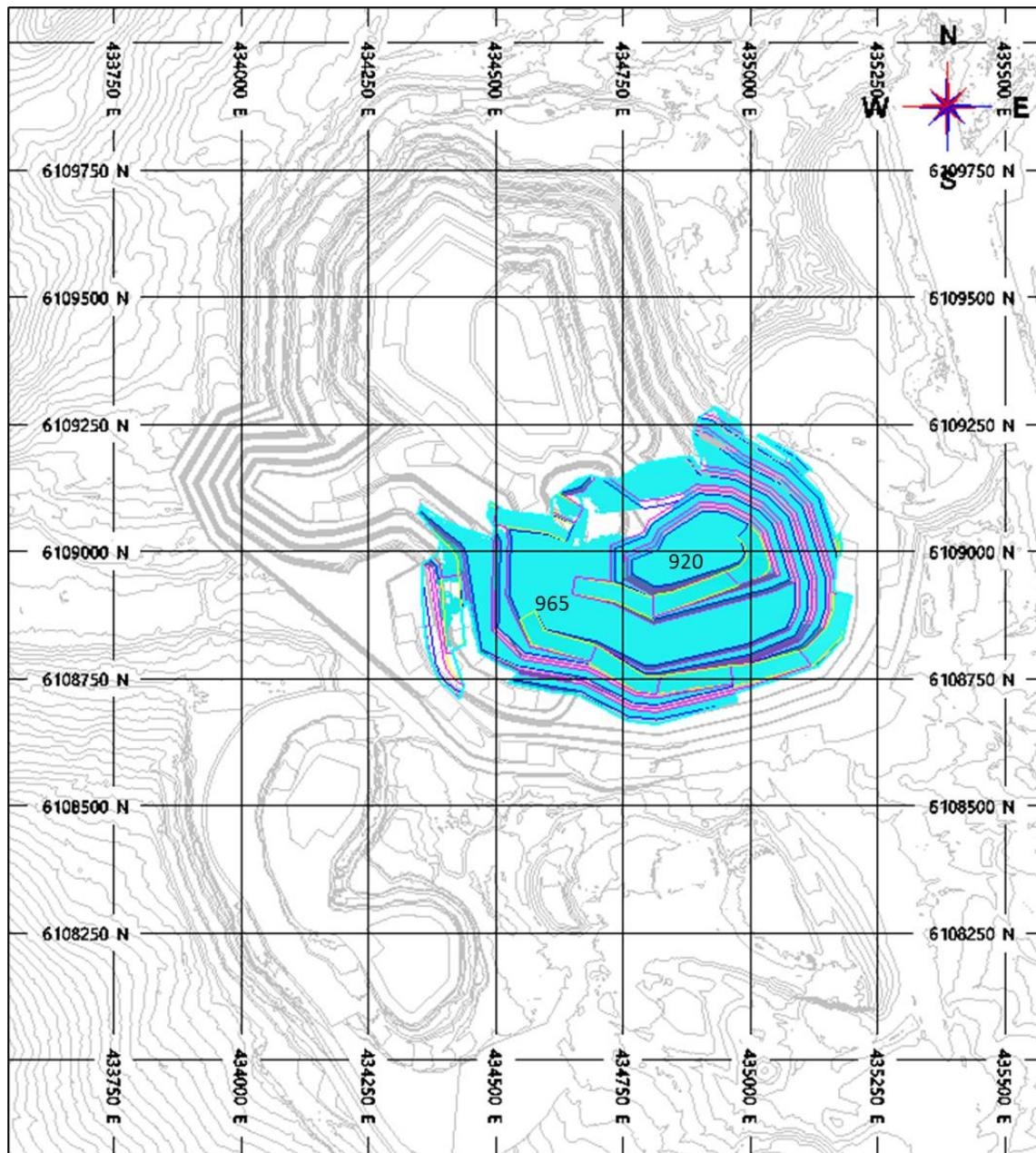


Figure prepared by AGP, 2022

Phase 7B

Phase 7B is designed to sort out access connections with MBX and SS while continuing to provide HGLC feed to the mill. It will tie into Phase 7A to provide continual pit access. This phase will merge with Phase 7A and push the pit bottom downwards from the 920-masl elevation to the 860-masl elevation. When complete, this area will be used for in pit dumping. The Phase 7B design is shown in Figure 16-6.

Figure 16-6: Phase 7B

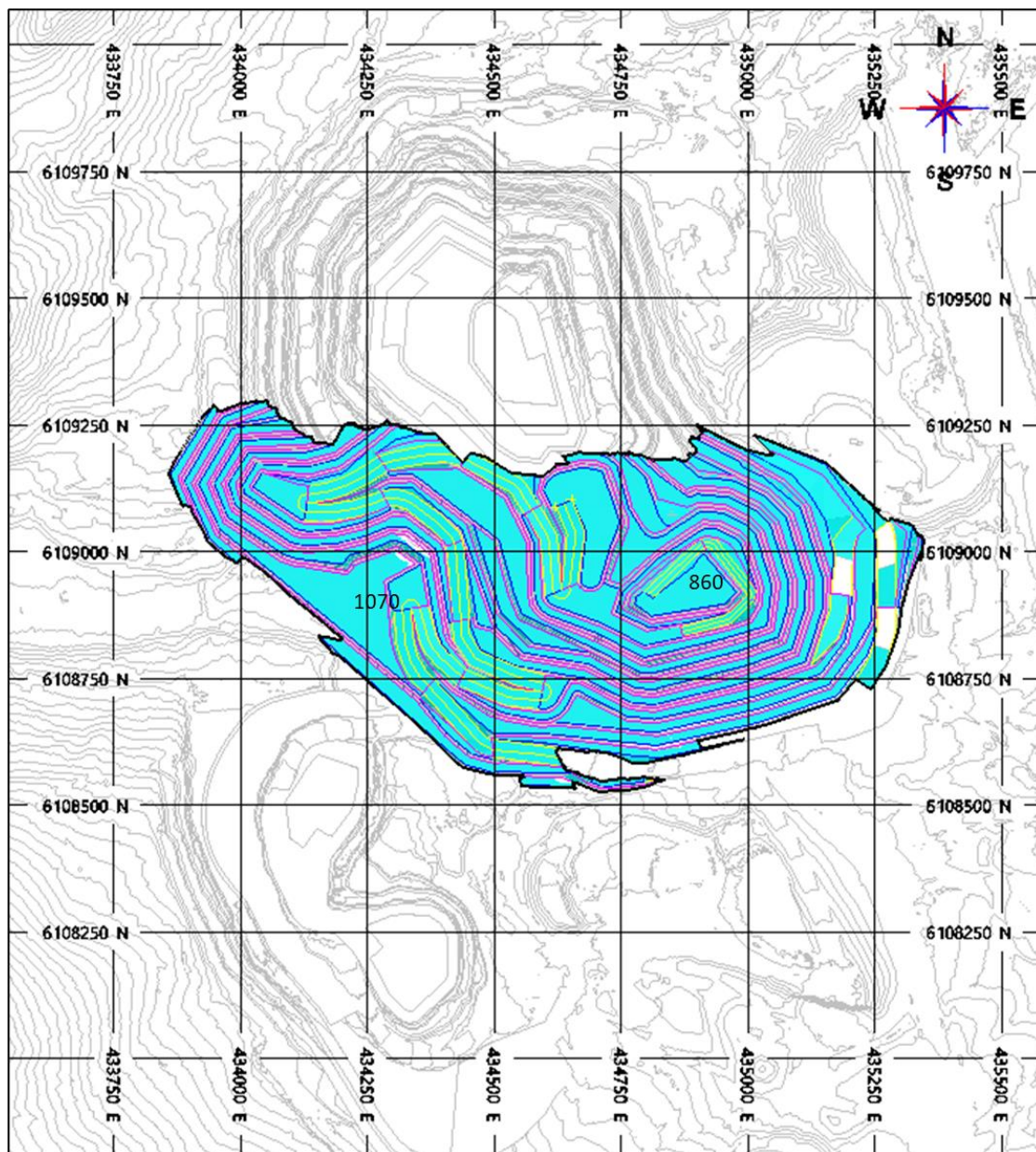


Figure prepared by AGP, 2022

Phase 9

Phase 9 is the phase that pushes the south end of SS to the ultimate limits and also ties in access to the MBX ramp system with the Phase 7A and 7B phases. This phase is intended to mine out the area quickly and provide short haul pit backfill for PAG material. The final depth of the phase is the 1010-masl elevation. The phase 9 design is shown in Figure 16-7.

Figure 16-7: Phase 9

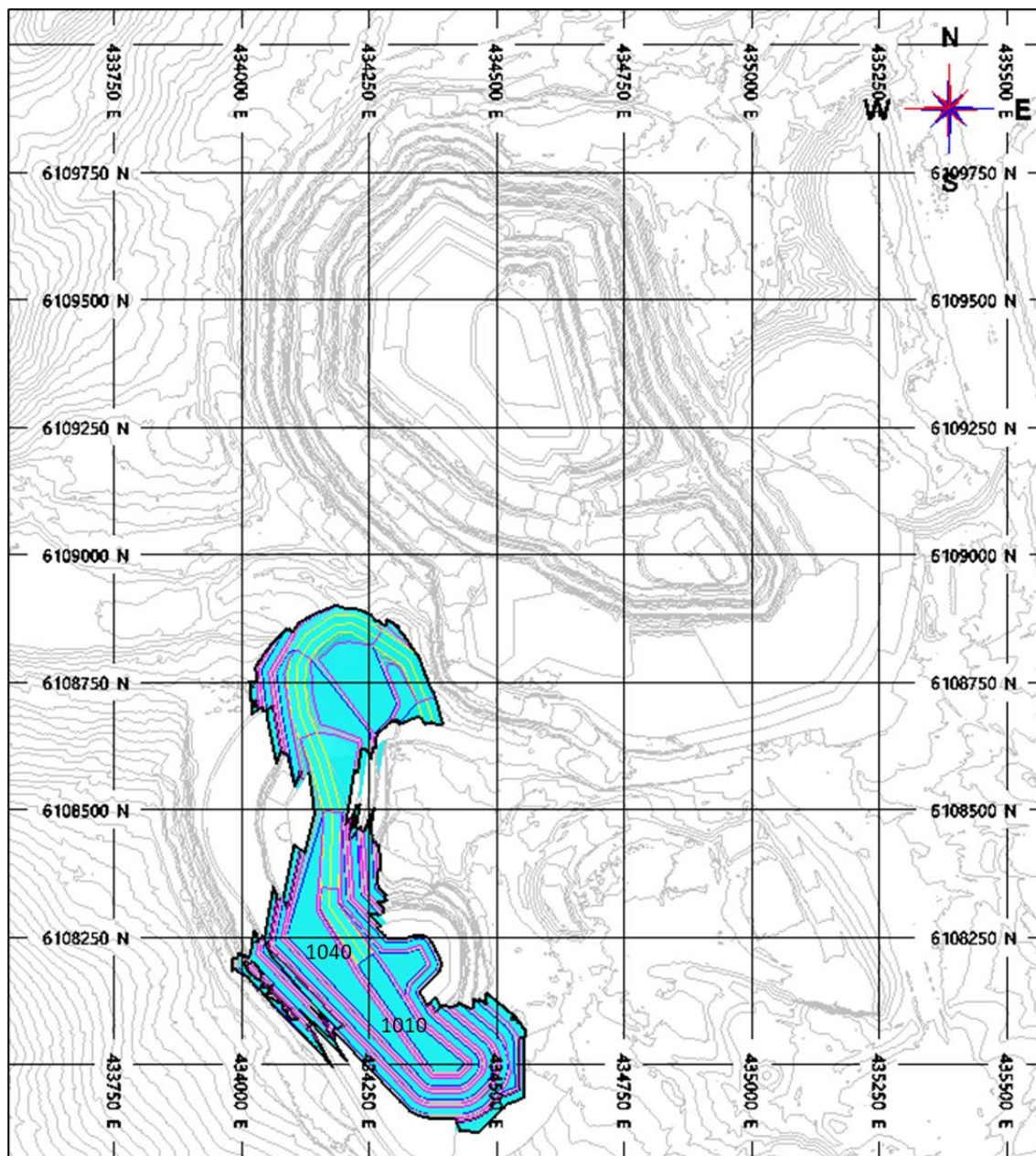


Figure prepared by AGP, 2021

Phase 10

Phase 10 completes the mining of the SS area of the Mount Milligan project. It is mined to a depth of 950 masl and ties into the Phase 7B access. Once complete, this area is also available for in-pit deposition of PAG material. The Phase 10 pit design is shown in Figure 16-8.

Figure 16-8: Phase 10

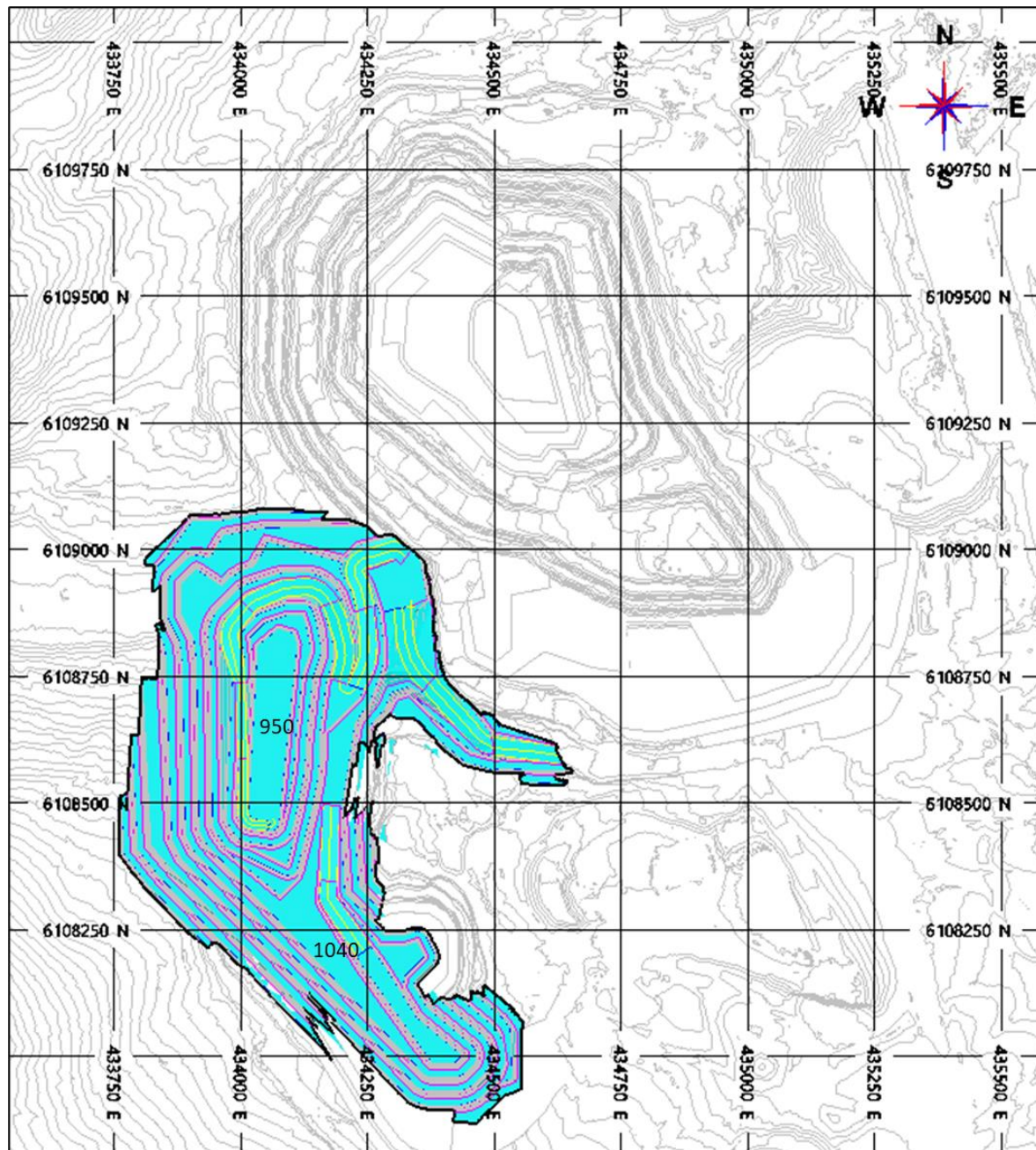


Figure prepared by AGP, 2022

Phase 11

Phase 11 is the final phase of the pit, mining the bottom in the NW corner to the 800 masl level matching the depth of Phase 5 in the center north. This phase makes use of the Phase 6 access along the north end. The main access ramp in this phase helps break up the slope for geotechnical purposes. The Phase 11 design is shown in Figure 16-9.

Figure 16-9: Phase 11

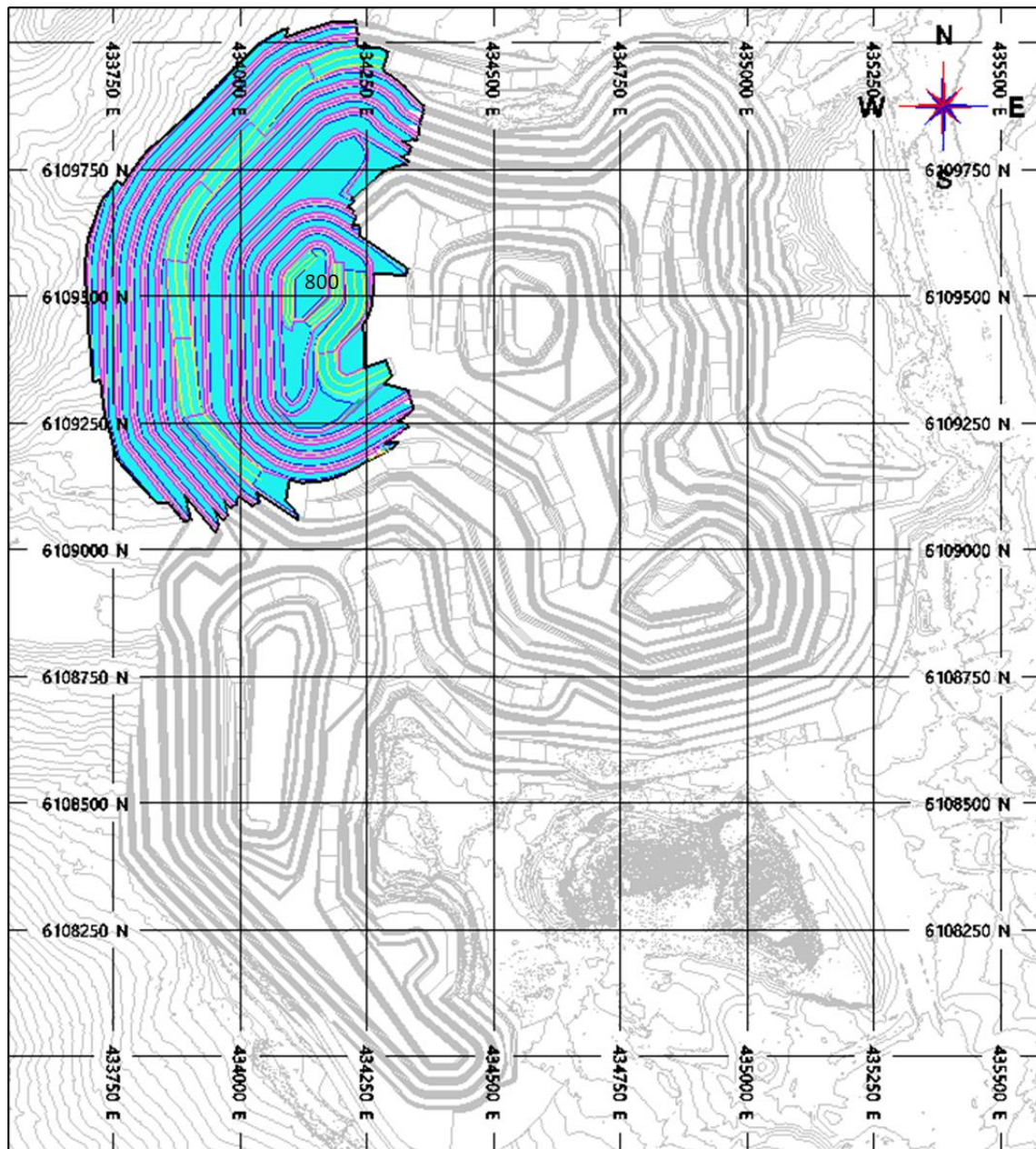


Figure prepared by AGP, 2022

16.7. WASTE ROCK STORAGE

Various rock types are present in the material mined within the final pit. All waste rock is to be stored in the tailings facility or dumped in the pit where and when possible. Waste rock and till from the open pit are stored or used in the construction of the TSF. This includes the Downstream, Upstream, Core and Filter portions of the TSF.

A total of 90 Mt of PAG material will be stored in the TSF. The remaining PAG will be stored in the pit below the 1040 masl. This is to ensure longer term storage beneath a final water level to prevent oxidation.

The waste rock storage plan is discussed further in Section 18.

16.8. MINE SCHEDULE

The 2022 LOM mine schedule considers pit mining from 2023 onward and stockpile reclaim (and includes a forecast stockpile balance at the end of 2022 of 2.7 Mt), delivers 220.9 Mt of mill feed grading 0.37 g/t gold and 0.18 % copper from 2023 – 2033.

The LOM plan is based on a forecast end of 2022 position and goes forward to 2033. The LOM plan mines 218.2 Mt of mill feed grading 0.37 g/t gold and 0.18% copper. Waste tonnage from the phase designs totals 199.7 Mt of which 143.9 Mt is PAG material and 18.4 Mt will be till. The LOM strip ratio is 0.92:1 (waste:ore). A forecast stockpile balance of 2.7 Mt at the end of 2022 is included in the mine schedule in addition to the 218.2 Mt from the pit phases.

The detailed planned mine schedule for 2023 onward is shown in Table 16-8 and Table 16-9. The LOM phase tonnages are shown in Table 16-10. Figure 16-10 shows the tonnage mined by phase by year for the LOM plan. Figure 16-11, Figure 16-12, and Figure 16-13 show the mill feed type and grades and contained metal accordingly over the LOM. Table 16-11 displays a summary of the LOM pit resource classifications for the mill feed that are used in the LOM plan.

The mine schedule assumes a maximum of 21.9 Mt/y of mineralized material will be sent to the process plant as feed.

Various scheduling guidelines were provided for the LOM schedule. In some instances, there was a slight relaxation of the parameters due to phase release variations. These parameters included:

- Feed Pyrite:Chalcopyrite ratio: maximum of 12
- Feed Copper Grade: 0.14% to 0.30 % copper
- Mercury Grade in Concentrate: cannot exceed 14ppm

The pyrite to chalcopyrite ratio in mill feed is expected to be exceeded in 2029 with a ratio of 13, but it will drop to only 6.4 the following year. No issues were encountered with the copper grade.

According to the LOM schedule, total stockpile capacity of approximately 9.9 Mt is reached in 2024. The stockpile balance declines from this point onwards until 2028. In 2029 to 2030 the stockpile again grows to a peak of 7.1 Mt to provide sufficient material to the plant as the pit phases are depleted. The stockpile will be depleted in 2033 at the end of the LOM plan.

Peak mining rate will be reached in 2023 at 54.3 Mt. For the period 2024 to 2028, the annual mining rate averages 48.1 Mt.

Table 16-8: Mine Schedule (mining summary and processed material)

Description		2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	Total
Mining Summary	Common Waste (Mt)	4.51	1.51	2.41	3.02	5.13	0.01	0	0	0	0	0	16.59
	NAG Waste (Mt)	0.32	1.25	0.72	0.4	0.51	0.41	1.86	1.88	1.82	3	0	12.17
	Oxide Waste (Mt)	3.68	1.02	1.97	0.74	1.17	0	0	0	0	0	0	8.58
	PAG Waste (Mt)	16.54	20.3	24.63	14.58	14.43	25.05	15.28	6.99	4.24	1.9	0	143.94
	Till Waste (Mt)	4.49	0.99	0.59	8.7	3.63	0.03	0	0	0	0	0	18.43
	Total Waste (Mt)	29.54	25.07	30.32	27.45	24.86	25.5	17.14	8.87	6.07	4.9	0	199.72
	Mined Ore (Mt)	24.77	26.81	17.06	18.85	22.32	22.28	25.78	22.78	21.67	15.86	0	218.18
	Au (g/t)	0.331	0.315	0.410	0.464	0.351	0.345	0.474	0.348	0.326	0.299	0	0.366
	Cu (%)	0.169	0.185	0.157	0.168	0.196	0.165	0.158	0.188	0.216	0.226	0	0.182
	VLTP	16.96	17.32	19	21.39	19.04	17.29	20.97	18.43	19.14	18.81	0	18.8
	Au cont. (M Oz)	0.18	0.18	0.16	0.20	0.17	0.17	0.28	0.17	0.15	0.10	0.00	2.06
	Cu cont. (M lbs)	74.2	88.8	47.9	57.3	78.7	65	72.6	76.6	84.3	64.7	0	710.1
	Mined Total (Mt)	54.3	51.9	47.4	46.3	47.2	47.8	38.4	32.4	32.6	19.6	0	417.9
Processed Material	Mill Feed (Mt)	21.9	21.9	21.9	21.9	21.9	21.9	21.9	21.9	21.9	21.9	1.9	220.9
	Au (g/t)	0.352	0.319	0.350	0.424	0.349	0.349	0.521	0.353	0.351	0.306	0.2331	0.366
	Cu (%)	0.177	0.203	0.147	0.162	0.191	0.164	0.176	0.183	0.209	0.206	0.1468	0.181
	VLTP	17.98	18.35	16.56	19.73	18.73	17.38	23.43	18.3	19.58	18.02	12.92	18.8
	PYCPY in Plant Feed	6.2	5.8	12	11.7	8.9	12	11.4	7.3	8.1	5.6	12	8.9
	Hg in Plant Feed (ppm)	14	18	18	15.6	6.4	10.2	8.6	8.4	12	10	14	12.1
	Au cont. (M Oz)	0.170	0.154	0.170	0.211	0.170	0.170	0.264	0.170	0.170	0.145	0.006	2.079
	Cu cont. (M lbs)	68.8	80	57	63.7	75.2	63.8	69.4	71.6	82.3	81	3.3	716.1

Table 16-9: Mine Schedule (stockpiles and material movement)

Opening Balance	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
Mass (Mt)	2.71	5.58	10.49	5.65	2.61	3.02	3.40	6.92	5.27	5.01	1.89	0
Cu Grade (%)	0.183	0.146	0.126	0.140	0.162	0.200	0.198	0.150	0.153	0.149	0.228	-
Au Grade (g/t)	0.386	0.274	0.284	0.411	0.689	0.658	0.603	0.594	0.800	0.890	0.642	-
Cu Contained (Mlbs)	10.96	17.92	29.12	17.40	9.29	13.34	14.83	22.87	17.79	16.45	9.48	-
Au Contained (kOz)	33.7	49.1	95.9	74.7	57.8	63.9	65.9	132.1	135.5	143.4	39.0	-

Note: A balance of 2.7 Mt of stockpile material at the end of 2022 is considered in this schedule for tonnages to balance.

Table 16-10: Annual Tonnes Mined by Phase – LOM Plan

Phase	Total Tonnage (Mt)										Total (Mt)
	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	
PHASE 5	-	-	-	8.44	24.17	6.51	21.39	6.54	1.13	7.53	75.7
PHASE 6	-	2.81	-	10.99	14.04	3.42	3.33	16.53	-	0.90	52.0
PHASE 7A	0.10	9.75	16.63	5.55	-	1.08	-	-	-	-	33.1
PHASE 7B	13.65	9.62	5.81	14.89	0.96	-	9.93	2.58	-	-	57.4
PHASE 9	20.61	-	-	-	-	-	-	-	-	-	20.6
PHASE 10	19.95	29.70	10.45	6.43	-	-	-	-	-	-	66.5
PHASE 11	-	-	14.49	-	8.00	36.77	3.76	6.77	31.51	11.18	112.5
Total	54.31	51.88	47.38	46.30	47.18	47.78	38.41	32.42	32.64	19.61	417.9

Figure 16-10: Annual Tonnes Mined by Phase

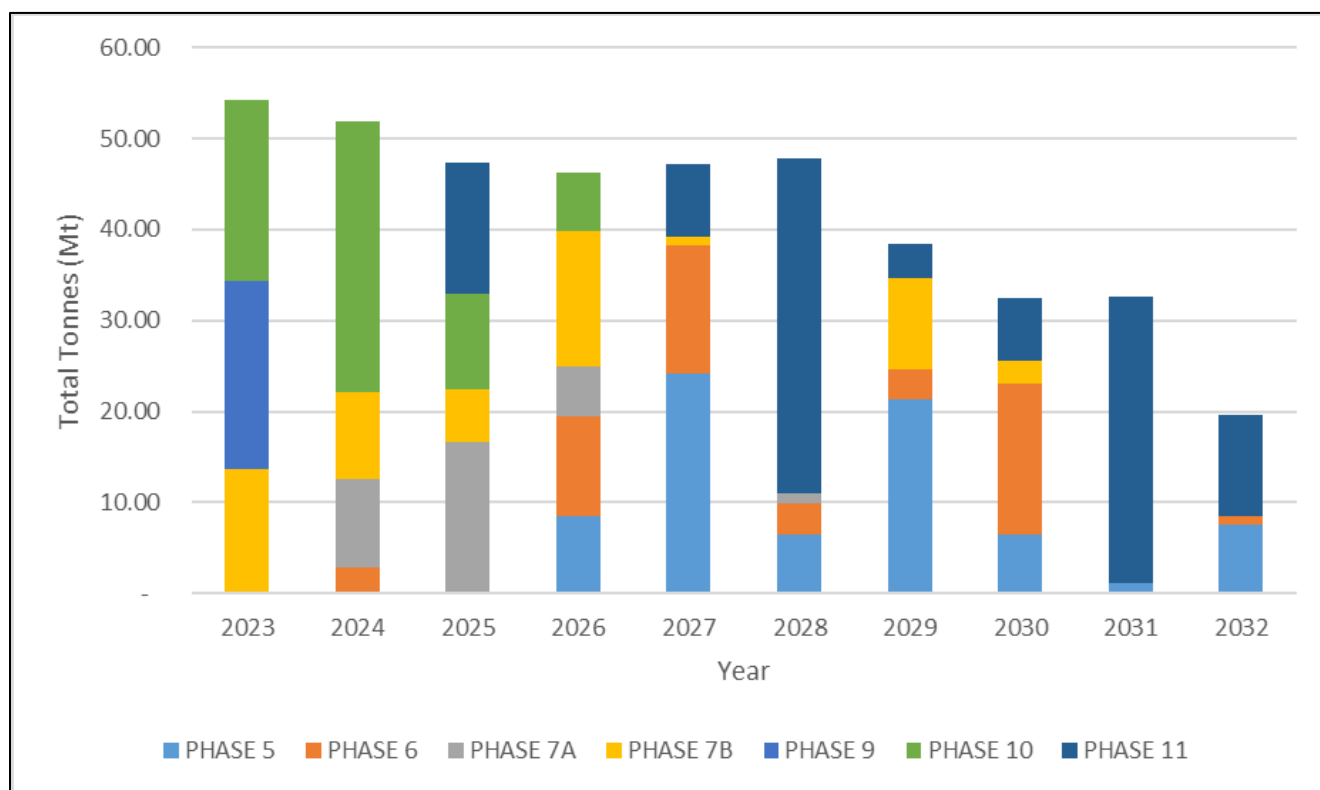


Figure prepared by AGP, 2022

Figure 16-11: LOM Mill Feed Tonnes and Grade

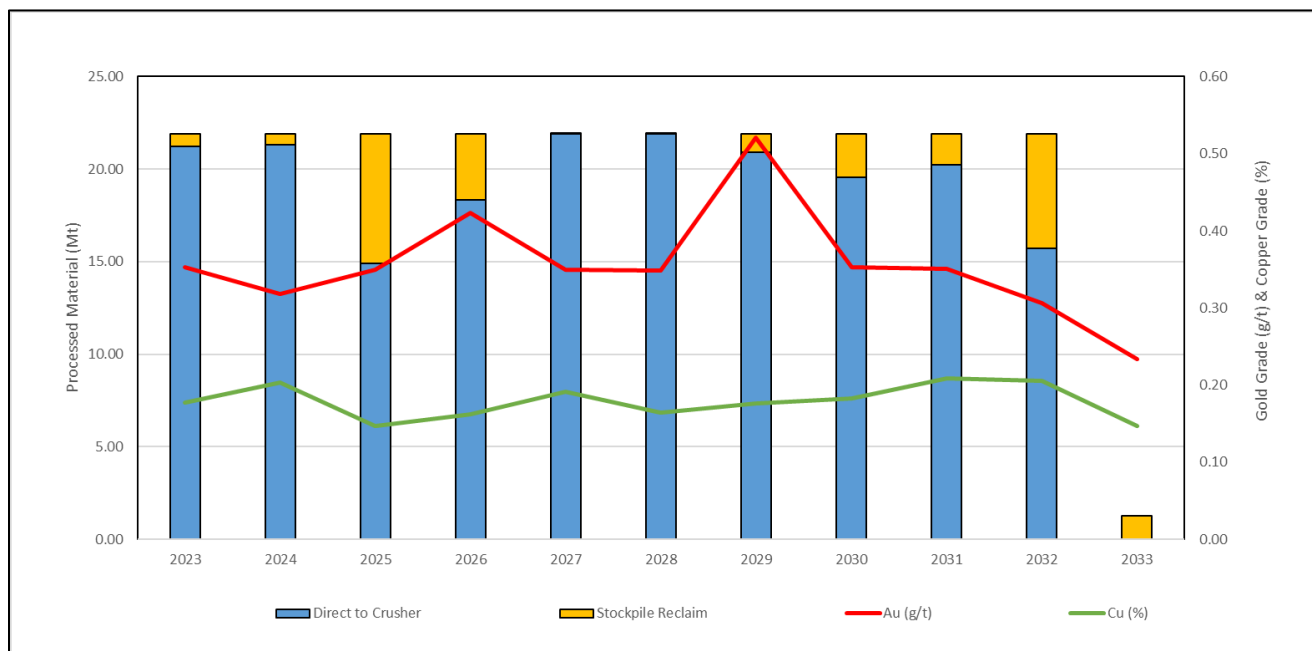


Figure prepared by AGP, 2022

Figure 16-12: Gold - Process Feed Grade and Contained Ounces

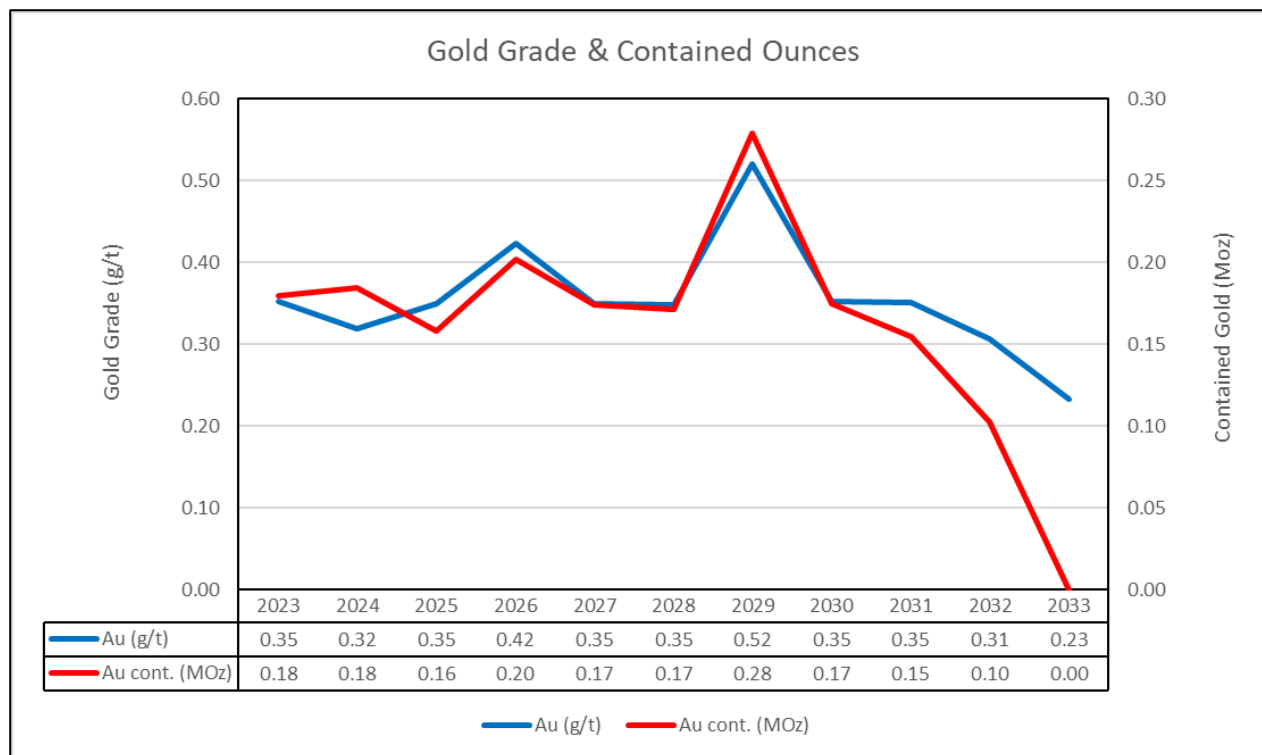


Figure prepared by AGP, 2022

Figure 16-13: Copper - Process Grade and Contained Pounds

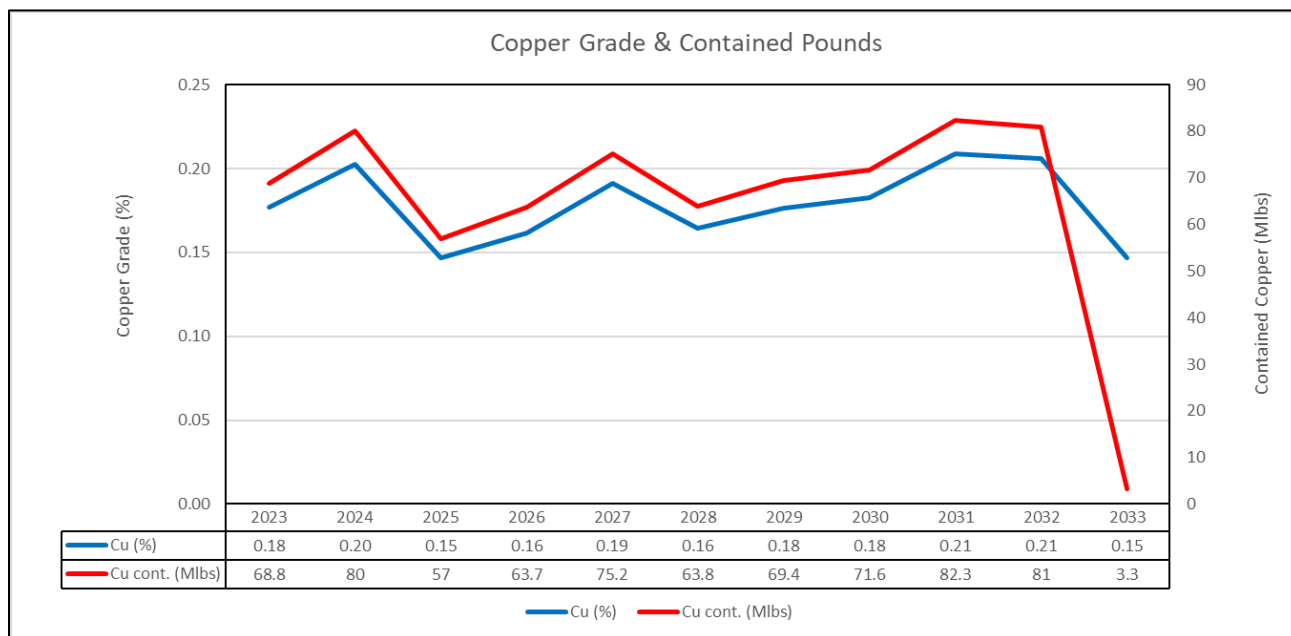


Figure prepared by AGP, 2022

Table 16-11: Reserve Summary of Scheduled Pit Phase Material (Year 2023+)

Reserve Class	Mill Feed (Mt)	Grade		Contained Metal	
		Au (g/t)	Cu (%)	Au (M oz)	Cu (M lbs)
Proven	62.9	0.37	0.20	0.5	226.6
Probable	155.3	0.37	0.17	1.3	483.5
Total	218.2	0.37	0.18	1.8	710.1

The mine schedule is described in detail below.

In Year 2023 the mining of Phase 9 is a priority to allow access to early in pit dumping space for PAG material. The PAG material would then be backfilled in the south end of the pit rather than in the TSF. Phase 7A and 7B have high contents of HGLC material but are mined in this period to allow access to the lower levels of the pit. Eight per cent of mined ore from the two phases is delivered direct to the mill while the remainder is stockpiled and will be blended into the feed in later years. Consequently, total mined tonnages are higher in 2023 and 2024 than later years. Phases 9 and 10 provide mineralized material with better grades and most of the till material required for TSF construction. The extra till is stockpiled for later use. Phase 9 is scheduled to be completed in 2023.

For 2024, mining continues in Phase 10 and for the next three years to provide the next in-pit dumping area as well as good grade material. Mining of Phases 7A and 7B continue in the lower levels with direct feed HGLC and some till material for the TSF. Phase 6 mining is initiated in 2024.

In Year 2025, Phase 10 continues and is the main source of feed for the mill. Phases 7A and 7B continue to advance with the tonnage from 7A starting to reduce. Phase 6 is idle, but Phase 11 is initiated and is the source of till for the TSF lifts in 2025.

At the end of 2026, Phase 10 is complete, which provides an area for in-pit dumping for the next three years. Phase 5 mining is initiated and is the source of till for the TSF this year. Phase 6 mining is restarted and provides ore to the mill while Phase 11 is idled. Phases 7A and 7B are mined providing blending material for the mill. Phase 7A is almost complete at the end of 2026.

In 2027, Phases 5 and 6 are the main source of mill feed. Phase 5 is also the main source of till material for the TSF. Phase 7B is the source of blending material of the HGLC material to manage the grade of mill feed. Phase 11 starts to provide ore tonnage to the mill with its restart.

At the end of 2028, Phase 7A is complete and 7B is idled. Phase 7A will be mined slowly due to a constraint on HGLC stockpile capacity. Phase 5 is the main source of till material, while Phase 6 provides ore to the mill. Phase 11 is the main source of feed to the mill.

In 2029, Phase 5 is the main source of material for the mill. Phases 11 and 6 will provide access and good material for blending. Phase 7B is mined to provide access and a source of HGLC material for blending.

In 2030 Phase 7B is completed and this area can now be used for in pit dumping. Phases 5 and 6 are the main ore and till sources for mining. Phase 11 is mined together with Phases 5 and 6.

In Year 2031, the ramps from Phase 7B provide access for Phase 5 and 6 on the lower levels. Phase 11 is the main source of mill feed.

For 2032, Phase 5 mining continues, providing access and some till. At this time the till stockpile has enough material stored to complete the TSF for the LOM plan. Phase 6 is completed, and Phase 11 is the main source of ore.

The mine plan is complete in 2033 with the completion of the mine phases and processing continues until depletion of the ore stockpiles.

16.9. MINING EQUIPMENT

The mining equipment currently in use is expected to be sufficient to meet the required LOM production schedule.

The drill fleet comprises two electric rotary blast hole drills (311 mm), one diesel blast hole drill (222 mm), and a smaller diesel blast hole drill (152 - 222 mm), preparing blast holes on 15-m benches. The blast holes are designed as 15-m holes with an additional 1.25 to 1.50 m of subdrill.

The blast patterns include production, buffer, and pre-shear holes. Current blasting practice uses a 70/30 emulsion/ANFO blend in the blast holes with a design 0.33 kg/t powder factor.

Loading is completed with two Cat 7495 41 m³ electric rope shovels, one Komatsu PC4000 22 m³ hydraulic excavator and two Cat 994 19 m³ front end loaders. The fleet is considered sufficient to meet the LOM plan loading needs.

The haulage fleet comprises thirteen Cat 793F haul trucks (229-t), two Cat 789C (181-t) trucks, and six Cat 740B articulated dump trucks. The articulated trucks are primarily employed for dam construction and project activities.

A typical fleet of support and ancillary equipment is present including six Cat D10T dozers, one Cat D8T and a Cat D6. Two Cat 834 rubber-tire dozers assist with pit and road maintenance functions. The grader fleet comprises two Cat 24, two Cat 16M, and one Cat 14M graders. Watering of the roads is provided by two Cat 777F water trucks and one Cat 740B water truck.

Additional support equipment, including lube and fuel trucks, compactors, and service trucks, are present at the mine.

16.10. GRADE CONTROL

Grade control is completed using blast hole drill cuttings. Every second blast hole is sampled and is assigned a unique sample number that relates to the blast hole ID. Samples are assayed for copper, gold, silver, sulphur, and iron (Cu, Au, Ag, S, and Fe). Final as-built data for blast hole locations are imported into the blast hole database from the fleet management system. Blast hole sample numbers are then entered into the blast hole database and assays are imported once received from the assay laboratory. Assays and as-built data are exported from the blast hole database and imported into Hexagon MinePlan software where the grades are interpolated into the ore control block model.

For waste material delineation, ABA data is derived from composites generated from six grouped adjacent blast hole samples. Composites IDs are flagged and submitted to the lab where pulps from each blast hole within the composites are combined and assayed for sulphur using S-Leco and Neutralization Potential (NP) methods with the Modified Sobek Method which uses sulphide sulphur. In addition to ABA data composites there are also assays done for mercury (Hg). ABA and Hg data provided by the lab is imported into the blast hole database, exported, and uploaded to Hexagon MinePlan software where this data is interpolated to the ore control block model.

The Ore Control Geologist uses the updated interpolated ore control block model to delineate ore control polygons for Ore and Waste within the Hexagon MinePlan software. Polygons are delineated to set operational inputs for grade (Cu, Au, Hg) as well as ABA data. Ore and Waste material types are assigned to each polygon and these polygons are sent to the fleet management system where they are outlined and tracked within the fleet management system. Polygons are also staked and flagged in the pit where the material type changes to allow for field visuals of grade control.

For selected ore blocks, blast movement of ore control polygons are tracked using BMT Blast Movement Monitors. The movement is estimated using BMT software and this movement assigned to the polygons with the polygons shifted appropriately to match the information. These are loaded into the fleet management system and staked in the pit at material contacts.

16.11. END OF PERIOD PLANS

Images of the estimated end of period positions of the pit phases are shown in Figure 16-14 to Figure 16-23.

Figure 16-14: End of Year 2022 – LOM Starting Position

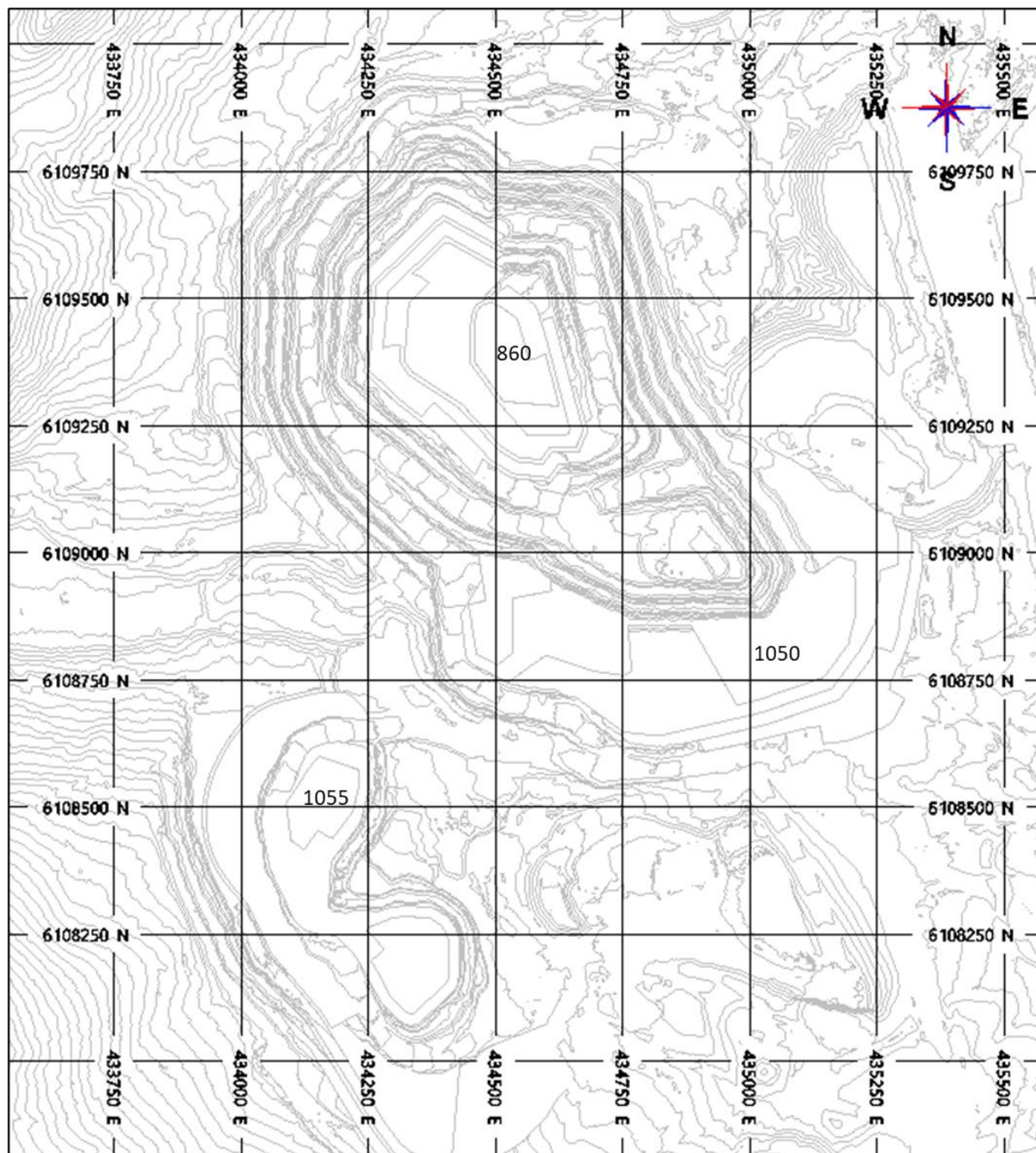


Figure prepared by AGP, 2022

Figure 16-15: End of Year 2023

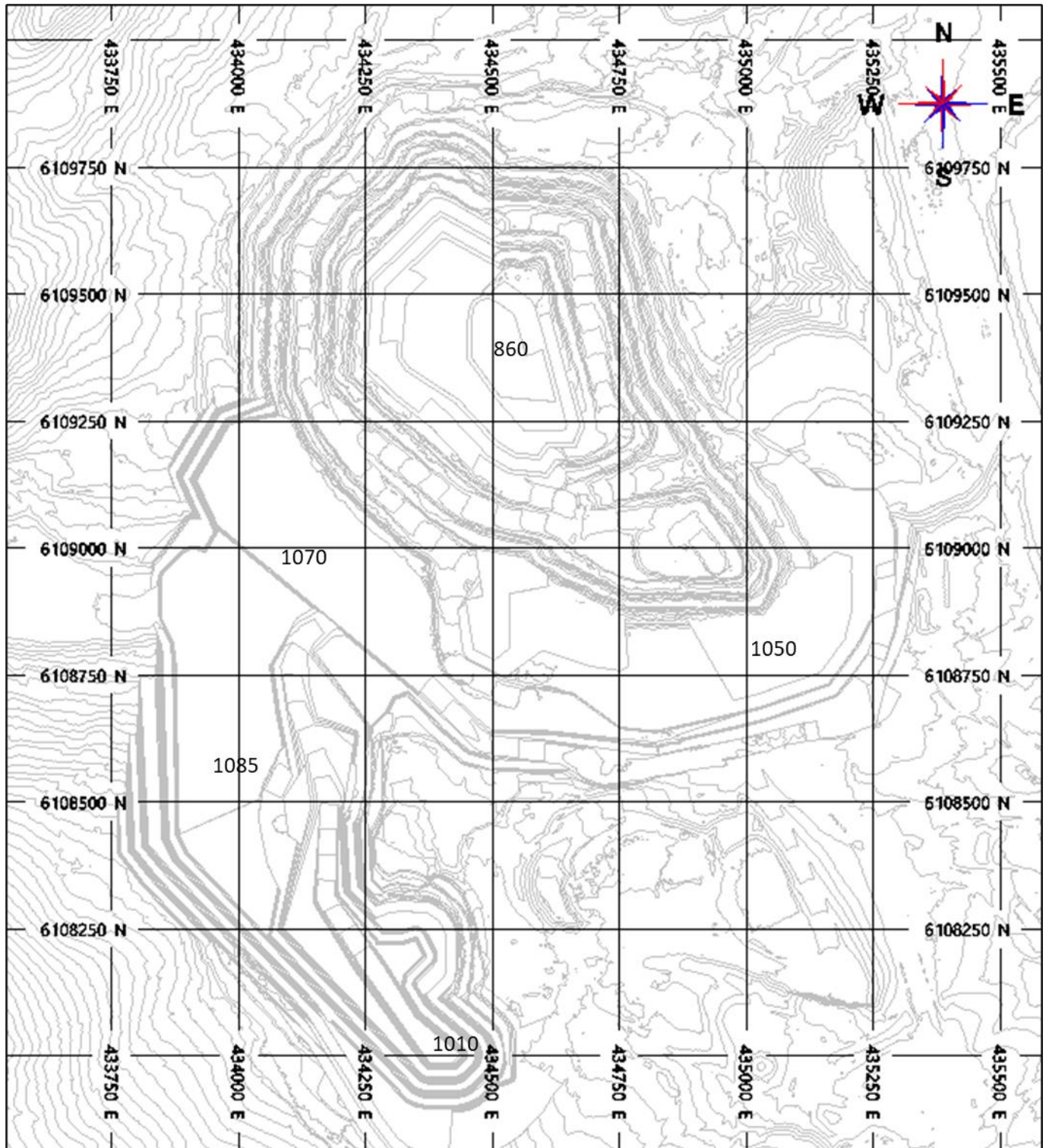


Figure prepared by AGP, 2022

Figure 16-16: End of Year 2024

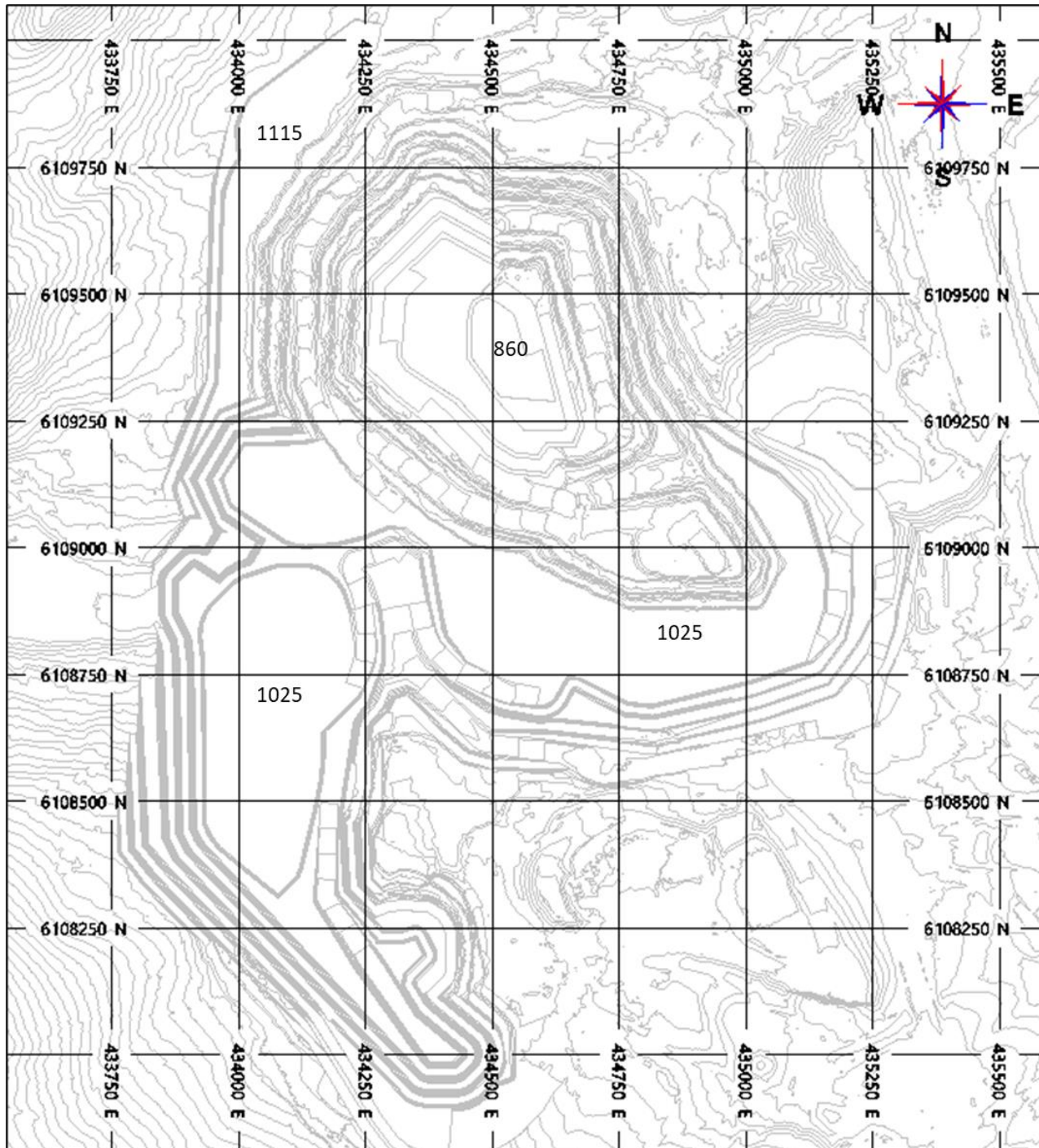


Figure prepared by AGP, 2022

Figure 16-17: End of Year 2025

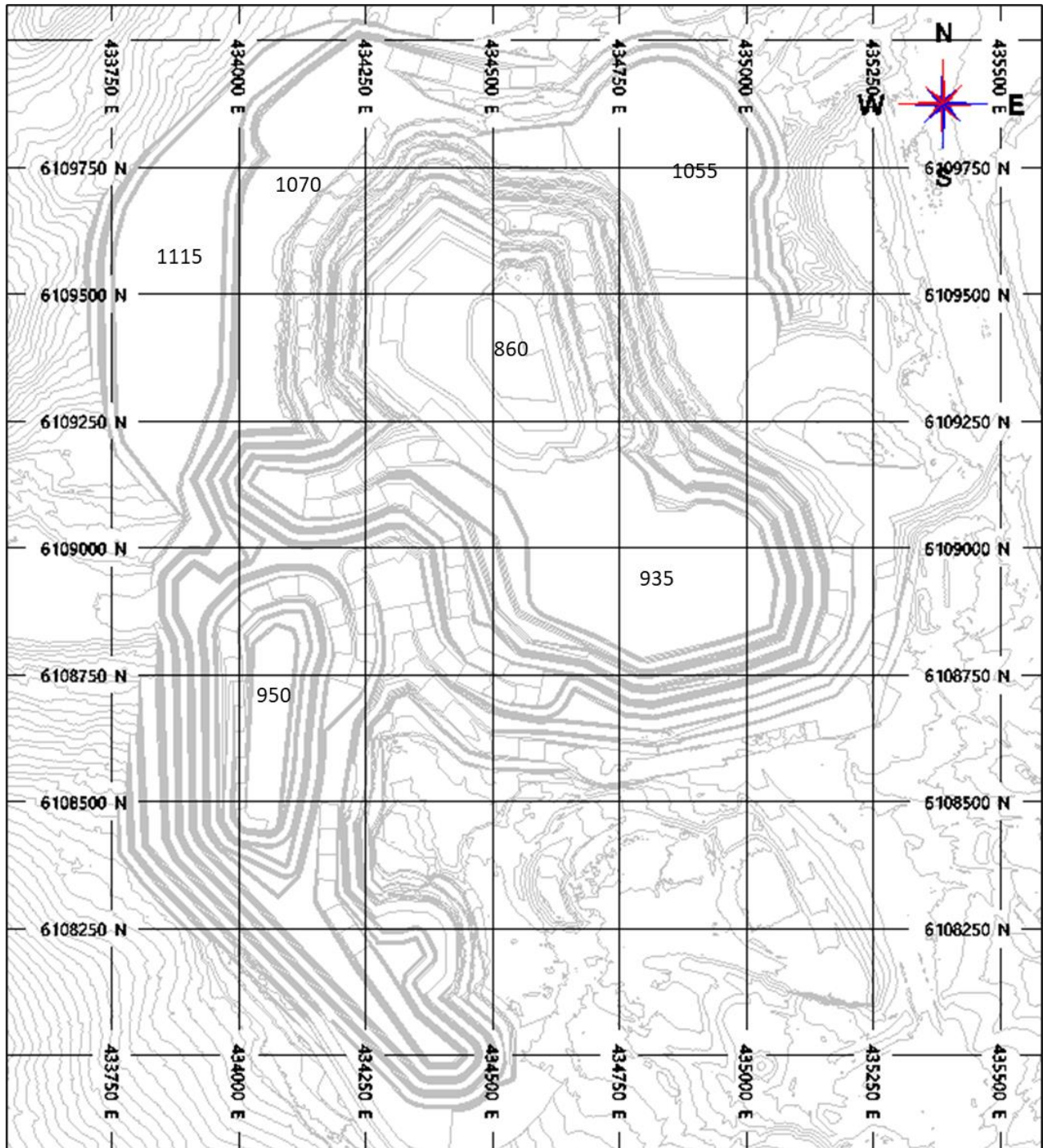


Figure prepared by AGP, 2022

Figure 16-18: End of Year 2026

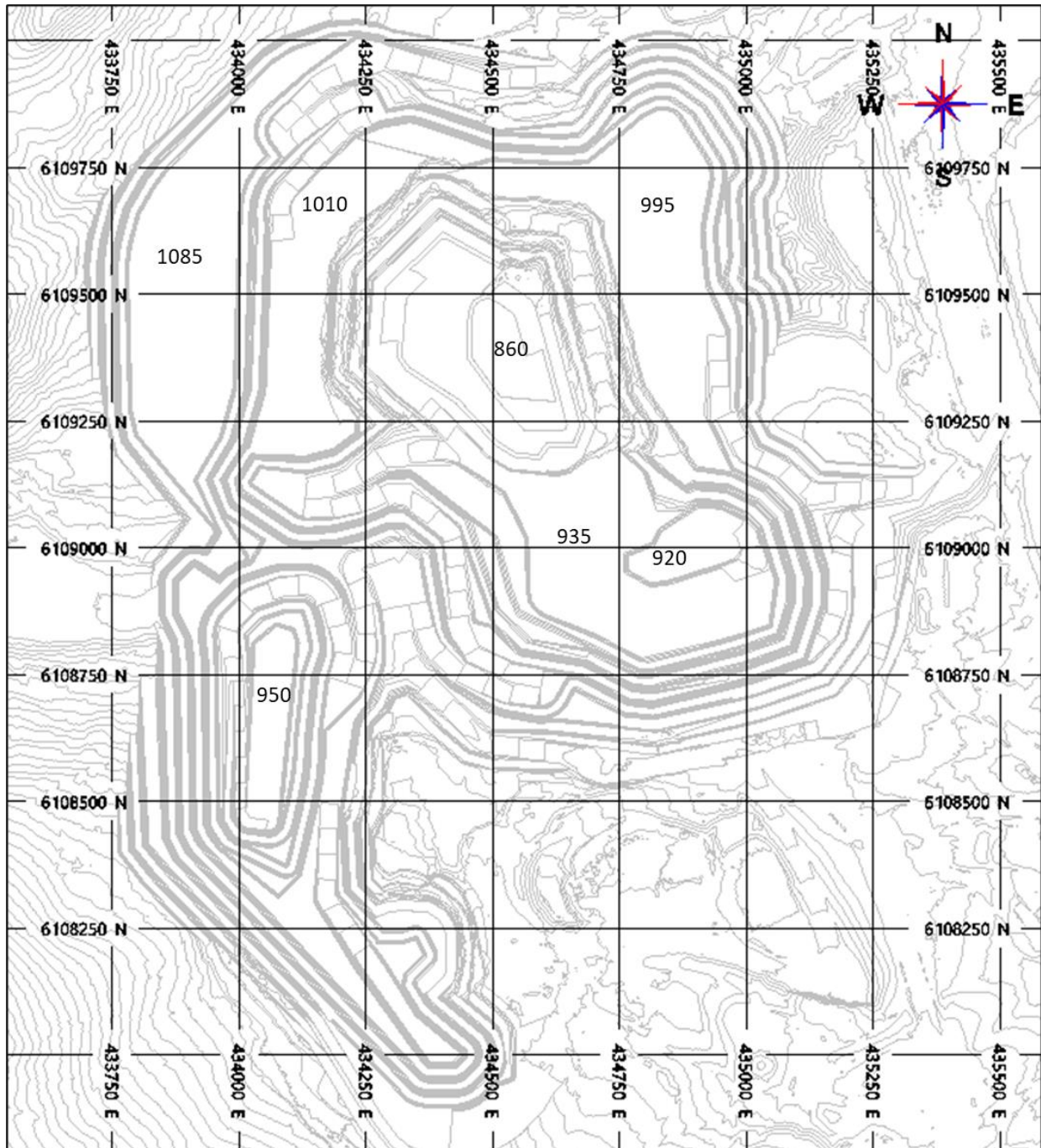


Figure prepared by AGP, 2022

Figure 16-19: End of Year 2027

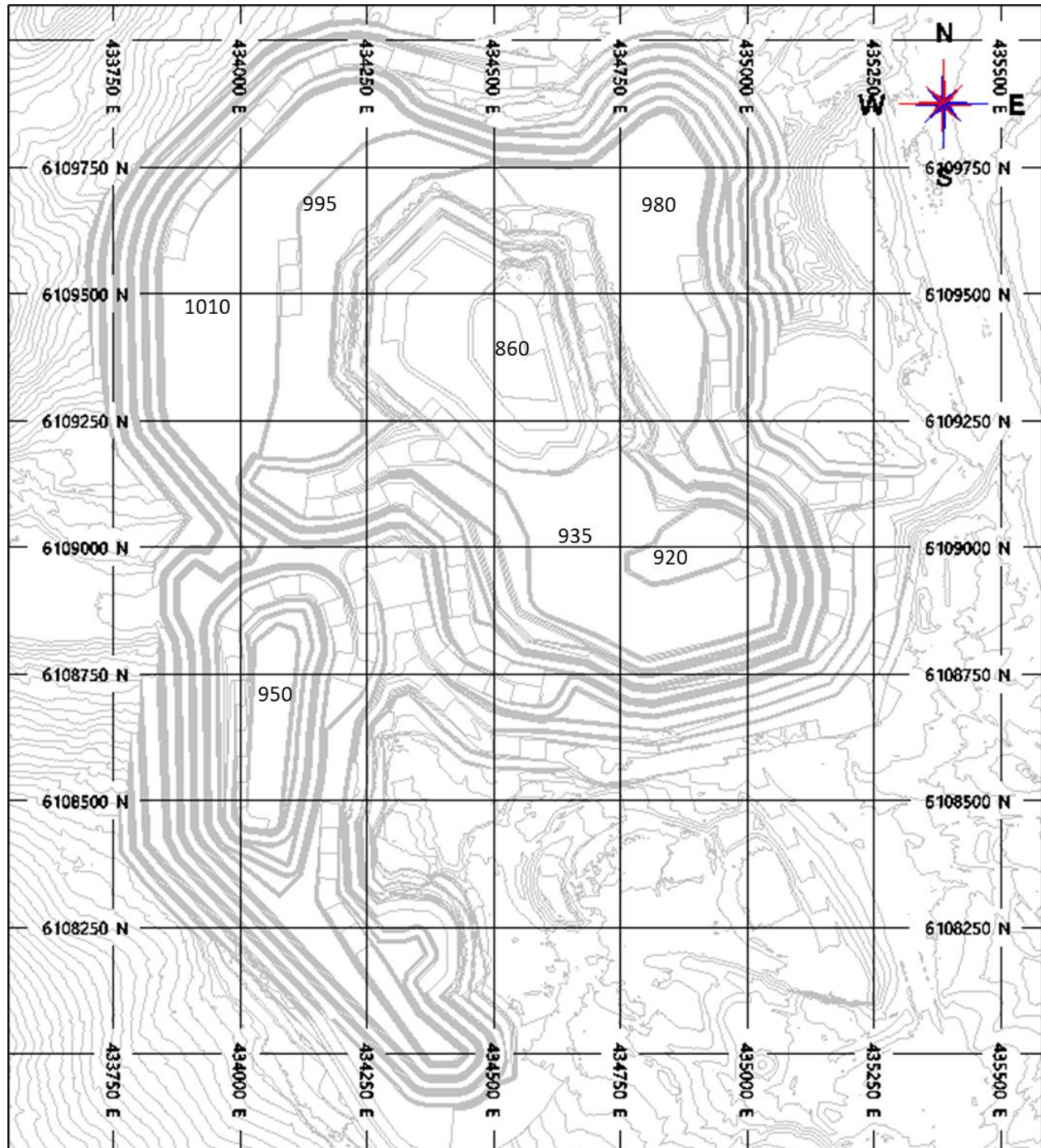


Figure prepared by AGP, 2022

Figure 16-20: End of Year 2028

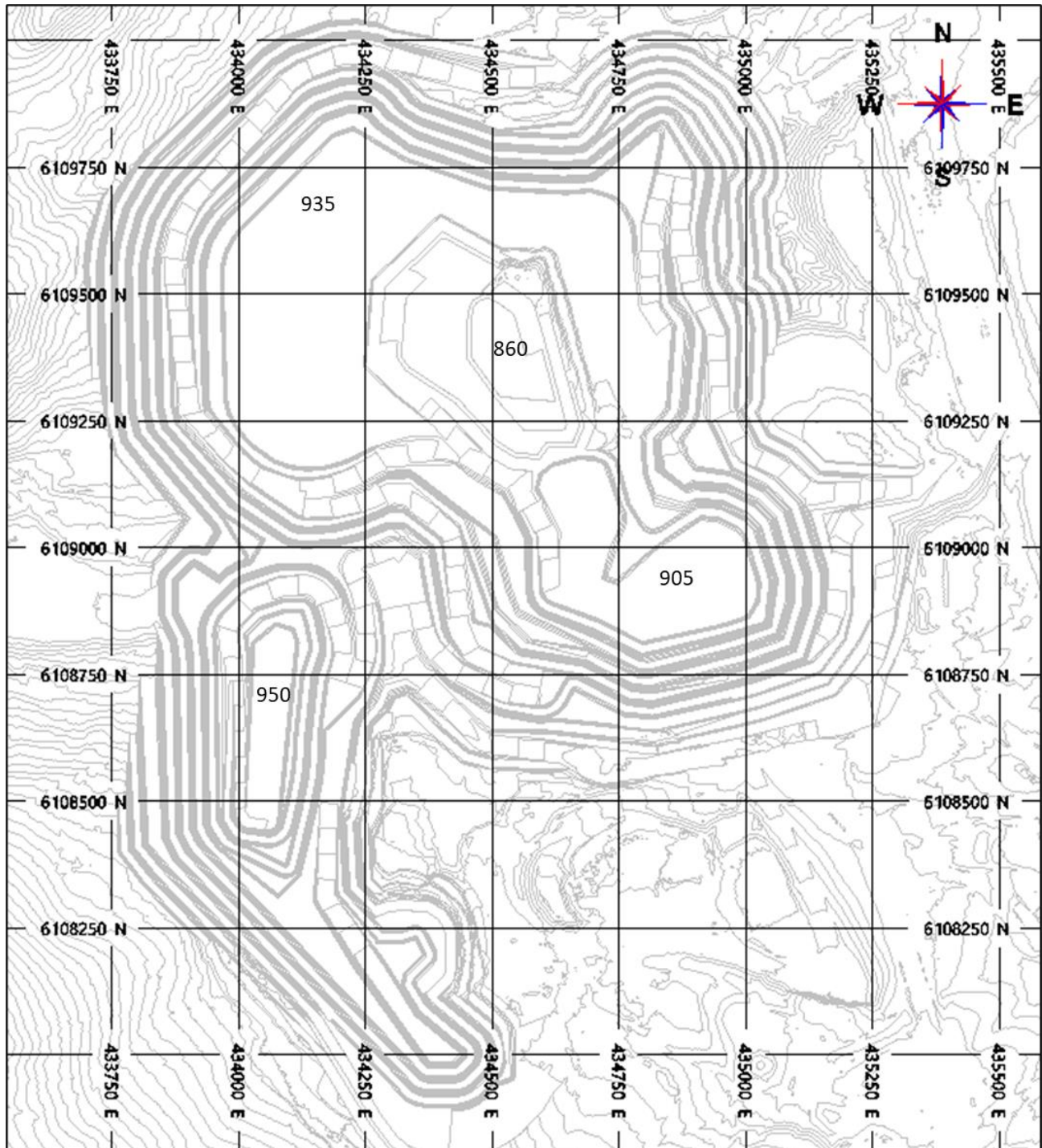


Figure prepared by AGP, 2022

Figure 16-21: End of Year 2029

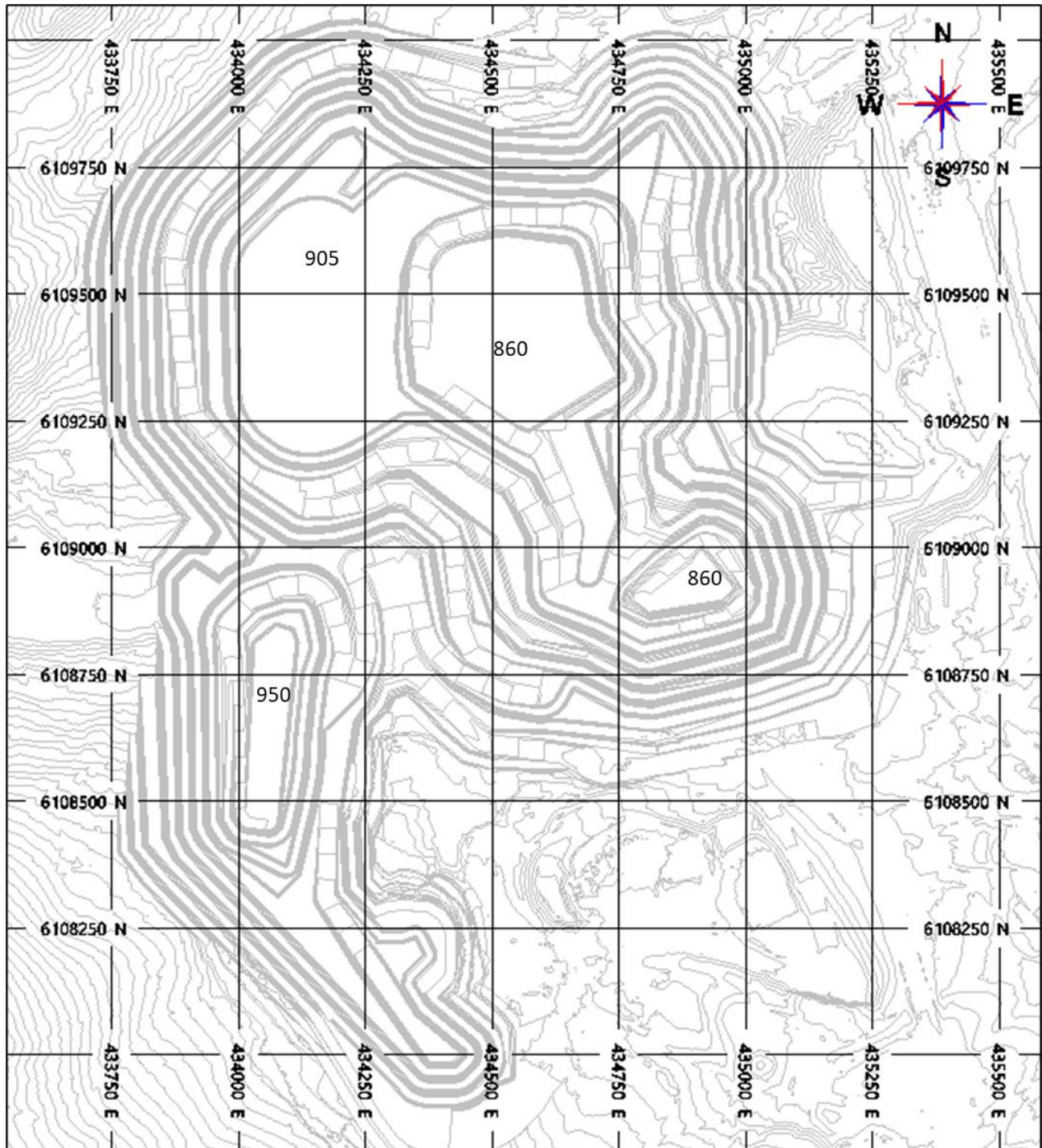


Figure prepared by AGP, 2022

Figure 16-22: End of Year 2030

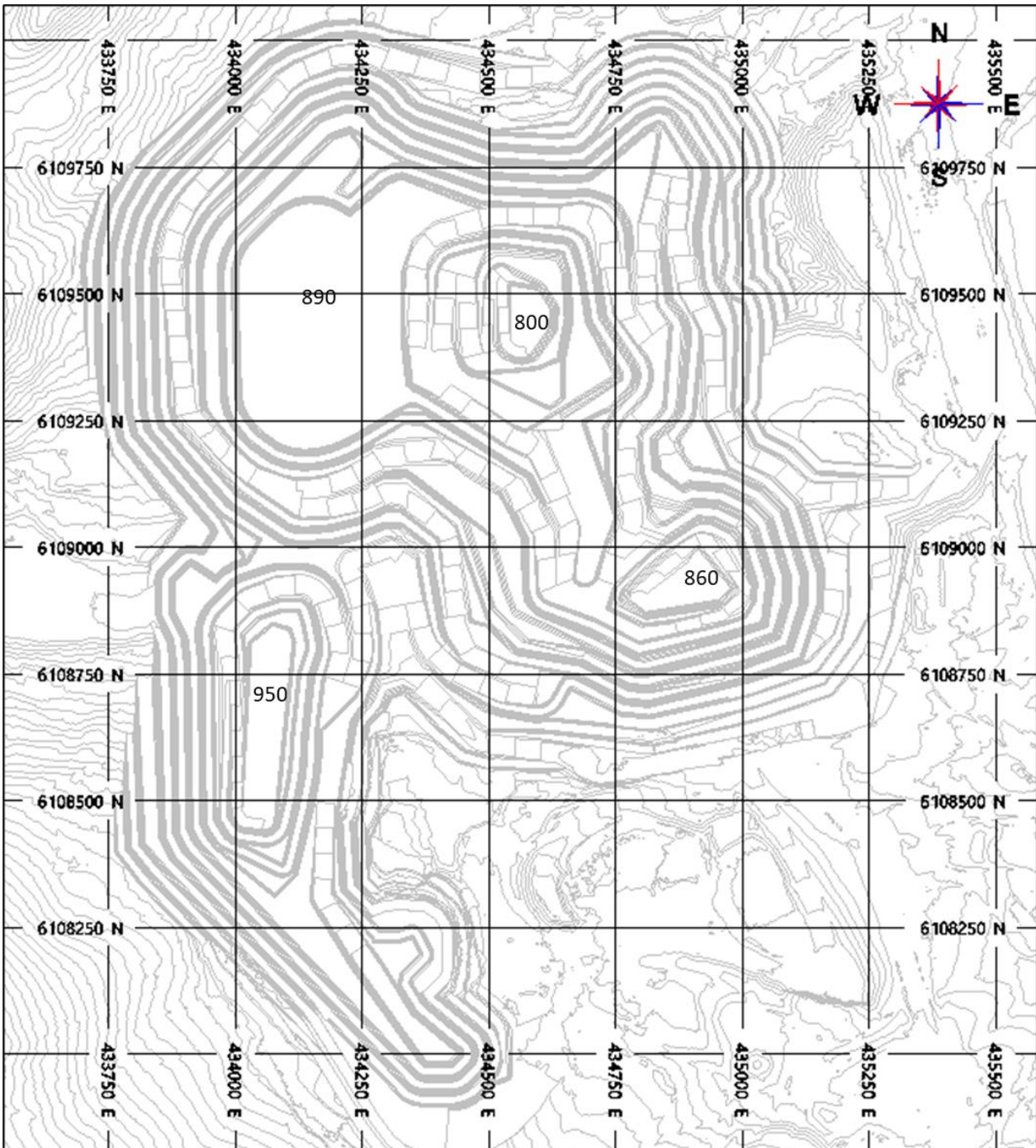
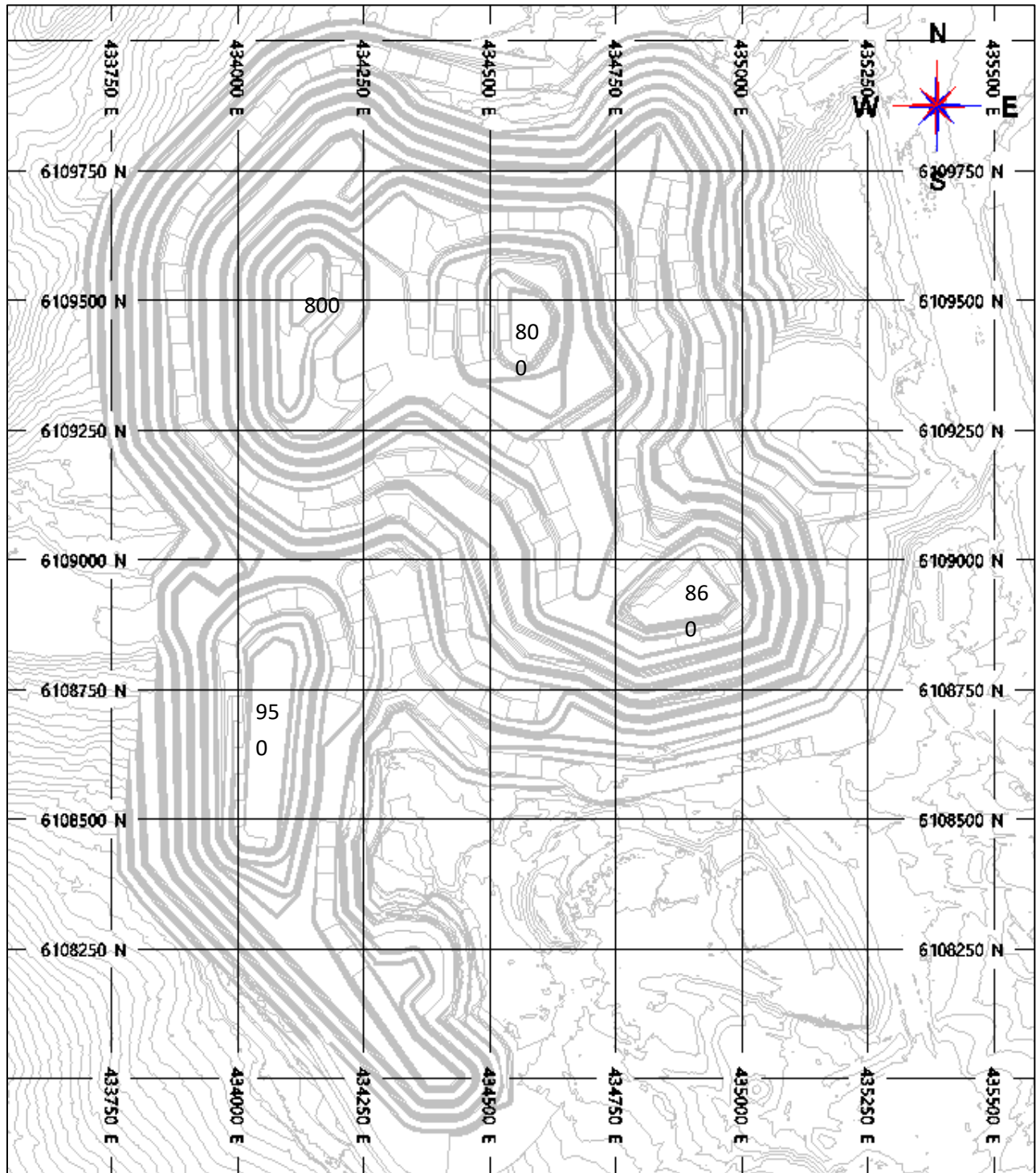


Figure prepared by AGP, 2022

Figure 16-23: End of Year 2031



Note: Figure prepared by AGP, 2022.

17. RECOVERY METHODS

17.1. SUMMARY

The Mount Milligan process plant was originally designed to process ore at a nominal rate of 60,000 tpd, producing a marketable concentrate containing copper, gold, and silver. A secondary crushing circuit, installed in 2016, together with process plant optimization projects, increased the capacity to a nominal rate of 62,500 tpd. Key process equipment includes:

- Primary crushing plant with a 1.524 m x 2.794 m (60 inches x 110 inches) gyratory crusher, powered by one 932 kW (1,250 Hp) motor
- Secondary crushing plant with two standard cone crushers prior to the grinding circuit, each powered by one 932 kW (1,250 Hp) motor
- SAG/ball mill/pebble crusher grinding circuit:
 - One 12.19 m x 6.4 m SAG mill with one 23.5 megawatt (MW) gearless motor drive
 - Two ball mills in parallel, each driven by two 6.5 MW (8,718 Hp) variable speed synchronous motors (26 MW total installed ball mill power), operated in closed circuit with two clusters of 10 x 33" hydrocyclones
 - Two short head cone crushers, each powered by one 746 kW (1,000 hp) motor
- Flotation circuits:
 - Rougher-scavenger flotation: two parallel trains of five 200 m³ tank cells
 - First cleaner flotation: two 100 m³ tank cells
 - Second cleaner flotation: four 30 m³ tank cells
 - Third cleaner flotation: two 30 m³ tank cells one 4 m x 12 m flotation column equipped with Cavitation Tube spargers
 - First cleaner scavenger flotation: five 100 m³
- Regrinding and gravity concentration circuits:
 - One 1,119 kW (1,500 Hp) vertical stirred mill (Tower Mill) operated in closed circuit with six 508 mm (20 inches) and two 381 mm (15 inches) hydrocyclones
 - Two 3,000 kW (4,020 Hp) horizontal stirred mills operated in closed circuit with twenty-six 254 mm (10 inches) hydrocyclones
 - One centrifugal gravity gold concentrator
- Dewatering and Filtration:
 - One 12.2 m diameter high-rate thickener
 - One concentrate pressure filter

ROM ore is crushed to a top size of 150 mm. Up to 100% of the gyratory crusher product is diverted to the secondary crushing circuit by a moveable gate. Prior to being introduced into the SAG and ball mill circuits, a variable percentage of the ore, depending on hardness, is sent to the secondary crushing circuit targeting a size of 80% passing (P_{80}) 60 mm, or finer, product. Oversize from the SAG discharge screens is recycled to the pebble crushing circuit, targeting a P_{80} of 12 mm, or finer, product. The design for the final feed to flotation from the ball mill circuit is a product size of P_{80} 200 μm (0.2 mm).

The rougher-scavenger flotation circuit produces a higher-grade rougher concentrate and a lower-grade scavenger concentrate. These concentrates are separately reground in their respective regrind mills and upgraded in three cleaner flotation stages to produce a final flotation concentrate targeting approximately 21.5% copper (Cu), and 30-40 gpt of gold (Au).

Gravity Recoverable Gold (GRG) is scalped from the rougher concentrate stream using a centrifugal concentrator prior to the cleaning circuits. The recovered gold is sent to the final flotation concentrate thickener. Final flotation concentrate is thickened and stored in the stock tank prior to filtering. The concentrate is pressure-filtered to a targeted moisture content of 8.5%, stockpiled, and then trucked to the rail loadout facility in Mackenzie, BC. The concentrate is then railed to North Vancouver, where it is loaded onto ships and sent to purchasers located around the Pacific Rim.

The rougher-scavenger tailings, containing mostly non-sulphide gangue minerals, is stored in the TSF, while the cleaner scavenger tailings, containing most of the sulphide gangue minerals, is stored in a separate area within the TSF. The latter is kept in a lined pond and underwater to prevent acid generation from the oxidation of the sulphide minerals. The simplified process plant flowsheet is shown in Figure 17-1.

[illegible]

Crushing

The originally installed coarse ore stockpile (COS) feed conveyor was split into two conveyors with a transfer station that allows the material to feed either the original coarse ore stockpile feed conveyor or be diverted to a new conveyor that feeds the secondary crushing circuit.

The secondary crushing circuit includes an inclined double deck vibrating screen (and two cone crushers) with all oversize material combined and conveyed to a surge stockpile. The screen oversize material is reclaimed from the surge stockpile and transferred to the secondary crushers targeting 45 mm top size, or finer, product. The product of the secondary crushers is combined with the screen undersize and transported by conveyor to the coarse ore stockpile.

The coarse ore stockpile has a live capacity of approximately 60,000 t.

Grinding

The grinding circuit comprises a SAG mill, ball mill, and pebble crushing (SABC) designed to process ore at a nominal rate of 2,717 t/h.

The discharge end of the SAG mill is equipped with 70-mm aperture grates to facilitate the removal of critical size material. The mill discharge is screened by two horizontal double deck vibrating screens with the oversize from both decks transferred to two pebble crushers targeting 12-mm P_{80} product. The crushed product is conveyed back to the SAG mill feed conveyor. The vibrating screen undersize, screened on a bottom deck aperture of 12.5 mm, is pumped by two centrifugal pumps, each with a standby pump, one for each ball mill hydrocyclone cluster. The hydrocyclone underflow streams feed the ball mills by gravity, while the hydrocyclone overflow streams feed the rougher flotation circuit by gravity.

Reagents, including slaked lime, a xanthate-based collector, and a dithiophosphinate-based promotor, can be added to the SAG mill feed chute, if necessary. Lime is used for pH control while the collector and promotor are bulk and selective collectors used in flotation, respectively.

Flotation

The rougher-scavenger flotation circuit includes two trains of five 200 m³ flotation cells. Each train has two rougher and three scavenger flotation cells. Rougher and scavenger flotation is carried out at natural pH and a slurry density of 35%-40% solids. PAX and A3409 are used as collectors while Polyfroth H28C is used as the frother.

The concentrates from the first two cells of each train (rougher concentrate) and the concentrates from the last three cells of each train (scavenger concentrate) are reground separately in their respective regrind circuits to produce an 80% passing size of 30-50 μm and 18-25 μm , respectively. The rougher and scavenger concentrates account for an overall mass of approximately 10%.

Two separate regrinding circuits are used for the rougher and the scavenger concentrates, respectively.

The rougher concentrate is reground to P_{80} 30-50 μm in the vertically stirred mill using steel ball media while the rougher-scavenger concentrate together with the first cleaner, second cleaner, and third cleaner flotation tailings are reground to P_{80} 18-25 μm in the horizontal stirred mills using ceramic ball media.

To recover coarse metallic gold particles, approximately 20% of the rougher concentrate regrind hydrocyclone underflow is diverted to a centrifugal gravity concentrator. The gravity concentrate is currently pumped directly to the copper concentrate thickener. Gravity concentrator tailings reports to the vertical regrind mill for further regrinding.

The reground concentrates undergo three stages of cleaning flotation to produce a final copper concentrate containing approximately 21.5% Cu and 30 to 40 g/t Au. The major equipment currently used in the cleaner flotation circuit includes:

- Two 100 m³ first cleaner flotation tank cells
- Four 30 m³ second cleaner flotation tank cells
- Two 30 m³ third cleaner flotation tank cells
- One 4 m x 12 m flotation column
- Five 100 m³ cleaner-scavenger flotation tank cells

The reground rougher concentrate is cleaned in the first cleaner *flotation cells* with the concentrate pumped to the second cleaner circuit. The second cleaner flotation concentrate is cleaned in a third cleaner column. The concentrate from the third cleaner column, which is the final concentrate, is pumped to the concentrate thickener. The third cleaner column tailings are recycled to the third cleaner tank cells as a scavenging circuit.

The tailings from the first, second, and third cleaner circuits are combined and pumped to the rougher-scavenger concentrate regrind circuit. The reground scavenger concentrate together with the combined cleaner tailings are cleaned in the cleaner-scavenger flotation cells. The concentrate from these cells is pumped to the first cleaner bank while the tailings are pumped to the TSF, where sulphide cleaner tailings are kept submerged to prevent oxidation of the sulphide minerals and the release of acid.

The reagents used in the rougher flotation circuit are also added to the cleaner flotation stages, at significantly lower dosages. Slaked lime is used to maintain the optimum pH in the cleaning circuit. For optimum pyrite rejection, the pH is maintained at approximately 11.3 in the cleaner circuit. Three SFRs were installed in May of 2022 to expand the cleaner flotation circuit capacity. These flotation cells will recover approximately 60% of the copper concentrate directly into the dewatering circuit, alleviating the existing cleaner flotation circuit of additional capacity. Shown in Figure 17-2 is a schematic for the installation in the overall circuit. This is further detailed in Item 13.

The diagram illustrates the proposed flotation circuit, showing the flow of material from the Flotation Feed through various stages of grinding and cleaning to the final concentrate. The circuit includes a new SFR Cleaner Scalper and several existing components like Cyclones, Tower Mill, IsaMills, and various cleaner stages. The flow is as follows:

- Flotation Feed** enters the **Rougher**.
- Rougher** output goes to **Rougher Scavengers** and **Cyclone**.
- Rougher Scavengers** output goes to **To TSF** and **Cyclone**.
- Cyclone** (left) output: **O/F** goes to **SFR Cleaner Scalper** (New); **U/F** goes to **Tower Mill Regrind**.
- Tower Mill Regrind** output goes to **1st Cleaners**.
- 1st Cleaners** output goes to **2nd Cleaners** and **3rd Cleaner Column**.
- 2nd Cleaners** output goes to **3rd Cleaner Column**.
- 3rd Cleaner Column** output goes to **3rd Cleaners** and **To Concentrate Dewatering and Storage**.
- 3rd Cleaners** output goes to **3rd Cleaner Column**.
- IsaMills Regrind** output goes to **Cyclone** (right).
- Cyclone** (right) output: **O/F** goes to **Cleaner Scavengers**; **U/F** goes to **1st Cleaners**.
- Cleaner Scavengers** output goes to **To TSF**.
- SFR Cleaner Scalper** (New) output goes to **1st Cleaners**.

Legend:

- New** (Red arrow)
- Existing** (Black arrow)

The final flotation concentrate is thickened to 60%-65% solids in a 12 m diameter high-rate thickener. The thickener underflow is pumped to the concentrate stock tank and fed to a 96 m² pressure filter. The filtered concentrate, containing approximately 8.5% moisture, is conveyed to the concentrate storage shed. The concentrate is transported by truck to a rail loadout facility located in Mackenzie, BC. The concentrate is then railed to North Vancouver where it is loaded onto ships and sent to purchasers located around the Pacific Rim.

Two tailings streams, the rougher-scavenger tailings and the cleaner-scavenger tailings, are deposited and stored in separate tailings storage areas within the TSF. The tailings pond supernatant is recycled to the process plant for re-use.

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Tailings embankment construction and management is further detailed in Item 18.

Reagents

The reagent preparation and storage facility is located within a spill containment area within the process plant building, designed to accommodate 110% of the content of the largest tank. The storage tanks and reagent systems are equipped with instrumentation and systems to enhance safety and control.

The collectors include a selective gold collector containing 30-40% sodium diisobutyldithiophosphate supplied by bulk tanker deliveries, and a solid type collector which is shipped to the mine site in supersacks and mixed on site. Polyfroth H28C is used as the frother and is shipped as liquid in bulk tankers. Flocculant, added to the thickener, is used for dewatering and is prepared in a wetting and mixing system. Quick lime, used for pH control, is delivered by bulk tanker trucks and prepared in a slaking system to produce a slaked lime slurry of 20% solids and distributed throughout the process plant.

New reagents are occasionally tested to enhance metal recovery and concentrate grading. These reagents are handled in accordance with applicable regulatory and safety requirements.

Assay and Metallurgical Laboratory

The onsite assay laboratory is equipped with necessary analytical instruments to provide all routine assays for the mine, the process plant, and the environment department. The most important of these instruments includes:

- Atomic Absorption (AA) spectrophotometers
- Mercury analyser
- Automatic titrator
- Fire assay furnaces
- Leco furnace

The metallurgical laboratory has equipment to conduct most necessary test work to monitor metallurgical performance and to improve the process flowsheet and efficiency.

Site Water and Air

Two separate water supply systems for fresh water and process water are provided to support the operation. Fresh and potable water is supplied to storage tanks from three ground water wells, located approximately one km south of the plant site.

Fresh water is used primarily for the following:

- Firewater for emergency use
- Reagent preparation
- Dust suppression
- Potable water supply (treated by chlorination and ultraviolet lamps)

The fresh-water tank, by design, is always full and provides at least two hours of firewater pumping volume in case of an emergency.

Process water consists of primarily reclaim water from the TSF, the copper concentrate thickener overflow, fresh make-up water from the Philip Lake & Meadows Creek pump stations, and water from the mine dewatering wells. Most of the water streams are pumped into the process water storage tanks except the thickener overflow which is recycled internal to the process plant.

Separate air service systems supply air to flotation, filtration, crushing and general plant services.

Power Consumption

The 2021 total site power consumption was 602,911 MWh. Just under 75% of the electrical power for the operation is consumed by the plant comminution circuits. Table 17-1 breaks down power consumption by area for 2021.

Table 17-1: 2021 Power Consumption (by area)

2021 Energy Consumption by Area		
³ Operational Grouping	Energy Consumption (%)	Energy Consumption (MWh/Operational Group)
Transmission Losses, Power System Services	2.1%	12,933
¹ Primary & Secondary Crushers, Stockpile Conveyor	2.1%	12,606
Pebble Crushing (Incl. Ancillary Equipment)	2.7%	16,183
SAG Mill Grinding	28.5%	171,546
Ball Mill Grinding	33.3%	200,999
Grinding Pumps	2.0%	12,358
Ancillary Grinding Equipment	3.6%	21,574
Regrinding	7.2%	43,245
Floatation Pumping & Air	1.5%	8,911
Ancillary Regrind & Floatation Equipment	5.0%	30,382
Reclaim Water	4.2%	25,611
Onsite Power Distribution	7.7%	46,564
⁴ Total	100%	602,911
Notes		
¹ All ancillary equipment within the Primary & Secondary Crusher areas are not metered. The consumption from the ancillary equipment in these areas has been captured in the 'Onsite Power Distribution' grouping. The only electrical loads considered in the 'Primary Crusher & Secondary Crushers, Stockpile Conveyor' grouping is the Primary Crusher motor, Secondary Crusher motors, and Stockpile Conveyor motors. ² 'Onsite Power Distribution' considers all electrical loads powered by means of the onsite distribution power lines including; - Onsite support infrastructure (Warehouse, Camp, Administrative Buildings, ect.). - All mining equipment (shovels, drills) and supporting infrastructure (truck shop, truck wash, ect.). - Fresh water pumping systems. - All ancillary equipment within the Primary & Secondary Crusher areas as mentioned in note 1. ³ The above data included a metering discrepancy of 0.96% compared to data provided in BC Hydro invoices. The metering discrepancy has been distributed equally to each operational grouping above. ⁴ Total energy shown as per data provided in BC Hydro invoices.		

17.3. PROCESS CONTROL AND INSTRUMENTATION

The plant control system consists of a Distributed Control System (DCS) with PC-based Operator Interface Stations (OIS) located in two separate control rooms: one in the primary crusher station and one in the main administration building, alongside the mine dispatch centre. The plant control rooms are staffed by trained personnel 24 hours per day.

18. PROJECT INFRASTRUCTURE

18.1. PROJECT INFRASTRUCTURE

Mount Milligan Mine is accessible by commercial air carrier to Prince George, British Columbia, then by vehicle from the east via Mackenzie on the Finlay Philip Forest Service Road and the North Philip Forest Service Road, and from the west via Fort St. James on the North Road and Rainbow Forest Service Road. Road travel to Mount Milligan Mine is 254 km from Prince George. The forestry-based communities of Mackenzie and Fort St. James are within daily commuting distance of the mine, and these communities are serviced by rail.

The infrastructure at Mount Milligan Mine includes a concentrator, a TSF and reclaim water ponds, as described elsewhere in this report, an administrative building and change house, a workshop/warehouse, a permanent operations residence, a first aid station, an emergency vehicle storage, a laboratory, and sewage and water treatment facilities. The power supply is provided by B.C. Hydro via a 91 km hydroelectric power line. Concentrate is transported by truck from the Project site to Mackenzie, transferred onto railcars of the Canadian National Railway, railed to existing port storage facilities of Vancouver Wharves in North Vancouver, and loaded as lots into bulk ore carriers. Concentrate is then shipped to customers via ocean transport.

18.2. TAILINGS STORAGE FACILITY

The TSF at the Mount Milligan Mine is designed to store tailings solids and PAG and oxide/weathered waste rock materials in designated areas. The TSF embankment is constructed as a centreline dam using open pit overburden and NAG waste rock materials. The TSF comprises two dams to include the Main Embankment and the WSB. The dams will eventually join as additional raises are completed. The highest portion of the TSF embankment is in the King Richard Creek valley and is approximately 62 m in height as measured from crest to downstream toe after Stage 8 construction is completed in 2022. The embankments are zoned earthfill structures, comprising a low permeability glacial till core zone, appropriate filter, and transition zones to mitigate piping of the core zone material. The TSF embankment core zone is keyed into the low permeability glacial till in the foundation.

The main TSF embankment is subdivided into segments designated as South, Southeast, Northeast, and North Dam. The South Dam is situated across the King Richard Creek valley; the Southeast/Northeast Dams are along the eastern plateau towards the Esker Lakes; the North Dam

is constructed through the esker deposit. The WSB is constructed along the western edge of the impoundment providing containment between the TSF and the Open Pit. The WSB has been extended towards the north and south and will continue to be extended until it connects into the Main embankment creating a continuous ring impoundment.

The maximum embankment height will eventually be 85 metres at the South Dam across King Richard Creek valley. The current constructed height at the South Dam is approximately 62 metres. Construction of each of the embankment stages is scheduled to correspond with material availability from the Open Pit and the projected rate of rise. Construction of the Stage 1 TSF and ancillary facilities commenced in December 2010 and was substantially completed in July 2013 to allow for operations to discharge tailings into the impoundment. Stage 8 is planned for 2022 with Stages 9-19 planned as annual raises based on the current mine schedule and planning. General arrangements illustrating the staged development together with a typical embankment cross section are shown on Figure 18-1 to Figure 18-4.

Figure 18-1: Stage 8 TSF General Arrangement

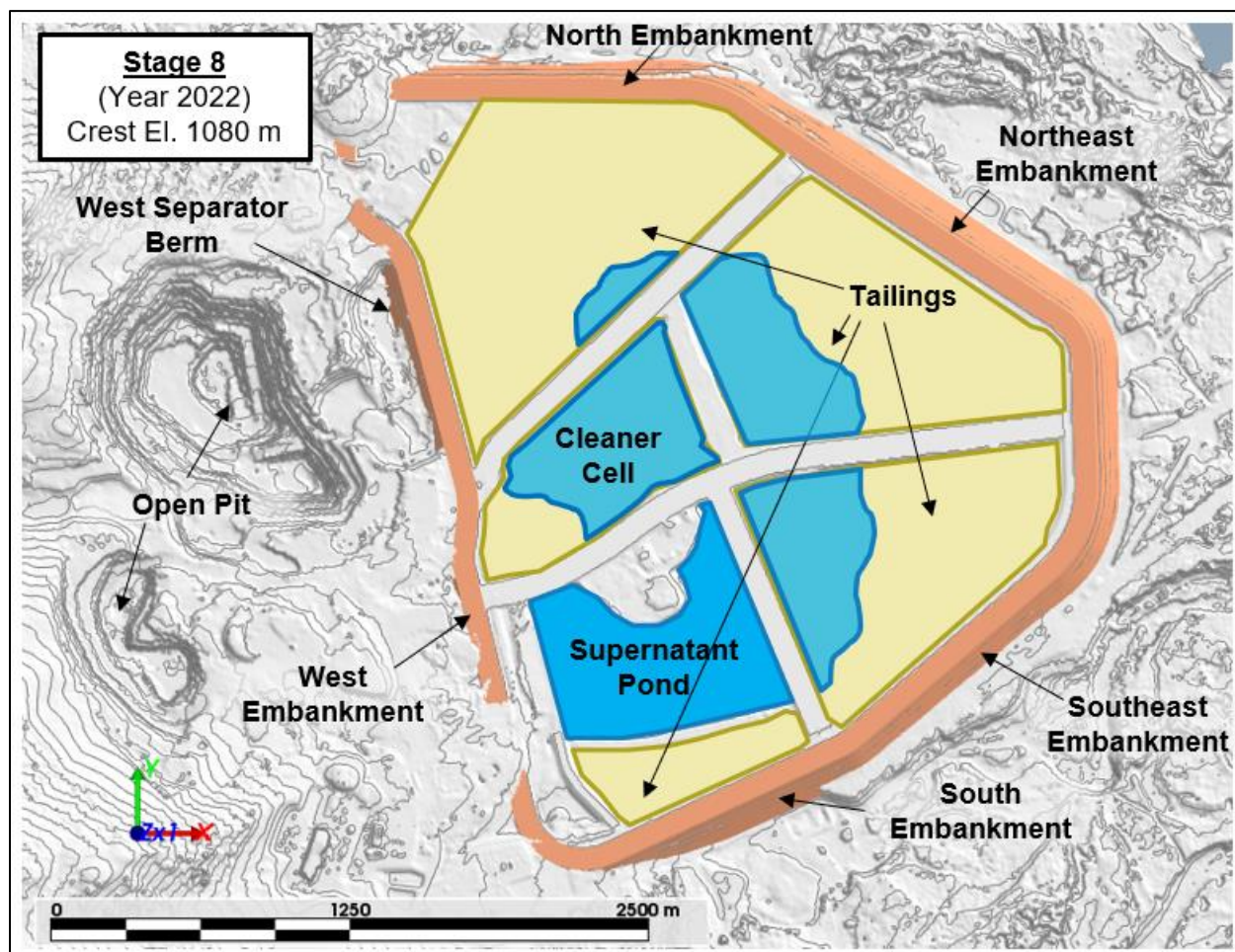


Figure 18-2: Stage 13 TSF General Arrangement

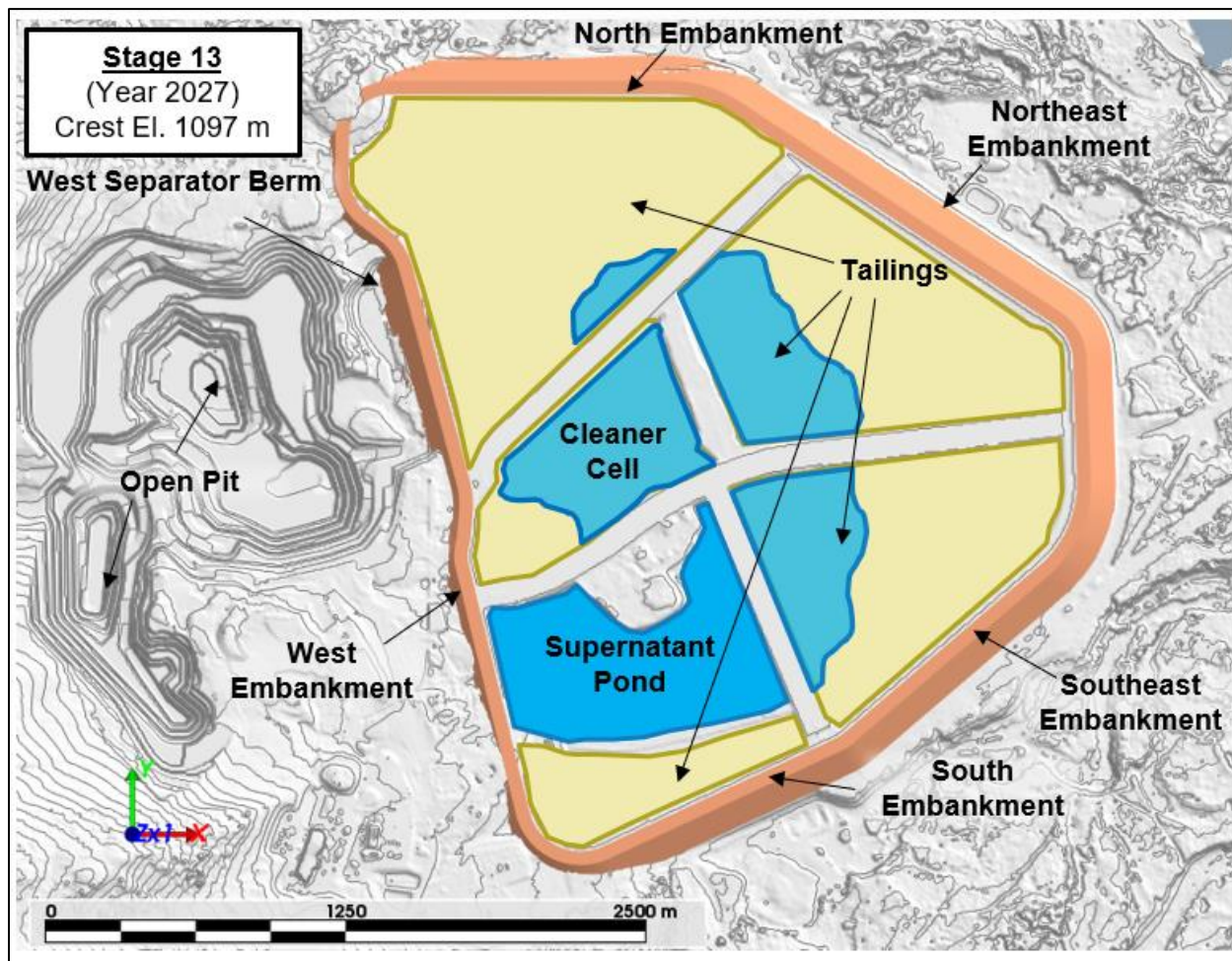


Figure 18-3: Stage 19 TSF General Arrangement

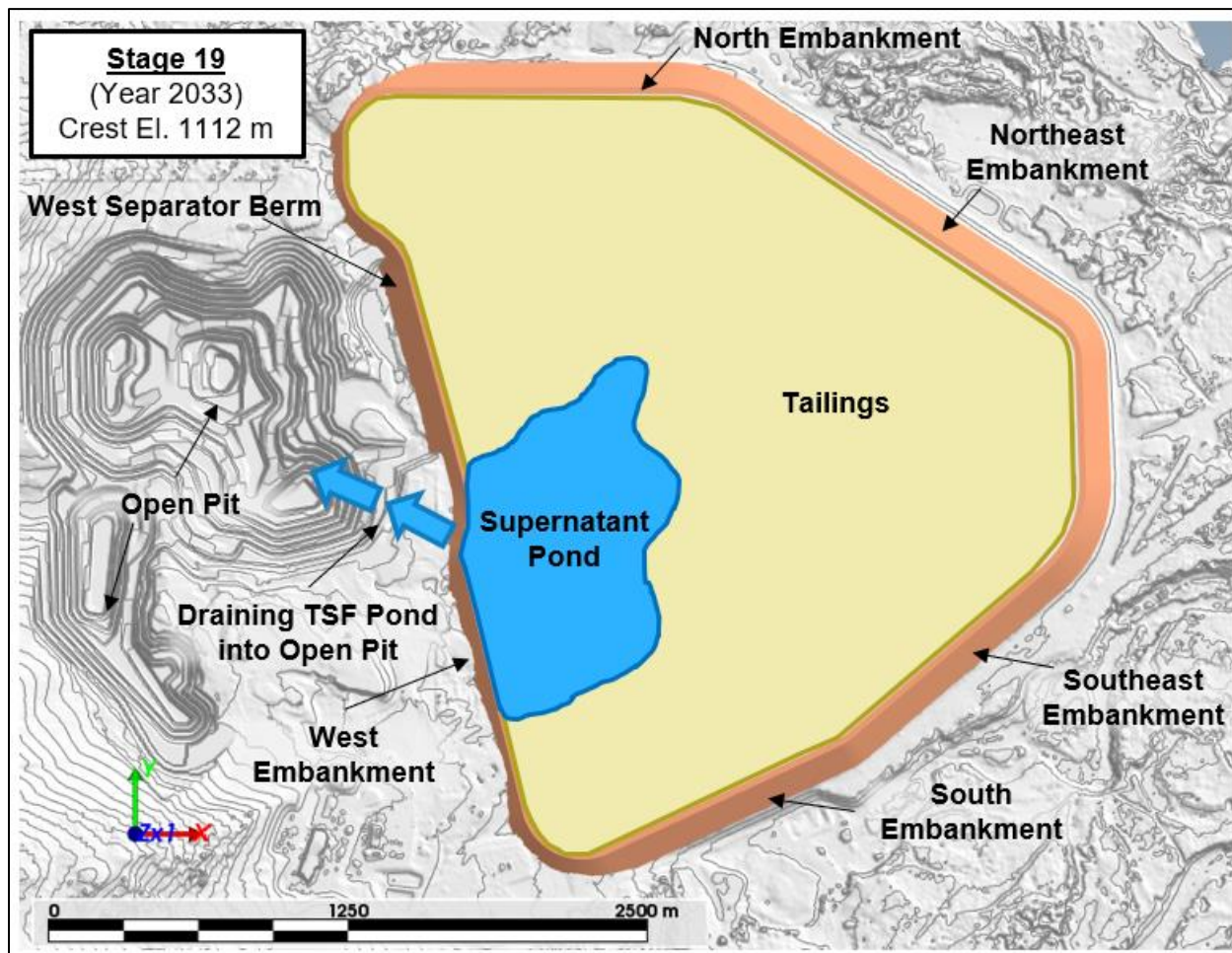


Figure 18-4: Typical TSF Embankment Section

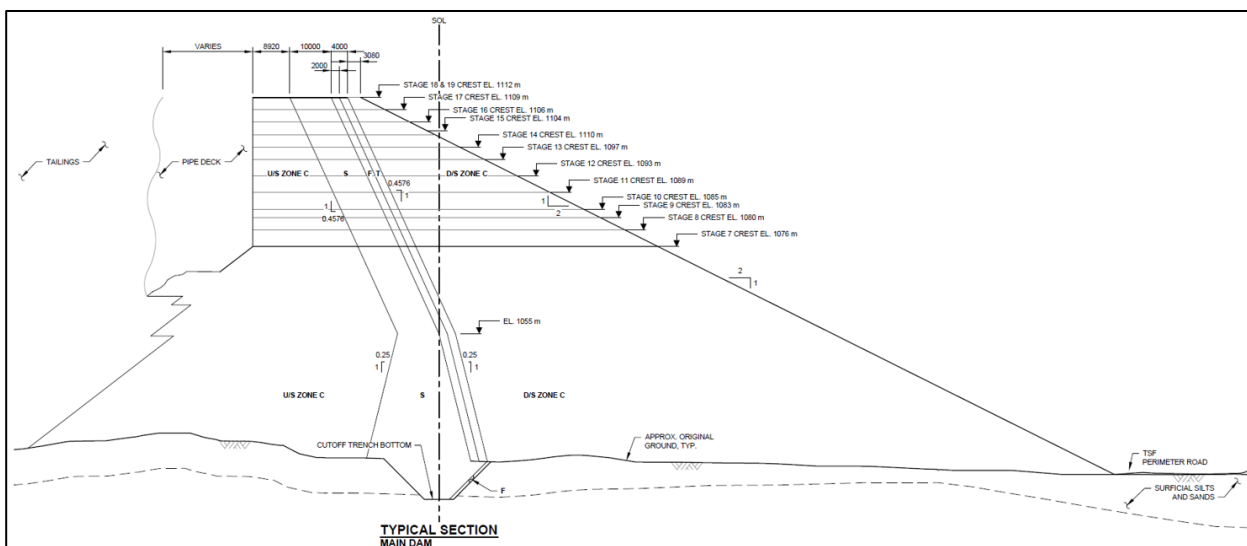


Table 18-1: TSF Embankment Staging and Construction Material Summary

Stage	Year	Crest Elevation	Embankment Construction Minimum Material Requirement				Total
			U/S Zone C	Core	Zone F & T	D/S Zone C	
		(m)	(m3)	(m3)	(m3)	(m3)	(m3)
8	2022	1080	0	0	0	0	0
9	2023	1083	621,918	291,005	95,111	1,360,020	2,368,053
10	2024	1085	403,342	186,423	76,557	856,801	1,523,122
11	2025	1089	783,106	368,508	172,931	1,582,546	2,907,091
12	2026	1093	739,825	374,448	169,946	1,375,754	2,659,974
13	2027	1097	659,625	406,130	136,893	1,178,831	2,381,478
14	2028	1101	587,629	412,925	149,330	929,284	2,079,168
15	2029	1104	400,856	289,220	135,247	529,172	1,354,495
16	2030	1106	248,305	192,309	90,334	275,319	806,267
17	2031	1109	342,902	287,603	133,007	294,462	1,057,974
18	2032	1112	295,575	299,998	104,710	171,731	872,015
19	2033	1112	0	0	0	0	0
Total (m3)			5,083,083	3,108,569	1,264,066	8,553,920	18,009,638

Scavenger tailings are discharged from pipelines into the TSF along the south, north, east, and southeast embankments as well as the WSB. Cleaner tailings are discharged from a separate pipeline into the cleaner tailings cell. The coarse fraction of the tailings is expected to settle rapidly and will accumulate closer to the discharge points, forming a gentle beach. Finer tailings particles will travel further and settle at a flatter slope adjacent to and beneath the supernatant pond. Beaches are developed to maximize storage volume and to control the location of the supernatant pond. Selective tailings deposition is used to maintain the supernatant pond away from the embankments to reduce seepage and ensure that reclaimed water is clear and accessible for reuse in the process plant. The tailings deposition strategy is expected to be modified towards the end of mine life. The intention is to shift the cleaner tailings discharge to the open pit for the last eight months of processing operations so that a scavenger cap of tailings can be placed over the cleaner tailings cell and the PAG cell, so that these deposits are capped with a layer that prevents oxygen ingress.

At closure the tailings pond will be drained to the open pit via a constructed overflow channel. Conditions for the establishment of a wetland above the western end of the cleaner cell will be considered with the objective of polishing water (if required) prior to running into the pit. Placement of organic material or topsoil will be considered to promote natural revegetation on the accessible portion of the scavenger tailing surface (i.e., beach adjacent to the tailings dam).

18.3. WATER MANAGEMENT

The Mount Milligan Mine site was designed to prevent direct discharge of surface contact water from the mine and associated infrastructure to the receiving environment. Natural topography and water

management structures including runoff collection/diversion ditches and sediment basins divert non-contact water away from the site and contact water to water management structures which direct the water towards the TSF. External water inputs to the site, within the site catchment, such as precipitation and runoff, are managed within the site and recycled for operational use (e.g., processing, dust control on TSF surface), where feasible.

Water required for ore processing is supplied to the mill from the TSF via the reclaim barge. The objective for water volume in the TSF is to maintain a minimum volume of approximately 5 Mm³ as well as to provide adequate storage for the environmental design flood without discharge during operations. A maximum free water volume of 10 Mm³ is permitted to be held within the TSF. Based on the LOM mine plan, the annual water demand for operation is between 8 Mm³ and 10 Mm³, which is a function of the ore processing rate, waste rock generation rate, tailings and evaporative losses and climatic conditions. Additional interstitial water within the placed tailings in the TSF is recovered to the free water pond from six pump towers located inside the perimeter of the TSF embankment dams.

Water external to the TSF is managed through several other water storage facilities and conveyance infrastructure. Four seepage collection and recycle ponds (located downstream of the TSF) collect seepage from the TSF, surface runoff downstream of the dam and act as temporary storage facilities for external make-up water pumped into the TSF (additional details on this provided in the subsection). Water from the seepage collection and recycle ponds is recycled and pumped back to the TSF on a regular basis. A secondary seepage collection system (the Meadows Creek Seepage Collection System) is utilized to collect seepage that may bypass the pathway to the seepage collection ponds, where it is pumped back to the seepage collection and recycle ponds. Several ongoing studies and monitoring activities are focused on characterizing and mitigating surficial and subsurface seepage pathways to the extent possible.

In 2022, as a result of seepage and ongoing work to enhance seepage collection, the Mount Milligan mine registered under the MDMER and is developing an environmental effects monitoring program in accordance with that regulation.

Water that collects in the mine pits (i.e., precipitation, runoff, and seepage) is regularly recycled to the TSF to minimize ponded water within the pits. Sewage effluent from the camp and other site buildings is treated through an aeration cell and secondary treatment cell before being discharged into the TSF.

The water management system is designed to minimize the use of external make-up water through:

- Maximizing capture of precipitation and runoff within the mine footprint and directing the captured water to the TSF
- Capturing freshet runoff from King Richard Creek to the TSF
- Pit dewatering to the TSF
- Seepage collection and recycling to the TSF
- Efficient recycling of process water

External Water Supply

The mine was originally approved to obtain process water from four water sources to operate the mill: the impoundment of King Richard Creek through the TSF starter dam and surface runoff from the mine site that flows into the TSF, direct precipitation onto the TSF, water pumped to the TSF from the Meadows Creek Water Supply Pond (MCWSP), and recycling of tailings water from the TSF. Due to costs, environmental impacts, and water supply limitations the MCWSP was not constructed. In late 2016, results of a bathymetric survey confirmed observations that water volumes in the TSF were critically low. Storage volumes in the TSF remained low throughout 2017 and TCM was forced to shut down the mill in December of the same year. To address the water shortage, the BC Environmental Assessment Office (EAO) approved Amendment #3 to EAC #M09-01 in January 2018 (EAO 2018), which allowed surface water withdrawals from Philip Lake 1 until October 31, 2018, and from Meadows Creek during the 2018 freshet. Despite these additional water sources, the mine continued to experience water shortages through 2018 and into 2019.

A site-wide water balance model was developed in 2018 and has been updated annually using GoldSim, a dynamic simulation software. The purpose of the model is to provide a tool to optimize and assess water management at the Mine which is done by estimating water volumes and inflow and outflow rates at each facility within the mine. The water balance model incorporates surface and groundwater inflows and outflows to the TSF and Open Pit. The model is based on the currently approved mine plan and schedule presented in the original EA application (Terrane Metals Corp. 2008), with an adjusted start of milling to 2013. The modelling period extends from 2020 through to post-closure, which is estimated to be 2175. The model focused on understanding make-up water requirements for the LOM. Make-up water is defined as water obtained from sources external to the facility. The make-up water is that volume required to maintain an operating pond for reclaim to the processing plant above the natural and operational inputs to the model (i.e., runoff, internal pumping). The total deficit includes the additional required make-up water from the external source (i.e., Philip Lake 1, Rainbow Creek, and groundwater withdrawals from the Meadows Creek Well Field and Lower Rainbow Valley Well Field).

In 2018 and 2019, TCM worked with the Major Mines Office (MMO), the EAO, the Mine Review Committee (MRC) and the Technical Advisory Committee (TAC), including government agencies (EAO, MMO, the BC Ministry of Energy, Mines and Petroleum Resources (now Ministry of Energy, Mines and Low Carbon Innovation; [EMLI]), the BC Ministry of Forests, Lands, Natural Resource Operations and Rural Development (FLNRORD), and the BC Ministry of Environment & Climate Change Strategy (MOECCS), and participating Indigenous nations (Nak'azdli Whut'en, Takla Nation, and McLeod Lake Indian Band), to obtain Amendments #4 through #7 to EAC #M09-01 and necessary permits to enable immediate-term and medium-term (to November 2021) surface and groundwater supply sources while investigations continued to identify viable LOM water supply source(s). The amendments also included extensive engagement conducted by TCM with participating Indigenous nations.

TCM is prioritizing groundwater as a LOM water supply source, where feasible and continues to undertake an extensive groundwater investigation program. However, current available groundwater sources and licensed volumes of groundwater are not capable of meeting the water supply demand for the LOM and thus, the mine cannot solely rely on groundwater supply. Additional external surface water sources will be needed as part of the LOM water supply solution.

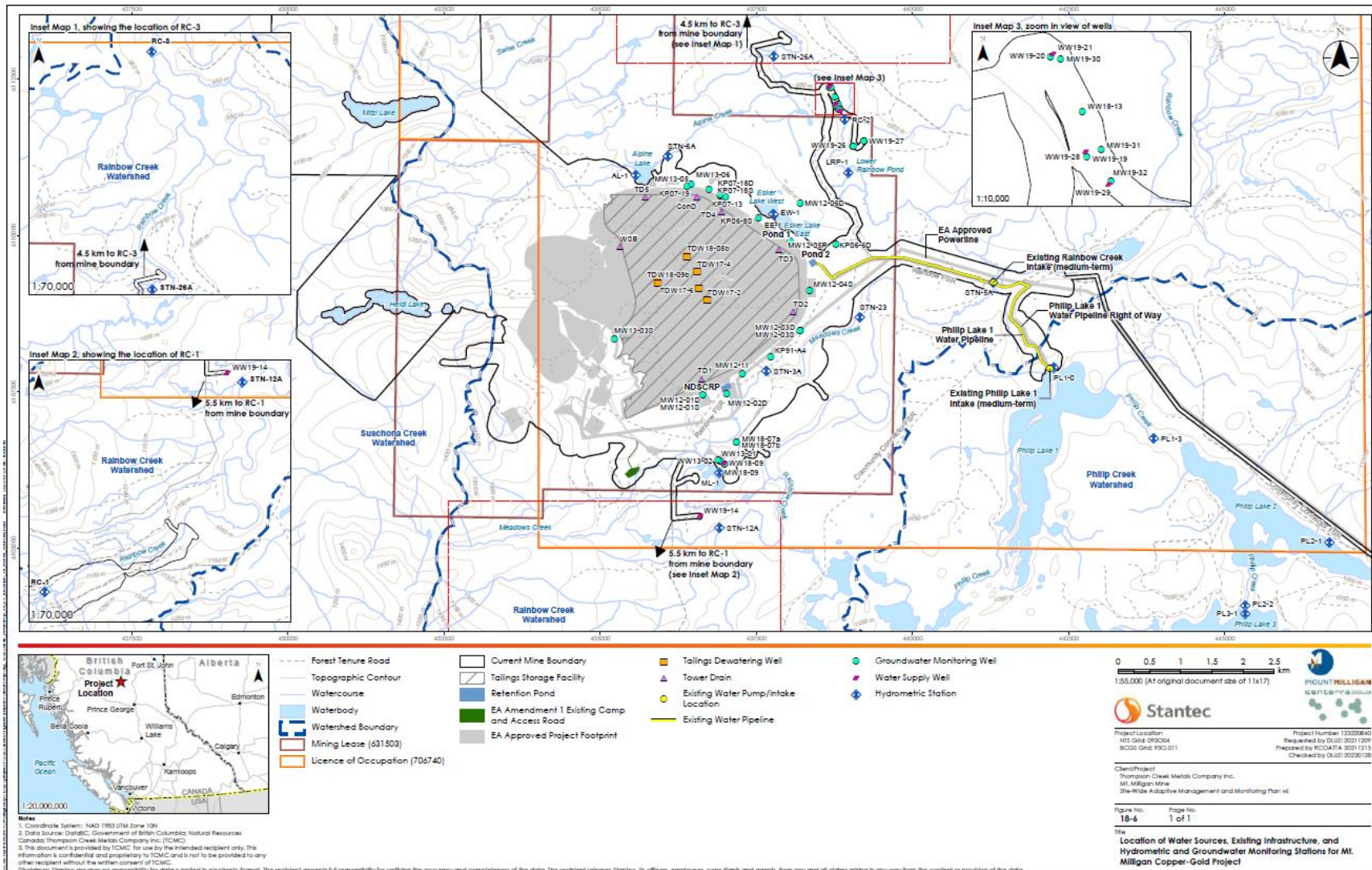
In November 2020, TCM applied to the EAO to amend EAC #M09-01 to extend the duration of the medium-term approvals under Amendment #6 (EAO 2019a; Philip Lake 1, Rainbow Creek, and Meadows Creek) by two years to November 30, 2023. This extension was requested to allow time for the completion of TCM's engineering studies, risk assessments and investigations, and for meaningful engagement with the TAC and Indigenous nations in support of an amendment request for a viable LOM water supply. The EAO issued the EAC amendment for the medium-term amendment extension on April 28, 2021 (EAO 2021). TCM is permitted to withdraw up to 7.0 Mm³ annually from Rainbow Creek and up to 3.5 Mm³ annually from Philip Lake 1. The currently permitted groundwater sources (the Meadows Creek Well Field and the Lower Rainbow Well Field) produce approximately 2.53 Mm³ and can provide sufficient water to limit impacts to production during most climatic conditions.

TCM currently holds *Water Sustainability Act* (WSA) licences to withdraw surface water from Philip Lake 1 and Rainbow Creek from April 1 to November 30 until November 30, 2023 (Water Licences 503422, 503497 and 503425, June, 2021). TCM is also permitted to withdraw water from Meadows Creek year-round (Water Licence 500927, February 1, 2019).

To ensure adequate water supply for the LOM, the mine is actively investigating additional long-term water sources. Ongoing investigation activities include hydrological and hydrogeological desk-top

analysis and field work, environmental impact assessment, community consultation and engineering evaluation. On January 6, 2022, TCM received the approval of an amendment to EAC #M09-01 to utilize LOM surface water withdrawals external to the TSF during the open water season (April 1 to November 30) from either the Nation River as a single surface water source or Rainbow Creek and Philip Lake 1 as a combined surface water source. Following additional engineering studies, cost optimization, and hydrological analysis, TCM is seeking Water Sustainability Act licence applications for the Rainbow Creek and Philip Creek option. This option, in addition to the currently permitted groundwater withdrawals, will be sufficient to maintain operations at current production targets for the approved LOM. Figure 18-5 provides the location of the current surface water and groundwater sources.

Figure 18-5: Locations of Existing Surface Water and Groundwater Sources



19. MARKET STUDIES AND CONTRACTS

19.1. MARKETABILITY

Mount Milligan Mine is strategically located for delivery to Asian custom smelters and the concentrate analysis is low in deleterious impurities such as arsenic, antimony, bismuth, chlorine, and fluorine. The presence of significant gold and payable silver values has been welcomed by custom smelters and has proven to be a positive factor when negotiating sales contracts, particularly with South Korea, The Philippines, and Japan. Historically, smelters located in China have offered less favourable terms for the recovery of gold and thus is a less attractive market for the Mount Milligan concentrates, however Chinese gold payables have improved in recent years. Delivery to North American smelters is also a possibility; however, the logistic costs are not as favourable as bulk transport to Asia so this would be an unlikely outcome.

19.2. CONTRACTS

Concentrate Sales

Copper/gold concentrate produced by the Mount Milligan Mine is sold to various smelters and off-take purchasers. TCM is currently party to three multi-year concentrate sales agreements for the sale of copper/gold concentrate produced at the Mount Milligan Mine. Pursuant to these agreements, an aggregate of approximately 120,000 tonnes in each of 2022 and 2023, 80,000 tonnes in 2024 and 60,000 tonnes in each of 2025, 2026, and 2027 will be sold.

Pricing under these concentrate sales agreements will be determined by reference to specified published reference prices during the applicable quotation periods. Payment for the concentrate will be based on the price for the agreed copper and gold content of the parcels delivered, less smelting and refining charges and certain other deductions, such as penalty elements within industry standards, if applicable. The copper smelting and refining charges are negotiated in good faith and agreed by the parties for each contract year based on terms generally acknowledged as industry benchmark terms. The gold refining charges are as specified in the agreements. The rates and charges are within industry norms.

Remaining concentrate produced at the Mount Milligan Mine will be sold under short-term contracts or on a spot basis. TCM may also choose to enter into another multi-year concentrate sales agreement, if appropriate, given the mine's expected concentrate production.

The commitments under the current multi-year concentrate sales agreements cover approximately 75% of the expected concentrate production in 2022 and are also less than the total expected production in 2023 and beyond. TCM intends to either extend the current multi-year agreements as the terms expire or may enter into additional multi-year sales agreements. To the extent that production is expected to exceed the volume committed under these agreements, TCM will sell the additional volume under short-term contracts or on a spot basis.

Stream Agreement with Royal Gold

Pursuant to the Stream Agreement with Royal Gold, TCM has agreed to sell to Royal Gold 35% of the gold produced and 18.75% of the copper production at the Mount Milligan Mine. Royal Gold pays \$435 per ounce of gold delivered and pays 15% of the spot price per metric tonne of copper delivered. When the Stream Agreement was originally entered into by TCM (prior to the Acquisition by Centerra), TCM received an upfront payment of \$781.5 million for the rights to receive future gold production. The Stream Agreement was amended by TCM as part of the Acquisition to reduce the percentage of gold production allocated to Royal Gold (from 52.25% to 35%) and to include 18.75% of the copper production from the Mount Milligan mine.

The Stream Agreement has an initial 50-year term, with automatic successive 10-year renewal periods.

TCM sells copper and gold concentrate from Mount Milligan Mine to customers and in connection with such sales, TCM purchases gold ounces and copper warrants in the market for delivery to Royal Gold in an amount based on a portion of the gold and copper content contained in the copper and gold concentrate sold to customers.

The Stream Agreement covers substantially the entire Property.

The Stream Agreement includes certain restrictions on assignment or transfer of the respective rights of both parties to the Stream Agreement.

20. ENVIRONMENTAL STUDIES, PERMITTING, AND SOCIAL OR COMMUNITY CONSIDERATIONS

Mining and mineral exploration activities are part of the social fabric of the Nechako plateau region where the Mount Milligan Mine is situated; an area containing several historical and operating mines as well as advanced exploration projects. The mine is 155 km northwest of Prince George, between the communities of Fort St. James and Mackenzie in central British Columbia.

The environmental assessment and permitting framework for mining in Canada, and British Columbia in particular, is well established, providing a comprehensive mechanism for reviewing major projects to assess potential impacts. The government authorizations held by Mount Milligan Mine followed rigorous and robust regulatory processes. These processes involved review by numerous ministries and agencies as well as indigenous nations and local communities.

The mine has been operating under these approvals since commencement with amendments pursued and received as required. The project plan incorporated design to limit its footprint and reduce environmental impacts. Engineering, construction, operation and management of mine facilities and components utilize criteria for responsible management to meet regulatory obligations. Environmental management plans and internal audits guide the compliance and monitoring programs at the mine. The mine continues to engage with indigenous nations, local communities, and stakeholders to share information and collaborate on environmental and/or social development initiatives.

20.1. ENVIRONMENTAL STUDIES

The regulatory framework for mining in British Columbia and Canada provides a rigorous mechanism for assessing the Mount Milligan Mine and its potential environmental and socio-economic impacts. The original environmental assessment and permit applications included an extensive list of environmental baseline studies and effects assessments.

These included comprehensive studies and assessments related to terrain and soils, air quality, noise, water quality and quantity, vegetation, wildlife, benthic invertebrates, periphyton, sediment quality, fish and fish habitat, visual and aesthetic resources, land use, archaeology, and heritage resources, social and economics, and human health. Potential environmental and socio-economic

effects and mitigations were fully considered in the environmental assessment and mine permitting processes and further reviewed as project plans were updated. Approvals were received based on the mitigations and management plans within the applications.

When the mine proposed to source surface water from either Nation River or a combination of Rainbow Creek and Philips Lake #1 for its long-term water supply, additional environmental flow needs studies were conducted in support of the application. The EA Certificate amendment application in 2021 concluded that with the implementation of proposed mitigation measures, the long-term water supply options would not result in changes to the characterization of residual or cumulative effects as outlined in the original EA application. The long-term water source will be located on a Licence of Occupation tenure held by Mount Milligan outside the permitted mine area.

The 2022 LOM mine plan (Item 16) is contained within the boundaries of the current permitted mine area. The production rate does not change. The TSF embankment will be raised to 1,112 m elevation. The pit outline will be extended, and some water management structures will be adjusted.

No significant new environmental studies are anticipated to be required. Incremental studies such as those required for archaeology or timber cruises prior to ancillary applications for licences to cut trees may be required. The new pit shell was reviewed by Knight Piésold and no additional geotechnical studies were determined to be required. As well, the same mineralogy is expected, and no new geochemistry studies are needed. As with current practice, sampling and lab analyses of waste rock and tailings will be undertaken to confirm geochemistry expected based on the geology model. A review of the site-wide water balance is planned in the context of the pit pushback and TSF dam raise.

20.2. WASTE DISPOSAL, MONITORING AND WATER MANAGEMENT

As noted in Item 18.2, the TSF at the Mount Milligan Mine is designed to store tailings solids and PAG and oxide/weathered waste rock materials in designated areas. The main TSF embankment is constructed in stages using annual raises throughout the LOM, from low permeability glacial till, overburden, and waste rock materials from stripping operations at the open pit and borrow areas within and near the TSF. The use of overburden and NAG waste rock for downstream TSF embankment construction eliminates the need for conventional waste rock dumps allowing the project footprint to be smaller.

Tailings from the mill was delivered by gravity to the TSF for as long as possible. Tailings pumping was commissioned in May 2022, with each delivery pipeline sized to carry up to 100% of the design

scavenger tailing production from the circuit. One of the three delivery pipelines is always available while allowing for maintenance work to be completed on the other two pipelines. Discharge into the TSF is and will be from valve controlled off-takes along the pipeline. A series of pumps transfer water from active mining bench sumps in the pit for discharge into the TSF. Two pumps and associated equipment were installed and commissioned in May 2022 to enable more control in tailings deposition.

The discharge plan of tailings into the TSF is described in Item 18.2.

Prior to construction and continuing in operations, surface water and groundwater sampling was implemented to monitor for potential impacts of the mining operation. An adaptive management plan including a Trigger Action Response Plan (TARP) was created and implemented, to manage groundwater pumping volumes if change attributed to these activities is noted in the environment. The Project distributes an annual water quality report that describes the monitoring program and results for a variety of parameters.

Environmental and other management plans are reviewed and updated as necessary and are submitted to applicable regulatory ministries for review and or approval as part of an adaptive management process.

20.3. PROJECT PERMITTING

The regulatory framework for mining in British Columbia and Canada is highly developed and prescriptive. Authorizations are required under several pieces of legislation and regulation. In addition, numerous policies and technical guidance documents exist and adherence to these is expected.

The project underwent several separate phases of mine planning and feasibility study over a 15-year period, including two separate EA processes. During those reviews, and in response to concerns from indigenous nations, regulators, and other stakeholders, the original 1993 mine plan was redesigned such that the footprint was reduced substantially.

Mount Milligan Mine received its EA certificate in March 2009, a *Mines Act* permit in September 2009 and *Environmental Management Act* (EMA) authorizations in July 2010. The mine has been operating under these approvals and subsequent amendments since commencement. *Mines Act* permit M-236 was amended in September 2021 approving an updated reclamation security. The most recent EA certificate amendment (Amendment #9) was granted in January 2022 for the LOM

water sourcing. The supporting surface water licence applications were submitted following the EAC Amendment #9 approval and are expected to be approved before the year end.

As discussed in the environmental studies Item (20.1), activities related to the updated mine plan are contained within the existing Permitted Mine Area. The long-term water source is located on a Licence of Occupation and other tenure held by Mount Milligan outside the permitted mine area. See Item 18.3 for more details on water management. Activities related to the 2022 LOM plan are not expected to change existing environmental considerations or necessitate changes to any approved mitigation measures.

It is anticipated that the 2022 LOM plan will involve an EAC and *Mines Act* permit amendment process similar to previous processes. Although it is not possible to ascertain the exact timeline, the process is expected to take more than six months after submission of application. And while it is not possible to guarantee that the 2022 LOM plan will be permitted, the mine has been successful in achieving all required authorizations to date.

20.4. SOCIAL OR COMMUNITY REQUIREMENTS

Within the federal and provincial regulatory frameworks, consultation with indigenous nations, local communities and other interested parties are required for new or amended mine project applications. Authorization-specific consultation strategies are developed and implemented by Mount Milligan.

The mine's Community Sustainability Committee formed in 2008 and comprises Indigenous Nations and local community representatives, such as councillors and municipal staff from Fort St. James, Mackenzie, Vanderhoof, and Prince George. This Committee is a mechanism for ongoing engagement allowing the mine to update the community on its activities and to hear of and resolve concerns, if any. The Committee is also used as a vehicle for disbursement of Community Funding and investments into the region. A Sustainability Management Plan for the mine, which was reviewed and approved by the Committee, remains in place.

A Socio-Economic Agreement with the McLeod Lake Indian Band, and an IBA with Nak'azdli Whut'en are in place. Both agreements commit the mine to the provision of financial payments. These amounts have been incorporated into the economic analysis in this Technical Report.

20.5. CLOSURE PROGRAM

As required by the BC *Environmental Assessment Act* and the BC *Mines Act*, proponents submit Reclamation Closure Programs for approval at the time of the initial application, and in the accordance with the mine permit M-236, every five years, thereafter.

The most recent M-236 Five Year/Conceptual Reclamation Plan (2019-2023) was submitted in 2019. Pursuant to the Reclamation Plan, the security amount was updated to reflect undiscounted and uninflated cost of reclamation of CA\$51.3 million. This Plan was approved by the Ministry of Mines and the permit amendment was issued in September 2021. The next Five Year/Conceptual Reclamation Plan is due October 2024.

The 2022 LOM plan will require an update of the Reclamation Closure Plan to account for the pit expansion, the increase of the TSF embankment height, and an updated site-wide water balance for post-mine closure. Additionally, the 2009 EAC contemplated a wetland in the reclaimed TSF at closure while the 2022 LOM mine plan proposes a dry cover as a mitigation measure.

21. CAPITAL AND OPERATING COSTS

The estimated future capital and operating costs for Mount Milligan are discussed and tabulated below in US dollars using an exchange rate of 1.30 US\$/CA\$; however, all cost estimates were developed in Canadian dollars. Please see section 22 of this document for sensitivity analyses.

Total operating and capital costs over the LOM are estimated at \$3,502 million, including \$990 million for mining costs, \$1,212 million for processing costs, \$492 million for administrative (G&A) costs and \$168 million for transportation costs, total selling and marketing costs are estimated at \$94 million, total treatment and refining charges are estimated at \$180 million and total capital expenditures are estimated at \$366 million, as shown in Table 21-1. Note: Mining costs for waste rock used in the tailings dam raise construction is initially accounted in operating costs, then transferred to the capital account for the infrastructure investment.

The total LOM capital expenditures required to extract the Mineral Reserves in the LOM plan is estimated at \$366 million, which includes capital equipment and component replacements, planned improvements to crushing equipment and site facilities, as well as water management, but excludes \$127 million TSF construction costs (included with mining opex). Non-acid generating waste rock is used for routine TSF raises, the cost of which is then capitalized to the TSF rather than as capitalized stripping. The current mine plan does not contemplate any growth capital.

The all-in sustaining cost per ounce sold, on a by-product basis, which includes sustaining capital and copper revenue credits, averages \$756/oz of gold for the period from 2022 to the end of the LOM. All-in sustaining cost per ounce sold, on a by-product basis, is a non-GAAP financial performance measure.

Table 21-1: LOM Capital and Operating Cost

Cost Summary (Total LOM)	Total \$M
Capital	366
Operating Cost	
Mining	990
Processing (including water supply)	1,212
Admin	492
Transportation	168
Selling and Marketing	94
Treatment and Refining	180
Total	3,502

21.1. MATERIAL ASSUMPTIONS

The following material assumptions have been used in the LOM plans, estimates of operating and capital costs:

- A gold price of \$1,500/oz,
- A copper price of \$3.25/lb,
- Exchange rate: \$1.00:CA\$1.30;
- Diesel fuel price assumption: \$0.92/litre delivered to site.
- Electrical Pricing of \$0.052/kWh.

The diesel price includes British Columbia taxes and freight to the onsite tank farms at Mount Milligan. Diesel fuel is sourced from the Prince George distribution terminal and correlates well with world oil prices.

21.2. CAPITAL COSTS

The LOM capital costs for the Mount Milligan mine are summarised in Table 21-2.

Table 21-2: LOM Capital Costs

Capital Category	Total (\$'000)
Mining Capital	
Mine Equipment Replacement	80,512
Equipment Component Replacement	150,642
Sub Total Mining	231,153
Processing Capital	
Water Management Capital	44,193
Process Plant Operating Capital	7,535
Process Plant Maintenance Capital	25,404
Sub Total Processing	77,132
Admin Capital	
Site Facility Improvement and Other Administration	847
Long-Term Agreements requiring capitalization (IFRS16)	57,892
Sub Total Admin	57,892
Total	366,177

The LOM capital spending has been included in the cash flow model.

The capital cost estimate assumes the replacement of nine and the addition of two 227t haul trucks and some auxiliary equipment. Major component rebuilds of the mobile fleet have been estimated based on expected operating hours per component.

Also included in the sustaining capital estimate is an external water pumping system and tailings line replacement.

21.3. OPERATING COSTS SUMMARY

Operating costs were developed from first principles, using site historic costs as a basis for calibrating the operating cost models, for an eleven-year mine life. This includes detailed estimates of personnel for all required roles/functions. Total LOM operating costs are summarized in Table 21-3.

Table 21-3: Operating Cost Summary

LOM Operating Costs Summary	LOM Cost (\$'000)	\$/tonne
Mining	990,177	2.13
Milling	1,212,159	4.99
Administration	491,775	2.03
Total	2,694,111	

Note: Unit mining costs are per tonne mined. All other unit rates are per tonne processed. Numbers may not sum precisely due to rounding.

21.4. MINE OPERATING COSTS

Key inputs for the mining costs are labour, maintenance, fuel, drill and blast consumables and mine general, and the costs are summarised by cost category in Table 21-4 and by activity in Table 21-5.

Table 21-4: Mining Costs by Category

Cost Category	LOM Cost (\$'000)	\$/tonne
Labour	340,239	0.73
Contractors	32,610	0.07
Professional Services	21,969	0.05
Power	11,132	0.02
Diesel	210,324	.45
Utilities/Other	1,867	.00
Maintenance	261,293	.56
Operating Materials	98,254	0.21
General/Other	12,487	.03
Total	990,176	2.13

Note: Unit mining rate is based on material mined.

Table 21-5: Mining Costs by Activity

By Activity	LOM Cost (\$'000)	\$/tonne
Hauling	310,824	0.67
Road and Dump	192,896	0.42
Drilling	39,717	0.09
Blasting	115,405	.25
Loading	85,339	0.18
Mine Admin	98,629	0.21
Mine Shops	125,573	0.27
Mine General	21,793	0.05
Total	990,176	2.13

Note: Unit mining rate is based on material mined.

Open pit mining for the Mount Milligan Mine will provide process plant feed at a nominal rate of 60,000 tpd.

The total LOM operating cost per tonne of material mined is \$2.13/tonne.

21.5. PROCESSING OPERATING COSTS

The processing costs are derived using a process plant feed rate of 60,000 tpd. Main inputs for the processing costs are electricity, grinding and reagents, labour, maintenance materials, water sourcing and contract labour. The costs are summarised by cost category in Table 21-6 and by activity in Table 21-7.

Table 21-6: Processing Costs by Category

Cost Category	LOM Cost (\$'000)	\$/tonne
Electricity	321,809	1.33
Grinding and Reagents	292,186	1.20
Labour	238,767	0.98
Parts and Spares	147,694	0.62
Liners	120,074	0.49
Contract Labour Services	68,029	0.28
Processing General	20,101	0.08
Professional Services	3,499	0.01
Total	1,212,158	4.99

Table 21-7: Processing Costs by Activity

By Activity	LOM Cost (\$'000)	\$/tonne
Grinding	614,398	2.53
Crushing	112,987	0.47
Water Sourcing	35,753	0.15
Flotation and Thickening	117,325	0.48
Processing Admin	67,798	0.28
Maintenance Shop and General	263,895	1.09
Total	1,212,158	4.99

Unit cost is calculated based on material processed.

Consumable costs were based on current contract pricing and historical consumption rates. Water management costs are included to extract process water from nearby sources based on historical experience and near-term expectations.

Plant maintenance has been factored in for mechanical, electrical and instrumentation together with an allowance for outside contractors to perform major shutdowns. Overall plant availability is estimated to be 94%.

The total LOM unitized material processing cost is \$4.99/tonne processed.

21.6. ADMINISTRATION COSTS

The administration costs for the operation include all site support, camp, bussing, insurances and general administration as well as an administrative office in Prince George. Administration costs are depicted by cost category in Table 21-8 and by activity in Table 21-9.

Table 21-8: Administration Costs by Category

Cost Category	LOM Cost (\$'000)	\$/tonne
Labour	38,609	0.16
Contractors	107,972	0.44
Professional Services	64,441	0.27
Power	29,575	0.12
Diesel	37,798	0.16
Utilities - Other	24,137	0.10
Environment	53,724	0.22
Maintenance	63,829	0.26
Operating Materials	39,649	0.16
General/Other	32,040	0.13
Total	491,775	2.03

Unit cost is calculated based on material processed.

Table 21-9: Administration Costs by Activity

By Activity	LOM Cost (\$'000)	\$/tonne
Property Taxes and Insurances	38,609	0.16
Site Admin, HR and IT	107,972	0.44
Camp Services	64,441	0.27
Purchasing/Warehouse	29,575	0.12
Crew Transportation	37,798	0.16
Safety	24,137	0.10
Environment	53,724	0.22
Site Facilities Maintenance	63,829	0.26
Sustainability	39,649	0.16
General	32,040	0.13
Total	491,775	2.03

Unit cost is calculated based on material processed.

The total LOM administration cost per tonne of material processed is \$2.03/tonne.

22. ECONOMIC ANALYSIS

The material assumptions used for the calculations presented in this Item have been stated in Item 21.

22.1. LOM PROCESSING

Table 22-1 provides an annual summary for the ore processing operation and resultant concentrate and metal production. Total ore tonnes processed is aligned with the Mineral Reserve estimate. Mineral Resources that are not Mineral Reserves do not have demonstrated economic viability and have neither been included in the Mine Plan nor the economic analysis. Metallurgical recoveries are based on recovery formulae discussed in Item 13. Payable metals are based on typical smelter terms.

Table 22-1: Processing Schedule

Item	Units	Total	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Ore Processed - Total	Mt	242.7	21.8	21.9	21.9	21.9	21.9	21.9	21.9	21.9	21.9	21.9	21.9	1.9
Ore Processed - Direct from pit	Mt	218.8	19.9	21.2	21.3	14.9	18.3	21.9	21.9	21.1	18.9	20.9	18.4	-
Ore Processed - From Stockpile	Mt	23.9	1.9	0.7	0.6	7.0	3.6	0.0	0.0	0.8	3.0	1.0	3.5	1.9
Cu feed grade	%	0.18	0.21	0.18	0.20	0.15	0.16	0.19	0.16	0.17	0.17	0.20	0.23	0.23
Au feed grade	g/t	0.37	0.43	0.35	0.32	0.35	0.42	0.35	0.35	0.35	0.35	0.35	0.44	0.64
Concentrate Produced DMT	ktonnes	1,666	170	144	167	118	131	157	133	139	139	167	186	16
Cu Recovery	%	80.2%	80.9%	80.0%	80.8%	78.8%	79.4%	80.4%	79.5%	79.8%	79.8%	80.8%	81.4%	81.5%
Cu Recovered	Mlb	790	80	68	79	56	62	74	63	66	66	79	88	8
Cu Payable Produced	Mlb	751	76	65	75	53	59	71	60	63	63	75	84	7
Au Recovery	%	68.0%	68.8%	67.7%	67.1%	67.6%	68.7%	67.6%	67.6%	67.7%	67.7%	67.7%	69.0%	71.1%
Au Recovered	Koz	1,976	209	168	151	166	205	166	166	167	168	168	215	28
Au Payable Produced	Koz	1,927	204	164	147	162	200	162	162	163	163	164	209	27

22.2. LOM CASH FLOW FORECAST

Using a gold price of \$1,500 per ounce, copper price of \$3.25/lb, and an exchange rate of CA\$:US\$ of 1.30:1.00, all the operating, transport, and capital cost forecasts have been used to estimate the net cash flow for the Mount Milligan Mine from 2022 to the end of 2033 to be \$640 M, as shown in Table 22-2. The after-tax NPV at a discount rate of 5% is \$486 M. RG on Table 22-2 refers to amounts per the Royal Gold stream.

Table 22-2: Cash Flow Summary (\$)

Description	Units	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	34-40	Total
Gold Sales	Ounces ('000's)	206	168	150	157	192	169	165	161	161	167	212	34	0	1,940
Gold Price	\$	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500		
Gold Sales	\$ Millions	309	252	225	236	287	253	247	242	241	250	318	50	0	2,910
Gold RG Share	\$ Millions	(77)	(63)	(56)	(59)	(71)	(63)	(61)	(60)	(60)	(62)	(79)	(13)	0	(723)
Smelter Charges	\$ Millions	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(0)	0	(10)
Net Gold Revenue	\$ Millions	231	188	168	176	215	189	185	181	181	187	238	38	0	2,177
Copper Sales	Lbs (Millions)	76	66	76	52	57	71	62	62	62	76	85	10	0	755
Copper Price	\$	3.25	3.25	3.25	3.25	3.25	3.25	3.25	3.25	3.25	3.25	3.25	3.25		
Copper Sales	\$ Millions	248	216	247	170	185	231	200	200	200	247	278	32	0	2,454
Copper RG Share	\$ Millions	(40)	(34)	(39)	(27)	(29)	(37)	(32)	(32)	(32)	(39)	(44)	(5)	0	(391)
Smelter Charges	\$ Millions	(17)	(15)	(17)	(12)	(13)	(16)	(14)	(14)	(14)	(17)	(19)	(2)	0	(170)
Net Copper Revenue	\$ Millions	191	167	190	131	143	178	155	155	155	190	214	25	0	1,893
Silver Revenue	\$ Millions	7	7	7	7	7	7	7	7	7	7	7	1	0	81
Total Revenue	\$ Millions	430	362	366	314	365	375	347	343	342	385	459	63	0	4,152
Operating Costs	\$ Millions	242	255	265	260	242	251	270	265	251	253	248	26	0	2,829
Royalties	\$ Millions	8	7	7	6	7	7	7	6	6	7	9	1	0	78
Capital Expenditures	\$ Millions	61	60	44	43	54	60	48	37	30	29	28	1	0	495
Lease Interest	\$ Millions	0	0	0	0	0	0	0	0	0	0	0	0	0	4
Working Capital	\$ Millions	38	-1	1	-2	0	1	-1	0	0	1	0	4	0	39.38
Reclamation	\$ Millions	0	0	0	0	0	0	0	0	0	0	0	0	39	39
Taxes	\$ Millions	5	2	2	1	2	2	2	2	2	3	4	0	0	28
Net Cash Flow	\$ Millions	75	39	46	6	59	53	22	32	52	92	171	31	-39	640
Cumulative Net Cash Flow	\$ Millions	75	115	161	167	226	279	301	333	385	478	648	679	640	

Offsite costs include delivery of copper concentrate to Pacific Rim Asian smelters. Concentrate from the mine site will be trucked to a storage and loadout facility at Mackenzie and transferred onto railcars for transport to port storage facilities at the Pembina loading facility in North Vancouver. TCM has leased 115 rail cars to ensure more reliable transport of concentrate to Vancouver.

22.3. SENSITIVITY ANALYSIS

Table 22-3 to Table 22-6 summarise sensitivities to gold price, copper price, change in cost estimates, and currency fluctuations to the United States Dollar, respectively.

Table 22-3: Sensitivity of NPV to Gold Price Changes

<i>NPV \$ millions</i>	<i>Sensitivity to Gold Price at 0%, 5%, and 8% Discount Rates</i>		
Discount Rate Gold Price (\$/ounce)	0%	5%	8%
-20%	277	200	168
-10%	459	343	294
\$1,500	640	486	421
+10%	822	630	547
+20%	1,004	773	674

Table 22-4: Sensitivity of NPV to Copper Price Changes

<i>NPV \$ millions</i>	<i>Sensitivity to Copper Price at 0%, 5% and 8% Discount Rates</i>		
Discount Rate Copper Price (\$/lb)	0%	5%	8%
-20%	244	174	146
-10%	442	330	283
\$3.25	640	486	421
+10%	838	642	558
+20%	1,037	798	696

Table 22-5: Sensitivity of NPV to the Impact of Change in Cost Estimates

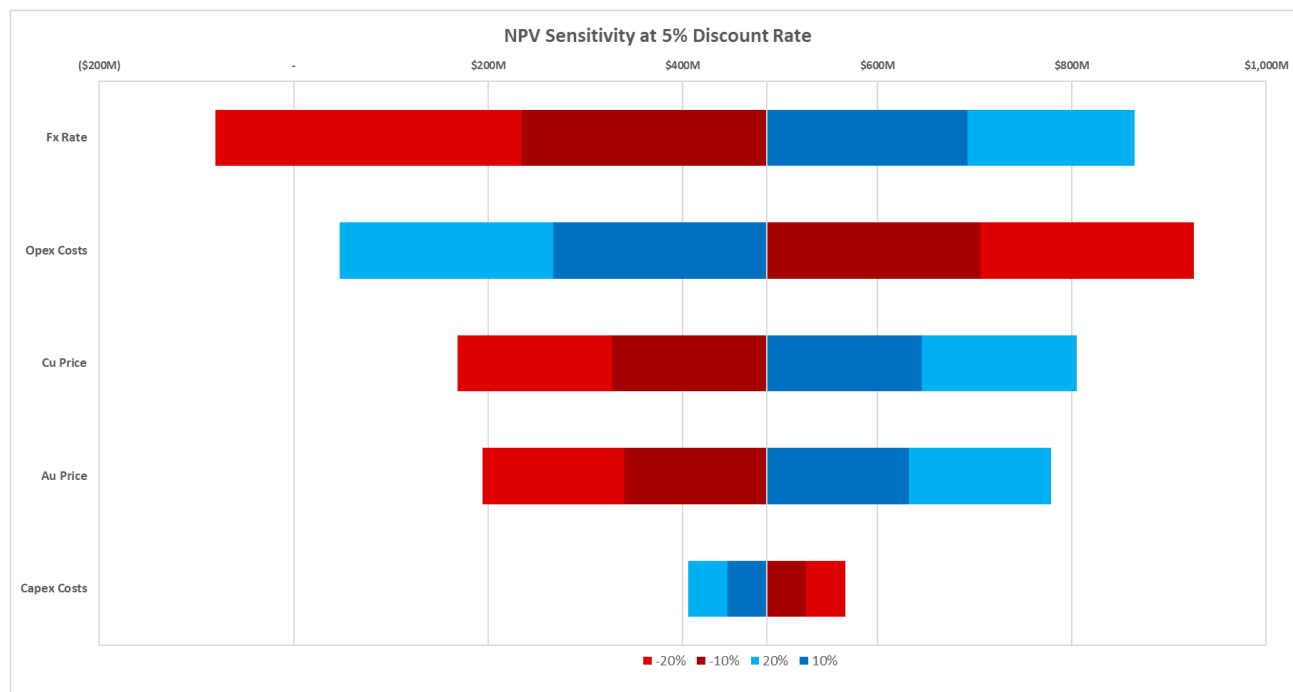
<i>NPV \$ millions</i>	<i>Sensitivities to costs at \$1,500/oz gold, \$3.25/lb copper and 5% Discount Rate</i>	
Variable	Operating Costs	Capital Costs
-20%	926	567
-10%	706	527
Base Case	486	486
+10%	267	446
+20%	47	406

Table 22-6: Sensitivity of NPV to the Impact of Change in Exchange Rate

<i>NPV \$ millions</i>	<i>Sensitivity to FX at 0%, 5% and 8% Discount Rates</i>		
Discount Rate FX (CA\$/US\$)	0%	5%	8%
-20%	(60)	(66)	(66)
-10%	329	241	205
1.30:1.00	640	486	421
+10%	895	687	598
+20%	1,107	854	745

Figure 22-1 provides a comparison of the sensitivity of NPV (at 5% discount rate) for the various parameters for +/-10% and +/-20% changes from the base case valuation for metal prices of \$1,500/oz gold and \$3.25/lb copper. As can be seen, the highest sensitivity is to OPEX and exchange rate changes, both for increases and decreases, with capex being the least sensitive parameter.

Figure 22-1: Comparison of Sensitivities



22.4. TAXATION

The determination of taxes involves significant estimation and judgment requiring a number of assumptions. The actual taxes payable will be subject to assessments by taxation authorities who may interpret tax legislation differently. The cash flow is based on management's best estimate of the probable outcome of these matters.

Corporation Taxes – Federal

Based on the pricing assumptions noted above, the mine is not expected to pay any federal income taxes at the statutory rate of 15% for approximately 11 years from 2022 to 2033 as there are sufficient deductions during that period to offset taxable income. These deductions include:

- Exploration and pre-production development expenditures allowed to be claimed discretionarily at 100% acceleration, limited to mine operating profit;
- Initial and sustaining capital expenditures generally allowed to be claimed discretionarily at 25%, limited to mine operating profit;
- Debt financing costs;
- Net operating loss carry forward allowed for up to 20 years; and
- Provincial mining taxes

Corporation Taxes – Provincial

The provincial corporate taxable income base is the same as the federal tax base. Based on the pricing assumptions noted above, the mine is not expected to pay any provincial income taxes at the statutory rate of 11%.

Mining Taxes – Provincial

The mine will be subject to the greater of two different taxes; either 2% tax on net current proceeds (net revenue less operating costs) or 13% tax on net revenue (net revenue less operating costs and capital expenditures). Additionally, to the extent the mine has previously paid the 2% tax on net current proceeds, this amount can be applied as a credit against the 13% tax on net revenue. Based on the pricing assumptions noted above, the mine is expected to pay the 2% net current proceeds tax for approximately 14 years from 2022 to 2035 as there are sufficient deductions and credits during that period to offset the 13% tax on net revenue. In lieu of allowing a deduction of debt financing costs, the net revenue can be reduced by an investment allowance which is earned on expenditures incurred to the extent they have not yet been deducted.

22.5. ENVIRONMENTAL LIABILITIES

The Project is operated in compliance with applicable environmental obligations arising from federal and provincial statutes and regulations that are necessary to the current operation of its business. The Mount Milligan Mine submitted the five-year revision to its Reclamation Plan in 2019 and will

submit another revision as per permit and regulatory requirements in 2024. The updated mine plan will require an updated version of the 5-year Closure Plan to be developed and submitted. Pursuant to the 2019 revision to the Reclamation Plan, the most recent estimate of the undiscounted and uninflated cost of reclamation is \$39 million.

23. ADJACENT PROPERTIES

There are no adjacent properties relevant to the assessment of the Project.

24. OTHER RELEVANT DATA AND INFORMATION

There is no additional relevant data or information that should be included in this Technical Report.

25. INTERPRETATION AND CONCLUSIONS

Based on the information contained herein, the Qualified Persons, as authors of this Technical Report, offer the following interpretations and conclusions.

25.1. GEOLOGY AND MINERAL RESOURCES

Mount Milligan is a roughly tabular, near-surface, alkalic copper-gold porphyry deposit currently being mined.

The procedures for drilling, sampling, sample preparation and analyses were determined to be appropriate for the type of mineralization and the estimation of Mineral Resources.

The effective date of the Mineral Resource Statement is December 31, 2021. Combined Measured and Indicated Mineral Resources, exclusive of Mineral Reserves, total of 189 Mt at 0.18% Cu and 0.30g/t Au containing 742 million pounds (lbs) of copper and 1.8 million ounces (oz) of gold and an Inferred Mineral Resource of 4.6 Mt at 0.07% Cu and 0.47g/t Au.

The 2022 resource estimate (see Table 14-14) saw a net increase of 63,955 kt of material resulting in 225 Mlb Cu and 420 koz Au of Measured and Indicated category above the previous resource estimate published in 2019 (see Table 14-15). The Inferred category saw a net loss of 3 Mlbs Cu and an increase of 15 koz Au.

Based on the verification of the drill hole database, and review of the geological interpretations, mineral resource block model, and 2021 reconciliation results, it is the QP's opinion that the supporting data and geological interpretations are representative of the Mt. Milligan deposit at the time of reporting, and that the 2021 Mineral Resource estimate has been estimated in conformity with CIM Estimation of Mineral Resource and Mineral Reserves Best Practices Guidelines (Nov 2019), and has been reported in accordance with the Canadian Securities Administrators' National Instrument 43-101.

25.2. MINING AND MINERAL RESERVES

Proven and Probable Mineral Reserves total 246.2 Mt at 0.18% Cu and 0.37g/t Au containing 996 Mlb copper and 2.9 Moz gold, effective December 31, 2021. This estimate was derived through the design of an ultimate pit, conforming to practical mining requirements, based on an optimized pit shell using metal prices of \$3.25/lb Cu and \$1,350/oz Au.

The Mineral Reserve estimate has been prepared using industry standard best practise methodologies with the classification of Proven and Probable Mineral Reserves conforming to CIM definitions and NI 43-101 requirements. The Mineral Reserves estimate is based on metal prices of \$1,350/oz Au and \$3.25/lb Cu, and a CA\$:US\$ exchange rate of 1.30.

Mining is carried out using a conventional drill-blast, load and haul approach. All waste rock is either used for TSF dam construction or stored within the TSF to comply with environmental approvals requirements.

Proven and Probable Mineral Reserves have been upgraded from 191 Mt at the end of 2019 to 246.2 Mt with the contained copper estimate up 4% to 996 Mlbs and gold estimate up 21.5% to 2.925 Moz relative to the 2019 Mineral Reserve statement despite depletion from mine production of 43.8 Mt.

25.3. MINERAL PROCESSING

Metallurgical recoveries are derived from operating results for the current process plant flowsheet configuration. Recovery improvements of 4% and 2% for gold and copper respectively have been incorporated into the LOM model.

This LOM update assumes that an average 60 ktpd process plant throughput will be achieved from 2022 onwards. While this has not yet been achieved over a calendar year, various operational and maintenance improvement initiatives have been completed or are in progress with this throughput goal as the ultimate objective; including improvements to the secondary crushing circuit, advanced process control and reclaim feeder improvements.

The tailings pumps were successfully installed and operational as of May 2022. These pumps will deposit rougher tailings into the TSF for the duration of the remaining LOM.

25.4. INFRASTRUCTURE

Tailings Storage

The TSF at the Mount Milligan Mine is designed to store tailings solids and PAG and oxide/weathered waste rock materials in designated areas. The TSF embankment is constructed as a centreline dam using open pit overburden and NAG waste rock materials. The TSF comprises two dams, including the Main Embankment and the WSB.

The maximum embankment height will eventually be 85 metres at the South Dam across King Richard Creek valley. The current constructed height at the South Dam is approximately 62 metres. Construction of each of the embankment stages is scheduled to correspond with material availability from the Open Pit and the projected rate of rise. Construction of the Stage 1 TSF and ancillary facilities commenced in December 2010 and was substantially completed in July 2013 to allow for operations to discharge tailings into the impoundment. Stage 8 is planned for 2022 with Stages 9-19 planned as annual raises based on the current mine schedule and planning.

Water Management

Mount Milligan Mine was designed to prevent discharge of surface contact water from the mine to the receiving environment. External water inputs to the site, within the site catchment, such as precipitation and runoff, are managed within the site and recycled for operational use, where feasible.

Water required for ore processing is supplied to the mill from the TSF via the reclaim barge.

To ensure adequate water supply for the LOM, the mine is actively investigating additional long-term water sources. Ongoing investigation activities include hydrological and hydrogeological desk-top analysis and field work, environmental impact assessment, community consultation and engineering evaluation. On January 6, 2022, TCM received the approval of an amendment to EAC #M09-01 to utilize LOM surface water withdrawals external to the TSF during the open water season (April 1 to November 30) from either the Nation River as a single surface water source or Rainbow Creek and Philip Lake 1 as a combined surface water source. Following additional engineering studies, cost optimization, and hydrological analysis, TCM is seeking Water Sustainability Act licence applications for the Rainbow Creek and Philip Creek option. This option, in addition to the currently permitted groundwater withdrawals, will be sufficient to maintain operations at current production targets for the approved LOM.

25.5. ENVIRONMENTAL AND SOCIAL

Mining and mineral exploration activities are part of the social fabric of the Nechako plateau region where the Mount Milligan Mine is situated. The government authorizations held by Mount Milligan Mine followed rigorous and robust regulatory processes. These processes involved review by numerous ministries and agencies as well as indigenous nations and local communities. A Socio-Economic Agreement with the McLeod Lake Indian Band, and an IBA with Nak'azdli Whut'en are in place. Both agreements commit the mine to the provision of financial payments.

The EA Certificate amendment application in 2021 concluded that with the implementation of proposed mitigation measures, the long-term water supply options would not result in changes to the characterization of residual or cumulative effects as outlined in the original EA application. The long-term water source will be located on a Licence of Occupation tenure held by Mount Milligan outside the permitted mine area. No major new environmental studies are anticipated to be required. Incremental studies such as those required for archaeology or timber cruises prior to ancillary applications for licences to cut trees may be required. A review of the site-wide water balance is warranted in the context of the pit pushback and TSF dam raise.

25.6. MINE CLOSURE

The most recent M-236 Five Year/Conceptual Reclamation Plan (2019-2023) was submitted in 2019. Pursuant to the revision to the Reclamation Plan, the security amount was updated to reflect undiscounted and uninflated cost of reclamation of CA\$51.3 million. This Plan was approved by the Ministry of Mines and the permit amendment was issued in September 2021. The next Five Year/Conceptual Reclamation Plan is due October 2024.

25.7. ECONOMIC ANALYSIS

Capital costs have been estimated for an eleven-year mine life and operating costs have been estimated for an average 60,000 tpd process plant feed rate. Operating cost estimates were developed from first principles using site historic costs as a basis for calibrating the estimates.

The base case economic assumptions are metal prices of \$1,500/oz Au and \$3.25/lb Cu, and a CA\$:US\$ exchange rate of 1.30.

The LOM all-in sustaining cost per ounce sold, on a by-product basis, before taxes is estimated at \$756/oz gold sold. This is a non-GAAP financial performance measure.

The NPV for the base case model is \$486 M after tax at a 5% discount rate.

25.8. RISKS AND MITIGATIONS

Throughout the various Items of this Report, risks are discussed. Table 25-1, on the following page, presents a summary of the six (6) key risks that could have a material effect on the Mt. Milligan Mine if not mitigated effectively.

The Mt. Milligan site is impacted by several additional risks, both specific to the operation as well as to the industry in general. The management of these and the identified key risks are undertaken as part of a broader risk management program detailed below.

Ongoing Management of Risk

The Mt. Milligan Mine participates in the Centerra Enterprise Risk Management (ERM) program which has been implemented to ensure risk-informed decision making throughout the organization. The program is based on leading international risk management standards such as ISO 31000 as well as industry best practice. Site management teams, at each operation and project, are responsible for identifying, assessing, treating, and monitoring risk. These risks and their mitigation action plans are recorded in a site risk register. Efforts are coordinated by appointed “Risk Champions” who facilitate the process to ensure consistency and continuity.

The Centerra Risk & Insurance Team, based at the Centerra corporate head office, is responsible for providing the requisite tools, guidance, oversight, and strategic direction for the ERM program. The risk management program at Centerra considers the full life of mine cycle from exploration through to closure. All aspects of the operation, the surrounding environment and stakeholders are considered when identifying risks. As such, the program encompasses a broad range including technical, financial, commercial, social, reputational, environmental, health and safety, political and human resources related risks.

A Risk Committee of Centerra’s Board of Directors provides oversight on the management of significant or critical risks through regular reporting by Management.

Table 25-1: Key Mount Milligan Risks

RISK RATING:	Priority Attention	Active Management Required	Diligent Monitoring	Monitor / Manage as Appropriate
Risk Category	Risk Description	Risk Rating	Mitigation Strategy	
Permitting & Approvals / Regulatory	Delay or non-compliance with permits resulting in fines, penalties, loss of permits, production interruptions and increased costs		- Developed and commenced implementing a formal plan outlining timelines and requirements. Dedicated resources to be hired to focus on permitting and compliance.	
Environment	Ineffective tailings dam management resulting in loss of life, production reductions or interruptions, reputational damage, and severe long-term environmental degradation.		- Third party independent annual reviews of dam structure and management practices. - Dam construction maintains the required freeboard for designed maximum inflow of water. - Follow requirements of the TSF, Operating, Maintenance, and Surveillance Manual.	
Social Performance: (Community Relations / Indigenous Peoples)	Relationships with local communities including indigenous groups resulting in possible operational disruption and/or delayed or denied permits.		- Implement an aligned and consistent framework for managing relationships with local stakeholders - Support new and existing training and community development programs that engage and provide benefits for local groups. - Mandatory indigenous cultural diversity training for all employees.	
Human Resources / Labour Relations	Increasing challenges attracting and retaining employees caused by tight job market resulting in employee turnover and loss of knowledge.		- Recruitment process outsourced end-to-end to reduce time-to-hire, cost-to-hire and reach a broader audience. HR structure revised to optimize existing HR capabilities. - Talent Management initiatives underway including talent calibrations, identifications of high-potentials and re-launch of the Centerra Competency Assessments and individual Development Plans. - Additional aspects of total compensation being considered for mid-level management.	
Financial	Costs are higher than estimated in the plan.		- Implemented comprehensive review of operating costs and key business drivers. - Implementing Capital Appropriation Request (CAR) policy focusing on project value, up-front engineering, and detailed scopes of work. - Developed and implemented a strategic sourcing plan for key commodities. - Implementing new Enterprise Resource	

Risk Category	Risk Description	Risk Rating	Mitigation Strategy
			Planning (ERP) system utilizing best practices and standardizing business processes.
Supply Chain (Contracts, Procurement, Logistics)	Inability to transport concentrate due to railroad operational issues, rail/port labour strikes and/or blockades.		- Monitoring of emerging issues. - Leased 115 rail cars and lids. - Expanded Mackenzie rail line to accommodate additional rail cars. - Added additional concentrate storage at site to compensate for service delays. - Signed contractual arrangements with railroad company to ensure service and rates. - Contingency plan developed including alternative storage and transportation.

26. RECOMMENDATIONS

The following recommendations are provided by the various Qualified Persons, segmented by subject matter. Recommendations for improvement projects should be costed and incorporated into Department Head budgets in the regular course of operations.

26.1. EXPLORATION

It is recommended that mineral exploration continue targeting both mineralization styles that have been identified with the Mount Milligan deposit, early stage gold and copper porphyry and late stage structurally-controlled high-gold-low-copper. Ongoing exploration programs should continue identifying, delineating, and testing target zones within the three strategic mineral exploration domains: at the deposit scale (Near-pit/infill), at the porphyry cluster scale (Brownfield outside the ultimate pit boundary but inside the mine lease), and for standalone deposits (Greenfield outside the mine lease).

In 2022, planned drilling at the Mount Milligan Mine focuses on zones where positive drilling results were returned in previous drilling campaigns from 2017-2021. Drilling programs continue to target gold and copper porphyry mineralization below and adjacent to the current ultimate open-pit boundary, as well as continue to test targets with potential for shallower mineralization peripheral to the current resource and reserve pits. Resource expansion and infill drilling is recommended on the western margin of the open-pit, i.e., in both the shallow Goldmark zone and the underlying DWBX zone, and on the eastern margin of the open-pit, i.e., in both the shallow Great Eastern Fault ("GEF") zone and underlying porphyry target associated with the recently discovered Great Eastern stock.

Additional greenfield exploration work is recommended to advance near surface porphyry targets in the Mount Milligan district and adjacent properties. Additional geophysics surveys recommended include airborne mobile Magneto-Telluride (MT) surveys and detailed ground IP surveys, regional geochemistry surveys including stream sediment sampling as well as soil and till sampling grids, regional data compilation, and ongoing helicopter supported diamond drilling programs.

Budgets for programs over the next three years should be on par with Centerra-TCM programs in recent years (2018-2021). Annually, these budgets have been in the range of \$5.5-8.5 million comprising diamond drilling (30,000 to 43,000 m), geophysics programs (15 to 60 line-km IP surveys and other surveys) and development of a comprehensive 3D exploration model and drilling database that compiles lithology, alteration, mineralization, structural, geochemical, and geophysical

information. Ongoing advancement of the 3D exploration models for both the Mount Milligan deposit and the Mount Milligan Greenfield district are recommended to improve understanding of the geometry of the deposit from fault block to fault block, to find extensions of known mineralization that could potentially be added to resources and reserves, and to guide future discoveries.

26.2. RESOURCE ESTIMATION

The QP recommends that Centerra evaluate the following potential changes to the block modelling methodologies:

- Evaluate the spatial continuity of Cu and Au grades independently for each Domain.
- Incorporate the 100 (shell) Domain into existing or reinterpreted Domains.
- Review the resource categorization methodology in conjunction with search estimate parameters.
- Apply more focus on structural trends in Domain wireframe reinterpretation.
- Analyze the distribution of potentially deleterious minerals for potential negative effects.

26.3. WATER MANAGEMENT

Continue to identify water sources for LOM operations and progress related approvals in consultation with Indigenous and regulatory parties.

Continue to monitor water supply, storage and use very closely to minimize potential process plant downtime due to lack of water.

26.4. MINE PLANNING AND OPERATIONS

Recommendations for improved mine planning and mine operations include:

- Evaluate using an optimized pit with a lower revenue factor as the basis for designing the ultimate pit, which is expected to reduce the strip ratio and possibly provide a more manageable balance between ore and waste tonnes. However, this is also likely to decrease the mined ore tonnes and may reduce the mine life.
- Re-work the mining schedule, including considerations for additional waste rock stockpiles, to provide adequate TSF dam building material for each construction season.
- Evaluate opportunities to further improve mining efficiencies, such as through the application of automation.

- Evaluate opportunities for in-pit waste and tailings storage.

26.5. PROCESSING AND METALLURGY

Recommended activities for process plant improvement are as follows:

- Continue optimizing SAG mill liner design to increase efficiency
- Continue to progress engineering of additional equipment to gain throughput and recovery benefits as below by:
 - Enclosing the banana screen in the secondary crushing circuit.
 - Installing two high shear flotation cells in front of the rougher flotation circuit.
 - Installing an additional horizontally stirred regrind mill
 - Installing two additional plates in the existing dewatering filter press
 - Upgrading the capacity of ball mill motors

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27.1. GLOSSARY OF UNITS, ABBREVIATIONS, AND SYMBOLS

Glossary of Units

Symbol	Definition
"	seconds (geographic)
'	foot/feet
'	minutes (geographic)
"	inches
#	number
%	percent
/	per
<	less than
>	greater than
µm	micrometer (micron)
a	annum/ year
Å	angstroms
asl	above sea level
BQ	36.5 mm diameter core
c.	circa
d	day
d/wk	days per week
dmt	dry metric tonne
fineness	parts per thousand of gold in an alloy
ft	feet
g	gram
g/cm ³	grams per cubic centimeter
g/dmt	grams per dry metric tonne
g/m ³	grams per cubic metre
Ga	billion years ago
ha	hectares
HP	horsepower
HQ	63.5-mm diameter core
kg/m ³	kilograms per cubic metre
kL	kiloliters
km	kilometer
km ²	square kilometers
koz	thousand ounces
kton	thousand tonnes
kV	kilovolt
kVA	kilovolt–ampere
kW	kilowatt
kWh	kilowatt hour
kWh/t	kilowatt hours per tonne
lb	pound
M	million
m	meter
m ³	cubic metre
m ³ /hr	cubic metre per hour
Ma	million years ago

Symbol	Definition
Mesh	size based on the number of openings in one inch of screen
mg/L	milligrams per liter
mi	mile/miles
MLbs	million pounds
Mm	million metres
mm	millimeter/millimetres
Moz	million ounces
mRL	meters relative level
Mton	million tonnes
Mtpa	million tonnes per annum
MW	megawatts
NQ/NQ2	47.6 mm size core
°	degrees
°C	degrees Celsius
oz	ounce/ounces (Troy ounce)
P	Passing, i.e., % passing through a screen
pH	measure of the acidity or alkalinity of a solution
pop	population
ppb	parts per billion
ppm	parts per million
PQ	85 mm diameter core
t	metric tonne
t/yr	tonnes per annum (tonnes per year)
tpd	tonnes per day
tph	tonnes per hour
tpod	tonnes per operating day
t/m ³	tonnes per cubic metre
TDS	total dissolved solids
TSS	total suspended solids
µm	micrometres
wt%	weight percent

Glossary of Abbreviations

Abbreviation	Definition
®	registered name
AAS	atomic absorption spectroscopy
AAL	Australian Assay Laboratories
AC	Aircore
Alcoa	Alcoa of Australia Ltd
Amdel	Amdel Laboratory
ANC	acid-neutralizing capacity
ANP	acid-neutralizing potential
ARD	acid-rock drainage
AuAA	cyanide-soluble gold
AuEq	gold equivalent
AuFA	fire assay
AuPR	preg-rob gold
AuSF	screen fire assay
AusIMM	Australasian Institute of Mining and Metallurgy
BFA	bench face angle
BLEG	bulk leach extractable gold
BLM	US Bureau of Land Management
BMCO	breakeven mill cut-off
BSCO	breakeven stockpile cut-off
C	Canadian
C.P.G.	Certified Professional Geologist
Capex	capital expenditure
CAF	cost adjustment factor
CDN	Canadian
CER	Consultative Environmental Review
CIL	carbon-in-leach
CIM	Canadian Institute of Mining, Metallurgy and Petroleum
CN _{wad}	Weak acid-dissociable cyanide
CRF	capital recovery factor
CRM	certified reference material
CST	cleaner scavenger tailings
CTOT	carbon total
Cu Eq	copper equivalent
CuCN	cyanide-soluble copper
E	east
EDA	exploratory data analysis
EIA	Environmental Impact Assessment
EIS	Environmental Impact Statement
EOM	end of month
EOY	end of year
EPA	Environmental Protection Authority
ERMP	Environmental Review and Management Program
FAusIMM	Fellow of the Australasian Institute of Mining and Metallurgy
GAAP	Generally Accepted Accounting Principles
Golder	Golder Associates Ltd.
Golder	Golder Associates Inc.
GN	mine grid north

Abbreviation	Definition
GPS	global positioning system
GRG	gravity recovery gold
H	horizontal
HC	high capacity
HPGR	high pressure grinding rolls
ICP	inductively-coupled plasma
ICP-AES	inductively-coupled plasma atomic emission spectroscopy
ICP-MS	inductively-coupled plasma mass spectrometry
ICP-OES	inductively-coupled plasma optical emission spectrometry
IRSA	Inter-ramp slope angle
IW	Impacted Water
JCR	joint condition rating
JORC	The Joint Ore Reserves Committee of The Australasian Institute of Mining and Metallurgy, Australian Institute of Geoscientists and Minerals Council of Australia
JV	joint venture
KV	kriging variance
L-G	Lerchs-Grossman
LC	low capacity
LOA	length overall
LOM	life-of-mine
LSK	large-scale kinetic
MAIG	Member of Australian Institute of Geoscientists
MAusIMM	Member of the Australasian Institute of Mining and Metallurgy
MIK	multiple-indicator kriging
MN	magnetic north
MPA	maximum potential acidity
MRF	Mine Rehabilitation Fund
MWMS	mine water management system
MWMT	meteoric water mobility testing
N	north
NAG	net acid generation/net acid generating
NAPP	net acid-producing potential
Newmont	Newmont Mining Corporation
NI 43-101	Canadian National Instrument 43-101 "Standards of Disclosure for Mineral Projects"
NOI	Notice of Intent
NN	nearest-neighbor
NNP	net neutralizing potential
NSR	net smelter return
NW	northwest
OK	ordinary kriging
Opex	operating expenditure
P.Eng.	Professional Engineer (CAN)
P.E.	Professional Engineer (US)
P.Geol	Professional Geologist (CAN)
P.G.	Professional Geologist (US)
PAG	potentially acid-generating
PLI	point load index
PoO	Plan of Operations
PSI	pounds per square inch
QA/QC	quality assurance and quality control

Abbreviation	Definition
QLT	quick leach test
QP	Qualified Person
RAB	rotary air blast
RC	reverse circulation
RDA	Residue Disposal Area
RM SME	Registered Member, Society for Mining, Metallurgy and Exploration
RMR	rock mass rating
ROM	run-of-mine
RPL	Environmental Monitoring Plan
RQD	rock quality designation
S	south
SAG	semi-autogenous grind
S&ER	Sustainability and External Relations
SE	southeast
SEIS	Supplemental Environmental Impact Statement
SG	specific gravity
SME	The Society for Mining, Metallurgy & Exploration
SME-RM	Registered Member of The Society for Mining, Metallurgy & Exploration
SMU	selective mining unit
SPET	State Plane East Truncated, Local Mine Grid
SRM	standard reference material
SS	sulfide sulphur
ST	scavenger tailings
STOT	sulphur total
SX-EW	solvent extraction–electrowinning
TF	tonnage factor
TN	true north
Topo	topography
UC	uniform conditioning
UG	underground
UHF	ultra-high frequency
USGS	United States Geologic Survey
US	United States
V	vertical
US\$	United States Dollar
VHF	very high frequency
VWP	vibrating wire piezometer
W	west
WD	waste dump
XRD	X-ray diffraction
WDX	waste dump expansion
WRF	waste rock formation
XRF	X-ray fluorescence

Glossary of Symbols

Symbol	Element
Ag	silver
Al	aluminum
As	arsenic
Au	gold
B	boron
Ba	barium
Be	beryllium
Bi	bismuth
C	carbon
Ca	calcium
CaCO ₃	calcium carbonate
CaO	calcium oxide
CaSO ₄ •2H ₂ O	calcium sulfide dehydrate
Cd	cadmium
Ce	cerium
Cl	chlorine
CN	cyanide
CO	carbon monoxide
Co	cobalt
Cr	chromium
Cs	cesium
Cu	copper
Fe	iron
FeOx	iron oxides
Ga	gallium
Ge	germanium
H	hydrogen
Hf	hafnium
Hg	mercury
In	indium
K	potassium
La	lanthium
Li	lithium
Mg	magnesium
Mn	manganese
Mn(OH) ₂	manganous hydroxide
MnO ₂	manganese dioxide
Mo	molybdenum
N	nitrogen
Na	sodium
Nb	niobium
NH ₃	ammonia
Ni	nickel
NOx	nitrogen oxide compounds
O ₂	oxygen
P	phosphorus
Pb	lead
Pd	palladium

Symbol	Element
Pt	platinum
Rb	rubidium
Re	rhenium
S	sulphur
Sb	antimony
Sc	scandium
Se	selenium